# Cohort B

# Academic Program Review Cohort B- Cycle 1

Computer Information Technology AAS & CERT

## Reflection and Mission

### PREVIOUS ACADEMIC PROGRAM REVIEW

Reflect on the past APR (if any) and respond to any recommendations that were made at that time. The reviewer should list recommendations, actions taken, and discuss implications or challenges resulting from those actions.

**Narrative:**

There is no evidence of any Three-Year Program Review or Annual Program Review conducted for the Computer Information Technology (CIT) Associate of Applied Science (A.A.S.) degree program.

There was an 'Academic Audit Self-Study Report' internally produced by the CIT department in January 2018. There was also an 'Audit Report' completed by a team of three auditors that were employed at three different Tennessee Board of Regents (TBR) Community College institutions.

**The 2018 Self-Study Report addressed A.A.S CIT degree program performance within the following key 'focal areas':**

- Learning Outcomes
- Curriculum and Co-Curriculum
- Teaching and Learning
- Student Learning Assessment
- Support

**The 2018 Self-Study Report provided the following 'Potential Improvement Initiatives':**

- Complete documentation of a standard process for CIT Quality Assurance and Continuous Improvement Process.
- Increase enrollment and completion of females in CIT degree program.
- Update all CITC course syllabi with TBR Common Curriculum Learning Outcomes
- Complete Quality Matters (QM) Course Alignment Worksheets for all CITC courses.
- Complete Quality Matters (QM) Audit for all CITC courses.

**The 2018 Audit Report provided the following 'Recommendations':**

- Internship program should remain in-place rather than adopting job shadowing
- Establishment of clear alignments between course sequencing and industry certification exams
- Provide more credit by exam opportunities for introductory classes.

**Assessment of 2018 Self-Study and Audit Report(s):**

- Completing an Annual Program Review (APR) is a first step towards building a Quality and Continuous Improvement Process. Establishing baseline assessment data that is mapped to degree program outcomes is on-going at this point.

- Distributed Education (DE) implemented Course Review Guidelines that closely align to Quality Matters (QM) Course Review Standards. ALL CITC courses within the current A.A.S. CIT degree curriculum have successfully passed through the DE Course Review process, and have an approved 'Model Course'.

- The CITC degree program does not include a required Internship course, because there were no active participating employers involved within the program. Students interested in internship/co-operative education opportunities can request for Work-Based Learning credit by consulting with the Work-Based Learning Coordinator and CIT Department Chair. Once an opportunity is established, the student then registers for the WKBL 2399 course. In addition to working within a team to complete a comprehensive project, the CIT Capstone course also provides students resume building opportunities along with potential employer networking opportunities.

- The current A.A.S CIT degree curriculum aligns to Cisco, CompTIA, and Microsoft vendor-supported certification pathways.

- Grant-funded certification exam vouchers were used to embed CompTIA Network+ and CompTIA Linux+ exams as the final examinations within the *CITC 1302: Introduction to Networking* AND *CITC 1332: Unix/Linux Operating System* courses.

**Evidence:**

- 2018 Academic Audit Team Report
- 2018 CIT Academic Audit Self-Study Report

### Program Mission Statement

What changes have the program made to the mission statement over the course of this cycle? Why were these changes made? Are any revisions planned?

**Narrative:**

The current mission statement for the Associate of Applied Science (A.A.S.) Computer Information Technology (CIT) degree program is defined within the VSCC Undergraduate Catalog and CIT AAS degree program website as follows:

*"The Associate of Applied Science degree in Computer Information Technology prepares students for positions in the workplace through the use of various systems, applications, languages and products. The program consists of a common core of General Education and Computer Information Technology major core courses, combined with subject-specific courses designed to address skills within the concentration area. These subject-specific courses focus on those skill-sets identified by employers as essential in today's workplace and, where appropriate, will prepare students for industry-recognized certification examinations. The concentrations are designed to ensure that students have the proper experience, training, and skills to begin work in the chosen field upon successful completion of the chosen program of study."*

No changes have been proposed to the CIT A.A.S., degree program's current mission statement.

### Alignment to Institution Mission

How does the mission of the program align with the mission of the institution?

**Narrative:**

**The current mission statement for the Associate of Applied Science (A.A.S.) Computer Information Technology (CIT) degree program is defined within the VSCC Undergraduate Catalog and CIT AAS degree program website as follows:**

*"The Associate of Applied Science degree in Computer Information Technology prepares students for positions in the workplace through the use of various systems, applications, languages and products. The program consists of a common core of General Education and Computer Information Technology major core courses, combined with subject-specific courses designed to address skills within the concentration area. These subject-specific courses focus on those skill-sets identified by employers as essential in today's workplace and, where appropriate, will prepare students for industry-recognized certification examinations. The concentrations are designed to ensure that students have the proper*

*experience, training, and skills to begin work in the chosen field upon successful completion of the chosen program of study."*

**The CIT program mission aligns to VSCC's institutional mission within the following key areas:**

- ***Quality and innovative education programs*** - Both the CIT A.A.S degree and CIT Technical Certificate credentials provide individuals with relevant skillsets required in current Information Technology (IT) entry-to-mid-level jobs. Several courses within the CIT program curriculum align to industry recognized certifications that are heavily valued in the IT workforce. These certification examinations provide a 'standard' for quality and avenue for 'stacked credentials' for students that graduate with not only a degree/certificate, but with industry-standard IT certifications.

- ***Building partnerships, strengthening internal and external community engagement*** - Several members of the CIT Advisory Committee come from IT employers within VSCC's service area(s), helping to establish an active consortium (or pipeline) for potential Work-Based Learning partners for students within the CIT program. Career and Technical Education (CTE) Coordinators at surrounding school systems are also members of the CIT Advisory Committee, and help provide a clear pathway for high-school students as they pursue Dual-Enrollment and Dual-Credit opportunities within the CIT degree program.

  CIT Faculty participate in high school and middle school tours and site visits, as well as facilitate student-ran organizations, such as the Cyber Security Club, who hosts cyber competitions, and speaker events, and seminars. The Technology Department and CIT programs collaborate closely with area school systems, job centers, and economic development officials in order to meet the needs of students and area employers and to secure external funds (grants) and other forms of support for this important and complex mission.

- ***Promoting diversity, cultural awareness, and economic development -*** The CIT program supports VSCC goal's to promote diversity and cultural awareness by fostering an academic environment that allows students from all backgrounds to contribute and participate as part of their learning experience at the college. This environment is shaped through a combination of classroom experiences and curricular activities.

- ***Prepare students for successful careers, university transfer, and meaningful civic participation in a global society -*** As stated on the VSCC CIT Department website:

  *"The A.A.S. CIT degree prepares students for positions in the workplace through the use of various systems, applications, languages and products. The program consists of a common core of General Education and Computer Information Technology major core courses, combined with subject-specific courses designed to address skills within the concentration area. These subject-specific courses focus on those skill-sets identified by employers as essential in today's workplace and, where appropriate, will prepare students for industry-recognized certification examinations. The concentrations are designed to ensure that students have the proper experience, training, and skills to begin work in the chosen field upon successful completion of the chosen program of study.*

- The information technology (IT) field offers an abundance of high quality entry-level and middle skills jobs paying a living wage and providing significant job security and career growth opportunities. IT is the career field most critical to our region's success in the digital, global economy. It is also a field where people with as little as a technical certificate and/or an associated industry certification can land a quality job. An associate's degree in the field, particularly if paired with an industry certification earned in our program or soon after graduation, is a ticket to prosperity and security.

- The department also offers an Associate of Science – Tennessee Transfer Pathway (A.S. – TTP) degree in Computer Science and another in Information Systems. The A.S. degree is for students planning to transfer seamlessly to a university when they graduate. The curriculum consists of general education courses with numerous advanced math classes to prepare for upper-division university courses. The curriculum for both programs also includes two or three courses in the student's chosen major."

# Curriculum and Student Learning Outcomes Assessment

### Curriculum Map

Review the Student Learning Outcomes and course(s) where each one is addressed. For each outcome, provide the assessment method and target level of achievement.

**Narrative:**

Program Outcomes for the the A.A.S. CIT degree and CIT Technical Certificate programs are defined by TBR. The outcomes to CITC/INFS/WEBT courses offered within both programs are also defined by TBR.

**There are 5 "Major Core" courses defined within the A.A.S. CIT degree program:**
CITC 1300 - *Beginning HTML and CSS
CITC 1301 - *Introduction to Programming and Logic
CITC 1302 - *Introduction to Networking
CITC 1303 - *Database Concepts
CITC 2190 - *Capstone Course in Computer Information

Based on review of Institutional Effectiveness (IE) Year-End Reports, and even the 2018 CIT Self-Study Audit Report, it seems that none of the "Major Core" CITC courses had been identified with measurable outcomes in alignment to the the CIT Program Outcomes, in respect to the type of assessments included in those courses. After reviewing the catalog of ALL CITC courses, a basic outcome mapping was put together that defines how each course's defined outcomes (or SLOs) are aligned to the CIT program outcomes (or PLO). The attached evidence '*CITC AAS Curriculum Mapping* ' demonstrates these alignments.

## Key Assessments to Help Establish Clear Evidence of Student Learning

To better establish 'clear' evidence of student learning, it has been decided to utilize (2) "key assessment" CITC courses that are offered within both the A.A.S CIT degree and CIT Technical Certificate programs. Both *CITC 1301: Introduction to Programming and Logic* and *CITC 1320: A+ Hardware and Software* were chosen due to how assessment rubric criteria defined within both courses are aligned to CIT program outcomes.

**CITC 1320 Key Assessment**
The key assessment identified in CITC 1320 is in the form of a comprehensive 'Team Capstone Project # 2' that aligns to CIT Program Outcomes 1, 2, 3, and 5. The assignment presents a problem statement and case scenario, and gives students a learning opportunity to work together (as a team) with the goal of upgrading a set of server computers, diagramming and connecting a simple network, and preparing a summary report. All of these deliverable require students to utilize both communication and critical-thinking skills.

**CITC 1301 Key Assessment**
CITC 1301 identifies 'Module 4 Programming Lab Assignment' as a "Key Assessment" that aligns to the CIT Program Outcome 4. The Module 4 Programming Assignment assesses students' knowledge of conditional logic using 'if / "else" conditional logic statements along

with a combination of logical operators and Boolean variables. The original rubric for the CITC 1301 Key Assessment is in the process of revision with a more-refined set of grading criteria, which would be better utilized for measurement of student learning.

Evidence of both CITC 1301 and CITC 1320 key assessment assignment rubrics are attached.

**Key Assessment Target Measurements**
Going forward into the next Academic Year, it is expected that that 80% of the students will achieve a score of 75% or higher on both CITC 1301 and CITC 1320 Key Assessments.

**Evidence:**

- CITC 1301 SLO Assessment Rubric
- CITC 1301 SLO Assessment Rubric_Revised
- CITC 1320 Team Capstone Project 2 Rubric
- CITC AAS Curriculum Mapping 2021-22

## EVIDENCE OF STUDENT LEARNING

Provide evidence of student learning and the program's effectiveness in reaching both program outcomes and course learning outcomes.

Please provide the following:
1. Summary of most recent assessment results.
2. Analysis of trends
3. Ideas for how those results could be used to improve student learning.

**Narrative:**

### Previous Evidence of Student Learning

Based on review of previous Institutional Effectiveness (IE) Year-End Reports and the 2018 CIT Self-Study Audit Report, although there were some defined program outcomes, none of these reports seem to define a consistent instrument used to measure assessment data for TBR-defined Program Outcomes of the A.A.S. CIT degree program.

Currently, there is a CIT Exit Exam that <u>all</u> CIT students are required to complete before they officially graduate. The CIT Exit Exam attempts to assess students' knowledge of the five major course courses defined within the A.A.S. CIT degree program:

CITC 1300 - *Beginning HTML and CSS
CITC 1301 - *Introduction to Programming and Logic
CITC 1302 - *Introduction to Networking
CITC 1303 - *Database Concepts
CITC 2190 - *Capstone Course in Computer Information

'Appendix 3' of the 2018 CIT Self-Study Audit Report indicates key assessment questions within the CIT Exit Exam that can used to help better measure CIT Program Outcomes. The problem with this measurement, however, is the fact that <u>none</u> of the CIT Program Outcomes identified in this report are derived from the TBR-defined program outcomes as indicated within the A.A.S CIT degree program. During the Fall of 2019, the CIT Exit Exam was revised with updated questions to reflect changes in the content taught within CITC core course courses, specifically CITC 1301 and CITC 1302 courses. Attached is the latest version of the CIT Exit Exam, as well as the 2019-20 and 2020-21 Exit Exam Results.

### Key Assessments to Establish Clear Evidence of Student Learning

As discussed in the previous 'Curriculum Map' section, it has been determined that evidence of student learning is best assessed using rubric-based assessments within courses that are defined within both A.A.S CIT degree and CIT Technical Certificate programs. CITC 1301 and CITC 1320 are the identified key assessment courses, and both courses use rubric-based key assessment(s) that can measure student learning from a Course Learning Outcomes (CLO) perspective, and, of course, Program Learning Outcomes (PLO) perspective. The target (or expectation) for the next upcoming Academic Year (AY) is that 80% of the students will achieve a score of 75% or higher on both key assessments.

### CIT Exit Exam Review and Re-Alignment

The CIT Exit Exam can still be an effective measurement of student learning, however, "key" exam questions need to be aligned to specific CLOs, and most importantly CIT PLOs. Questions from each course should also be re-reviewed to keep consistent with content and materials taught in the Major Core courses, such as CITC 2190, which is now a 1-credit hour course that will focus on career exploration and team-and-project based learning objectives.

**Evidence:**

- Appendix 3 CIT Self-Study Audit Report
- CIT 1920 Exit Exam Item Analysis
- CIT 2021 Exit Exam Item Analysis
- CIT EXIT EXAM -Revised Fall 2019

## Qualitative and Quantitative Program Information

### STUDENT, GRADUATE, AND ALUMNI SATISFACTION

Use student and graduate surveys, course evaluations, and any other appropriate evidence needed to explore the qualitative satisfaction results from students and graduates to provide information to the items below.

Discuss student, graduate, alumni perceptions of the program, faculty, career projection, and any other program recommendations given "during this survey".

Summarize any noted themes, trends, areas for improvement, identified strengths, and implications for the program for both students and alumni.

**Narrative:**

### Graduate Survey Results

Students that graduated from the A.A.S. CIT Degree program completed a Graduate Survey during 2020 and 2021.

Based on the data taken from both surveys, the largest percentage of CIT graduates that completed the surveys felt that the A.A.S CIT degree impacted their life by helping them gain "entry into the job market" (42% in 2020, and 36% in 2021). Gradating students also felt that the A.A.S. degree helped "improved job mobility opportunities", and "salary increase".

**2020 Graduate Survey Results**
**(Areas of Impact)**

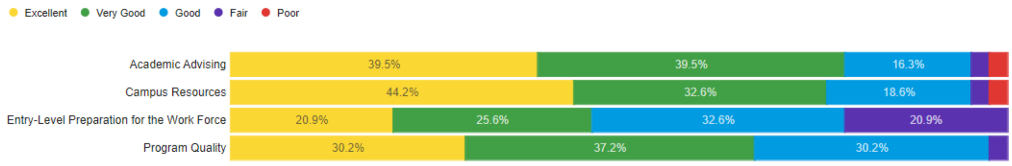

**2021 Graduate Survey Results**
**(Areas of Impact)**

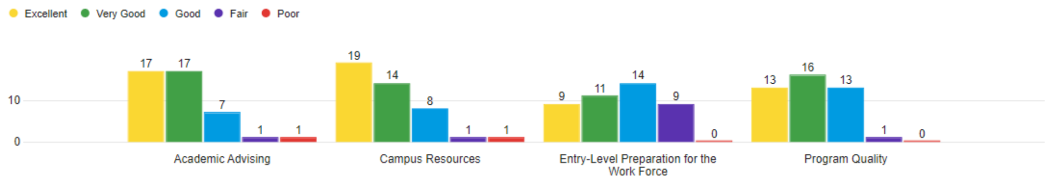
What impact(s) did the Computer Information Technology A.A.S. have for you? Mark all that apply.


When looking at the quality of the degree program itself, both 2020 and 2021 survey data indicated that the majority of CIT graduates that completed the survey rated the "Program Quality" of the CIT degree program as "Very Good" (37.2% in 2020 and 31.6% in 2021). The majority of CIT graduates that completed the survey also felt that the A.A.S. CIT degree program provided them with an "Entry-Level Preparation for the Work Force", as this particular was rated "Good" (32.6% in 2020 and 36.8% in 2021);

## <u>2020 Graduate Survey **Results**</u>
## ("Program Quality" and "Entry-Level Preparation for the Work Force")

Rate the following OVERALL areas:

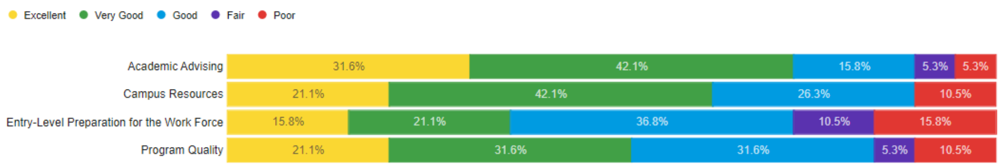


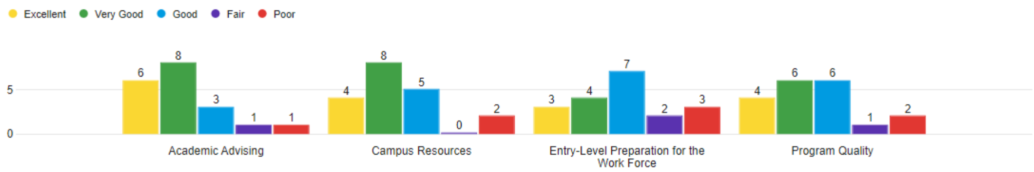




| Field | Min | Max | Mean | Standard Deviation | Variance | Responses | Sum |
|---|---|---|---|---|---|---|---|
| Academic Advising | 1.00 | 5.00 | 4.12 | 0.92 | 0.85 | 43 | 177.00 |
| Campus Resources | 1.00 | 5.00 | 4.14 | 0.95 | 0.91 | 43 | 178.00 |
| Entry-Level Preparation for the Work Force | 2.00 | 5.00 | 3.47 | 1.04 | 1.09 | 43 | 149.00 |
| Program Quality | 2.00 | 5.00 | 3.95 | 0.83 | 0.70 | 43 | 170.00 |

## <u>2021 Graduate Survey **Results**</u>
## ("Program Quality" and "Entry-Level Preparation for the Work Force")

Rate the following OVERALL areas:




| Field | Min | Max | Mean | Standard Deviation | Variance | Responses | Sum |
|---|---|---|---|---|---|---|---|
| Academic Advising | 1.00 | 5.00 | 3.89 | 1.07 | 1.15 | 19 | 74.00 |
| Campus Resources | 1.00 | 5.00 | 3.63 | 1.13 | 1.29 | 19 | 69.00 |
| Entry-Level Preparation for the Work Force | 1.00 | 5.00 | 3.11 | 1.25 | 1.57 | 19 | 59.00 |
| Program Quality | 1.00 | 5.00 | 3.47 | 1.19 | 1.41 | 19 | 66.00 |

**RECRUITMENT, ENROLLMENT, RETENTION, AND GRADUATION: FACT BOOK DATA**

Attach screenshots from the Fact Book of the areas below.

1. Headcounts
2. FTE Comparison
3. Student Primary Location
4. Retention Statistics
5. Graduation Statistics
6. Demographics

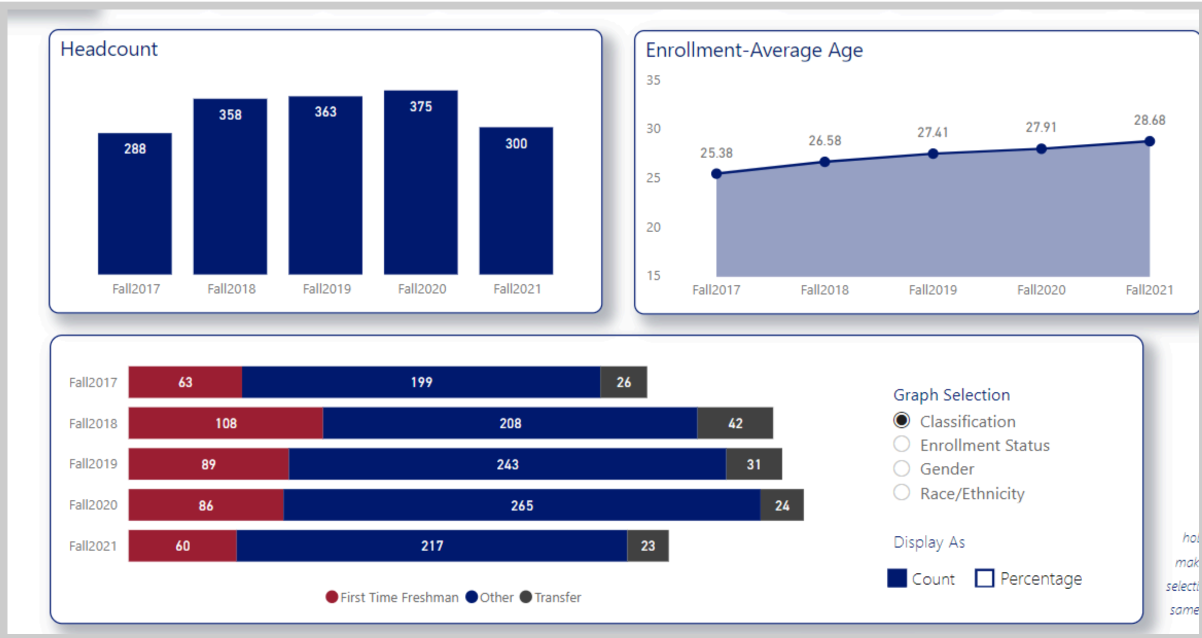
Using the Fact Book data listed below, analyze the follow:

Recruitment
Enrollment
Retention
Graduation

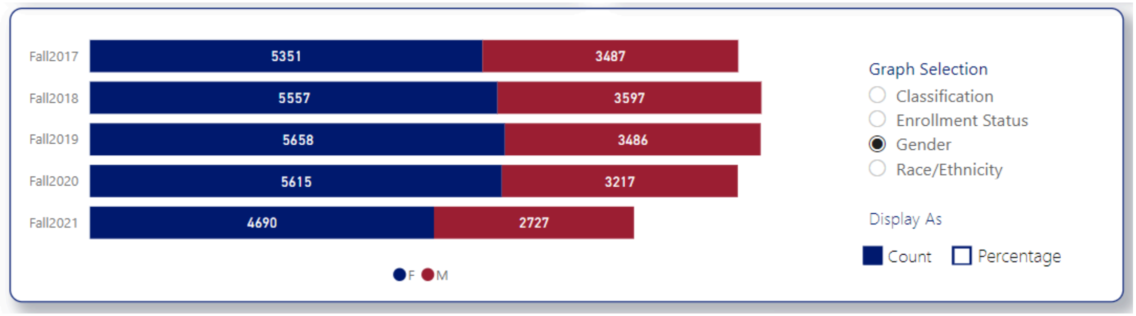
The reviewer could discuss marketing and recruitment efforts in the community and local high schools; analyze enrollment, retention, and graduation trends; and discuss potential reasons, strengths, plans for improvement, and any other implications deemed appropriate.

**Narrative:**

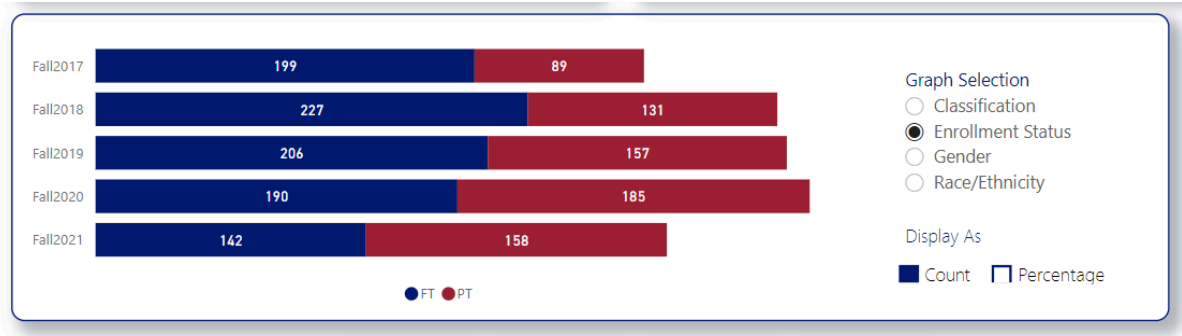
**Enrollment By Count**: Down 20% from Fall 2020; Up only slightly from Fall 2017; FTF declining sharply; Avg. age rising steadily (now at 29)




**Enrollment by Gender for the college overall - 37% male**

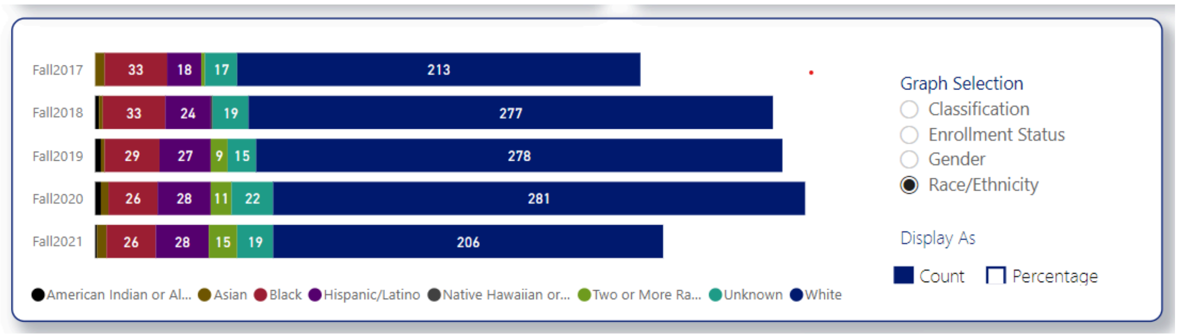



**Enrollment by Classification:** 69% of CIT AAS students attended FT in Fall 2017 -- down by almost a third to 47% in Fall 2021

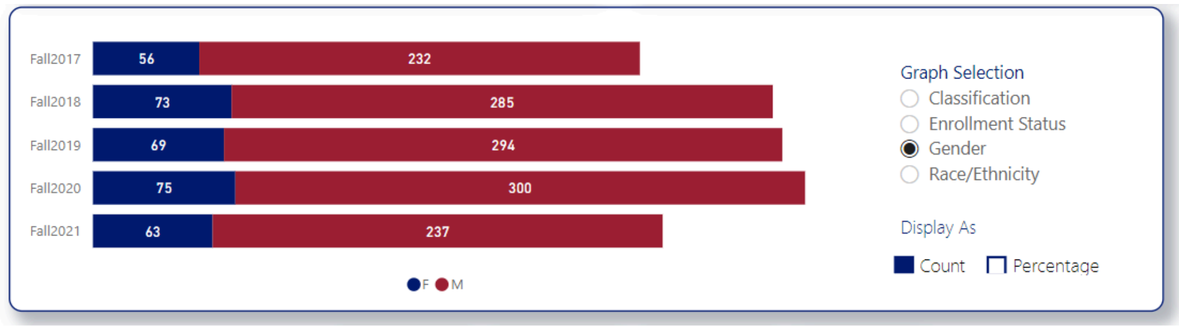



**Enrollment by Age:**

**Enrollment by Race/Ethnicity**: White students make up 69% of CIT AAS enrollment; Students of color in CIT AAS meet/exceed their representation levels in the overall VSCC population

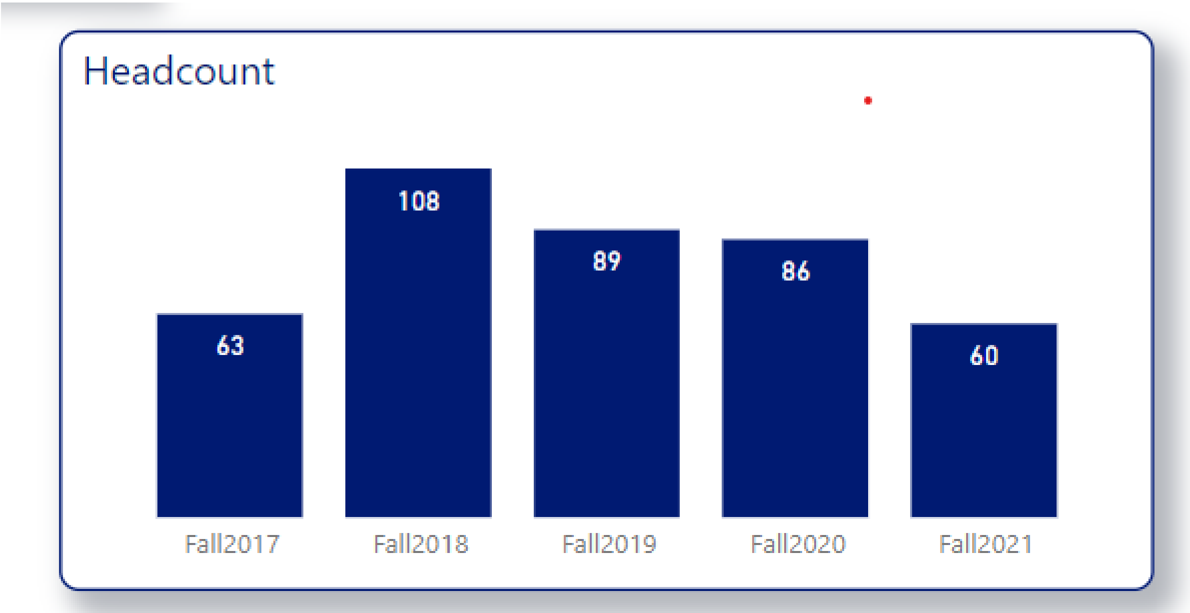



**Enrollment by Gender for CIT AAS**: 79% Male

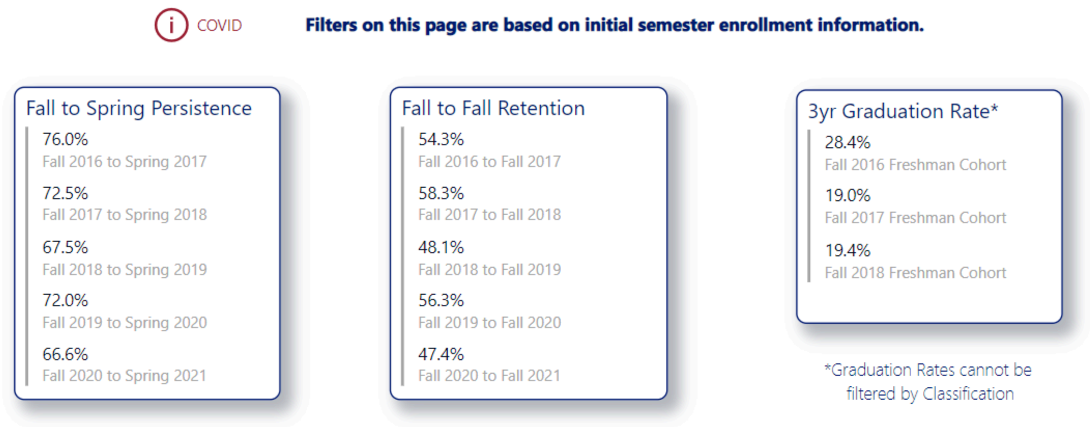



**First Time Freshmen in CIT AAS:** The number of FTF entering the CIT AAS programs has dropped 44% since the peak in Fall 2018. Vol State and other TBR CCs saw a large, brief spike in FTF enrollments in Fall

2018. CIT AAS and other STEM programs that attract mostly male students have been particularly hard hit by the Covid pandemic and the effects of rising wages in plentiful unskilled jobs.




**Persistence, Retention, and 3-Year Graduation Rate for CIT AAS**




**Graduation Numbers for CIT AAS**: Up 111% since AY16-17; up 49% vs AY18-19

| Degree | 16-17 | 17-18 | 18-19 | 19-20 | 20-21 |
|---|---|---|---|---|---|
| Associate of Applied Science | 36 | 58 | 51 | 54 | 76 |
| **Total** | **36** | **58** | **51** | **54** | **76** |

| Concentrations | 16-17 | 17-18 | 18-19 | 19-20 | 20-21 |
|---|---|---|---|---|---|
| Administration and Management | 1 | 4 | 2 | 4 | 8 |
| Cyber Defense | 6 | 11 | 18 | 21 | 35 |
| Networking | 13 | 18 | 12 | 7 | 10 |
| Programming | 16 | 23 | 16 | 19 | 21 |
| Systems Analyst | | 2 | 3 | 3 | 2 |

| Majors | 16-17 | 17-18 | 18-19 | 19-20 | 20-21 |
|---|---|---|---|---|---|
| Administrative Prof Technology | 6 | | | | |
| Advanced EMT | 53 | 45 | 58 | 86 | 48 |
| Allied Health Career | 1 | | 1 | | |
| Business | 45 | 49 | 47 | 61 | 80 |
| Coding Reimbursement | 15 | | 1 | | |
| Computer Information Tech | 36 | 58 | 51 | 54 | 76 |
| Criminal Justice | | | 9 | 33 | 57 |
| Criminal Justice Law Enfrcemnt | 60 | 49 | 37 | 23 | 1 |
| Dental Assistant | 16 | 12 | 19 | 14 | 12 |
| Diagnostic Medical Sonography | 10 | 16 | 16 | 15 | 13 |
| Early Childhood Education | 22 | 22 | 14 | 25 | 39 |
| Emergency Medical Technician | 79 | 103 | 106 | 135 | 95 |
| Entertainment Media Production | 10 | 12 | 22 | 12 | 18 |
| Fire Science | 4 | 9 | 36 | 42 | 59 |
| Fire Science Technology | 29 | 12 | 1 | 2 | |
| General Business Admin | 10 | 1 | 2 | | |

**Computer & Information Sciences Associate Degrees Awarded by Institution**

| **Institution** | **2014-15** | **2015-16** | **2016-17** | **2017-18** | **2018-19** | **2019-20** | **2020-21** | **2021 CIT AAS degrees awarded as % of Institution's 2020-2021 Associate's Degrees** |
|---|---|---|---|---|---|---|---|---|
| Pellissippi State CC | 183 | 142 | 106 | 205 | 200 | 190 | 167 | 7.6 |
| Nashville State CC | 66 | 53 | 43 | 46 | 33 | 76 | 74 | 6.1 |
| Jackson State CC | 15 | 21 | 19 | 30 | 20 | 35 | 41 | 5.5 |
| **Volunteer State CC** | **14** | **25** | **57** | **58** | **51** | **53** | **76** | **4.4** |
| Northeast State CC | 32 | 34 | 32 | 43 | 37 | 55 | 51 | 3.5 |
| Columbia State CC | 37 | 15 | 10 | 16 | 21 | 19 | 32 | 3.1 |
| Cleveland State CC | 13 | 14 | 15 | 10 | 25 | 32 | 32 | 2.8 |
| Southwest Tennessee CC | 40 | 36 | 44 | 57 | 36 | 22 | 29 | 2.6 |
| Dyersburg State CC | 12 | 8 | 5 | 7 | 7 | 9 | 16 | 2.3 |
| Roane State CC | 2 | 3 | 8 | 5 | 8 | 9 | 16 | 1.4 |
| Chattanooga State CC | 27 | 33 | 20 | 24 | 26 | 29 | 18 | 1.2 |
| Motlow State CC | 1 | 2 | 1 | 5 | 0 | 10 | 13 | 1 |
| Walters State CC | 19 | 20 | 15 | 26 | 29 | 15 | 12 | 0.9 |

*https://www.tbr.edu/policy-strategy/data-and-research

Recruitment:

- Many more Tennessee high schools are now adding CIT CTE courses to their curriculum - bodes well for an upturn in enrollment by traditional, college age students if we can promote access to on ground, cohort-like learning opportunities
- We believe scheduling our on ground sections more intentionally and promoting cohort-like learning opportunities with optional "bootcamp-like" labs for completing homework will help us attract more students in the 23-and-under age category and help inspire them to attend full-time and persist and succeed at higher rates.
- The Governor's GIVE Grant I was supposed to fund the Greater Nashville Technology Council to do lots of things to promote IT careers and associated education pathways. Covid prevented them from executing many of those. We expect they will be able to execute them in Spring and Fall 2022. We believe that may benefit (slightly) our efforts to reverse the current trend of declining enrollments.

Enrollment

- Headcount declining sharply in last two years -- Covid plus males foregoing college at even higher rates than girls and CIT AAS and other STEM fields that attract mostly male student populations are being particularly hard hit
- Demographics swinging decidedly toward older students attending in part-time status

Retention

- Retention has been near the overall rate for the college. Student success is a challenge in these rigorous programs with only 55% of all students classified as "college ready." To increase our contributions to the region's tech workforce, the CIT AAS program not only needs to attract more students but also needs to attract a larger number of better qualified students. To do so, the college will have to provide the Technology Department/Division/Programs the program management and promotion capacity to work with others to promote interest in the profession, publicize our program and their ROI, and deliver them innovatively.

Graduation

- Graduation numbers for the CIT AAS programs have skyrocketed. This was accomplished by improving teaching and making courses and degrees more accessible by developing a quality, "model course" for every course in the curriculum, thereby making the degrees highly accessible online. The launch of the Tennessee Re-Connect program in 2018 brought many more adult students into this and many other Vol State programs. That, too, has influenced the number of students coming out of the programs recently.
- The final table inserted above shows that Vol State produced the second most CIT AAS degrees of any TBR community college in AY2020-2021. Vol State CIT AAS programs rank fourth among TBR CCs for CIT AAS degrees awarded as a percentage of total Associate's degrees awarded from each institution (4.4% in AY2020-2021).

# Supporting Program Information

## FACULTY USAGE AND DEVELOPMENT (OPTIONAL)

Review faculty teaching loads, adjunct to full time ratios, faculty development opportunities, faculty end of course evaluations, faculty surveys, and any other evidence deemed appropriate. The reviewer could discuss the impact of teaching, the ratio of adjunct to full time staffing, end of course evaluations relative to faculty, and any noted themes identified from the review.

**Narrative:**

Previously summarized demands for entry- and middle-skills tech employees is rising both in the greater Nashville area and around Cookeville in the eastern side of our service area The workforce gap is growing rapidly, nationally and locally, in this critical, strategic field. The program requires FT CIT faculty to develop, manage, continually update, and teach the robust portfolio of 31 courses and three concentrations and a technical certificate that we offer, advise and support 400 students (in CIT and Computer Science AS - TTP programs), build strategic relationships with employers and community partners, team with and engage feeder school systems and their students, and satisfy routine college tasks of academic administration and shared governance. The college also needs FT CIT faculty to teach on ground, day-time courses in order to attract more traditional students and give them the best chance of success and completion. At present, the program has 11 FT faculty, 1 of whom is in a temporary, grant-funded position. Four have course releases -- the department chair (with a two-course release); a faculty member managing our grants; a Distance Education fellow supporting the college's development of model courses; and a FT faculty member in a temporary, grant-funded position in the UC - tasked to support student outreach and recruitment to help boost the visibility of our programs there. The resulting teaching capacity associated with each concentration or cluster of courses and campus location are listed, below:

- 2.8 FT Faculty Specializing in Programming - all at the Gallatin campus (one serves as our DE fellow)
  - Programming concentration has our highest enrollment and has the highest number of concentration-unique courses in that pathway (8)
  - All three transfer Computer Science courses in the Comp Sci AS - TTP program are taught by one of these faculty members
  - Computer Science enrollments and retention rates have been rising significantly in recent years
- 1.8 FT Faculty Specializing in Cyber Defense - both at the Gallatin campus

- One of these faculty members has a one course release this AY to oversee/lead our participation in the GIVE Grant (ends Dec 2022).
- 4.2 FT Faculty that teach Networking and other key, high-demand intro courses (A+ Hardware and Software)
    - The department chair and temporary, and the two FT CIT instructors in Livingston (one is temporary, grant funded) are among them
- 1 FT Faculty member in Gallatin who is a CIT generalist, managing our INFS 1010 course (required by every Business and Technology student in every program, ~1,400 students), Spreadsheet Applications, Project Management, and Introduction to Database Concepts courses.

Other notable observations include:

- The 31 courses required for our three CIT AAS concentrations and Computer Science AS - TTP program and 11 FT faculty (one temporary) produces an average course management/update/development/refresh workload of ~3 courses for every FT CIT faculty member. Because the technologies are constantly changing and industry certifications to which these courses are aligned is constantly changing, course development and course management responsibilities for CIT courses are quite significant.
- The percentage of CIT program sections taught by adjunct faculty has been declining the last two years as enrollments in CIT programs and courses began declining in 2019. The college's Fall 2019 decision to remove/drop "Digital Literacy" from core gen ed graduation requirements (which had required all students to take INFS 1010 or test out of it) also reduced the department's dependence on adjunct faculty. We expect department initiatives will stop the enrollment decline in coming academic years and with sufficient resources for outreach, recruiting, and promotion, we expect enrollments to begin rising in response to more area high schools offering CIT CTE courses and pandemic infection rates beginning to wane.
- The department is particularly thin in FT faculty qualified to teach transfer courses required for the Computer Science program which is offered by this same department. Only three of our FT faculty meet SACS qualifications for teaching transfer courses. One of them is not a Programming instructor and those three courses are best taught by a faculty member specializing in Programming. The second of the three FT instructors qualified to teach the transfer courses has never done so and already carries an average course prep load of ~4 courses each semester. This highlights the challenge of attracting and retaining professionals with a master's degree in a qualifying field is a threat to these programs and our ability to deliver both our CIT AAS program and its three concentrations as well as the Computer Science AS - TTP program.
    - The program faces a similar threat/challenge with the CIT AAS - Networking concentration, where a cluster of three courses aligned with the Cisco Certified Network Administrator (CCNA) certification are the heart of the program. Only one of our FT faculty members is presently certified to teach these three courses. The FT faculty member in the temporary, grant-funded position is beginning the training required to be certified to teach those courses. He should be qualified to teach the first one by Fall 2022 and qualified to teach all three by Spring 2023. We also hope to have one of our new FT CIT faculty members in Gallatin is now beginning to take these courses in Spring 2022 with the same, desired certification timetable. These challenges are representative of the significant professional development workload required of FT CIT faculty members in this department.
- Program faculty have heavier average course prep workloads than found in many departments of the college.
- With courses aligned to key industry certifications, some faculty have to complete training required by industry or proprietary certification bodies just to teach those courses. Industry standards, best practices, and technologies are constantly evolving in IT, requiring CIT faculty to invest significant effort in training and professional development related to their field every year. Despite this, their participation rates in Teaching and Learning Center offerings relative to the department's FT faculty numbers is very comparable to that of the college, overall.
- Serving the CIT workforce development needs of the Upper Cumberland region is a significant strategic challenge for this program.
    - Putnam County (Cookeville) has a moderate population density but surrounding counties are quite rural and sparsely populated
    - Post-secondary attainment levels outside of Putnam County are quite low and some counties in the region are classified as at risk
    - Because Tennessee Tech University has strong Computer Science/Engineering programs and a low cost of living, Cookeville has attracted tech companies looking to capitalize on the strong supply of IT/Computer grads with Bachelor's degree credentials and above. However, those employers are reporting enormous struggles finding and hiring entry-level and middle skills IT technicians. They

are looking to us to do that for them and we do not have sufficient enrollments to do so and produce very, very few CIT graduates from that part of our service area.
    - Compounding the challenge of a low population density in the region, TCAT Livingston and Vol State have based overlapping/competing CIT offerings at facilities two minutes apart in Livingston, a 30-minute drive from the more densely-populated sector of the region, Cookeville, where on ground CIT offerings would have a much better chance to generate much better enrollments.
    - With insufficient enrollments for on ground sections to make, FT CIT faculty assigned there have been teaching online sections. An inability to provide reliable access to on ground CIT courses is likely an additional barrier to growing CIT AAS enrollments there and meeting CIT workforce needs of the region.

## CLASS OFFERINGS (OPTIONAL)

Address scheduling effectiveness and space utilization. The reviewer could review faculty to student ratios at each location and online and discuss the impact relative to faculty and student success.

**Narrative:**

In 2019 and 2020, CIT AAS faculty worked with their chair and the dean to review enrollment and persistence numbers and decided to right-size the portfolio of concentrations, reducing them from five down to three by consolidating three very similar, lower-enrollment concentrations into one (Networking). The portfolio of concentrations has now stabilized at three. Faculty also reviewed and redefined the curriculum for each concentration to improve our alignment with key industry certifications most sought after by employers. They not only defined the curriculum for each concentration but worked to optimize the sequencing of courses to maximize the number of course requirements common across the three concentrations and where they fall in the pathway in order to concentrate demand for each course across the portfolio of programs. The goal was to expand students' modality, location, and time choices by doing so. We also began applying the best practice of prescriptive scheduling, offering required courses in either Fall or Spring semester but not both. This, too, helps concentrate enrollment demand.

We also began offering at least one section of all first-year CIT AAS courses every summer to provide a "safety net"/buffer for any CIT AAS students that had to drop or failed a first-year course. This also helps Part-Time (PT) students progress through the program more expeditiously and allows students who start the program in a spring semester complete all first year CIT course requirements before Fall semester.

In Fall 2021, we introduced a new imbedded Technical Certificate in Computer Information Technology (CIT). The 18-credit hour, intermediate post-secondary credential can be earned in two semesters of either FT or PT study. The courses are imbedded in all three concentrations.

Since many of our students attend part-time, we worked hard to develop a sequencing map for part-time students. This was particularly challenging given the highly clustered nature of the curriculum. However, after many hours of effort, it was developed and shared with faculty and supporting FT advisors. We have attached a copy of that file, as well.

Classroom access in the Mattox Building in Gallatin is sufficient for current and projected enrollments. With the shift to more prescriptive scheduling, the department is becoming more intentional about how on ground sections are scheduled. Beginning Fall 2022, the program will schedule for and promote "bootcamp-like" learning/tutoring opportunities in the 8:00 - 9:25 window and 12:45 - 2:10 window. We hope to promote these aggressively, particularly to new First Time Freshmen as a means of inspiring more to attend FT and to register for on ground sections. There is sufficient classroom capacity at the Livingston campus to offer any on ground course that enrollment might allow. Unless Robertson County Schools can somehow market IT careers sufficiently to grow significant interest in dual enrollment offerings at the Springfield campus, we do not think demand for on ground sections there will be sufficient to offer on any ground sections there even in the mid-term.

With the Spring 2021 schedule, we began "pairing" on ground sections of CIT classes at the Livingston campus with a synchronous, "Zoom" online section in the same time slot. Our goal is to have the Livingston-based CIT instructor teach students registered for the on ground section and those registered for the Zoom session in the same time slot, simultaneously. By "pairing" these sections, we hope to reach enrollments sufficient for the on ground sections to "make." Such creative scheduling practices will be necessary to enable on ground offerings of CIT courses in the Upper Cumberland.

**Evidence:**

- CIT AAS Curriculum - Fall 2021
- Pathways for Part-Time CIT AAS Students

## CO-CURRICULAR ACTIVITIES (OPTIONAL)

Discuss co-curricular activities available for students in the program. These might include clubs, internships or other Work Based Learning opportunities, service learning, guest speakers, or any other activities designed to enhance student experience in the program.

**Narrative:**

With a portfolio of 31 courses and only 11 FT faculty to deliver them at two sites, Gallatin and Livingston (attempted), faculty in the department have heavy course development/refresh, course prep, and professional development loads. This leaves little capacity to offer, promote, and manage co-curriculum programs. However, with the curriculum in these young programs (first launched in 2016) now maturing and stabilizing a bit and model courses now developed and certified for every offering, our co-curricular efforts are expanding. Today, these include:

- A Cyber Defense Club
- An annual forum each fall in conjunction with Cyber Security Day
- A "Capture the Flag" cyber defense competition in AY2020-2021

<u>GOING FORWARD</u>:
Program faculty aspire to participate in SkillsUSA competitions in the future, potentially as early as Spring 2022.

## INDUSTRY TRENDS AND INNOVATIONS (OPTIONAL)

Examine the program demand within the labor market including local, regional, and/or national. The reviewer could examine information from industry advisory boards and information gathered from creditable industry sources as well as from external accrediting bodies. The reviewer could also discuss any existing or emerging trends within the field as they relate to your program and employment of your student's post-graduation as well as the outlook for the field in the next 5 years. This summary could also identify best practices, trends, and innovation in the field and discuss implications for the program.

**Narrative:**

<u>INDUSTRY TRENDS</u>:

- <u>Exploding Workforce Requirements/Expanding Workforce Gap</u>: Economic Modeling (Emsi) projects that middle-skills tech jobs and IT technician positions for which our programs prepare students will increase 14% this decade. The Tennessee LEAP report listed IT and computer science jobs as one of the top six career clusters in their workforce development requirements report. Economic development leaders at the state, in Nashville, Gallatin, and Cookeville have all stated that tech sector employers are their number one targets for industry recruitment due to the quality of jobs they offer, the smaller facility footprint such employers provide, and the prestige they bring to our region. These jobs and associated employers pay higher wage rates with smaller industrial footprints, making them top targets for economic development agencies seeking to boost per capita income and the quality of jobs in their community. Nashville and its suburbs are becoming internationally recognized as a Tech Hub, drawing more and more tech employers to the region. Companies announcing major moves and/or expansions in or near our service area include NTT Data, Amazon, Google, Facebook/Meta, Oracle, Dell, and SAIC.
- <u>Digitization and Automation of the Global Economy</u>: The modern, digital economy is greatly expanding both the number of jobs that require formal IT training and the percentage of those jobs as a total portion of the economy. The large tech workforce requirements of newcomers will be in addition to the rising number of tech workers required by employers already present, both in and out of the tech sector. Employers in every sector are ramping up the pace of their digitization and automation efforts in response

to the labor shortage that will not go away any time soon. Education institutions in the region will have to prioritize Computer Science and CIT programs if they hope to come anywhere close to satisfying the workforce requirements of the region. At their Dec 2021 meeting, Vol State CIT Advisory Committee members expressed strong praise for the design of Vol State CIT programs as well as strong concern about the number of graduates coming out of the state's education institutions and the growing workforce gap in this field.

- Path to Prosperity: The Computer Information Technology field provides job opportunities and strong earning power to entry- and middle-skills tech professionals with as little as an Associate's degree or even a technical certificate. Wage and salary growth in this field is likely to continue soaring in the years ahead as the supply of such professionals and the demand for such workers grow farther and farther apart. This field is an ideal place for community colleges to lift students from less advantaged backgrounds and their families into economic prosperity and security.
- Employers' Desire/Preference for Industry Certifications: Employers in the tech sector increasingly look for/prefer applicants with industry certifications that validate their skills proficiency level. There are a soaring number of entry-level and middle skills IT jobs that provide quality work and a living wage. However, many employers look to industry certifications as proof of an applicant's proficiency level and capabilities. To prepare Vol State students to compete for and excel in the *better* middle-skills, tech jobs, we will need to boost students' commitment levels, academic aspirations, and understanding of the jobs marketplace and market value of various credentials and skill sets.
- Competing Education and Training Programs: Because of the large and growing workforce gap, more and more non-traditional and proprietary training programs are popping up in IT -- from those offered by the larger tech companies themselves (Google, Amazon, Cisco...) to those provided by non-profit agencies such as Nashville Software School. The ever increasing number of providers and product/program variations (CompTIA networking certification vice Google vice Cisco...) actually threatens to fragment demand and undermine the viability and quality of traditional providers by fragmenting the relatively small number of students who elect to pursue an education/training program in IT and other STEM fields. With so many new, market entrants emerging every day, organizations stretching their own product/program lines too long and thin may cannibalize their own share of the market and threaten the viability of some of their own core programs. Community colleges will need to make very good, strategic decisions and execute them well in order for their credit- and degree-bearing programs in this field to survive and thrive in this highly competitive, increasingly crowded market. They have an important role to play but they must do so wisely and invest in in appropriately for these programs to survive and succeed.

## EMPLOYMENT AND TRANSFER FORECASTS (OPTIONAL)

Discuss where students go after graduation and what is being done to enhance the likelihood of success or assist students in reaching those goals. The reviewer could discuss employment outlook, obtainment, curriculum alignment for transfers, and the employment demands of the industries where students will work.

**Narrative:**

EMPLOYMENT OPPORTUNITIES FORECAST:

Producing a sufficient supply of IT-qualified workers is seen as a key to the economic growth strategies of mid-state communities. Nashville, Gallatin and Putnam County, in particular, are prioritizing recruitment of technology-based employers. Those employers report finding entry- and middle-skills CIT workers particularly challenging. We believe our CIT AAS programs are strategically important to the economic health and future strength of our service area since greater Metropolitan Nashville and the Upper Cumberland are becoming major centers for IT employment. Facebook’s recent decision to put a data center in Gallatin and SAIC’s large investment in Cookeville are just a few examples of the rich employment opportunities in this field.

A 2019 article in the *Nashville Business Journal* quoted a study by *RentCafe* ranking Nashville as the third fastest growing tech region in the US with a 22.7% increase in tech jobs during the preceding three years. In their 2018 LEAP Occupational Analysis, the Center for Economic Research in Tennessee (CERT) listed IT as one of the six occupations with the highest employer demand. The same report documented that there were 46,099 tech job postings in the preceding year, more than 12 times the number of college degrees awarded in the field in the same time frame. MTSU’s, “State of Middle Tennessee Tech 2018 Workforce Report” predicted a 16% growth rate for IT jobs for the 2017-2022 time frame.

IT integration services giant, SAIC, has a major business unit in Cookeville and they have reported significant challenges finding personnel to fill their entry- and middle-skills IT positions in the Upper Cumberland. Sumner County economic development officials have told Vol State leaders that IT and tech employers are the number one target for their employer recruitment/economic development efforts. They believe the region's success landing Facebook has positioned the region to land more such tech-based employers but indicate the region's ability to produce more IT-qualified workers is the most important decision criteria for targeted companies.

According to 2020 data from www.onetonline.org, professionals in this job category, SOC code 15-1232, Computer User Support Specialist earning at the 25th percentile make annual salaries of $36,150 in our service area. Those at the median earn $45,170. A sample of job titles included Help Desk Technician, Desktop Support Specialist, Technical Support Specialist and IT Technician. Jobs in this cluster are particularly appropriate for graduates of community colleges, as 51.4% of all personnel in these roles have reported educational attainment of an associate's degree or less (nationally). The Emsi report rated job posting demand in our service area as **High**. An Emsi Q1 2021 Data Set report for the SOC code 15-1232, Computer User Support Specialists shows that jobs in this field are projected to grow 14.3% in our service area for the period of 2020-2030.

TRANSFER OPPORTUNITIES/FORECASTS:
Bachelor's degree-awarding institutions are accepting more and more CTE program credits into degree completion programs. Additionally, Western Kentucky University accepts all *courses in the Vol State CIT AAS program for their Bachelor's Degree in Cyber Defense.

Both Tennessee Tech and Middle Tennessee State University offer bachelor's degree programs in professional studies that will accept all or most credits earned in Vol State CIT AAS programs. This enables students who "flip the curriculum" and take tech education courses at the front of their college journey to earn a bachelor's degree for a ticket to middle management and beyond.

The number of non-traditional, proprietary IT training programs is exploding, providing CIT AAS graduates opportunities to continue their professional development and broaden their credentials and expand their earning power.

## Recommendations Action Plan, Review, and Feedback

### SUMMARY AND ANALYSIS

The reviewer should discuss what went right, what went wrong, and the implications of those results.

**Narrative:**

SUMMARY AND ANALYSIS:
This report provides valuable insights into the performance of this programs, the perspective with which we should view it, and its potential to make invaluable contributions, going forward, to the interests and needs of area students and employers.

PERFORMANCE - WHAT WENT RIGHT: Several very positive trends were documented in this report, including:

- Completion: CIT AAS degrees awarded went up 44% in AY2020-2021
    - Vol State was #2 of 13 among TBR community colleges in CIT AAS degrees awarded in AY2020-2021
    - CIT AAS degrees accounted for more than 4% of associates degrees awarded by the college in AY2020-2021
- Access:
    - FT faculty developed new model courses for about a third of their 31-course portfolio in just two years
        - Today, every course is available online, greatly benefitting the working adults that comprise >50% of the students

    - Secured Perkins Grant funding for a temporary, FT CIT instructor to double FT CIT instructors and capacity for on ground instruction in the Upper Cumberland
        - This new faculty member also has course releases to support program promotion and student recruitment
- Curriculum: Faculty made many wise, strategic decisions in the last two years
    - Right-sized and reshaped the portfolio of CIT AAS programs, consolidating three very similar, loosely structured concentrations into one -- yielding a portfolio of three CIT AAS concentrations much more appropriate for the enrollment levels possible in the region and the interests of area students and employers.
    - Created a new, 18-credit hour, imbedded CIT Technical Certificate that can be earned in just two semesters of part-time study.
    - Redesigned the curriculum to align it with industry certifications and skills employers seek in entry- and middle-skills IT programmers and technicians. Students in the Networking and Cyber Defense programs now take courses that align with nine sought after industry certifications
- Facilities:
    - College leaders submitted a proposal for a major, capital improvement project to renovate and update the Mattox Building. If approved, this would greatly enhance the learning environment and aesthetics of our on ground instructional location.

<u>PERSPECTIVE - WHAT WENT WRONG AND FACTORS AFFECTING IT</u>:

- Even as degrees awarded have been sharply rising and programs have been rapidly maturing, enrollments of First Time Freshmen have been declining at rates even faster than that of the college overall.
    - Historically, males make up ~80% of CIT AAS enrollments and male high school graduates have been foregoing/delaying post-secondary education at significantly higher rates than female students in the Covid era.
    - With the large and growing workforce gap with entry- and middle-skills IT workers, more and more competing providers are offering IT training programs. This is likely fragmenting enrollment more than it increases it, making it hard for legacy programs to achieve enrollment levels needed to maintain healthy, quality programs and provide them in the range of modalities and location sites needed to serve diverse students with diverse learning preferences and needs.
        - Competition to attract and serve working adults looking to upskill or make a career change is most intense and increasing most rapidly. These students currently make up more than 50 of CIT AAS enrollment.
    - The increasingly competitive higher education and training marketplace is perhaps most crowded and most competitive in the IT arena.
        - To make these programs successful and contribute anywhere near the number and quality of IT workers we should for the region, the college will have to make really good strategy decisions that require deep analysis of intended and unintended consequences of each action and invest more resources to enable market analysis, program adaptation and marketing, partnership development (with feeder school systems and employers), grant writing to secure non-traditional resources, student engagement, and career transition/job placement support.
    - Area CTE directors and employers report that today's high school students, their families, and secondary teachers have little understanding of post-secondary career (CTE) programs in IT or the associated careers, professional duties, and earning power of these credentials and associated industry certifications. Students and families who do understand it generally have higher education attainment levels and higher post-secondary aspirations and resources. Most of those students choosing to pursue a STEM major do so at a regional or national university.
        - These challenges make it particularly hard to recruit, retain, and graduate a significant number of "traditional" students from community college and TCAT CIT programs. Yet, this is the subset of the potential CIT student population where community colleges have some potential strategic advantage in the marketplace.
    - Location: The decision years ago to offer CIT AAS courses at the Livingston campus (30 minutes from the population center of the region and largest concentration of students with Tech/IT interest, Cookeville and two minutes from another TBR institution offering directly competing programs) has hurt CIT AAS enrollments in the UC, despite the sizeable number of tech employers and tech jobs in that region. It remains to be seen whether recent grant-funded interventions and a forthcoming articulation agreement with TCAT Livingston will be sufficient to overcome the location challenge.
- Learning:
    - Like the CIT AAS itself, course and program learning assessment instruments are very young and maturing and require additional attention in coming months and years.

- Less than 10% of students taking industry certifications imbedded in Fall 2021 courses (with grant dollars) earned those certifications. Most scored at levels below that required for certification. Score distribution needs to be further evaluated. Faculty and leaders should reflect on feasible changes that could boost pass rates.
  - Unlike selective admissions and licensing-based programs such as Nursing and Health Science, passing these exams is not a standard of success/quality for community college IT programs. However, scores on these nationally normed, industry-created exams does provide an important reference point. We expect these numbers to rise significantly over time if we get resources to engage students earlier to attract more who are better qualified and influence their academic aspirations and life decisions that significantly affect their academic success.

<u>POTENTIAL - ANALYSIS AND RECOMMENDATIONS</u>: Though recent trends in CIT AAS enrollments and declining admissions of new first time freshmen will surely reverse rising completion rates in the near-term, with sufficient prioritization, sound strategies, and sufficient staff capacity added and logically deployed for program promotion and program management, the CIT AAS program could make enormous contributions to the college's workforce contribution in this vital field, the continued economic growth of the region, and the strategic interests of the college. Evidence supporting this conclusion includes:

1. The portfolio of CIT AAS concentrations is now logically sized and shaped and well suited to meet the needs of all our stakeholders.
   - Though it will require constant reevaluation and refinement in response to never-ending technology changes and marketplace developments, our program/product offerings are well positioned for the near-term.
2. The Technology Department now has a strong cadre of full-time faculty with IT knowledge levels and specializations that align very well with the programs/concentrations we are providing and enrollment patterns across them.
   - The Department now has two FT CIT faculty in the Upper Cumberland (one temporary, grant-funded), the minimum number needed to have any chance at developing a viable enrollments and on ground access to two of the three CIT AAS concentrations.
3. Sumner County Schools, Wilson County Schools, and Putnam County Schools have greatly expanded the computer science and CIT courses they teach in their high schools (college prep and CTE) as well as the number of high schools where they offer them. Thus, many more secondary school students are developing an aptitude and stronger interest in IT-related careers.
   - These courses at nearby high schools also provide a channel for engaging prospective CIT AAS students earlier and more impactfully to attract more traditional students, attract more better qualified traditional students, and influence the decisions they make that affect their potential for post-secondary success.
4. Our state government, federal agencies, corporations, and non-profits are all very concerned about and emphasizing and investing in support of career education programs - particularly those in strategically-critical fields like IT.
   - Most of the investments are in the form of grant monies fenced for STEM education fields (such as IT and computer science) and for career programs (such as CIT AAS). However, while these funding opportunities are significant, rising and likely to continue doing so, the Business and Technology Division needs more administrative capacity to pursue, capture, and manage these resources and meet other, program promotion and program management missions. Failure to make these investments not only misses out on valuable, non-traditional resources needed for our programs to thrive, it defers those funds to other providers with whom we compete for enrollments. It is not capture these dollars and benefit or maintain status quo. It is capture these dollars and benefit or lose competitive advantage and strategic opportunity to others and have our own programs decline.
5. We have recently added numerous members to our CIT Advisory Committee from upper-management and executive levels of larger, tech companies and tech-dependent employers, along side CTE leaders from area school systems, and regional economic development officials.
   - This provides us the connections and relationships with which we could strengthen pathway alignments, promotional teamwork, and employer investments (scalable, program-compatible work-based learning; post-program pipelines to employment; and financial).
   - Though other Vol State units foster and support relationships with schools, employers, and economic development agencies, to turn relationships with these potential partners into strong, synergistic teamwork requires significant, division- and program-level engagement for which we are not currently resourced.
6. We have developed and socialized a viable plan for a 12-month "*FastPath*" to a CIT AAS degree.

    - Students most interested in career programs prefer shorter paths to strong earning power and a rewarding career. More and more providers are marketing micro-credentials and a fast path to a job. To attract more and better qualified students into the CIT AAS program, we must develop, make viable, and market shorter pathways to this credential.
    - Earning credit for as few as two, first-year CIT AAS courses while in high school would position new FTF to complete the CIT AAS degree (and sit for associated industry certification exams) just 15 months after their high school graduation. Completing two CIT AAS courses and three or more gen ed courses in high school would position them to earn their CIT AAS degree only 12 months after graduating high school. Making this a reality and assertively marketing it could attract many more students into this program and many more, better qualified students into it. These actions would produce many more graduates from the CIT AAS program with much stronger IT skill levels and job readiness.
    - Besides enabling and inspiring HS students to earn credits for two CIT courses while in high school, the plan also would require students to enroll FT the summer immediately after they graduate high school through the following Spring (12 months). The opportunity to earn a CIT AAS degree and qualifying to sit for as many as nine industry certification exams within a year of HS graduation should be very attractive relative to trying to work near-full time while taking 12-15 hours of courses per semester.
        - Financial aid to cover tuition and fees for enrollment in the bridge summer semester (immediately following HS graduation) is a gap in the plan. However, we assess that employers or grants would be willing to provide the funds to scholarship qualifying students to attract more highly qualified students into these pathways and get them through them successfully at higher rates and faster tempos.
    - The program could and should seek support of area school systems to offer dual enrollment courses on ground at select, area high schools and then do so - either with current or additional (grant-funded) FT faculty. In this way, we would increase the visibility of our programs, boost our FTE, and "prime the pump" for our own "FastPath" program.
    - More effectively promoting existing dual credit opportunities and looking for additional (and legitimate/appropriate) new dual credit opportunities for CIT classes could help.
    - The soon-to-be signed articulation agreement with TCAT Livingston could help, since they teach CIT courses on ground to large numbers of students enrolled in CIT CTE courses at Putnam County high schools.
7. With a portfolio of programs and pathways that align well with each other and with employers' needs, we are well-positioned to *strategically* and simultaneously engage a select group of tech and tech-dependent employers to propose the creation of a pool of similarly structured, clinicals-like, college compatible, paid internships for qualifying second year CIT AAS students.
    - Informing them of our current enrollment challenges and likely workforce contribution declines and our potential to do radically better with their support (in the form of these internship *programs*), we would likely find many employers more than happy to partner with us in this initiative.

**Evidence:**

- CIT VSCC Insider Article

## TWO YEAR RECOMMENDATION ACTION PLAN

Discuss the next steps for the program by providing a program action plan for the next two academic years. Recommendations must include at least two actions focused on student learning and one action focused on student success. The program action plan must include a timeline for each recommendation.

Feedback and approval will be provided. Refer to the APR timeline for approval deadlines.

**Narrative:**

**Student Learning Item 1: Schedule on ground sections of first year CIT courses to facilitate *de facto* cohort-based learning and draw more students into on ground sections for increased hands-on learning,**

**stronger student-to-student connections, closer connections to the faculty and department, and increased proximity to college support programs and support staff.**

- The 2020 move to prescriptive, highly sequential, horizontally-aligned curriculum pathways concentrates enrollment demand and enables us to offer more modality options and on ground sections.
- This enables us to assertively promote de facto or cohort-like learning for stronger learning communities and greater participation rates in co-curricular programming.

Timeline:

- Early-Feb 2022
  - Meet with faculty to solidify the specific scheduling decisions for on ground sections at the Gallatin campus for Fall 2022 and Spring 2023
  - Distribute press release through PA publicizing this scheduling innovation and associated learning benefits for students.
  - Distribute recruiting materials promoting CIT programs and de facto cohort/cohort-like learning opportunities to students at area high schools, leveraging partnerships with area CTE directors.
- Mid-Feb 2022
  - Meet with employers to explore feasibility of offering commonly-structured and -scheduled, paid internships to qualified second-year CIT AAS students
  - If a plan/program materializes, market that in conjunction with program and career field promotion
- Spring- Summer 2022
  - Send texts, emails, and letters with program and career field information to to all Vol State applicants and visitors expressing interest in CIT and/or Computer Science
- Sum 2022
  - Host Program forums in conjunction with each Campus Connect to educate CIT enrollees and prospects about pathways and career options in CIT
- Fall 2022
  - Host an early semester "Welcome Event" for CIT AAS majors to reinforce messages and promote optimal learning choices
- CY2023 and CY 2024
  - Repeat the same cycle but apply lessons learned from previous year(s)
  - ideally commit to and assertively promote "cohort-like" on ground learning opportunities for Year 2 classes in Gallatin for AY2023-2024
  - Ideally find/stimulate sufficient enrollment demand to begin offering cohort-like, on ground learning opportunities for the Networking and Cyber Defense concentrations in the Upper Cumberland by AY2023-2024

**Student Learning Item 2: Schedule and assertively promote optional, instructor-supported, on-ground lab time before and after two-course blocks of CIT classes offered as part of the on-ground cohort opportunity in Gallatin**. This bootcamp-style learning opportunity is a common learning and scheduling tactic in skills-focused computer and IT classes.

<u>TIMELINE</u>: The actions and schedule for this initiative will parallel that of the cohort scheduling initiative listed as Item 1, above.

- Early Feb 2022
  - Meet with faculty and commit to specific scheduling model and classrooms/lab assignments for Fall 2022 and Spring 2023
  - Promote the bootcamp-style learning opportunities in conjunction with initatives to promote the de facto cohort opportunities in item 1, above.
- Spring 2022
  - Brief employer partners about this initiative in Feb/Spring meetings and inquire about possible employer-provided incentives to boost student utilization of "bootcamp" lab time.
- Spring - Fall 2022
  - Educate prospective and incoming CIT students about the "bootcamp style" learning time and the benefits of participation.
- CY2023 and CY2024
  - Review Fall 2022 participation rates, learning and retention impacts and opportunities for program refinement.
  - Identify them and publicize the initiative again in subsequent recruiting cycles and student engagement campaigns

**Student Success Item 1: Redevelop the model course for INFS 1010, Computer Applications to integrate key Student Success material from FYEX classes into INFS 1010 assignments. Make more of the assignments team-based to foster student-to-student connections and working as teams.**

- The breadth and depth of learning required for entry-level and middle skills CIT/tech jobs and certifications doesn't allow room for FYEX in CIT AAS or BUS AAS curriculum
- INFS 1010, Computer Applications is an early-pathway course required for every BUS and CIT/TECH major (AAS, TCs, and AS - TTP)
  - This makes it an ideal course to introduce or reinforce key student success principles and foster student connectedness

Timeline:

- Fall 2022
  - Dean and Chair meet with key faculty and instructional design specialists to discuss the idea and get feedback on its feasibility
  - Dean and Chair formally charter interdisciplinary team to work the project and redesign the model course in Spr 2022
- Spr 2022
  - Interdisciplinary team (CIT and BUS faculty; instructional specialists in DE; FYEX course managers...) redesign INFS 1010
- Sum 2022
  - CIT Faculty beta test a prototype of the revised course
- Sum 2022
  - Course manager provides course orientation to adjunct faculty who will teach the revised version of the course for the first time in Fall 2022
- Dec 2022/Jan 2023
  - Course manager revises/"tweaks"  INFS 1010 model course in response to end-of-course student evals and post-semester feedback from faculty who taught it.
- CY 2023-2024
  - Division/Tech Department continues to deliver the revised version of the course universal to the curriculum of every B&T program student (~1,500)

**Student Success Item 2: Simultaneously meet with senior leaders from larger tech- and tech-dependent employers from the region to pitch the creation of a pool of similarly-structured, college- (schedule- and workload-) compatible, recurring, paid internships or apprenticeships for qualifying second year CIT AAS students.**

- Most students already enrolled in the CIT AAS program are already working full-time or nearly full-time. Jobs are plentiful and most are trying to take advantage of high wage rates in unskilled jobs at the same time they work their way to a CIT AAS degree. Thus, we have seen very low student interest in part-time job postings from area employers that are publicized in the middle of or just prior to the start of a semester.

Conversely, prospective and future traditional students entering or contemplating the CIT AAS program would likely have significant interest in part-time, college-friendly, paid IT internships available to them (with long lead/notice times) during the second-year of their CIT AAS studies. Developing and systematically promoting a pool of such opportunities as an optional but alluring element of our program available to qualifying students could produce many strategic dividends:

- attract more students and better students to the program - particularly traditional students who lack but need significant work experience in white collar roles
- positively affect students' early-program, work-school balance decisions
- inspire increased academic commitment and excellence to compete for the opportunities
- support more successful transitions to work for CIT AAS graduates and boost the reputation of our program and enhance student recruitment
-

To produce internships for CIT AAS students at a scale that would meaningfully impact student recruitment, academic performance, retention, and completion, we need to work intentionally and collaboratively with employers to develop a "program" of such opportunities that could be promoted in aggregate and would roll over systematically from semester-to-semester, year-to-year. Because "traditional" students attending full-time (and therefore most likely to be interested in such opportunities) are only a fraction of each cohort of graduates, we estimate that we could have a very significant impact on workforce production and student recruitment with a pool of only 20-25 or so opportunities in a pilot year or two. If the program had the impact on student recruitment and completion we desire, we might need to scale it up further or the student competition for the opportunities would just increase -- either way we go would be good for the program, the employers, and for workforce production.

TIMELINE:

- Feb-Mar 2022
    - Division hosts meeting with leaders from tech- and tech-dependent employers and economic development leaders with interest
    - Division provides overview of CIT AAS program/concentrations and discuss workforce needs and gaps
    - Present vision for a CIT AAS "FastPath" and for a pool of similarly structured, paid, college-friendly, IT internships for students in the second year of the pathway
    - Explore employer interest in these ideas as responses to the workforce gap and program recruitment and workforce production challenges
- Mar-Apr 2022
    - Follow up meetings with employers, college support units, and other stakeholders/partners and supporting agencies
- Apr 2022
    - Publicize the Fall 2022 pilot of any new pool of paid internships that the employers might elect to create
- Summer 2022
    - Reach out to returning CIT AAS students to promote the opportunities and support applicant screening and placement
- Fall 2022
    - First cohort of interns begin work with their IT employers
    - Business and Technology Dean and Tech Dept Chair work with partner employers to assess success and logical refinements to the program
        - Promote successes to other tech- and tech-dependent employers and seek to scale up the program by Fall 2023
- Spring 2023
    - Repeat actions listed above to promote the opportunity and help employers identify program participants for AY 2023-3024

## DEAN(S) APPROVAL AND FEEDBACK.

The Dean(s) will provide feedback and approval on the completed review and action plan.

**Narrative:**

The Department Chair, Johnny Maynard and I collaboratively wrote this report and deliberated and reviewed one another's input into the report. We both agree this report accurately depicts the current state of the CIT AAS program, where it has been, how far it has come, and sound strategies to improve it, going forward. Thus, naturally, I approve it. We look forward to having others review this report and provide us their feedback.

ANDREW W. WHITE, Dean
Business and Technology Division

## AVPAA & VPAA Review and Feedback

The Vice President of Academic Affairs and Assistant Vice President of Academic Affairs will provide feedback and approval on the completed review and action plan. VPAA and AVPAA should consider institutional planning and budgetary details when providing their review.

## PRESENTATION FEEDBACK

The APR will be presented to a group of peers. Feedback retrieved from the presentation will be summarized and included.

# Action Plan Status Reports

## YEAR 2 ACTION PLAN REVIEW

The reviewer should review progress on action plan over the past year. Discussion should include challenges and successes. If any changes are planned due to the results, those should be discussed and an updated action plan should be included in the narrative or attached as evidence.

**Narrative:**

| YEAR 2 | | | | | |
|---|---|---|---|---|---|
| | Initiative | Achieved Milestones | Challenges | Changes if applicable | Progress Toward Performance Indicator(s) |
| 1 | **Student Learning Item 1:** Schedule on ground sections of first year CIT courses to facilitate *de facto* cohort-based learning and draw more students into on ground sections for increased hands-on learning, stronger student-to-student connections, closer connections to the faculty and department, and increased proximity to college support programs and support staff. PLO 1 | Beginning in AY 2022-23, first year on-campus classes were scheduled within daytime timeslots that would encourage student cohorts. | Not enough enrollment in second-year on-campus classes. The majority of students that were in the second half of their degree were mostly online at this point. | No changes in timeline. | 40% of CITC courses within the AAS CIT degree curriculum were scheduled as daytime on-campus class offerings. |
| 2 | **Student Learning Item 2:** Schedule and | Select on-campus courses in the curriculum offered additional on-campus | Not all FT faculty agreed to offer additional lab times to support | No changes in timeline. | 15% of the CITC courses within the |

| | | | | | |
|---|---|---|---|---|---|
| | assertively promote optional, instructor-supported, on-ground lab time before and after two-course blocks of CIT classes offered as part of the on-ground cohort opportunity in Gallatin PLO 5 | lab sessions outside the regularly scheduled classes.<br><br>The courses included -- Cisco CCNA classes (CITC 1323, 1324, 2321) and CompTIA A+ and CompTIA Network+ classes (CITC 1320 and CITC 1302). | their classes outside of the individual office hours they were already required to provide.<br><br>Only two FT faculty members agreed to provide additional lab time schedule after their regularly scheduled class periods. This was to help encourage test prep for industry standard IT certifications (CompTIA and Cisco) and also preparation for SkillsUSA-related competition events that select students elected to participate in. | | AAS CIT curriculum offered on-campus lab support outside of the required class time period. |
| 3 | **Student Success Item 1:** Redevelop the model course for INFS 1010, Computer Applications to integrate key Student Success material from FYEX classes into INFS 1010 assignments. Make more of the assignments team-based to foster student-to-student connections and working as teams. | A new FYEX course is in the process of being finalized which may include team-based assignments/activities. | Due to changes in faculty and leadership personnel within the B&T division, the INFS 1010 course was <u>not</u> modified to include Student Success materials that were originally intended for a FYEX course. | A new FYEX course is in the process of being finalized which may include team-based assignments/activities. | The INFS 1010 course was <u>not</u> modified to include Student Success materials that were originally intended for a FYEX course. |
| 4 | **Student Success Item** | Met with leadership within the Workforce | The VP of Economic | This outcome was not achieved. | This outcome |

| | | | | | |
|---|---|---|---|---|---|
| | **2:** Simultaneously meet with senior leaders from larger tech- and tech-dependent employers from the region to pitch the creation of a pool of similarly structured, college- (schedule- and workload-) compatible, recurring, paid internships or apprenticeships for qualifying second year CIT AAS students. | and Economic Development division at VSCC 2-3 different occasions to discuss Work-Based Learning (WBL) opportunities and building a consortium group of local-area employers that would commit to offering rotating WBL/Internship opportunities for CIT students every academic year. | Development Workforce never did follow-up with any support with assisting in building WBL/Internships.<br><br>In addition, there were leadership changes at the VPAA and Dean-level, which affected this initiative as well. | | was not achieved. |

## YEAR 3 ACTION PLAN REVIEW

The reviewer should review progress on action plan over the past year. Discussion should include challenges and successes. The reviewer should discuss the actions future and continuous in the program.

**Narrative:**

| YEAR 3 Progress | | | | | |
|---|---|---|---|---|---|
| | Initiative | Achieved Milestones | Challenges | Performance Indicator Result | Initiative Achieved? Yes/No |
| 1 | Incorporate CompTIA A+, Network+, and Security+ certification exams as the final exam component within key CITC certification-aligned courses. PLO 5 | Certification exams were embedded as the Final Exam component in the following CITC courses: CITC 1302, CITC 1320, CITC 1332, CITC 2321, and CITC 2326.<br><br>The certification exam vouchers were funded using (3) different grant sources. | Difficulty scheduling exams at low-capacity Pearson VUE on-site Testing Center during Final Exam week.<br><br>Limited grant funding in terms of dollar $$ amount eligibility and availability. | | YES |
| 2 | Schedule first year CITC courses on-campus to continue | | | | |

| | | | | | |
|---|---|---|---|---|---|
| | cohort student groups. PLO 1 | | | | |
| 3 | | | | | |
| 4 | | | | | |

| YEAR 3 Initiative Analysis | | | | |
|---|---|---|---|---|
| | Initiative | How has this initiative supported student learning or student success? | Will there be any further action taken on this initiative? If yes, explain. | How will this initiative inform program decisions in the future? |
| EX. | Increase the achievement of PLO #1 by developing and implementing an online tutorial resource to complement teaching the XYZ assignment in VSCC1234 | | None | The assignment success rate for pilot sections was 80% PLO #1 achievement of pilot sections was 86% |
| 1 | Incorporate CompTIA A+, Network+, and Security+ certification exams as the final exam component within the CITC 1320, CITC 1302, and CITC 2326 courses. | | | |
| 2 | Schedule second year CITC courses on-campus to continue cohort student groups. | | | |
| 3 | | | | |
| 4 | | | | |

# Academic Program Review Cohort B- Cycle 1

Health Sciences AAS

## Reflection and Mission

### PREVIOUS ACADEMIC PROGRAM REVIEW

Reflect on the past APR (if any) and respond to any recommendations that were made at that time. The reviewer should list recommendations, actions taken, and discuss implications or challenges resulting from those actions.

**Narrative:**

This is the first Academic Program Review (APR).

### Program Mission Statement

What changes have the program made to the mission statement over the course of this cycle? Why were these changes made? Are any revisions planned?

**Narrative:**

The mission statement of the Health Sciences Associate of Applied Science (HS AAS) options degree is as follows: The College offers an Associate of Applied Science degree as a career ladder for persons who have successfully completed their certificate program of study in Dental Assistant, Diagnostic Medical Sonography, EMT-Paramedic, or Sleep Diagnostics.

The only change that is warranted at this time for consistency of terminology is changing Dental Assistant to Dental Assisting.

### Alignment to Institution Mission

How does the mission of the program align with the mission of the institution?

**Narrative:**

Volunteer State Community College is a public, comprehensive community college offering quality, innovative educational programs, support, and services. Vol State is committed to building partnerships, strengthening internal and external community engagement, and promoting diversity, cultural awareness, and economic development to prepare students for successful careers, university transfer, and meaningful civic participation in a global society.

The mission statement aligns with the institution's mission to offer quality, innovative programs, build partnerships, and prepare students for successful careers.

## Curriculum and Student Learning Outcomes Assessment

### Curriculum Map

Review the Student Learning Outcomes and course(s) where each one is addressed. For each outcome, provide the assessment method and target level of achievement.

**Narrative:**

See attached evidence for an example of curriculum mapping for the Diagnostic Medical Sonography program.

**HS AAS Outcomes**

1. Develop the knowledge and skills required of a health sciences professional. [DAST1530, DAST1610, SONO2036C, SONO2010, SONO2000, EMPS2412, PSG120, PSG131, PSG132]
2. Utilize the language of medicine to enhance effective communication. [DAST1530, DAST1610, SONO2036C, SONO2010, SONO2000, EMPS2412, PSG132]
3. Demonstrate critical thinking skills in the clinical environment. [DAST1530, DAST1610, SONO2036C, SONO2010, SONO2000, EMPS2412, PSG120]
4. Exhibit proficiency in the licensure and/or certification. [DAST1530, DAST1610, SONO2036C, EMPS2412, PSG120]

**Dental Assisting: Learning Outcomes**
Upon completion of this program, the graduate will have developed the skills, knowledge and abilities to accomplish the following:

1. Student will demonstrate the basic skills to be a competent dental assistant. [DAST1530 & DAST 1610] [HS AAS PLO #1]
2. Students will define the terminology unique to dental assisting. [DAST1530 & DAST 1610] [HS AAS PLO #2]
3. Students will illustrate comprehension of professional ethical standards for dental assisting. [DAST1530 & DAST 1610] [HS AAS PLO #1]
4. Students will apply and demonstrate critical thinking, problem solving and decision-making skills within the parameters of dental assisting. [DAST1530 & DAST 1610] [HS AAS PLO #3]

**Diagnostic Medical Sonography: Learning Outcomes**
Upon completion of this program, the graduate will have developed the skills, knowledge and abilities to accomplish the following:

1. Demonstrate knowledge in appropriate didactic areas. [SONO2036C] [HS AAS PLO #4]
2. Integrate knowledge into competency-based clinical experience. [SONO2036C] [HS AAS PLO #1]
3. Apply skills to exceed the standards of the profession. [SONO2036C, SONO2010, SONO2000] [HS AAS PLO #1]
4. Analyze clinical data. [SONO2036C, SONO2010, SONO2000] [HS AAS PLO #3]
5. Write effectively in more than one format as part of the writing across the professional curriculum initiative. [SONO2036C, SONO2010, SONO2000] [HS AAS PLO #2]
6. Provide excellent quality care to enrich and support the community through active engagement. [SONO2036C, SONO2010, SONO2000] [HS AAS PLO #1]
7. Engage and maintain personal professional relationships. [SONO2036C] [HS AAS PLO #1]

**Emergency Medical Technology- Paramedic Learning Outcomes**
Upon completion of this program, the graduate will have developed the skills, knowledge and abilities to accomplish the following:

1. The student will integrate and synthesize the multiple determinants of professional roles and responsibilities of the emergency medical services system as well as emergency medical services system operations as a paramedic in the pre-hospital emergency care setting. [EMPS2412] [HS AAS PLO #1]
2. The student will integrate and synthesize the multiple determinants of medical and legal considerations of operating as a paramedic in the pre-hospital emergency care setting. [EMPS2412] [HS AAS PLO #3]
3. The student will integrate and synthesize the multiple determinants of pharmacological interventions available to treat patients as a paramedic in the pre-hospital emergency care setting. [EMPS2412] [HS AAS PLO #4]

4. The student will recognize, classify and translate the multiple determinants of public health as a paramedic in the pre-hospital emergency care setting. [EMPS2412] [HS AAS PLO #1]
5. Students will safely and effectively perform the psychomotor skills relative to medical and trauma care of age-related patients associated with care as a paramedic in the pre-hospital emergency care setting. [EMPS2412] [HS AAS PLO #3]

**Sleep Diagnostic Technology: Learning Outcomes**
Upon completion of this program, the graduate will have developed the skills, knowledge and abilities to accomplish the following:

1. Graduates will become competent polysomnographic technologists with training in the fundamentals of polysomnographic technology, laboratory fundamentals, and clinical skills as outlined in, but not limited to, the American Association of Sleep Technologists (AAST) Curriculum Outline. [PSG120] [HS AAS PLO #3 & #4]
2. Graduates will have achieved both academic and clinical competency in cognitive, psychomotor, and affective learning domains using the following organizations as sources of standards: American Association of Sleep Technologists, Board of Registered Polysomnographic Technologists, and the American Academy of Sleep Medicine. [PSG120] [HS AAS PLO #1]
3. The students' accomplishments of their academic and clinical objectives will prepare them with entry level skills as Polysomnographic Technicians. [PSG120, PSG131, PSG132] [HS AAS PLO #1 & #2]
4. Successful completion of the program will help prepare the graduate to take the Board of Registered Polysomnographic Technologists Registry Exam. [PSG120] [HS AAS PLO #4]

**Evidence:**

- DMS Program Outcome Mapping 2021

## EVIDENCE OF STUDENT LEARNING

Provide evidence of student learning and the program's effectiveness in reaching both program outcomes and course learning outcomes.

Please provide the following:
1. Summary of most recent assessment results.
2. Analysis of trends
3. Ideas for how those results could be used to improve student learning.

**Narrative:**

Paramedic National Registry The College is committed to developing graduates who effectively meet the needs of the workplace. Associate of Applied Sciences degree options in Health Sciences (HS AAS) is offered as a career ladder for persons who have successfully completed a technical certificate program of study. The degree is designed to assist employees in the health sciences field with their general and continued education. The technical certificate programs of study include Dental Assisting, Diagnostic Medical Sonography, Emergency Medical Technology (EMT) - Paramedic, and Sleep Diagnostics Technology.

The HS AAS degree requirements are listed:
**General Education Requirements - Credits (15 - 17)**

- Humanities and/or Fine Arts **Credits: (3)**
- Natural Sciences/Math (BIOL 2010 Preferred - See Advisor) **Credits: (3-4)**
- ENGL 1010 - ^English Composition I **Credits: (3)**

- One additional General Education course from any of the following categories: Communication, Humanities/Fine Arts, Math, Natural Science or Social/Behavioral Sciences **Credits: (3-4)**
- PSYC 1030 - ^Introduction to Psychology **Credits: (3)**
- **OR** PSYC 2130 - ^Lifespan Development Psychology **Credits: (3)**

**Professional Core Course Requirements - Credits (15)**

- ALHS 2312 - *Contemporary Issues in Healthcare **Credits: (3)**
- BUSN 1380 - *Supervisory Management **Credits: (3)**
- BUSN 2370 - *Legal Environment of Business **Credits: (3)**
- Electives: Choose from the list below or from a MATH, AHC, ALHS, Natural Science course or the student may apply Dental Assisting Technical Certificate credits earned in excess of 30 credit hours as electives **Credits: (6)**
- AHC 115 - *Medical Terminology **Credits: (3)**
- BIOL 2020 - ^Human Anatomy and Physiology II **Credits: (4)**
- MATH 1130 - ^College Algebra **Credits: (3)**
- PHYS 1030 - ^Survey of Physics **Credits: (4)**

**Total Degree Requirements - Credits (60-62)**

It is recommended that prospective students follow recommended schedules for the specific technical certificate and to consult with an academic advisor to schedule the remaining courses. Students may be awarded up to 31 - 32 hours of credit based towards the HS AAS degree upon initial certification in the programs.

Each of the programs provide an exceptional education that when partnered with the HS AAS degree will benefit the graduate. The programs are fully accredited by their respective accrediting bodies thereby undergoing stringent processes for reviewing effectiveness from an external perspective. The bodies utilize collaborative, peer-reviewed processes to evaluate educational quality. They require obtainment of evidence-based standards that reflect what a student needs to know and be able to do to successfully function within the profession upon graduation.

**External Evaluation**

To ensure accountability, annual surveys or reports are submitted by the program directors. The compilation of the reports provides an opportunity for faculty to assess quality and reflect on the previous year. Programs must undergo the reaccreditation process every several years at various intervals depending on the program. The process includes completion of a self-study and culminates with a site visit by the accreditors.

The self-study is an in-depth evaluation and recording of the educational journey of the program since the previous accreditation cycle. An example of an areas that is included is institutional effectiveness, which evaluates everything from planning and assessment to financial and community support of the program. The program itself is assessed with regards to admissions, matriculation, curriculum, instruction, and assessment.

Additional areas of evaluation are support services, clinical affiliations, resources such as classroom, library, and computer access as well as support from business and industry, advisory committees (see meeting notes in evidence section) administration, faculty, and staff. Job placement and obtaining licensure, registry, and credentials are also assessed. The programs are accrediting by the following organizations:

***Dental Assisting:*** The Commission on Dental Accreditation (CODA),
https://coda.ada.org/en

***Diagnostic Medical Sonography:*** The Commission on Accreditation of Allied Health Education Programs in cooperation with the Joint Review Committee on Education in Diagnostic Medical Sonography

https://www.caahep.org/
https://www.jrcdms.org

***Emergency Medical Technology – Paramedic:*** The Commission on Accreditation of Allied Health Education Programs upon recommendation with the Commission on Accreditation of Educational Programs for the Emergency Medical Services Professions

https://www.caahep.org/
https://coaemsp.org/

***Sleep Diagnostic Technology:*** The Commission on Accreditation of Allied Health Education Programs in cooperation with the Commission on Accreditation of Educational Programs in Polysomnographic Technology

https://www.caahep.org/About-CAAHEP/Committees-on-Accreditation/Polysomnographic-Technology.aspx

To continue to deliver a high-quality education, program faculty meet regularly to discuss student matriculation, course structuring, student and program evaluation, clinical objectives, job placement, and registry and credential attainment. Program directors and College administrators review the status of each program to ensure that competencies are consistent with stated goals and standards. Additionally, respective advisory committees and clinical personnel provide input related to overall program effectiveness.

Program directors must also ensure compliance with the Southern Association of Colleges and Schools Commission on Colleges (SACSCOC). It is the College's institutional accrediting body. In addition to ensuring quality, the organization desires to improve effectiveness at institutions of higher learning by requiring compliance of specific standards which cover similar categories as the programmatic accreditation bodies. Examples of categories of standards that are addressed with SACSCOC are educational policies, procedures, and practices, student achievement, and qualifications of faculty members.

**Internal Evaluation**
Internal evaluation of each program's effectiveness is evaluated by Academic Audits that are facilitated by the Tennessee Board of Regents (TBR). The audit is a faculty-driven model of ongoing self-reflection, collaboration, teamwork, and peer feedback. Academic Audit Outcomes:

1. A faculty-centered methodology that supports institutional effectiveness expectations.
2. A thoroughly documented process that meets the requirements for Quality Assurance Funding
3. A process that sustains continuous quality improvement of teaching and learning

The audits are typically completed every three-years and require completion of a self-study, peer review, and a site visit. At the end of the audit, the auditor team provides feedback which includes commendations, affirmations, and recommendations for continued improvement. (https://www.tbr.edu/academics/academic-audit).

Through collaboration with the Office of Institutional Effectiveness, Research, Planning and Assessment, programs are further assessed through Academic Program Reviews (APR). The APR is a comprehensive assessment designed to ensure the programs are providing quality and innovative education to students. Completion of the APR requires assessment, reflection, and actions over a 3-year cycle.

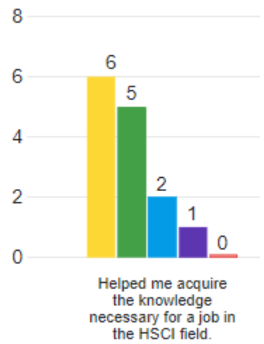
The evaluation of the following areas: job placement rates, licensure/certification passage rates, employer and graduate satisfaction ratings are of particular interest to all stakeholders. Information from those areas provide insight about the quality of each individual program connected to the HS AAS degree and the services of the college. The Academic Program Review will examine the results of Graduate and Employer survey results as well as data from IERPA to assess the effectiveness of the degree.

Evidence of student learning after obtaining the HS AAS degree consists of the following questions obtained from the 2020 Graduate Surveys:

● Strongly Agree ● Generally Agree ● Neutral ● Generally Disagree ● Strongly Disagree

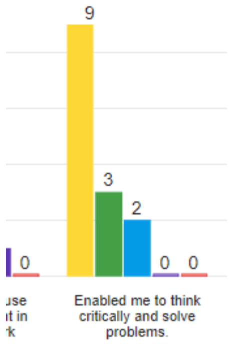
**1. Graduates will acquire the knowledge necessary for a job in the health care field. 4.14 (Graduate Survey, page 7)**

| Field | Min | Max | Mean | Standard Deviation | Variance | Responses | Sum |
|---|---|---|---|---|---|---|---|
| Helped me acquire the knowledge necessary for a job in the HSCI field. | 2.00 | 5.00 | 4.14 | 0.91 | 0.84 | 14 | 58.00 |




**2. Graduates will be able think critically and solve problems. 4.50 (Graduate Survey, page 7)**

| Field | Min | Max | Mean | Standard Deviation | Variance | Responses | Sum |
|---|---|---|---|---|---|---|---|
| Helped me acquire the knowledge necessary for a job in the HSCI field. | 2.00 | 5.00 | 4.14 | 0.91 | 0.84 | 14 | 58.00 |
| Enabled me to think critically and solve problems. | 3.00 | 5.00 | 4.50 | 0.73 | 0.54 | 14 | 63.00 |




**3. The employee demonstrates the ability to work independently as evidenced by Employer Survey results (page 2).**

| Field | Poor | | Needs Improvement | | Fair | | Good | | Excellent | | Total |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 4. The ability to work independently. | 0.0% | 0 | 0.0% | 0 | 0.0% | 0 | 25.0% | 3 | 75.0% | 9 | 12 |

**4. Graduates will demonstrate appropriate work ethic as evidenced by Employer Survey results (page 2).**

| Field | Poor | | Needs Improvement | | Fair | | Good | | Excellent | | Total |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 6. Appropriate work ethic. | 0.0% | 0 | 0.0% | 0 | 0.0% | 0 | 25.0% | 3 | 75.0% | 9 | 12 |

**5. Board examination passage and state registry rates will reach a threshold of 90% in all degree option programs.** One of the primary goals of all programs is to prepare graduates to successful pass certification and registry exams. Examination agencies publish content outlines. Program faculty crossmatch the information with learning outcomes and objectives from accreditation agencies and employers. This information can serve as a gap analysis for faculty to assist with the development and revision of learning objectives particularly when revisions in the exams occur.

To become a Registered Dental Assistant (RDA) graduates from the program must apply for registration through the Tennessee Department of Dentistry and pass an ethics course. The chart below represents the number of graduates who obtained the RDA designation. During the program, specialty certification courses were implemented to strengthen the students' knowledge base and to increase employability. One such course is the certification course in coronal polishing. The link to the content outline is noted. Review national examination matrices for the remaining programs through the accompanying links.

| | |
|---|---|
| Dental Assisting | Certification Course in Coronal Polishing _ Content Outline pages 19-22 |
| Diagnostic Medical Sonography | Sonography Principles and Instrumentation Exam - Content Outline |
| EMT - Paramedic | Paramedic Registry Exam - Content Outline |
| Sleep Diagnostics | Registered Polysomnography Technician Exam - Content Outline |

As indicated below, although the 90% threshold average was met in the Diagnostic Sonography and Paramedic programs. The average passage rate for Dental Assisting is 89% over the 3-year period. This is primarily due to the 79% passage rate in 2018. Since then, the curriculum has been revised and additional certifications were implemented. Note a steady increase in the program in subsequent years. The Sleep Diagnostics program is well below the 90% threshold for passage rates. This could be due to a variety of reasons and will be examined in detail in the 2nd-year APR.

**PASSAGE RATES**

*Registry # National Board Examination

| Program | 2018 | 2019 | 2020 | 2018 - 2020 |
|---|---|---|---|---|
| Dental Assistant* | 15/19 = 79% | 13/14 = 93% | 12/12 = 100% | 40/45 = 89% |
| Diagnostic Medical Sonography# | 16/16 = 100% | 16/16 = 100% | 12/13 = 92% | 44/45 = 98% |
| EMT - Paramedic# | 41/45 = 91% | 43/48= 90% | 37/41 = 90% | 121/134 = 90% |
| Sleep Diagnostics# | 6/9 = 78% | 9/11 = 82% | 6/11 = 55% | 21/31 = 68% |

In summary, evidence of student learning with the HS AAS degree option is demonstrated by positive internal and external accreditation results, maintenance of standards, the production of high outcomes, the responses to graduate surveys, and the excellent board examination passage rates. The faculty use methodical processes for developing measurable, clear, and relevant outcomes based on accreditation standards and objectives. Through combining and analyzing the resulting data, program faculty can identify strengths and gaps in instructional sequence and content and make modifications as needed.

Evidence:

- Accreditation Report - Paramedic
- Annual Reports
  - Dental Assisting
  - Diagnostic Medical Sonography
- Surveys
  - Employer _ HS AAS 2020
  - Graduate _ HS AAS 2020
  - Graduate and Employer Response Report _ HS AAS 2020
- Minutes of Advisory Committee Meetings
  - Dental Assisting
  - Emergency Medical Services (includes Paramedic)
  - Sleep Diagnostic Technology

**Evidence:**

- Accreditation Report _ Paramedic
- Annual Survey _ Dental Assisting
- Annual Survey _ Diagnostic Medical Sonography

- Minutes 2020 Dental Advisory Committee
- Minutes 2020 Diagnostic Sonography Advisory Committee
- Minutes 2020 Emergency Medical Services Advisory Committee
- Minutes 2020 Sleep Diagnostic Advisory Committee
- Surveys _ Employer 2018_ HS AAS
- Surveys _ Employer 2019 _ HS AAS
- Surveys _ Employer 2020_ HS AAS
- Surveys _ Graduate 2018_ HS AAS
- Surveys _ Graduate 2019 _ HS AAS
- Surveys _ Graduate 2020_ HS AAS

## Qualitative and Quantitative Program Information

### STUDENT, GRADUATE, AND ALUMNI SATISFACTION

Use student and graduate surveys, course evaluations, and any other appropriate evidence needed to explore the qualitative satisfaction results from students and graduates to provide information to the items below.

Discuss student, graduate, alumni perceptions of the program, faculty, career projection, and any other program recommendations given "during this survey".

Summarize any noted themes, trends, areas for improvement, identified strengths, and implications for the program for both students and alumni.

**Narrative:**

The Health Sciences Technical Certificate programs within the HS AAS degree have strong support from the College, clinical affiliates, communities of interest, current students, and graduates. Course evaluations for current students are captured and analyzed through the College's EvaluationKIT software to aid instructors in improving course content and teaching practices. Graduate survey tools are collected and evaluated to measure graduate satisfaction with the assistance of the IERPA department. The programs implement comprehensive assessments and strategies to continuously improve effectiveness.

Faculty members routinely pursue continued education to ensure teaching excellence and the overall success of the programs. Continued education is pursued, externally, through respective professional and accreditation organizations. From an internal perspective, they participate in educational offerings through Vol State's Teaching and Learning Institute for topics related to High Impact Practices such as service and experiential learning.

The diverse pedagogical approaches employed by faculty include, but are not limited to, group facilitation, lecture, utilization of technology, clinical simulations, and pragmatic laboratory activities. In addition to volunteer activities, students participate in service-learning activities during clinical rotations. In the clinical realm, learning is enhanced through direct patient care while students are also providing service to the community. The varied approaches afford students the opportunity to experience success in a variety of formats.

Although each technical certificate program had a robust submission of course evaluations, one weakness noted is the low number of responses for the HS AAS 2020 Graduate and Employer Surveys. The number of graduate responses was 14 and there were 12 employer responses. Some potential roadblocks to survey completion are IERPA not knowing a graduates preferred contact method or current employer. The program directors typically know the best way to get in contact with graduates, where they are employed, and the name of their direct supervisors. A recommendation is that program directors more closely collaborate with IERPA to increase the number of responses.

Graduates appreciate the significant value of the HS AAS options degree as reflected in the following comments from anonymous surveys:

- "Highly recommend this program, and I am so thankful I pursued it!" (2020)
- "I was a paramedic for 30 years prior to finishing my degree. The courses provided by Vol State are great for entry level EMTs and paramedics. I would recommend all paramedics finish their degree to help them advance in their career. If more medics and EMTs have degrees, it is easier to negotiate for more money." (2019)
- "Upon hire, I was able to perform nearly all exams required as a professional and I was able to clearly communicate findings to radiologists." (2018)

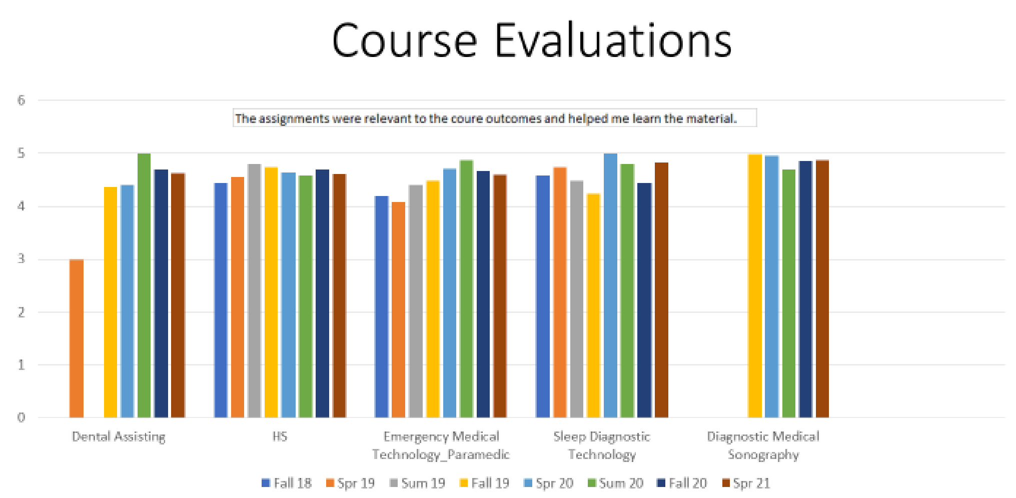
According to page 8 of the Graduate Survey, course content, working with peers, and the faculty are listed as top strengths of the Health Sciences AAS degree. Students' perception of faculty efficacy can be noted in course evaluation responses for the question ***"The assignments were relevant to the course outcomes and helped me learn the material".***

The following Likert scale was utilized:

- 5 = Strongly Agree
- 4 = Agree
- 3 = Neither Agree or Disagree
- 2 = Disagree
- 1 = Strongly Disagree

The 3-year average rating for the question with each discipline, including the HS AAS, is listed below.

| Discipline | 3-Year Average (Fall 18' to Spring 21') |
|---|---|
| Dental Assisting | 4.35 |
| Diagnostic Medical Sonography | 4.87 |
| Emergency Medical Technology - Paramedic | 4.5 |
| **Health Sciences (HS)** | **4.63** |
| Sleep Diagnostic Technology | 4.64 |



Another satisfaction tool is assessing how many graduates have obtained obtain gainful employment. With each technical certificate program having an excellent reputation in the healthcare community of Middle Tennessee, employment has not been an issue. Graduates are well-received within their individual employment settings and have strong employment outcomes as demonstrated by the placement data below:

| Placement Rates | | | | |
|---|---|---|---|---|
| | **2018** | **2019** | **2020** | **2018 - 2020** |
| Dental Assistant | 14/19 =73% | 13/14 = 93% | 12/12 =100% | 39/45 =87% |
| Diagnostic Medical Sonography | 16/16 = 100% | 16/16 = 100% | 12/13 = 92% | 44/45 = 98% |
| Paramedic | 45/45=100% | 48/48 = 100% | 41/41 =100% | 134/134=100% |
| Sleep Diagnostics | 14/23 = 61% | 27/31 = 87% | 16/23 = 70% | 57/77 = 74% |

In summary, the HS AAS degree option has been shown to be effective based on student and graduate satisfaction. Listed strengths include the faculty, accomplishment of learning outcomes, and improvement in career opportunities and gainful employment. It supports the mission of the College by being an option in innovative, quality programs and promoting economic development to prepare students for successful careers. Students are educated in high quality programs that prepare them to be competent and proficient graduates. Upon graduation, gainful employment is obtained to meet the demands of the workforce. In the future, each program director will collaborate more closely with the IERPA department to increase the quantity of survey responses. Satisfaction for all stakeholders will continue to be monitored.

Evidence:

- Course Evaluations – 2018 to 2021
    1. Dental Assisting
    2. Diagnostic Medical Sonography
    3. Emergency Medical Technology – Paramedic
    4. Health Sciences
    5. Sleep Diagnostic Technology
- Surveys
    1. Graduate – HS AAS 2020
    2. Employer – HS AAS 2020
    3. Graduate and Employer Response Report – HS AAS 2020

**Evidence:**

- Course Evaluations_ Dental Assisting 2018 to 2021
- Course Evaluations_ Diagnostic Medical Sonography 2018 to 2021
- Course Evaluations_ EMT Paramedic 2018 to 2021
- Course Evaluations_ Health Sciences 2018 to 2021
- Course Evaluations_ Sleep Diagnostic 2018 to 2021
- Surveys _ Employer 2020_ HS AAS
- Surveys _ Graduate 2020_ HS AAS
- Surveys _ Graduate and Employer Response Report 2020_ HS AAS

## RECRUITMENT, ENROLLMENT, RETENTION, AND GRADUATION: FACT BOOK DATA

Attach screenshots from the Fact Book of the areas below.

1. Headcounts
2. FTE Comparison
3. Student Primary Location
4. Retention Statistics

5. Graduation Statistics
6. Demographics

Using the Fact Book data listed below, analyze the follow:

Recruitment
Enrollment
Retention
Graduation

The reviewer could discuss marketing and recruitment efforts in the community and local high schools; analyze enrollment, retention, and graduation trends; and discuss potential reasons, strengths, plans for improvement, and any other implications deemed appropriate.

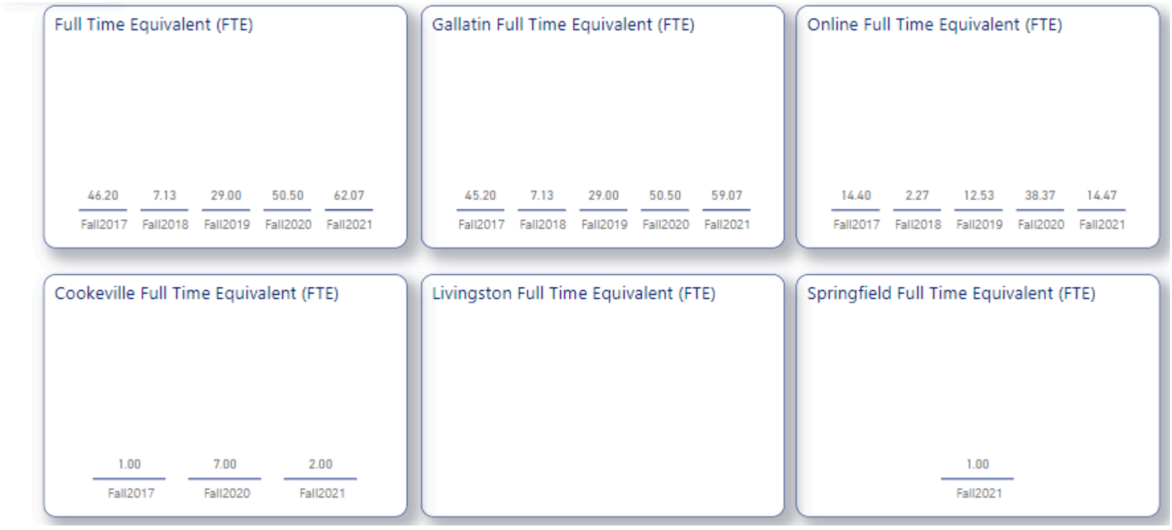
**Narrative:**

The following data was obtained from Vol State's Fact Book on 11/30/2021. It reflects students and graduates who declared the HS AAS as a degree from Fall 2018 to Spring 2021. They are blocked out in blue and contained details about programs unrelated to the HS AAS degree.

**HS AAS Program Headcounts**

| Degree/Program/Concentration | Fall2017 | Fall2018 | Fall2019 | Fall2020 | Fall2021 |
|---|---|---|---|---|---|
| **Associate of Applied Science** | **1256** | **1392** | **1460** | **1460** | **1335** |
| [redacted] | | | | | |
| **Health Science** | **39** | **7** | **36** | **59** | **78** |
| | 20 | 4 | 2 | 1 | |
| Dental Assistant | 2 | 1 | 4 | 3 | 1 |
| Dental Assisting | | | | 15 | 15 |
| Diagnostic Medical Sonography | | | | 1 | 17 |
| EMT Paramedic | 1 | | 18 | 29 | 36 |
| Sleep Diagnostic Technology | 16 | 2 | 12 | 10 | 9 |

**FTE Comparison**



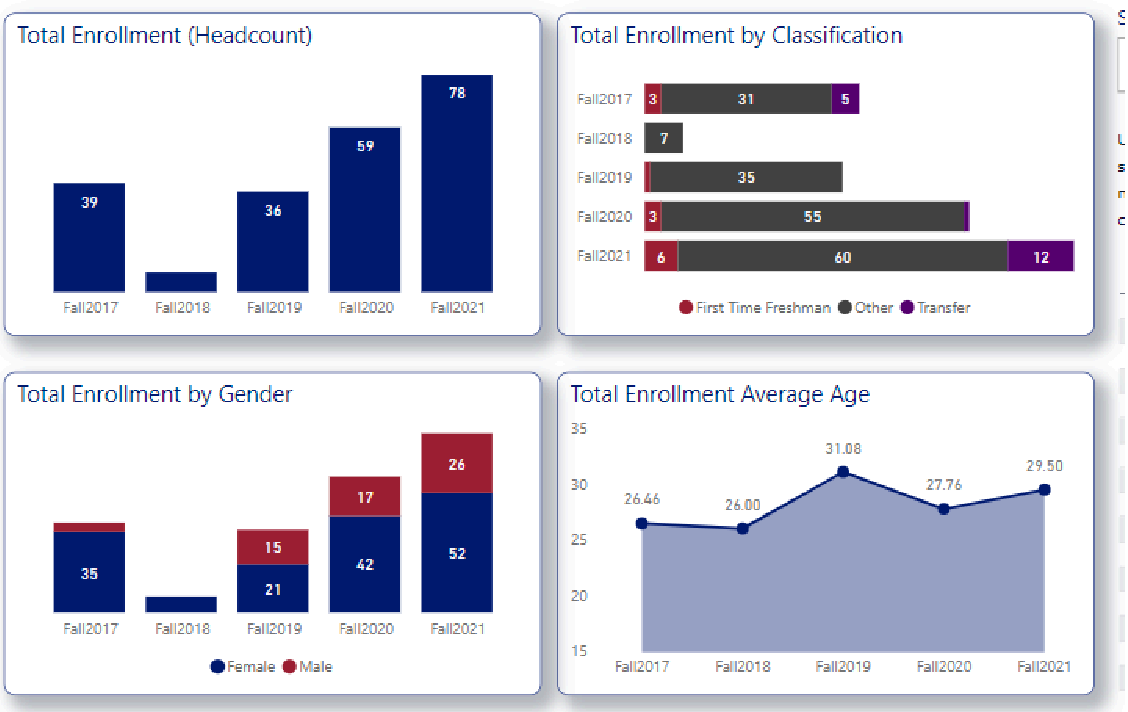

Residency

| Resident | Fall2017 | Fall2018 | Fall2019 | Fall2020 | Fall2021 |
|---|---|---|---|---|---|
| In State | 36 | 7 | 35 | 59 | 75 |
| Out-of-State | 3 | | 1 | | 3 |
| **Total** | **39** | **7** | **36** | **59** | **78** |

| State/County | Fall2017 | Fall2018 | Fall2019 | Fall2020 | Fall2021 |
|---|---|---|---|---|---|
| Sumner County | 14 | 3 | 11 | 19 | 26 |
| Davidson County | 4 | | 3 | 5 | 10 |
| Montgomery County | 2 | | 6 | 7 | 4 |
| Wilson County | 1 | 1 | 2 | 6 | 7 |
| Robertson County | 4 | | 3 | 2 | 5 |
| Rutherford County | | | 3 | 2 | 5 |
| Putnam County | 2 | | | 4 | 2 |
| Cheatham County | 1 | | 2 | 1 | 3 |
| Macon County | 1 | 1 | 1 | 1 | 1 |
| White County | | | | 3 | 2 |
| Hamilton County | 2 | | | | 1 |
| Jackson County | | 1 | | | 2 |
| Shelby County | 1 | | | 2 | |
| Van Buren County | 1 | | | 1 | 1 |
| Cumberland County | | | 1 | 1 | |
| KENTUCKY | 1 | | 1 | | |
| Smith County | | | 1 | | 1 |
| Trousdale County | 1 | | | 1 | |
| Bradley County | 1 | | | | |
| Clay County | | | | 1 | |
| Coffee County | | | | 1 | |
| DeKalb County | | | | | 1 |

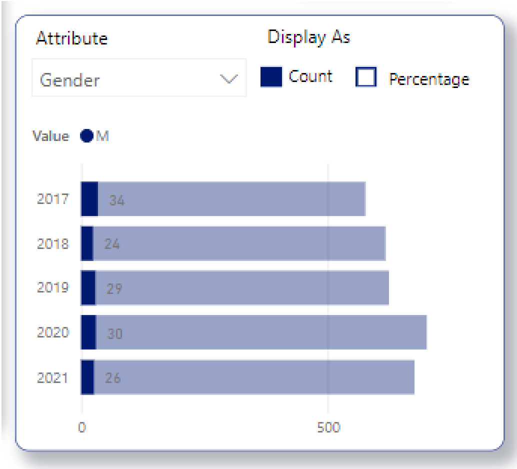
| | | | | | |
|---|---|---|---|---|---|
| DeKalb County | | | | | 1 |
| Dickson County | | | | | 1 |
| Fentress County | | | | | 1 |
| GEORGIA | | | | | 1 |
| Gibson County | | | 1 | | |
| Greene County | | | | | 1 |
| Humphreys County | 1 | | | | |
| INDIANA | | | | | 1 |
| Logan County | 1 | | | | |
| Maury County | | | | 1 | |
| NORTH CAROLINA | 1 | | | | |
| Scott County | | 1 | | | |
| Simpson County | | | | | 1 |
| Stewart County | | | 1 | | |
| Tipton County | | | | 1 | |
| Williamson County | | | | | 1 |

**Student Primary Location and Demographics**




**Demographics based on Gender and Age**

When reviewing the total enrollment, note that the average enrollment age represents what is traditionally considered a non-traditional student or adult learner. There have been more female students than males for the past three years. Most of the male graduates are from the EMT - Paramedic Program, as evidenced by the following graph.




The Fact Book defines persistence as students who enroll in the fall semester and return the following spring or graduate within that academic year. Fall-to-spring persistence for the HS AAS degree averaged 75% from 2018 to 2021.

**Fall to Spring Persistence**

NA
Fall 2016 to Spring 2017

61.5%
Fall 2017 to Spring 2018

66.7%
Fall 2018 to Spring 2019

83.3%
Fall 2019 to Spring 2020

75.0%
Fall 2020 to Spring 2021

**Fall to Fall Retention**

NA
Fall 2016 to Fall 2017

0.0%
Fall 2017 to Fall 2018

0.0%
Fall 2018 to Fall 2019

66.7%
Fall 2019 to Fall 2020

22.2%
Fall 2020 to Fall 2021

Since 2018, 102 graduates have earned the HS AAS degree, as evidenced by the following graph:

| Degree | 16-17 | 17-18 | 18-19 | 19-20 | 20-21 |
|---|---|---|---|---|---|
| Associate of Applied Science | 59 | 50 | 50 | 34 | 18 |
| **Total** | **59** | **50** | **50** | **34** | **18** |

Graduates are further divided by program of study.

| Concentration | Number of Graduates (2018-2021) |
|---|---|
| Dental Assistant | 15 |
| Diagnostic Medical Sonography | 17 |

| | |
|---|---|
| Emergency Medical Technology - Paramedic | 64 |
| Sleep Diagnostic Technology | 35 |

| Concentrations | 16-17 | 17-18 | 18-19 | 19-20 | 20-21 |
|---|---|---|---|---|---|
| Dental Assistant | 14 | 10 | 6 | 5 | 2 |
| Dental Assisting | | | | | 4 |
| Diagnostic Medical Sonography | 10 | 8 | 9 | 8 | |
| EMT Paramedic | 21 | 13 | 29 | 15 | 7 |
| Sleep Diagnostic Technology | 14 | 19 | 5 | 6 | 5 |

Based on the collected data, there are specific categories of students who could possibly benefit from the College's Office of Student Services. Marketing that focuses on available support services should provide emphasis on non-traditional students. Prospective male students should be solicited even though health careers at the community college level tend to be more female-dominated industries.

## Supporting Program Information

### FACULTY USAGE AND DEVELOPMENT (OPTIONAL)

Review faculty teaching loads, adjunct to full time ratios, faculty development opportunities, faculty end of course evaluations, faculty surveys, and any other evidence deemed appropriate. The reviewer could discuss the impact of teaching, the ratio of adjunct to full time staffing, end of course evaluations relative to faculty, and any noted themes identified from the review.

### CLASS OFFERINGS (OPTIONAL)

Address scheduling effectiveness and space utilization. The reviewer could review faculty to student ratios at each location and online and discuss the impact relative to faculty and student success.

### CO-CURRICULAR ACTIVITIES (OPTIONAL)

Discuss co-curricular activities available for students in the program. These might include clubs, internships or other Work Based Learning opportunities, service learning, guest speakers, or any other activities designed to enhance student experience in the program.

### INDUSTRY TRENDS AND INNOVATIONS (OPTIONAL)

Examine the program demand within the labor market including local, regional, and/or national. The reviewer could examine information from industry advisory boards and information gathered from creditable industry sources as well as from external accrediting bodies. The reviewer could also discuss any existing or emerging trends within the field as they relate to your program and employment of your student's post-graduation as well as the outlook for the field in the next 5 years. This summary could also identify best practices, trends, and innovation in the field and discuss implications for the program.

### EMPLOYMENT AND TRANSFER FORECASTS (OPTIONAL)

Discuss where students go after graduation and what is being done to enhance the likelihood of success or assist students in reaching those goals. The reviewer could discuss employment outlook, obtainment, curriculum alignment for transfers, and the employment demands of the industries where students will work.

## Recommendations Action Plan, Review, and Feedback

### SUMMARY AND ANALYSIS

The reviewer should discuss what went right, what went wrong, and the implications of those results.

**Narrative:**

The HS AAS option degree has been shown to be effective through the results of surveys from graduates and employers and during discussions with faculty and advisory board committees and positive standings with all accrediting bodies. As part of this review, students were surveyed through course evaluations and the responses were assessed. The students' perceptions regarding the value of the degree itself is of particular significance. Students overwhelmingly (4.6 out of a 5.0 Likert scale) agree that the assignments provided by faculty were relevant to the course outcomes and helped them learn the material.

The faculty consistently deliver high quality education with support from the College and community stakeholders. In the graduate survey, alumni expounded on the value of the degree for career advancement and increased competence in chosen healthcare fields. Since 2018, 89% (226/255) graduates have obtained state registry and credentials thereby providing an opportunity to join the growing workforce. Approximately 91% (274/301) graduates have obtained employment in the last three years. The employer survey revealed that 100% (12/12) employers indicated that graduates were adequately prepared for their positions and would hire another graduate from the HS AAS program if given the opportunity.

Recruitment is an area that needs specialized attention to assist the College with optimizing its mission statement of offering quality programs, promoting diversity, and preparing students for successful careers. The current headcount of students pursuing the degree is 78. Although that number is higher than it has been since 2018, recruitment efforts should be escalated. Most of the students reside in Sumner County which is not surprising since Vol State is the only public community college in the service area.

According to the US Census Bureau, Women hold 76% of all healthcare jobs https://www.census.gov/library/stories/2019/08/your-health-care-in-womens-hands.html and as expected the total enrollment by gender indicates that most students identify as female. The College can address the lack of diversity by marketing to other genders and races. The total enrollment age remains steady at approximately 29-years-of-age. We are attracting non-traditional students who have needs that are different from the 18-year-old college student. Are we meeting the needs of the adult learner? Do we have enough flexibility in course offerings to make it possible for them to be successful in our programs?

In summary, the HS AAS degree is fulfilling its mission statement by being a career ladder to those who have successfully completed technical certificate programs for Dental Assisting, Diagnostic Medical Sonography, EMT-Paramedic, and Sleep Diagnostic Technology. Students receive stellar educations and are eligible to obtain registry and credentials to make them job ready. Employers are pleased with the graduates and provide gainful employment at increased rates. The ultimate beneficiary is the community who receive health care from our graduates.

## TWO YEAR RECOMMENDATION ACTION PLAN

Discuss the next steps for the program by providing a program action plan for the next two academic years. Recommendations must include at least two actions focused on student learning and one action focused on student success. The program action plan must include a timeline for each recommendation.

Feedback and approval will be provided. Refer to the APR timeline for approval deadlines.

**Narrative:**

- **Student Learning Recommendation:** Assess components of the following professional core courses for the HS AAS degree in spring of 22:
  - ***ALHS 2312 –***Contemporary Issues to Healthcare
    - AY 2022-2023- Revise learning objectives to include higher levels of Bloom's Taxonomy.
    - AY 2023-2024
  - ***AHC 115 –*** Medical Terminology
    - AY 2022-2023- Collaborate with health sciences program directors to create short videos discussing physiological systems into eLearn shells. The goal is to introduce a different way to familiarize students to medical terminology from the health care professional perspective.
    - AY 2023-2024

- **Student Success Recommendation**: Improve Graduate Satisfaction Survey response rates.
  - ***AY 2022-2023-*** Collaborate with IERPA in the solicitation of completed graduate and employer surveys.
  - ***AY 2023-2024-***

- **Student Success Recommendation**: Improve recruitment efforts of the HS AAS degree.
  - ***AY 2022-2023-*** Assist in the development of marketing efforts targeting under-represented demographics in Fall 2022. Promote the HS AAS options degree to persons who complete one of the technical certificates in Spring.
  - ***AY2023-2024-***

## DEAN(S) APPROVAL AND FEEDBACK.

The Dean(s) will provide feedback and approval on the completed review and action plan.

### AVPAA & VPAA Review and Feedback

The Vice President of Academic Affairs and Assistant Vice President of Academic Affairs will provide feedback and approval on the completed review and action plan. VPAA and AVPAA should consider institutional planning and budgetary details when providing their review.

## PRESENTATION FEEDBACK

The APR will be presented to a group of peers. Feedback retrieved from the presentation will be summarized and included.

# Action Plan Status Reports

## YEAR 2 ACTION PLAN REVIEW

The reviewer should review progress on action plan over the past year. Discussion should include challenges and successes. If any changes are planned due to the results, those should be discussed and an updated action plan should be included in the narrative or attached as evidence.

**Narrative:**

**YEAR 2 ACTION PLAN REVIEW**

The reviewer should review progress on action plan over the past year. Discussion should include challenges and successes. If any changes are planned due to the results, those should be discussed, and an updated action plan should be included in the narrative or attached as evidence.

| YEAR 2 | | | | | |
|---|---|---|---|---|---|
| | Initiative | Achieved Milestones | Challenges | Changes if applicable | Progress Toward Performance Indicator(s) |
| 1 | Revise learning objectives to include higher levels of Bloom's Taxonomy for ALHS 2312 | A new edition of the textbook has been adopted and the SLO's have been reviewed. | This course does not have a lead instructor who is tasked with this initiative. A faculty member is needed to complete this project. | No changes | In process |
| 2 | For the AHC 115 course, collaborate with health sciences program directors to create short videos discussing physiological systems into eLearn shells. The goal is to introduce a different way to familiarize students to medical terminology from the health care professional perspective. | The medical terminology coordinator has created a list of topics based on the course schedule. | Not having a coordinator has created challenges. Now, we have one and she will lead this project. | No changes | In process |
| 3 | Collaborate with RASI in the solicitation of completed graduate and employer surveys. | The process for survey dissemination has been reviewed. | Lack of adequate staffing in RASI has delayed this initiative. | No changes | In process |
| 4 | Assist in the development of marketing efforts targeting under-represented demographics in Fall 2022. Promote the HS AAS options degree to persons who | This milestone was not met by Fall of 2022. Currently, the marketing | Lack of adequate staffing in Marketing/PR to assist with this initiative. | No changes | In process. An example of a proposed banner reflecting multiple |

| | | | | | |
|---|---|---|---|---|---|
| | complete one of the technical certificates in the spring. | team has created videos and collected images with multiple populations. | | | populations is saved in the evidence section. |

## Year 1 Academic Program Review - Two Year Recommendation Action Plan

- **Student Learning Recommendation:** Assess components of the following professional core courses for the HS AAS degree in the Spring of 22:
  - ***ALHS 2312 –***Contemporary Issues to Healthcare
    - AY 2022-2023- Revise learning objectives to include higher levels of Bloom's Taxonomy.
    - AY 2023-2024- Review the outcomes associated with the new textbook edition and the faculty with complete revisions by the Fall of 2024.
  - ***AHC 115 –*** Medical Terminology
    - AY 2022-2023- Collaborate with health sciences program directors to create short videos discussing physiological systems into eLearn shells. The goal is to introduce a different way to familiarize students with medical terminology from the healthcare professional perspective.
    - AY 2023-2024- Assign topics to the directors. Collect completed videos and add them to the eLearn shells in the Fall of 2024. At least 75% of all programs will be represented.
- **Student Success Recommendation**: Improve Graduate Satisfaction Survey response rates.
  - ***AY 2022-2023-*** Collaborate with IERPA in soliciting completed graduate and employer surveys.
  - ***AY 2023-2024-*** The dean will assess survey submission rates. Contact graduates and employers to increase the obtainment of feedback.
- **Student Success Recommendation**: Improve recruitment efforts of the HS AAS degree.
  - ***AY 2022-2023-*** Assist in developing marketing efforts targeting under-represented demographics in Fall 2022. Promote the HS AAS options degree to persons who complete one of the technical certificates in the spring.
  - ***AY2023-2024-***. Purchase and display program literature and banners that reflect multiple populations by the Fall of 2024.

**Evidence:**

- 2024VS_11x21-TableBanner_HealthSci_FINAL

## YEAR 3 ACTION PLAN REVIEW

The reviewer should review progress on action plan over the past year. Discussion should include challenges and successes. The reviewer should discuss the actions future and continuous in the program.

**Narrative:**

**YEAR 3 ACTION PLAN REVIEW**

**The reviewer should review progress on action plan over the past year. Discussion should include challenges and successes. The reviewer should discuss the actions future and continuous in the program.**

Narrative:
The first initiative was focused on revising the learning objectives for ALHS 2312 to align with higher-order thinking skills from Bloom's Taxonomy. Although a new textbook was received and an instructor was assigned to lead the revision process, progress was slower than expected due to the instructor's limited time and the course falling outside of her regular workload. As a result, the goal of having 70% of Student Learning Outcomes (SLOs) written at or above the "Understand" level of Bloom's Taxonomy wasn't achieved. This initiative remains important and will continue, as the goal is to enhance critical thinking skills.

The second initiative aimed to enhance the AHC 115 course by embedding short instructional videos created by Health Sciences program directors. These videos are meant to introduce physiological systems from the perspective of healthcare professionals to help students better understand medical terminology. So far, one program has submitted a video that has been successfully integrated into the AHC 115 eLearn shell, but additional submissions were delayed due to time constraints. The original goal of 50% program participation has not been met yet. However, this initiative should continue as it will be another way to engage students with different learning styles.

The third initiative has been met. It involved collaboration with RASI to improve the process of soliciting graduate and employer survey responses. The survey plan was explained to all directors.

The fourth initiative was aimed at developing marketing strategies targeted at special populations, specifically to promote the Human Services AAS Options degree to students completing technical certificates. Although new staff were hired in the Office of Marketing and Public Relations to support this initiative, delays occurred due to earlier staffing shortages. A strategy meeting is scheduled for May 2025 to discuss the next steps. This initiative remains a priority, as it plays an important role in increasing enrollment and improving access for special populations.

In summary, while only one initiative has been fully completed, individuals are positioned to reach the goals of the remaining initiatives which have potential for successful completion with continued effort and institutional support. Each of these initiatives contributes to success. However, it's important to note that initiatives without clear ownership are unlikely to be prioritized, given the many other competing demands. Assigning responsibility to specific individuals or units will be key to ensuring successful completion moving forward.

| YEAR 3 | | | | | |
|---|---|---|---|---|---|
| | Initiative | Achieved Milestones | Challenges | Performance Indicator Result | Initiative Achieved? Yes/No |
| 1 | Revise the learning objectives for ALHS 2312 to incorporate higher-order thinking skills by aligning them with the upper | A new textbook was received at the start of Spring 2025. An instructor has been assigned | The instructor lacked dedicated time during the semester to complete the | At least 70% of Student Learning Outcomes should be written at or | No |

| | | | | | |
|---|---|---|---|---|---|
| | levels of Bloom’s Taxonomy. | to revise the objectives and consult with the course developer at RSCC, as this is a common core course. Work on this initiative is ongoing. | revisions, and the course is not part of her regular teaching responsibilities. | above the "Understand" level of Bloom’s Taxonomy. | |
| 2 | The Health Sciences directors will develop brief lecture videos into AHC 115 eLearn shells via the coordinator. These videos aim to present physiological systems from a healthcare professional’s perspective to enhance student understanding of medical terminology. | A video was submitted by the respiratory care director. It has been embedded into the AHC 115 eLearn shell. The initiative remains in progress. | While supportive of the initiative, the remaining program directors have not yet allocated time to develop their videos. | 50% of Health Sciences programs will contribute videos for medical terminology topics. | No |
| 3 | Collaborate with RASI to improve the solicitation of graduate and employer survey responses. | The Dean met with RASI staff to refine and implement the data collection process. This initiative is complete. | None reported. | 100% of program directors will be notified of the survey solicitation process. | Yes |
| 4 | Assist in the development of marketing strategies targeting special populations, with a focus on promoting the Human Services AAS Options degree to technical certificate completers. | New staff were hired in the Office of Marketing and Public Relations to support this initiative. The Dean and Marketing/PR Director will meet in Summer 25' to finalize strategies. | The Office was previously understaffed and unable to prioritize this initiative. | 70% of marketing content will include messaging tailored to special populations. | No |

# Academic Program Review Cohort B- Cycle 1

Ophthalmic Technician AAS

## Reflection and Mission

### PREVIOUS ACADEMIC PROGRAM REVIEW

Reflect on the past APR (if any) and respond to any recommendations that were made at that time. The reviewer should list recommendations, actions taken, and discuss implications or challenges resulting from those actions.

**Narrative:**

This is the first Annual Program Review for this program.

### Program Mission Statement

What changes have the program made to the mission statement over the course of this cycle? Why were these changes made? Are any revisions planned?

**Narrative:**

The mission of the Ophthalmic Technician Program at Volunteer State Community College is to prepare students to enter the workforce at an intermediate level for ophthalmic technicians. The program also instills a desire and commitment in students to become proficient not at the minimum level to graduate, but at the highest level of their ability. In addition, the program prepares students to demonstrate their service to others in a caring, knowledgeable, and skillful manner.

No revisions have been made to the program's mission statement.

### Alignment to Institution Mission

How does the mission of the program align with the mission of the institution?

**Narrative:**

Volunteer State Community College Mission Statement
Volunteer State Community College is a public, comprehensive community college offering quality, innovative educational programs, support, and services. Vol State is committed to building partnerships, strengthening internal and external community engagement, and promoting diversity, cultural awareness, and economic development to prepare students for successful careers, university transfer, and meaningful civic participation in a global society.

Graduates from the program will support the college's mission by providing skilled expertise to the ophthalmology community, including private practice settings, hospital-based practice, military eye clinics, and educational practice settings around the world. The program assists both traditional and non-traditional students to achieve a degree in a profession that will be in demand for years to come. In addition, the Ophthalmic Program provides meaningful civic participation through a variety of service learning projects.

## Curriculum and Student Learning Outcomes Assessment

### Curriculum Map

Review the Student Learning Outcomes and course(s) where each one is addressed. For each outcome, provide the assessment method and target level of achievement.

**Narrative:**

The program learning outcomes for the Ophthalmic Program will be assessed in OPHT 2310 Clinical Applications I and OPHT 2230 Clinical Applications III. The program designated the learning outcomes to be assessed in the first and third semester of the program to track the improvement of knowledge and skills from the beginning of the program to the end, just prior to graduation The learning outcomes will assess both the knowledge and skill required for an ophthalmic technician to pass their certification exam and to be considered good employees by the ophthalmology practices that employ them.

The manner in which the learning outcomes will be assessed is through testing of the students in the clinical setting during the first and third semesters. The program's clinical coordinator will observe students in their clinical setting while they perform a patient exam using the knowledge and skills taught to them during the current or previous semesters. In order for a student to be successful in performing a patient exam, students will have to use proper communication skills, document test findings in the proper manner, solve problems, work within their scope of practice, and physically perform the testing with accuracy. Program faculty hope to observe that students are meeting the mentioned requirements at an average level in the first semester and at an above average level in the third semester.

**Evidence:**

- Ophthalmic Technology AAS Curriculum Mapping_ (6)

## EVIDENCE OF STUDENT LEARNING

Provide evidence of student learning and the program’s effectiveness in reaching both program outcomes and course learning outcomes.

Please provide the following:
1. Summary of most recent assessment results.
2. Analysis of trends
3. Ideas for how those results could be used to improve student learning.

**Narrative:**

Program Learning Outcomes:

1. Graduates will utilize and apply their knowledge of ophthalmic technology to solve problems in a clinical setting.
2. Students will demonstrate the knowledge and skills needed to assume positions as entry-level technicians in a variety of eye-care settings.
3. Graduates will perform ophthalmic skills at the technician level with competency and safety as defined by national certification and accreditation standards set forth by JCAHPO and ICA.
4. Graduates of the program will be able to perform patient testing and demonstrate proper documentation in the patient record.
5. Graduates will demonstrate proficiency with the terminology and abbreviations used within the field of ophthalmic technology.
6. Graduates will demonstrate and apply their knowledge of the rules and regulations of the boards and agencies involved in the field of ophthalmic technology.

Assessment of student and program learning outcomes is currently based on data from the Certified Ophthalmic Technician (COT) exam scores and course and graduate surveys. The data from these items will be used to determine if curriculum and programmatic changes made in the fall of 2020 are beneficial and have allowed for student success. In addition, the program will use the results of its assessments in its key assessment courses OPHT2310 and OPHT2230 to determine the effectiveness of its curriculum and teaching methods.

One of the changes made in the fall of 2020 was a change to the program's curriculum, which increased the credit hours in some of the courses that students have historically struggled with the most, and in which past course and graduate survey feedback suggested that credit hours be increased. Because the 2020-2021 academic year was a large departure from the manner in which the courses are normally taught, it is difficult to assess how thoroughly successful these changes were in 2020-2021. The program will use the 2021-2022 academic year to assess Introduction to Ophthalmic Disease, Ophthalmic Optics, Retinoscopy and Refractometry, and Ocular Motility for increased course success and retention of material. Program faculty will look at the scores on the comprehensive final exams and scores on the COT exam to determine if the changes have had the desired effects.

The Certified Ophthalmic Technician (COT) exam is a 2 part test needed by graduates in order to obtain certification. Part 1 of the COT exam is a 200 question multiple choice exam covering 22 content areas. It is a requirement that students take part 1 of the exam prior to graduating from the program. Part 2 of the COT exam is a simulated skill exam. Students have up to 2 years after passing part 1, to take part 2. Both parts of the test must be passed before a graduate can earn their Certified Ophthalmic Technician credentials.

The program has had 100% passage of part 1 of the Certified Ophthalmic Technician exam for the last 2 years. The improved pass rates represent a dramatic increase over the results from 2019 when the program and national pass rates were down significantly due to a considerable change in the certification exam. The class of 2019 had a 44% passage rate on part 1 of the COT exam. The program made adjustments in various courses at that time to improve the passage rates for 2020. Official passage rates for the Class of 2020 are attached to this report. Official passage rates will not be provided for the Class of 2021 until later in 2022, however, students must submit their results from the Pearson VUE testing center, prior to graduation. Therefore, program faculty already have proof that all students in the most recent class, passed part 1 of the COT. The VSCC Ophthalmic Program passage rates are consistently higher than the national average and demonstrate that the program is providing a high equality education to its students. Although the program has enjoyed great passage rates on its most recent examinations, program faculty believe that improvements can always be made.

The exam report submitted by each student upon completion of part 1 of the COT breaks down the scores in each of the 22 content areas. Each year the program director evaluates the results and chooses 1-2 content areas to focus on improving in the coming year. For the 2020-2021 academic year, program faculty concentrated on improving the biometry (ultrasound) COT results. To achieve the goal of improving the scores on this content area, faculty developed extra assignments, brought in guest lecturers with extensive experience in biometry, and utilized webinars on the content areas. The results of the extra effort made in this area was a 15% increase in the biometry content scores from the Class of 2020 to the Class of 2021.

For the 2021-2022 academic year, program faculty have chosen to concentrate on improving the scores on the contact lens section of the COT. Success on the contact lens section of the COT requires students to apply a great deal of the math concepts they learned in various ophthalmic classes in previous semesters. Math tends to be a significant weakness for many students. As a result of the math challenges faced by many students, the program increased the course credit hours in its math course, Ophthalmic Optics, as well as its Retinoscopy/Refractometry course, as it is heavily math based. In conjunction with the curriculum changes, faculty have developed new clinical assignments relating to contact lenses and are working to obtain more clinical experience in this area for students. The goal of the course changes, and additional content activities, is to see the contact lens scores on the COT improve by at least 15% for the Class of 2022.

Starting with the 2021-2022 academic year, the Ophthalmic Program will assess the course and program learning outcomes in its key assessment courses, OPHT 2310 Clinical Applications I and OPHT 2230 Clinical Applications III. The program chose clinical courses for key assessment courses as they require students to utilize and transfer class and lab knowledge and skills across all ophthalmic courses, in order to perform competently as an ophthalmic technician. Program faculty hope to see that all students achieve an "average" score on the first semester assessment and "above average" on the same assessment areas and skills on the 3rd semester assessment. If areas of weakness are determined by assessment data, program faculty will concentrate on strengthening activities to improve assessment outcomes. Assessment Rubrics are provided as evidence.

**Evidence:**

- Class of 2019 COT Passage Rates
- Class of 2020 COT Passage Rates
- Class of 2021 COT Passage Rates
- OPHT 2230 Employee Characteristic SLO Assessment Rubric Rubric Detail 12-15-2021 152306
- OPHT 2230 Knowledge SLO Assessment Rubric Rubric Detail 12-15-2021 152318
- OPHT 2230 Skill Demonstration SLO Assessment Rubric Rubric Detail 12-15-2021 152328
- OPHT 2310 Employee Characteristics SLO Assessment Rubric Rubric Detail 12-15-2021 152341
- OPHT 2310 Knowledge SLO Assessment Rubric Rubric Detail 12-15-2021 152347
- OPHT 2310 Skill Demonstration SLO Assessment Rubric Rubric Detail 12-15-2021 152353

# Qualitative and Quantitative Program Information

## STUDENT, GRADUATE, AND ALUMNI SATISFACTION

Use student and graduate surveys, course evaluations, and any other appropriate evidence needed to explore the qualitative satisfaction results from students and graduates to provide information to the items below.

Discuss student, graduate, alumni perceptions of the program, faculty, career projection, and any other program recommendations given "during this survey".

Summarize any noted themes, trends, areas for improvement, identified strengths, and implications for the program for both students and alumni.

**Narrative:**

**Graduate Survey**
Graduate surveys completed for students graduating in the spring of 2020 yielded positive scores. These surveys are the latest set of surveys available to the program. The average mean score on the 4 main sections of the survey is 4.75 out of 5.00. The score for "program quality" was 5.00 out of 5.00. The lowest section score on the survey was a 4.20 out of 5.00 on the academic advising section. The advising question includes advising for the entirety of the students time at Volunteer State, and not just the time that Ophthalmic Program faculty were assigned as advisors. The program director attempts to meet with every student interested in the Ophthalmic Program, in person to provide a tour of the lab and in-person discussion. The program director will continue to place importance on advising and will engage the division completion advisors to improve the scores on the advising section. The International Council of Accreditation does not have set thresholds for the various sections on required surveys.

The Ophthalmic Program revised its curriculum with an effective start date of fall of 2020. Feedback received from students on previous graduate surveys reflected the students' desire to be taught more information about diseases. Program faculty also noted that the program needed to increase and reinforce topics on ocular and systemic disease, therefore this need was addressed by changing the curriculum. The Introduction to Ophthalmic Disease, Basic Ophthalmic Pharmacology, and Ocular Motility courses were increased by 1 credit hour each. Each of these courses covers topics on ophthalmic and systemic diseases. The goal of the changes was to bolster the education of students in the area of disease. Program faculty will assess the overall score on the Certified Ophthalmic Technician (COT) exam and the comprehensive final exam scores in each course for the effectiveness of the changes.

**Employer Survey**
Employer surveys were conducted on the Class of 2020, however, only 3 surveys were returned. It is difficult to make significant assessments with such a small return. One survey can have a significant impact on the results with a small return. Faculty will work with personnel in the Institutional Effectiveness, Research, Planning, and Assessment (IERPA) office to improve the return results. Faculty sent placement information to IERPA for the Class of 2021 and will stress the importance of survey returns with employers. During the most recent program advisory committee meeting, program faculty discussed the importance of returning the employer survey during its annual advisory committee meeting.

The results of the employer survey yielded a mean average of 4.18 out of 5.00 on the 4 sections surveyed. The feedback received from one employer revealed that the employer would like students to have more knowledge of the retina. As a general practice, the program ensures that all students rotate through a retina specialty clinic during their year in the program. Program faculty believe that increasing the credit hours in the courses in which disease topics are taught, will enhance students' knowledge with regards to the retina. To enhance learning, new clinical activities were developed for specialty areas such as the retina. Providing additional clinic activities will require students to utilize their time effectively and enhance their knowledge when in specialty clinics.

**Student Resource Survey**
The Student Resource Survey was sent to 13 students. Eight students returned the survey for a return rate of 62%. Students rated the program a 4.88/5.00 for the overall quality of the resources supporting the program. Resources include faculty, class, lab, library, equipment, and clinical resources.

The lowest scores on the survey were in the areas of clinical instruction and physician interaction. The pandemic had significant impacts on the healthcare community. Ophthalmology offices were closed for an extended period of time in the spring to summer months of 2020. As a result, when clinics reopened, it resulted in a significant increase in the workloads of doctors and technicians. The increased workload lasted through the fall semester, when more clinic shutdowns were feared. Program faculty work hard to ensure students have the best clinical experience by routinely meeting with students to discuss positive and negative aspects of their ongoing clinical experience, and to solve any identified problems. During the 2020-2021 academic year, faculty utilized a new clinic assignment tool to ensure that students were receiving the practice and supervision that is required for success. If a regular and routine review of the clinic assignment demonstrated that students were not performing the number and type of skills required for competency, the program clinical coordinator was armed with specific data that could be used in discussions with clinical instructors and students.

**Program Resource Survey**
Program resource surveys were completed by advisory board members, VSCC faculty, and the program's various clinical instructors. The overall score across all areas surveyed was 4.67 out of 5.00. The program scored 4.86 out of 5.00 on the classroom, lab, and resources available to students. The clinical resources available to students scored a 4.64 out of 5.00.

The feedback expressed in the survey included:

- Increase the number of clinical rotation sites
    - The program has 12-15 clinical sites available to it each semester. Multiple sites have multiple locations and or take multiple students, therefore, the program has more than

enough sites to meet its maximum student capacity. The program will continue to monitor its need for additional clinic sites.

- Physician Interaction
  - To increase student interaction with physicians, the program arranges guest lectures from ophthalmologists in the community. In addition, program faculty consistently stress the importance of physician interaction during its annual advisory committee meeting. Students are also routinely encouraged to seek instruction and knowledge from physicians while in clinic. When able, students are encouraged to follow the physician's scribe (and scribe themselves after they learn this skill) to allow them to be part of the exam process for which the physician is involved.

**Evidence:**

- 2018 Graduate surveys
- 2019 Graduate surveys
- 2020 Employer Survey
- 2020 Graduate surveys
- Program Personnel Resource Surveys
- Student Resource Survey

## RECRUITMENT, ENROLLMENT, RETENTION, AND GRADUATION: FACT BOOK DATA

Attach screenshots from the Fact Book of the areas below.

1. Headcounts
2. FTE Comparison
3. Student Primary Location
4. Retention Statistics
5. Graduation Statistics
6. Demographics

Using the Fact Book data listed below, analyze the follow:

Recruitment
Enrollment
Retention
Graduation

The reviewer could discuss marketing and recruitment efforts in the community and local high schools; analyze enrollment, retention, and graduation trends; and discuss potential reasons, strengths, plans for improvement, and any other implications deemed appropriate.

**Narrative:**

Enrollment and retention have been a challenge for the program for several years. Beginning with the 2016 cohort, which became the class of 2017, college administration required the program to change its start date from fall to a summer. This change was made with the thought that TN Promise would not cover the program if it didn't start its cohort in the summer. Program faculty voiced serious concern about this change, to no avail. As predicted and voiced, the change was detrimental to the program in a few ways.

By changing the start date, students had only 2 semesters in which to complete their prerequisite classes. Not only was the program required to move up its start date, shortening the length of time that students had to complete prerequisite courses, it was also told to add prerequisite courses to make it full-time for 5 semesters. Over the years, many students used the summer before the fall cohort began, to complete 1 or 2 remaining prerequisites. This was no longer an option with the program starting in the summer. Many students were prevented from applying to the program because required prerequisite courses like Anatomy and Physiology I, were full before students could register in the fall. With only the fall and spring semesters available to take BIOL 2010 and 2020 before application for a summer start, something as simple as not being able to register for BIOL 2010 in the fall can put a student a year behind. The program lost several applicants every year because they did not have 1 or 2 of their prerequisites completed. Students were then forced to wait an entire year before applying to the program. Based on experience, if a student is not accepted into a program, and therefore must wait a year to reapply, the likelihood of them reapplying decreases significantly.

Once the program start date was changed to the summer, it could no longer be used as a back-up option for the students who did not get accepted into the 2 largest health science programs on campus, Physical Therapist Assistant (PTA) and Radiologic Technology (RADT). The PTA program starts in the summer and the RADT program starts in the fall, therefore these students were no longer able to apply to the OPH Program as a second option, when not accepted into these programs. Before the semester start change, the program accepted several students each year who chose the Ophthalmic Program as a back-up option if not admitted into PTA or RADT. The VSCC Factbook only provides information dating back to 2016, therefore a table titled "Ophthalmic Headcount" was formulated to demonstrate the impact that the semester change had on program enrollment and retention. The table can be viewed by looking at the document attached to this section.

Retention became an issue when the start date changed from fall to summer for a couple of reasons. First, the program was forced to accept students with lower GPAs to ensure the program enrolled enough students. Students who are not academically strong generally struggle with health science programs. Program faculty were able to achieve success with some academically weaker students, but others were not able to maintain the grades and rigors of the program, even with extra intervention. The second reason for retention struggles after the semester start change is because at the same time the program start date changed, the program's accrediting body made changes. The accreditation change required the program to increase its clinical hour requirements from 760 hours to 960 hours. This significant addition of clinical hours, along with a shorter summer semester, meant that students would have to be in class and clinic almost all day, 5 days per week, for the entire summer. During the years in which the program changes were in affect, students who struggled academically or financially quit during or after the first semester at higher rates. A student is much more likely to quit in or after their first semester than their second or third, as they are far more invested the longer they remain in a program. Student retention would have been far better if this difficult summer semester was the third and final semester of the program, versus the first semester. If the program had remained a fall start with the increased clinical hours, the schedule would have still been full and demanding but students would have had some days off and periodic breaks (Labor Day, Fall Break, Thanksgiving) to help.

As of the Class of 2020, the program has gone back to a fall start and has been able to decrease its clinical hour requirements to 750, due to new accreditation changes. These changes resulted in a class size similar to that prior to the semester start change.

Program marketing is extremely important for the Ophthalmic Program. The Ophthalmic Program has a couple of unique challenges not faced by other health science programs. One challenge of the program is that it is the only Ophthalmic Program in the state. Students often search various college and university websites looking for degrees and professions. Students are more familiar with programs that show up on multiple websites and unfamiliar with programs that are less common or easy to find. Another unique challenge that the program has that other programs do not, is the lack of understanding of the profession. Many people have never had an eye exam and are therefore not familiar with eye conditions. Students who have had an eye exam are familiar with optometry, not ophthalmology. While these 2 professions both care for the eyes, the technicians that work in each, have very different roles and responsibilities and this concept is not understood.

Ophthalmic technicians must learn photography, ultrasound, and surgical assisting, amongst many other skills. Optometry technicians will not perform any of these skills. Most people don't understand that there is a difference. To combat these challenges, program faculty spend a great deal of time with marketing and education. For many years, program faculty have hosted medical terminology students in the ophthalmic lab to teach them about eye terminology with hands-on activities. Many ophthalmic students have come to the program as a result of attending one of these sessions and getting to see and work with the equipment. In addition, program faculty visit anatomy and physiology classes to educate students on the program and profession.

In addition to marketing to VSCC students, program faculty also market the program to high school students in various ways. Faculty participate annually in between 5-10 high school visits per year. High school visits give faculty an opportunity to discuss the profession of ophthalmic technology while providing hands-on ophthalmic learning activities, led by ophthalmic students. The visits not only benefit the program as a marketing tool, but also benefit the students in allowing them to demonstrate their knowledge and grow their communication skills as they educate the high school students. In addition to the regular high school visits, program faculty have provided ophthalmic workshops at the HOSA (Health Occupations Student Association) competition held on campus each spring. Program faculty have also visited several local high schools to teach the ophthalmic content contained within their health science classes. These various activities help program faculty create awareness of this unique program and profession.

The program creates the majority of its awareness through both the campus and high school events discussed. As a result of program faculty not being able to participate in these events during 2020-2021, applications to the program were down this year. Program faculty have restarted their pre-COVID marketing strategies to increase applications for the 2022-2023 school year.

**Evidence:**

- Program Statistics-grad, retention

## Supporting Program Information

### FACULTY USAGE AND DEVELOPMENT (OPTIONAL)

Review faculty teaching loads, adjunct to full time ratios, faculty development opportunities, faculty end of course evaluations, faculty surveys, and any other evidence deemed appropriate. The reviewer could discuss the impact of teaching, the ratio of adjunct to full time staffing, end of course evaluations relative to faculty, and any noted themes identified from the review.

### CLASS OFFERINGS (OPTIONAL)

Address scheduling effectiveness and space utilization. The reviewer could review faculty to student ratios at each location and online and discuss the impact relative to faculty and student success.

### CO-CURRICULAR ACTIVITIES (OPTIONAL)

Discuss co-curricular activities available for students in the program. These might include clubs, internships or other Work Based Learning opportunities, service learning, guest speakers, or any other activities designed to enhance student experience in the program.

**Narrative:**

The Ophthalmic Program incorporates several high impact practices (HIP) and co-curricular activities into its curriculum. The biggest HIP embedded in the Ophthalmic Program is its partnership with the Hendersonville Rotary Club and the shared medical mission in Guatemala. Ophthalmic students and the ophthalmic program director have partnered with the Hendersonville Rotary Club for approximately 13 years. The project requires that the entire cohort of students obtain donated glasses, measure the prescriptions, and organize the prescriptions so they may be used in the eye clinic in Guatemala. Once the glasses have been prepared, they are taken to Guatemala where 5-6 of the program students will spend 4 days working in an eye clinic. Students and faculty service between 500-700 patients in the 4 day clinic. Not only do the students get to use the skills they are acquiring during their time in the program, they gain knowledge of Rotary as an international charitable organization, gain an understanding of the challenges in a third world country, and strengthen their critical thinking skills.

In the past, program faculty analyzed the scores on the COT exam for those students participating in the entire Guatemala project versus those that helped ready the glasses but did not travel to Guatemala. Faculty found that scores in key areas of the COT were consistently higher for those students working in the Guatemala clinic due to the great deal of clinical practice and the requirement of critical thinking skills as they applied to the clinical tasks. In addition to the ophthalmic benefits gained from this project, students have gained cultural and global awareness, that has led several graduates to participate in medical mission projects after leaving the program.

Another HIP incorporated into the Ophthalmic Program is the integration of program students with the variety of students visiting the VSCC campus. Many high school students, as well as some VSCC science students, visit the ophthalmic lab to learn about the eye. The program includes its students in this process to provide an opportunity for students to practice and reinforce their clinical and communication skills as they teach the visiting students about the eye equipment and various ocular conditions. These opportunities also serve as great marketing tools, as visiting students feel comfortable asking program students many questions.

In addition to the program's HIP opportunities, students do work-based learning. Students must obtain a minimum of 750 hours in various ophthalmology specialty clinics, prior to graduation. Work based learning is an invaluable piece of the student's education and allows them to reinforce the concepts and skills taught in class and lab.

The program uses guest lecturers in courses within the program. Program faculty are responsible for teaching students how to use approximately 60 pieces of equipment and perform approximately 80-100 different tests spanning all ophthalmology specialties. As a result of the vast equipment used in the profession, and the continual change and development of new equipment, program faculty utilize guest lecturers to provide the high level of didactic education needed for skill development on certain pieces of equipment. Many of the specialty equipment guest lecturers are graduates of the program. Program faculty believe that using program graduates to teach material, encourages current students and demonstrates that there are rewards for hard work while in the program and after graduation. In addition to the utilization of program graduates as specialty guest lecturers, the program is fortunate to have several ophthalmologists, covering various specialties, provide free guest lectures for students.

It is the belief of program faculty that the various activities listed in this document create well-rounded, highly educated, and skillful technicians.

## INDUSTRY TRENDS AND INNOVATIONS (OPTIONAL)

Examine the program demand within the labor market including local, regional, and/or national. The reviewer could examine information from industry advisory boards and information gathered from creditable industry sources as well as from external accrediting bodies. The reviewer could also discuss any existing or emerging trends within the field as they relate to your program and employment of your student's post-graduation as well as the outlook for the field in the next 5 years. This summary could also identify best practices, trends, and innovation in the field and discuss implications for the program.

**Narrative:**

There is a great need for trained ophthalmic technicians in Tennessee and across the U.S.. A discussion takes place at every Ophthalmic Advisory Committee meeting regarding the need for trained technicians. Many of the program's clinic sites hire multiple graduates each year. Frequently, students are offered jobs during their first semester clinical rotation, as clinics try to ensure that they can hire knowledgeable and skilled technicians. Vanderbilt Eye Institute started its own training program this year due to the lack of trained technicians. Program faculty get requests consistently from all over the country, from clinics wanting to hire graduates.

Tennessee, especially Middle Tennessee, employs one of the higher numbers of ophthalmic technicians in the country. See attached document. The job forecast for ophthalmic technicians is projected to grow by 14% between 2020-2030 per CareerOneStop.org. With the U.S. population ageing, the number of people suffering from eye disease will continue to grow. In addition, around 700,000 people in the U.S. undergo elective corrective eye surgeries like LASIK, each year. The large numbers of people with significant eye disease means that ophthalmologists see anywhere from 30-60 patients per day. The only way to efficiently care for such a large number of patients is to have competent and skilled ophthalmic technicians as part of the eye team.

**Evidence:**

- Labor Market Information

### EMPLOYMENT AND TRANSFER FORECASTS (OPTIONAL)

Discuss where students go after graduation and what is being done to enhance the likelihood of success or assist students in reaching those goals. The reviewer could discuss employment outlook, obtainment, curriculum alignment for transfers, and the employment demands of the industries where students will work.

## Recommendations Action Plan, Review, and Feedback

### SUMMARY AND ANALYSIS

The reviewer should discuss what went right, what went wrong, and the implications of those results.

**Narrative:**

The area that the program will concentrate on most heavily in the next few years is enrollment and retention. The program will work to return to its normal enrollment with the marketing efforts discussed in the annual program review. In addition, the program will track its retention as a result of the curriculum changes that moved the start date back to the fall and lowered the clinical hour requirements by 200 hours. In addition, course outcomes will be monitored for those courses that increased their credit hours. This year program faculty will specifically evaluate Introduction to Ophthalmic Disease, Ocular Motility, Retinoscopy and Refractometry, and Ophthalmic Optics courses for improved knowledge and retention. The effectiveness of the changes to these courses will be monitored using comprehensive final exam scores, assessing the distribution of course grades, and evaluating the success of students on the COT exam.

Program faculty will continue to use the most recent results of the COT exam to strengthen areas within the curriculum, until the program undergoes the assessment of its course and program learning outcomes in the Summer of 2022. The program will create additional lab, class, and clinical activities in content areas on the COT exam, which demonstrated lower scores. In addition, the program will utilize the results of the program and student learning outcomes assessment finalized in August 2022, to further improve student learning outcomes.

## TWO YEAR RECOMMENDATION ACTION PLAN

Discuss the next steps for the program by providing a program action plan for the next two academic years. Recommendations must include at least two actions focused on student learning and one action focused on student success. The program action plan must include a timeline for each recommendation.

Feedback and approval will be provided. Refer to the APR timeline for approval deadlines.

**Narrative:**

**Student Learning Action Plan**
1. Program faculty will assess the effectiveness of changes to its curriculum. Program faculty will evaluate 2 factors in 4 of the courses that underwent curriculum changes in the Fall of 2020. The 4 courses that will be evaluated for better student learning outcomes are, Introduction to Ophthalmic Disease, Ocular Motility, Retinoscopy and Refractometry, and Ophthalmic Optics. The 2 factors that will be assessed in the individual courses to determine student success are:

- Grades on comprehensive final exams: Each course requires students to take a comprehensive final exam covering all topics taught during the semester. By evaluating this exam, program faculty will determine if the increase in class and instructor time created improvement in learning and retention of knowledge. The evaluation will take place during the 2021-2022 academic year at the end of the course.
- Passage rates on the COT as a whole and individual content areas: Faculty will evaluate the passage rate on the COT exam to determine if the curriculum changes are maintaining high passage rates. In addition, faculty will evaluate the various content area scores for success on content related to the listed courses. The effectiveness of the curriculum changes in the 4 courses listed above on the COT exam will take place at the end of August 2022 after the students have completed and reported the scores for part 1 of the COT exam.

In addition to assessing the above mentioned individual courses for their outcomes, the program will assess the overall effectiveness of the curriculum changes by assessing course and program learning outcomes in its key assessment courses. If students are benefiting from the curriculum changes across all courses that were changed, the course and program learning outcomes assessed clinically should reflect expected scores.

- Course learning outcomes 1 and 2 will be assessed clinically to determine if students are meeting the outcome goals of transferring their knowledge and skills from the classroom and lab to the clinical setting. Faculty will look to see that students score as "average" during the first semester and "above average" in their third semester.
- Program outcomes 1, 4, and 5 will be assessed in OPHT 2230 to determine if the curriculum changes have been effective. Faculty will be looking to determine if students scored "above average" on the above mentioned program outcomes in OPHT 2230.

2. Program faculty will assess the contact lens section of the COT exam for improvement in contact lens scores. Program faculty have developed new clinical activities to aid students in using their contact lens knowledge and to push them to use their clinical time wisely to improve their knowledge and skills in this area. The 2022 scores on the content lens section of the COT exam will be assessed for improved scores over previous years.

**Student Success Action Plan**
The Ophthalmic Program will concentrate over the next 2 years on increasing enrollment and retention. . In order for the program to be successful in the area of retention, it must attract many students so it can enroll a cohort of students who are academically strong enough to handle the challenges of a full didactic course load, 750 clinical hours, and balance life and financial challenges. The program will return to its normal marketing events to increase enrollment of academically ready and capable students and improve retention of those enrolled students. Program faculty will interact with high school visitors, meet with health science teachers from the various high schools in Sumner County, participate in Health Science Discovery Days, and will visit various medical terminology and anatomy and physiology classes on the VSCC campus. The program will determine the success of this action plan with application and enrollment numbers for the next cohort of students, starting the program in fall of 2022. Based on its marketing efforts, the program will continue with its strategies for the 2022-2023 school year, while exploring alternative marketing strategies if needed.

The program will also work with IERPA to improve it Employer Survey return results.

### DEAN(S) APPROVAL AND FEEDBACK.

The Dean(s) will provide feedback and approval on the completed review and action plan.

**Narrative:**

A comprehensive review of the Ophthalmic APR has been conducted. Dr. Cornish has a thorough understanding of the strengths and weaknesses of the program. She has a solid plan for continuous improvement and assessment of outcomes. The College will collaborate with program faculty with efforts to increase exposure and enrollment of the program.

### AVPAA & VPAA Review and Feedback

The Vice President of Academic Affairs and Assistant Vice President of Academic Affairs will provide feedback and approval on the completed review and action plan. VPAA and AVPAA should consider institutional planning and budgetary details when providing their review.

**Narrative:**

The Office of Institutional Effectiveness and Assessment has reviewed this report and will begin supporting the report champion in preparing for the presentation of this review to the Institutional Effectiveness Committee.

### PRESENTATION FEEDBACK

The APR will be presented to a group of peers. Feedback retrieved from the presentation will be summarized and included.

## Action Plan Status Reports

### YEAR 2 ACTION PLAN REVIEW

The reviewer should review progress on action plan over the past year. Discussion should include challenges and successes. If any changes are planned due to the results, those should be discussed and an updated action plan should be included in the narrative or attached as evidence.

**Narrative:**

| YEAR 2 | | | | | |
|---|---|---|---|---|---|
| | Initiative | Achieved Milestones | Challenges | Changes if applicable | Progress Toward Performance Indicator(s) |
| 1 | Program faculty will assess the effectiveness of changes to its curriculum. PLO 2 | VIA Assessments were developed and placed into eLearn for future assessment of curriculum and program initiatives. | The heavy and intensive didactic, lab, and clinical schedule is a challenge for many students who must also work and or take care of a family. These challenges play a significant role in students' success despite curriculum changes.<br><br>VIA Assessments were not in place for the 2021-2022 academic year. | Consideration being given to model course development for online and hybrid delivery to help students with time management.<br><br>Use Employer Satisfaction Survey results for assessing PLOs against curriculum changes. | • 3 of the 4 courses showed an increase in the comprehensive final exam scores over the previous 2 cohorts using the old curriculum. OPHT 2327 demonstrated a significant increase in scores. One of the 4 courses showed test scores slightly lower than the previous 2 cohorts on the old curriculum.<br>• 6 out of 7 students passed the multiple-choice certification exam on the first attempt. One student passed on the second attempt.<br>• Employer Surveys for the Class of 2022 reported a mean score of 4.84/5.00 on the knowledge base section of the survey demonstrating high scores for PLO #1. |

| | | | | | |
|---|---|---|---|---|---|
| | | | | | • Employer Surveys for the Class of 2022 reported a mean score of 4.92/5.00 on the clinical proficiency section of the survey demonstrating high scores for PLO #4. |
| 2 | Program faculty will assess the contact lens section of the certification exam for improvement in contact lens scores. PLO 1 | Contact lens activity developed to ensure students engaged with pertinent concepts while in clinic. | The area of contact lens knowledge and skill is one of the areas in which students have the least exposure clinically. The majority of the program clinic affiliates do not work with contact lenses. It is a constant challenge to ensure students get contact lens exposure. | Continue to develop contact lens activities to utilize in clinic and to offer contact lens opportunities to students to help reinforce didactic knowledge. | The mean score on the contact lens section of the exam showed a mean score of 5.6 out 6.00 as compared to a national mean score of 3.78 for the contact lens section. Mean score averages on this section from 2017-2020 were 5.0. |
| 3 | The program will increase enrollment and retention. | Faculty began participating in select pre-pandemic marketing events. | The heavy and intensive didactic, lab, and clinical schedule is a challenge for many students who must also work and or take care of a family. These challenges play a significant role in retention rates.<br><br>GPA and BIOL 2020 are | Decrease the GPA for application to the program. Remove BIOL 2020 as a required program and replace with COMM 2045 or 2025.<br><br>Discuss changes with the Ophthalmic Advisory Committee to discuss changes to the | AY 22-23: The program enrolled 10 students and graduated 10. The program therefore had 100% retention. Enrollment needs to increase. |

| | | | | | |
|---|---|---|---|---|---|
| | | | barriers for some students pursuing the program. | program that may alleviate some of the barriers to student enrollment and success. | |

**Evidence:**

- VSCC_HealthSciences_DiscoveryDays SP22

## YEAR 3 ACTION PLAN REVIEW

The reviewer should review progress on action plan over the past year. Discussion should include challenges and successes. The reviewer should discuss the actions future and continuous in the program.

**Narrative:**

| YEAR 3 Progress | | | | | |
|---|---|---|---|---|---|
| | Initiative | Achieved Milestones | Challenges | Performance Indicator Result | Initiative Achieved? Yes/No |
| 1 | Program faculty will assess the effectiveness of changes to its curriculum PLO 2 | The program made curriculum changes in the form of updating the credit hours for 4 courses where the required course content was heavier than the course credit hours supported. | The 2023–2024 cohort, in particular, demonstrated notable gaps in academic readiness and preparation. One of the courses being assessed allows for a student exemption from the final exam if they have an "A" average on all tests leading to the final exam. The number of exemptions can skew the results. | The 2023-2024 cohort scored lower in 3 of 4 courses and higher in 1 course. This cohort scored lower than pre-curriculum change students across almost every course in the program. The 2023-2024 cohort achieved 100% passage on the the multiple choice portion of the Certified Ophthalmic Technician exam. The VSCC Ophthalmic Program graduates achieve higher mean scores in each of the 22 content areas than the means scores achieved by students from all other Ophthalmic Programs in the US and Canada, which also demonstrates the success of the VSCC | Yes |

| | | | | | |
|---|---|---|---|---|---|
| | | | | curriculum.<br><br>Employer Surveys for the Class of 2023 reported a mean score of 4.56/5.00 on the knowledge base section of the survey demonstrating success with curriculum changes and PLO #1.<br><br>Employer Surveys for the Class of 2023 reported a mean score of 4.63/5.00 on the clinical proficiency section of the survey which demonstrates success with curriculum changes as well as PLO #4. | |
| 2 | Program faculty will assess the contact lens section of the certification exam for improvement in contact lens scores. PLO 1 | Program faculty ensured every student had clinical experience with contact lens concepts and students continued to complete clinical activities designed to ensure student engagement with the contact lens concepts they learned. | The topic of contact lens remains an area of ophthalmology where students experience limited clinical engagement. Program faculty are constantly striving to create more educational opportunities for students in this area. | The Certified Ophthalmic Technician Exam Performance Report for 2021-2022, 2022-2023, and 2023-2024 have shown higher mean scores for VSCC students in the area of contact lens as compared to students from all other ophthalmic training programs. | Yes |
| 3 | The program will increase enrollment and retention. | The program retained 100% of its students throughout the 3 semesters of the program. | Lack of marketing. The Ophthalmic Program is the only program of its kind in TN, which leads to under exposure if not marketed. In addition, the | 2021-2022: Retention: 78%, Enrollment: 9 students<br>2022-2023: Retention: 100%, Enrollment 10 students<br>2023-2024: Retention: The program lost 1 student due to insufficient academic | Yes- retention<br>No- enrollment |

| | | | | | |
|---|---|---|---|---|---|
| | | | program struggles with understanding of the profession.<br><br>Full time nature of the program and student financial or personal challenges. | performance.<br>Enrollment: 7 students.<br>The number of Introduction to Ophthalmic Technology students has tripled over the last year, which is expected to lead to an enrollment increase. | |
| 4 | | | | | |

| YEAR 3 Initiative Analysis | | | | |
|---|---|---|---|---|
| | Initiative | How has this initiative supported student learning or student success? | Will there be any further action taken on this initiative? If yes, explain. | How will this initiative inform program decisions in the future? |
| EX. | Increase the achievement of PLO #1 by developing and implementing an online tutorial resource to complement teaching the XYZ assignment in VSCC1234 | | None | The assignment success rate for pilot sections was 80%<br>PLO #1 achievement of pilot sections was 86% |
| 1 | Program faculty will assess the effectiveness of changes to its curriculum | This initiative has allowed students to increase their skill level and competency in OPHT 2326/2326L and 2327/2327L. This initiative has created better alignment between required content and credit hours for OPHT 2221 and 2223. | None | This initiative will support faculty in making needed curriculum changes in the future and will provide a framework for an assessment tool to determine the effects of changes. |

| | | | | |
|---|---|---|---|---|
| 2 | Program faculty will assess the contact lens section of the certification exam for improvement in contact lens scores. | This initiative has ensured that students apply didactic contact lens knowledge in the clinical setting.  Data on the Certified Ophthalmic Technician for the past 3 years shows students scoring higher on this section of the exam as compared to students from all other ophthalmic programs in the US and Canada. | No other formal action will be implemented on this initiative. | This initiative will continue to an indicator of program success.  Program faculty will continue to assess the outcomes in this content area to strengthen its initiatives and to increase student success. |
| 3 | The program will increase enrollment and retention. | This initiative allowed the program director to explore and implement actions within the program that would increase enrollment and retention.  Changes were made to the program prerequisite requirements and entering GPA.  In addition, several model courses were developed to allow course delivery methods that can aid non-traditional students, students with demanding work schedules, and students with families that need flexibility in their schedules. | Yes.  The program will continue to explore opportunities to increase enrollment. | This initiative will aid the program in its course delivery and course scheduling and other enrollment and retention initiatives. |
| 4 | | | | |

**Evidence:**

- OPH Performance Report 2022-2023

# Academic Program Review Cohort B- Cycle 1

Sleep Diagnostics Technology CERT

## Reflection and Mission

### PREVIOUS ACADEMIC PROGRAM REVIEW

Reflect on the past APR (if any) and respond to any recommendations that were made at that time. The reviewer should list recommendations, actions taken, and discuss implications or challenges resulting from those actions.

**Narrative:**

This is the first Academic Program Review.

### Program Mission Statement

What changes have the program made to the mission statement over the course of this cycle? Why were these changes made? Are any revisions planned?

**Narrative:**

The program is designed to prepare the student for employment in a Sleep Disorder Center and to qualify students to sit for the international RPSGT credentialing examination. Link between Institutional Mission and Departmental Mission: 1. Satisfying the community workforce demand for qualified polysomnographic technologists. 2. Providing a pathway for students to enter into a stable, successful, and rewarding career. 3. This certificate program can be supplemented with a general education core to complete an A.A.S. degree and prepare students for the university level.

*No mission statement revisions have been made.*

### Alignment to Institution Mission

How does the mission of the program align with the mission of the institution?

**Narrative:**

Volunteer State Community College is a public, comprehensive community college offering quality, innovative educational programs, support, and services. Vol State is committed to building partnerships, strengthening internal and external community engagement, and promoting diversity, cultural awareness, and economic development to prepare students for successful careers, university transfer, and meaningful civic participation in a global society.

Link between Institutional Mission and Departmental Mission: 1. Satisfying the community workforce demand for qualified polysomnographic technologists. 2. Providing a pathway for students to enter into a stable, successful, and rewarding career. 3. This certificate program can be supplemented with a general education core to complete an A.A.S. degree and prepare students for the university level.

## Curriculum and Student Learning Outcomes Assessment

### Curriculum Map

Review the Student Learning Outcomes and course(s) where each one is addressed. For each outcome, provide the assessment method and target level of achievement.

**Narrative:**

**DESCRIPTION OF THE PROFESSION**

Polysomnographic technologists use sleep technology as part of a team, under the general supervision of a licensed physician, by applying a unique body of knowledge and methodological skills involving the education, evaluation, treatment, and follow-up of sleep disorders in patients of all ages. The polysomnographic technologist performs polysomnography and tests such as the Multiple Sleep Latency Test, Maintenance of Wakefulness Test, Actigraphy, and others used by a physician to diagnose and treat sleep disorders. These tests involve recording, monitoring, and analyzing EEG (electroencephalography), EOG (electrooculography), EMG (electromyography), ECG (electrocardiography), and multiple breathing variables including capnometry and oximetry during sleep and wakefulness. Testing procedures may involve application and adjustment of therapeutic modalities such as supplemental oxygen or positive airway pressure and include application of techniques, equipment, and procedures that are safe, aseptic, preventive, and restorative. Interpretive knowledge is required to recognize and respond to respiratory, cardiac or behavioral events that may occur during testing procedures. Technologists provide supportive services related to the ongoing treatment of sleep-related problems. The professional realm of this support includes patient instruction on the use of devices for the treatment of breathing problems during sleep and helping individuals develop sleeping habits that promote good sleep hygiene. (This description was copied directly from the CoaPSG's Standards and Guidelines for the Accreditation of Educational Programs in Polysomnographic Technology)

**PROGRAM OUTCOMES**

**Program Outcome #1: Students will become competent polysomnographic technologists with training in the fundamentals of polysomnographic technology, laboratory fundamentals, and clinical skills as outlined in, but not limited to, the Association of Polysomnographic Technologists Curriculum Outline.**

This outcome encompasses a large amount of both didactic knowledge and practical skills necessary to function on a professional level in a sleep laboratory as a polysomnographic technician. It is meant to include basic laboratory skills in regard to patient hook-up, equipment functionality and calibration, and the American Academy of Sleep Medicine (AASM) recommended protocol for initiating and running a polysomnogram, multiple sleep latency test, maintenance of wakefulness test, or home sleep apnea testing procedure. Students will receive instruction and guidance on patient interactions, quality control, and infection control. Students' basic laboratory knowledge and skills will be assessed after receiving instruction and guidance from their clinical instructor during the PSG 120C course in the campus sleep laboratory. East TN students will receive this instruction in a clinical affiliate sleep center in that region.

PSG 120C is a hands-on lab course that is designed to expose students to the equipment they will be utilizing on the job as a polysomnographic technologist. Students learn the international 10-20 system of head measurement, proper placement of all sensors to a patient the various devices and equipment used in a sleep lab, the functionality of each, basic equipment calibrations, and safety protocols. Students will also be assessed on their readiness for employment which will include professional dress, punctuality, and the completion of all required clinical documentation for the fall clinical rotations. There are certain skills that are considered essential to the field and some are life skills needed to secure and maintain employment. This course will assess those skills utilizing the attached rubric.

**Program Outcome #2: Clinical Competencies**

**Students will have achieved both academic and clinical competency in cognitive, psychomotor, and affective learning domains using the following organizations as sources of standards: Association of Polysomnographic Technologists, Board of Registered Polysomnographic Technologists, and the American Academy of Sleep Medicine.**

The accrediting body that oversees the structure, format, and success of the Sleep Diagnostics Technology Program at VSCC is the Committee on Accreditation for Polysomnographic Technologist Education (CoaPSG), a division of The Commission on Accreditation of Allied Health Education

Programs (CAAHEP). They set technical standards for any polysomnography education in the four above-mentioned categories, which can be viewed at the following address: https://www.caahep.org/CAAHEP/media/CAAHEP-Documents/StandardsandGuidelinesPSGMay2011rev71921.pdf . Since this is a healthcare program, and patient safety is a concern, there are also minimum student physical capabilities needed to function in a sleep lab. In addition to the practical knowledge to operate laboratory equipment, students would need the physical dexterity required to perform CPR, turn a patient on their side in the case of a seizure, visualize waveforms on screen, and hear a patient's distress call. A physical exam is part of the required clinical documentation, and students are required to be CPR certified in order to attend clinical rotations and qualify to sit for the registry exam after graduation.
Though VSCC does not require the following for entry into the program, many of our clinical affiliate sites will also require that students be vaccinated for certain illnesses and be able to pass a background check and drug screen. The TN polysomnography licensure law also requires the passage of a background check before students can obtain a state license to practice polysomnography.
The attached PSG 120C grading rubric will be utilized to assess this outcome.

**Program Outcome #3: Entry Level Skills**
**Students' accomplishments of their academic and clinical objectives will prepare them with entry-level skills as polysomnographic technicians.**
Some polysomnographic skillsets require experience to master well. However, employers have been increasing expectations of graduates in recent years. Part of this has occurred as a result of the passage of the TN Polysomnography Act which took final effect in the year 2010. It created our state licensure requirement. That state licensure mandated that all polysomnography techs working in TN labs be registered by the Board of Polysomnographic Technologists and be graduates of a CAAHEP accredited polysomnography educational program. Prior to that time, they were free to hire trainees and teach them on the job. This increase in standards has also increased the expectations of employers. Therefore, more attention has been given to tasks that were not previously expected of new graduates. In addition to the skills mentioned in the first two outcomes, students are also expected to be able to satisfactorily analyze, or score patient data and compile that data into a report that the physician will use in diagnosing the patient. The scoring of patient data is an advanced task that incorporates a large portion of the knowledge students have gained throughout the program to carry out accurately. Therefore, that task will be assessed individually using online scoring comparison exams from the American Academy of Sleep Medicine. It is referred to as inter-scorer reliability testing. The exams are purchased and utilized as a means of assessing how students are progressing. This online service sends individualized reports of student success/failure. Students will be compared with the AASM gold-standard scorer and must be within 85% accuracy to receive a passing score. It is worth mentioning that the AASM Gold-standard scorer has been criticized for his inaccuracy, but the AASM has hired four new gold-standard scorers to serve as scoring experts for comparison. It is also worth noting that these exams vary greatly in difficulty, depending on the patient data that is selected for each month. This service is provided online and the AASM forwards the report to VSCC. This service also complements the online format of the program well. Data scoring is also a task that graduates will be required to do as an ongoing quality assurance measure on the job. All scorers are required to pass an inter-scorer reliability exam periodically in order to continue scoring actual patient data.
In addition, students will be assessed on their data analysis skills by utilizing a critical thinking assignment in the PSG 132 Course. Students will be asked to analyze some patient data and narrate their reasoning as to how they arrived at the conclusions they did. The grading rubric for this activity is also attached. Students are graded on their completeness and how well they can support their conclusions with evidence. Feedback is given in the online grading environment, but we also go over this assignment together in our final class meeting.

**Program Outcome #4: Board Preparation**
**Successful completion of the program will help prepare the graduate to take the Board of Registered Polysomnographic Technologists Registry Exam.**

One of the primary goals of the program is to help students prepare students to pass their board exams after graduation. All program courses as well as their CPR certification are part of that preparation. Registry exam passage rates are also an accreditation standard and are included on the program's annual accreditation report, which would also require an action plan if the passage rate ever fell too low.

**Evidence:**

- PSG 132 Data Interpretation Exercise Rubric Detail
- PSG120 Lab Rubric Rubric Detail 11-08-2021 125042 (1)

## EVIDENCE OF STUDENT LEARNING

Provide evidence of student learning and the program's effectiveness in reaching both program outcomes and course learning outcomes.

Please provide the following:
1. Summary of most recent assessment results.
2. Analysis of trends
3. Ideas for how those results could be used to improve student learning.

**Narrative:**

### Summary and Analysis of Outcomes Assessments

- **Program Outcome #1: Students will become competent polysomnographic technologists with training in the fundamentals of polysomnographic technology, laboratory fundamentals, and clinical skills as outlined in, but not limited to, the Association of Polysomnographic Technologists Curriculum Outline.**
- **Program Outcome #2: Students will have achieved both academic and clinical competency in cognitive, psychomotor, and affective learning domains using the following organizations as sources of standards: Association of Polysomnographic Technologists, Board of Registered Polysomnographic Technologists, and the American Academy of Sleep Medicine.**
- **Program Outcome #3: Students' accomplishments of their academic and clinical objectives will prepare them with entry-level skills as polysomnographic technicians.**

Program Outcomes #1, #2, and #3 were measured with the course outcomes below. Measures were chosen because they are the best representations of the skills needed for success in the field after graduation. Therefore, successful results on these are representative of successful fulfillment of program outcomes 1-3.

**1. Demonstrate critical thinking skills in the analysis of recorded sleep data. This goal is assessed via utilizing the data interpretation exercises section in the Registry Review Course-PSG132.**
This goal has been consistently met. However, due to its importance and informative value, this is one that we will continue to monitor.
It is expected that a minimum of 80% of students in this course will average greater than 12 points on the rubric. The goal was achieved again by the 2021 cohort. The median score was 100% with a standard deviation of 6.98%. Since our goal was met. These results show a continued increase in consistency among students on the outcome. We will shoot for a goal of 80% or more of the class earning a minimum of 14/20 on the rubric which computes to a percentage score of 70% or better.
No additional funds are needed.
2018: Average Score = 93.3% ; Median = 95%; Standard deviation = 7.59%
2019 Average Score = 87.45%; Median = 92.5%; Standard Deviation = 13.31
2020 Average Score = 88.28%; Median of 90%; Standard Deviation = 8.1%.
2021 Average Score = 96.53%; Median = 100%; Standard Deviation = 6.98%

**2. Familiarity with Polysomnographic instruments and ancillary equipment. Students need to understand how each piece of equipment works as well as why and when to use it. Assessed during PSG 120C.**

Familiarity with Polysomnographic instruments and ancillary equipment. Students need to understand how each piece of equipment works as well as why and when to use it. This will be assessed via student demonstration of understanding in the laboratory during PSG 120C. This outcome has historically been assessed via utilizing the clinical check-off sheets. Students are required to demonstrate proficiency in the VSCC campus laboratory before continuing on to clinical affiliate sites. 100% success was again achieved for the 2019 -2021 cohorts. We are committed to working with each student at their pace until they each achieve this goal. Any student not achieving a satisfactory result on an initial attempt will be retested after further instruction. Continue to accept nothing less than a satisfactory demonstration by 100% of the students in the cohort. No additional funds are needed. This goal has been consistently met, it will continue to be monitored. However, a new rubric will be utilized for the assessment of this outcome beginning with the 2022 cohort. That rubric is attached.

**3. Students will demonstrate their knowledge of the American Academy of Sleep Medicine (AASM) scoring rules and ability to distinguish among sleep stages and recognize micro-arousals in EEG patterns, different types of scorable respiratory events, Limb movements and PLMs. PSG 131**

This outcome is assessed during the PSG 131 Sleep Scoring course each fall semester. Students will be compared with the AASM gold-standard scorer and must be within 85% accuracy to receive a passing score. Experienced scorers are expected to achieve an 85% match or better when compared to the AASM Gold-standard scorer in order to continue at this task. If our students can achieve this before entering the workforce, it should be considered a great accomplishment. For that reason, we will consider 50% or more of the class scoring over 85% as a successful result. This service is funded via TAF revenue. No additional funds are needed at this time.

2018 - 100% passage - Goal well exceeded
2019 - 24/31 = 77.4% passage - Goal achieved
2020 - 92.3% passage of the exam - goal achieved

Although this goal has been consistently met, it is a job function that will continue to be monitored as an ongoing assessment.

These results are extraordinarily good considering that the national average on the exam among professionals is only an 82.6% agreement on sleep stages alone. That is below the 85% cut-off score needed to pass the test. This means that our students must score better than an average experienced professional to pass the exam. I could not find a value for what percentage score above 85% nationally for comparison. However, a research analysis conducted by Rosenberg & Van Hout (2013) on scoring consistency presented in the Journal of Clinical Sleep Medicine suggested a lower threshold. The conclusion was stated as follows:

"Returning to the issue that prompted the development of the AASM ISR program, center managers may wish to set a relatively low-cut point (such as 75%) that reflects basic stage scoring competence. A consistent score above this threshold could be used as a criterion for entrance to the pool of scorers. It is not recommended that a high threshold, such as 95%, be used as a criterion for promotion or bonuses. Such a criterion would be in excess of the agreement between expert scorers and is likely to occur only as a result of chance." (para. 24)

- **Program Outcome #4: Successful completion of the program will help prepare the graduate to take the Board of Registered Polysomnographic Technologists Registry Exam.**

Program outcome #4 is that the technical certificate will help to prepare students to take the RPSGT exam after graduation. Every course is designed with registry exam in mind. The RPSGT exam has a history of being notoriously difficult in comparison to other allied health fields. The average nationwide passage rate in 2010 was 62% which was a significant improvement over previous years. Therefore the threshold was set at a 65% passage rate. However, The VSCC program has always been above the threshold of results suggested by the CoAPSG. That being said, there is still some room for improvement. The RPSGT Passage rates for VSCC graduates over the past three years are 78%, 82% and 55% respectively so far. Only about half of the 2020 class has taken the exam at this time. The 2020 year was affected by a pandemic-related temporary closing of RPSGT testing sites and some sleep centers. This number will continue to evolve for a while yet.

**Reference**

Rosenberg, R. S., & Van Hout, S. (2013). The American Academy of Sleep Medicine inter-scorer reliability program: sleep stage scoring. *Journal of clinical sleep medicine: JCSM: official publication of the American Academy of Sleep Medicine*, *9*(1), 81–87. https://doi.org/10.5664/jcsm.2350

**Evidence:**

- PSG120 Lab Rubric Rubric Detail 11-08-2021 125042

## Qualitative and Quantitative Program Information

### STUDENT, GRADUATE, AND ALUMNI SATISFACTION

Use student and graduate surveys, course evaluations, and any other appropriate evidence needed to explore the qualitative satisfaction results from students and graduates to provide information to the items below.

Discuss student, graduate, alumni perceptions of the program, faculty, career projection, and any other program recommendations given "during this survey".

Summarize any noted themes, trends, areas for improvement, identified strengths, and implications for the program for both students and alumni.

**Narrative:**

### Historical Perspective

The Sleep Diagnostics Program at VSCC has grown in the number of graduates as well as the amount of geographic coverage area since its beginning. The first cohort in 2004 produced a total of seven graduates. Over the past several years, the cohort size has grown to around 30 students annually.

When our state license law was written, it stipulated that there had to be education coverage in all regions of the state in order to make education a requirement for licensure. At that time, A-step study modules combined with work internships were being used by some to qualify for the board exam. For those other pathways to be denied, the licensure bill said that there must be two CAAHEP accredited polysomnography programs in East TN, one in Middle TN, and one in West TN. ETSU and Roane State Community College had programs in East TN, VSCC covered Middle TN, but there were no polysomnography schools in West TN. The TN Sleep Society President approached our program director to ask if we could expand this program to West TN in order to satisfy the education coverage stipulation in that law. Because of the program's mostly online format, we did that by hiring an adjunct instructor to teach the two clinical courses out of a clinical affiliate lab in that area. The state board of health approved the expansion as the fourth program in TN, and the law took final effect in July of 2010.

Since then, Miller-Motte began a program in Clarksville TN, but it did not survive accreditation issues. ETSU's program also dissolved around the same time of reaccreditation. Then Concorde Technical College in Memphis started a program that is still in operation. Those changes left Roane State, VSCC, and Concord. We were also approached by the COO of the Ballad Health System about filling the gap left by ETSU's program departure. Therefore, we did a similar expansion in the East TN region by teaching the two clinical courses out of a hospital there.

VSCC's expansion not only provided the necessary education coverage to satisfy our state license law, but it continues to provide employees to labs throughout TN and beyond. Since our inception, we have had students attend the Sleep Diagnostics Technology program from 13 different states.

These expansions do bring with them challenges, however. It has been challenging to market in those areas and to grow those sections. The registry exam for the field is also challenging for many. Therefore, the program has to walk a narrow line in regard to student selections. On one hand, we need to provide employee coverage, but on the other hand, we still must meet accreditation standards by only admitting students who show the aptitude for learning the material. Accreditation standards require annual reports of placement rates, graduation rates, registry exam passage rates, graduate and employer survey results, and resource survey results from students and program personnel. Program personnel includes teaching staff, the medical director, and our advisory board.

We are proud that our program has thrived and expanded while many of our competitors have closed due to an inability to maintain accreditation standards.

## Discussion of Survey Data

The Sleep Diagnostics Technology Program has never received a bad overall rating on any survey measure. However, there are a few items that we have addressed over the past few cycles. One of those comments was that the East TN clinical instructor's technical knowledge needed improvement. That instructor became registered via a respiratory care add-on track rather than going through a school of polysomnography, so that was a weak spot in her education. Therefore the program director has been working with her to improve her theoretical knowledge. She was also given access to our summer PSG 110 instrumentation course that should also help her to understand some of the questions her lab students might ask her.

Another comment that has been suggested a few times is that we should move the sleep scoring course to the summer semester instead of waiting for the fall. That has been considered before, and it was determined that it is best to leave it in the fall. The reasoning was that students need to understand how the equipment and sensors function in generating the waveforms on screen before attempting to interpret the data produced. That being said, we have added more scoring training toward the end of the summer PSG 120C course to provide a bit of a head start to the fall scoring class. That was implemented when the summer semester was expanded to a 15-week schedule. The program director will continue to work with the East TN clinical instructor to ensure that this is being done in a uniform way for all clinical students.

One area of current concern is in the return ratio of some of the student questionnaires utilized for the annual accreditation report. Though the overall ratings have always been good, there is a minimum threshold percentage of students who are required to be surveyed. That has fallen a little short on a couple for the past two cohorts. Those two years are the only years since the program's origin that this has occurred. On the accreditation report, we suggested that the department work more closely with IERPA staff to ensure that enough surveys are returned and enough data is collected to satisfy the accrediting body. We explained that the IERPA department has a new V.P. and is undergoing many changes, but that we would continue to work on improving communication with them.

Also, it is important to understand that all students who go through the program courses graduate with a certificate in Sleep Diagnostics Technology. That is regardless of whether they are certificate majors or A.A.S. in Health Sciences Degree majors. Sometimes the survey results separate the two majors. Therefore, when viewing cohort results, it may be necessary to combine the survey results of the certificate majors with the A.A.S. majors to reveal the true number of certificate students. The annual accreditation report includes data from certificate graduates only. All students must graduate with the technical certificate in Sleep Diagnostics Technology because the certificate is the only one of the two that is accredited through the CAAHEP. That is what qualifies students to sit for their registry exam after graduation. This should explain why our graduate numbers are higher than our enrollment numbers for the certificate majors.

**Evidence:**

- 2018 Cohort Employer Survey Results
- 2018 Cohort Graduate Survey Results
- 2019 Cohort Graduation Satisfaction Survey Results
- 2019 Employer Survey Results
- 2020 Graduate Survey Results
- Graduate Surveys Returned
- Program Resource Surveys by Student 2019

## RECRUITMENT, ENROLLMENT, RETENTION, AND GRADUATION: FACT BOOK DATA

Attach screenshots from the Fact Book of the areas below.

1. Headcounts
2. FTE Comparison
3. Student Primary Location
4. Retention Statistics
5. Graduation Statistics
6. Demographics

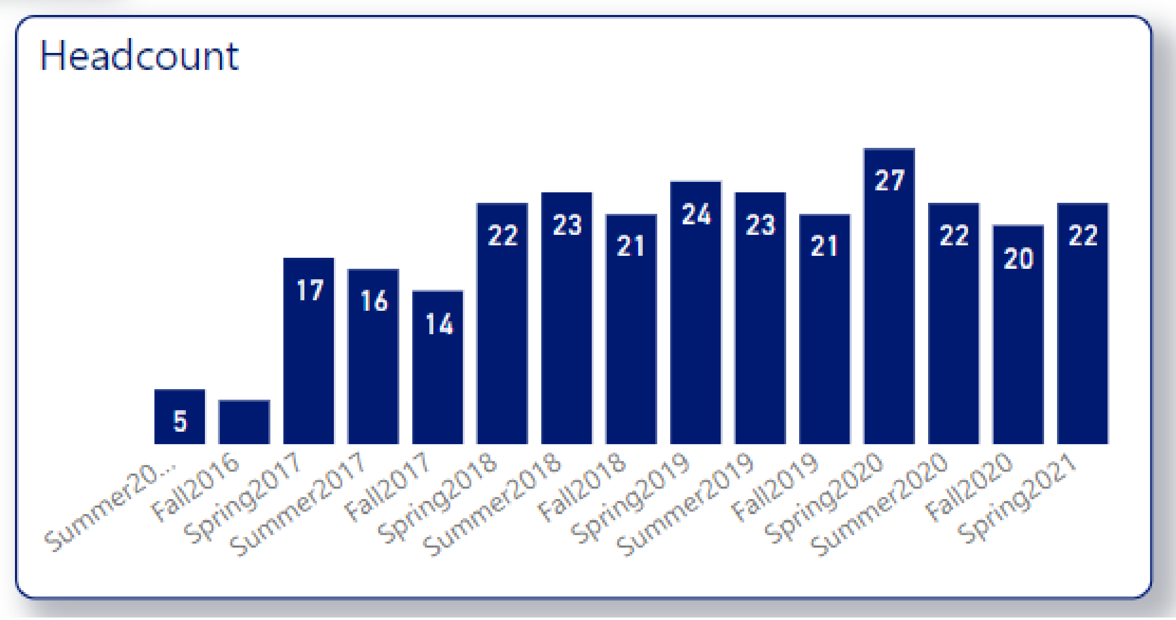
Using the Fact Book data listed below, analyze the follow:

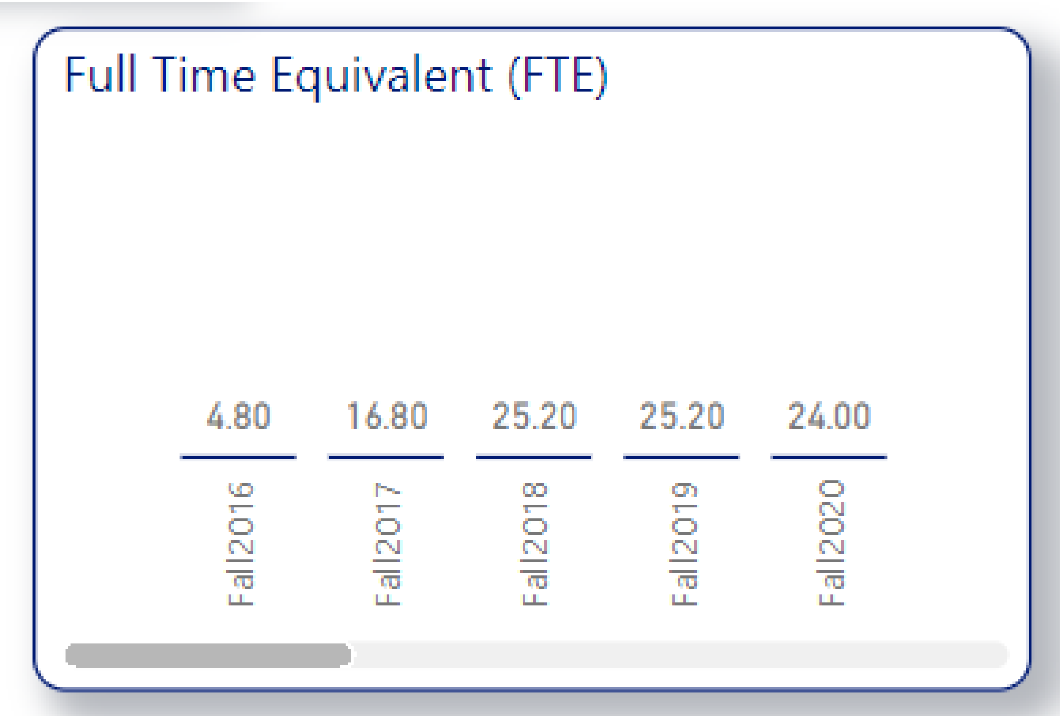
Recruitment
Enrollment
Retention
Graduation

The reviewer could discuss marketing and recruitment efforts in the community and local high schools; analyze enrollment, retention, and graduation trends; and discuss potential reasons, strengths, plans for improvement, and any other implications deemed appropriate.

**Narrative:**

Charts like the first two below are misleading because they only show students with a major in the Sleep Diagnostics Technology Certificate. Again, all Sleep Diagnostics AAS in Health Sciences majors also received the certificate. Those two numbers should be added together for cohort information. For example, the fact book shows a total of 14 students attending in the fall of 2017. However, there were 32 who actually were in the fall 2017 cohort who graduated with a Sleep Diagnostics Technology Certificate. There were 37 enrolled for the spring of 2017. That combined information can be seen in the chart at the bottom of the screen, which is a more accurate representation of the cohort numbers. It is also worth noting that there is no single data chart that can be retrieved from the Fact Book data that includes all students in the cohort or that includes an accurate FTE count for the faculty teaching these program courses.


Headcount
5
17
16
14
22
23
21
24
23
21
27
22
20
22
Summer20...
Fall2016
Spring2017
Summer2017
Fall2017
Spring2018
Summer2018
Fall2018
Spring2019
Summer2019
Fall2019
Spring2020
Summer2020
Fall2020
Spring2021


Full Time Equivalent (FTE)
4.80
16.80
25.20
25.20
24.00
Fall2016
Fall2017
Fall2018
Fall2019
Fall2020

| 2016 - 2020 Retention Average | | | | |
|---|---|---|---|---|
| 152 | 27 | 65% | 82.24% | 83.7% |
| **Total Students** | **Dropped Out** | **Threshold** | **Percentage Retention** | **3-Year Average** |

| **Graduation Year** | **2020** | **2019** | **2018** | **2017** | **2016** |
|---|---|---|---|---|---|
| # Initially Enrolled | 31 | 34 | 27 | 37 | 23 |
| # Added to Class | 0 | 0 | 0 | 0 | 0 |
| Attrition Due to General Education Courses | 0 | 0 | 0 | 0 | 0 |
| Attrition Due to Non-Academic Reasons | 7 | 3 | 3 | 5 | 5 |
| Attrition Due to Professional Courses | 1 | 0 | 1 | 2 | 0 |
| # Dropped Out | 8 | 3 | 4 | 7 | 5 |
| # In-Progress or Stopped Out | 0 | 0 | 0 | 0 | 0 |
| # Graduated | 23 | 31 | 23 | 32 | 16 |
| Retention Percentage | 74.19% | 91.18% | 85.19% | 81.08% | 78.26% |

**Retention Detailed Analysis**

**Retention Action Plan**

The Sleep Diagnostics Program has tripled in enrollment since its inception. This has been partly due to its expansions into other regions of TN and beyond. The online format has aided in our capability to do this. Last year, the Covid-19 pandemic seemed to have a significant impact on retention. We lost several students to illness and other related factors. We have backed away somewhat from the West TN area due to the existence of another program in Memphis. However, the East TN, and Chattanooga area, in particular, has made up for it and seems to be growing in interest. The number of applications submitted for the 2021 cohort was the highest ever, so there seems to be no decline in student interest regardless of the overall slow down in enrollment campus-wide.

The demographics of the program cohorts have shown lots of diversity, and a significant portion of our cohort is consistently made up of non-traditional students.

| Sleep Graduate Demographics | | | | | | |
|---|---|---|---|---|---|---|
| Grad Year | 2020 | 2019 | 2018 | 2017 | 2016 | 2015 |
| # of Grads | 23 | 30 | 23 | 32 | 16 | 23 |
| Male | 3 | 3 | 3 | 7 | 2 | 8 |
| Female | 20 | 27 | 20 | 25 | 14 | 15 |
| Age less than 25 years | 10 | 9 | 7 | 9 | 5 | 8 |
| 25 to 35 years | 10 | 7 | 7 | 13 | 6 | 6 |
| 36 to 45 years | 1 | 7 | 5 | 5 | 3 | 4 |
| 46 or more years | 2 | 7 | 4 | 5 | 2 | 5 |
| Unknown | 0 | 1 | 0 | 0 | 0 | 0 |
| African American | 3 | 4 | 6 | 6 | 4 | 5 |
| American Indian | 0 | 0 | 0 | 0 | 0 | 0 |
| Asian | 1 | 0 | 0 | 1 | 0 | 0 |
| Caucasian | 18 | 25 | 16 | 24 | 12 | 15 |
| Hispanic | 1 | 0 | 1 | 0 | 0 | 1 |
| Other | 0 | 0 | 0 | 1 | 0 | 2 |

# Supporting Program Information

## FACULTY USAGE AND DEVELOPMENT (OPTIONAL)

Review faculty teaching loads, adjunct to full time ratios, faculty development opportunities, faculty end of course evaluations, faculty surveys, and any other evidence deemed appropriate. The reviewer could discuss the impact of teaching, the ratio of adjunct to full time staffing, end of course evaluations relative to faculty, and any noted themes identified from the review.

**Narrative:**

It is important that reviewers of this academic program review understand the nature of the program format. In particular, our summer clinical practice I course is taught as a laboratory course with an instructor. Our campus sleep lab was not designed to hold the entire cohort at once. This was intentional on the part of the program director as a way to limit waste. Most of the program courses are offered via an online format. Because students would not be on campus all year round, it was decided that it would be more efficient to utilize less space. Therefore, due to the small capacity of our campus lab and a desire to provide better service to our students, the cohort is divided into smaller groups for lab training. There are generally three to four separate middle TN groups and an additional East TN group that is taught by an adjunct in the Bristol TN area. In recent years, there has been an increase in enrollment from the Chattanooga and southern part of TN. Although most of those students are technically East TN residents, they generally come to Gallatin for their lab training since it is a shorter commute. The point is that the same instructor may be teaching several groups per week but only getting credit for one because they are registered for the same course.

Secondly, any increase in enrollment numbers compounds the work of clinical instructors exponentially and not only because of having to split them into more groups. More students means much more clinical documentation, more time in locating and scheduling clinical sites, more late night phone calls, and more information to gather for accreditation reports. The sleep diagnostics technology students do clinical rotations on third shift. Local sleep labs are not always open every day of the week, so matching the students' schedules with lab schedules while only sending one student to each lab at a time is an incredible challenge. Most sleep lab control rooms are not large enough to accommodate more than one student at a time. So far, the program has had students from 13 different states. When students register from regions we haven't served before, it also requires much more effort to locate new clinical affiliates in those locations.

Lastly, it is important to understand that the entire cohort's information cannot be retrieved in a single report from the Fact Book data. Our certificate majors are always separated from the A.A.S. in Health Sciences majors even though they are in the same cohort. Those reports are pulled based on student majors. Therefore it can be very misleading to pull a report for Sleep Diagnostics Technology certificate majors. The A.A.S. majors also complete the certificate even though it may not be listed as their current major. That is why our cohort graduate numbers for the certificate will always be larger than the number of certificate majors that can be viewed in any one report. Therefore, cohort information is best gathered from graduate numbers rather than majors.

It is also critical to understand the significance of our East TN and West TN expansions. Those expansions were necessary to satisfy the state polysomnography act that established our state licensure requirements. Without our presence in East TN, the law's requirement of graduation from a CAAHEP accredited program like this one would go away. That would be a huge blow to our enrollment because it would allow applicants for a state license to practice polysomnography to qualify through alternative pathways. Therefore, the East TN presence is crucial. Even though the East TN clinical group is sometimes small in numbers, those students also take the rest of the online courses in the same course sections as the rest of the cohort.

## CLASS OFFERINGS (OPTIONAL)

Address scheduling effectiveness and space utilization. The reviewer could review faculty to student ratios at each location and online and discuss the impact relative to faculty and student success.

**Narrative:**

The Sleep Diagnostics Technology Program is a cohort based program, which accepts a new cohort of students each spring semester. All program courses are required to be taken in succession. Below is a brief summary of the professional program courses included within the Certificate in Sleep Diagnostics Technology. Selection preference is granted to applicants who have completed a medical terminology course. Coursework in Human anatomy and Physiology is also highly encouraged but not required for entry.

**Courses/Semester-by-Semester Sequence**

**Spring Semester: Physical Aspects of Polysomnography**

**PSG 101* Anatomy and Physiology of Sleep Disorders (3)** A study of the Anatomy and Physiology that is specific to sleep disorders medicine. The areas to be covered will include; structure of the nervous system, macro-and micro-anatomy including the upper and lower airway, the respiratory and cardiac system, circadian rhythm biology, sleep architecture and the physiology of REM.
**PSG 102* Classification of Sleep Disorders (3)** A study of the 84 sleep disorders with emphasis on disorders evaluated during a Polysomnographic study. Upon completing the course, a student will have an understanding of the importance of documentation in the sleep center setting by interpretation of patient history and physical assessment.

**Summer Semester: Technical Application of Polysomnography**

**PSG 110* Sleep Polysomnography Instrumentation (3)** A course of study covering three major items: basic electronics, Analog PSG's and Digital PSG's. Emphasis placed on computer systems, equipment calibration, signal pathways, relationships between analog and digital equipment and troubleshooting. PREREQUISITES: PSG 101*, PSG 102* or permission of Program Director.
**PSG 120C Clinical Practice I (4)** Students will receive instruction and clinical supervision on clinical application of electrode placement utilizing the 1020 method, proper patient preparation, montages, protocols, impedance checks, calibrations, troubleshooting, vital sign assessment, proper documentation and note taking. PREREQUISITES: PSG 101*, PSG 102* or permission of Program Director.
*Upon completion of the Summer semester, the student should be able to set-up and run a sleep study with minimal amount of guidance.

**Fall Semester: Data Analysis of Polysomnography**

**PSG 130* Data Management in Polysomnography (4)** A study of the mathematical equations used in a sleep report with focus on sleep report review, sleep documentation and editing and reviewing physician's final report. Also covered is the proper flow of patients into an accredited sleep disorders center. Sleep center management covered; time permitting. PREREQUISITES: PSG 110* and PSG 120C* or permission of Program Director.
**PSG 131* Sleep Scoring (4)** A study of the sleep staging of adults and pediatrics, respiratory scoring, scoring of PLMS, scoring of REM density, scoring of EKG arrhythmias and recognition of atypical EEG patterns. PREREQUISITES: PSG 110* and PSG 120C* or permission of Program Director.
**PSG 132* Sleep Registry Review (4)** This course is designed to prepare the student for the NBRPT examination. It includes test preparation and review. PREREQUISITES: PSG 110* and PSG 120C* or permission of Program Director.
**PSG 133C* Clinical Practice II (6)** Supervised clinical application of electrode placement utilizing the 1020 method, proper patient preparation, montages, protocols, impedance checks, calibrations, trouble-shooting, vital sign assessment, proper documentation and note-taking. PREREQUISITES: PSG 110* and PSG 120C* or permission of program director. 120 clinical hours per term.

## CO-CURRICULAR ACTIVITIES (OPTIONAL)

Discuss co-curricular activities available for students in the program. These might include clubs, internships or other Work Based Learning opportunities, service learning, guest speakers, or any other activities designed to enhance student experience in the program.

### INDUSTRY TRENDS AND INNOVATIONS (OPTIONAL)

Examine the program demand within the labor market including local, regional, and/or national. The reviewer could examine information from industry advisory boards and information gathered from creditable industry sources as well as from external accrediting bodies. The reviewer could also discuss any existing or emerging trends within the field as they relate to your program and employment of your student's post-graduation as well as the outlook for the field in the next 5 years. This summary could also identify best practices, trends, and innovation in the field and discuss implications for the program.

### EMPLOYMENT AND TRANSFER FORECASTS (OPTIONAL)

Discuss where students go after graduation and what is being done to enhance the likelihood of success or assist students in reaching those goals. The reviewer could discuss employment outlook, obtainment, curriculum alignment for transfers, and the employment demands of the industries where students will work.

## Recommendations Action Plan, Review, and Feedback

### SUMMARY AND ANALYSIS

The reviewer should discuss what went right, what went wrong, and the implications of those results.

**Narrative:**

The results of all three course outcome measures have exceeded the goals set so far.

**1. Demonstrate critical thinking skills in the analysis of recorded sleep data.**
This goal has been consistently met. These results show increased consistency among students on the outcome. We will shoot for a goal of 80% or more of the class earning a minimum of 14/20 on the rubric which computes to a percentage score of 70% or better. No additional funds are needed.

**2. Familiarity with Polysomnographic instruments and ancillary equipment. Students need to understand how each piece of equipment works as well as why and when to use it.**
This goal has been consistently met, but it will continue to be monitored. However, a new rubric will be utilized for a more quantitative assessment of this outcome beginning with the 2022 cohort.

**3. Students will demonstrate their knowledge of the American Academy of Sleep Medicine (AASM) scoring rules and ability to distinguish among sleep stages and recognize micro-arousals in EEG patterns, different types of scorable respiratory events, Limb movements, and PLMs.**

So far, the results for this outcome measure have exceeded expectations. Our results have been comparable to those of experienced technologists in the field. Although this goal has been consistently met, it is a job function that will continue to be monitored as an ongoing assessment. This is partly due to the fact that these scoring rules continue to be updated annually. It is also an essential job function that requires critical thinking and the application of a culmination of program knowledge learned. The sleep diagnostics faculty will monitor current guidelines and parameters, and keep courses updated accordingly. No other changes are needed at this time.

**Overall Program Summary Analysis:**
The past three years have been challenging for the Sleep Diagnostics Technology Program. The pandemic caused some sleep clinics to close their doors temporarily. There was also a CPAP machine recall by the largest manufacturer, Philips/Respironics which also caused temporary halting of lab

titrations in some areas. Through it all, students received reasonably normal instruction and were able to complete all clinical rotation requirements. The design of our program was better suited for navigating the challenges presented by the Covid-19 pandemic than most others. However, we did have to get a little creative in performing lab instruction via the Zoom app for the first 6 weeks of the 2020 summer PSG 120 Clinical Lab course that is normally taught in person in our campus sleep lab. Though all three years produced good graduate cohorts, some effects of the pandemic were not immediately apparent. One of those effects was a slight decrease in retention. This was not due to academics but was more related to life circumstances and illnesses. It was also challenging to find clinical affiliates willing to accept students during the pandemic. However, through lots of scheduling creativity and cooperation of clinical affiliates, we were able to accomplish a relatively normal clinical rotation schedule. This was complicated further for the recent 2021 cohort due to some students choosing not to receive the Covid-19 vaccine. However, we were able to find enough clinical affiliate sites willing to take them, and those students were able to complete their rotations as well.
During a time when the overall enrollment numbers for the college are down, the Sleep Diagnostics Technology Program has not experienced a similar effect. In fact, the number of applicants for our 2022 cohort was higher than ever before. We also continue to add new service locations. While this has been challenging for the program director to manage, regarding clinical affiliates, it is a positive trend for the college. The most significant limitation to this growth is the way our out-of-state tuition rates and e-rates are calculated. Currently, the rules are such that an out-of-state student can qualify for the e-rate only if they take all online courses for a semester. If they take one course that requires them to meet somewhere in person (even if it is at one of our clinical affiliate sites), they are charged the out-of-state rate for the entire semester, even if most of those courses are online. The program has reached out to TBR officials regarding this and has received positive comments of support for altering this rule, but it is yet to happen. Those students are taught by the same faculty, so it is just an unnecessary loss of opportunity and revenue for VSCC.

## TWO YEAR RECOMMENDATION ACTION PLAN

Discuss the next steps for the program by providing a program action plan for the next two academic years. Recommendations must include at least two actions focused on student learning and one action focused on student success. The program action plan must include a timeline for each recommendation.

Feedback and approval will be provided. Refer to the APR timeline for approval deadlines.

**Narrative:**

### Current and Future Course Learning Assessments

The following learning outcomes will continue to be assessed. They are representative of a large amount of knowledge needed to complete each activity. Together, they represent a skill set needed to function as a professional polysomnographic technologist.

1. **Instruments and Ancillary Equipment -** Students learn the international 10-20 system of head measurement, proper placement of all sensors to a patient the various devices and equipment used in a sleep lab, the functionality of each, basic equipment calibrations, and safety protocols. Students will also be assessed on their readiness for employment which will include professional dress, punctuality, and the completion of all required clinical documentation for the fall clinical rotations. This outcome will be assessed near the end of the PSG 120C course, which is a hands-on laboratory course that is designed to pair well with the didactic learning in PSG 110. Students will be assessed via the attached rubric.

2. **Problem-solving/ critical thinking improvement -** Students will use critical thinking/problem-solving skills to analyze patient data. Students will be assessed on their data analysis skills by utilizing a critical thinking assignment in the PSG 132 Course. Students will be asked to analyze some patient data and narrate their reasoning as to how they arrived at the conclusions they did. The grading rubric for this activity is also attached. Students are graded on their completeness and how well they can support their conclusions with evidence. Feedback is given in the online grading environment, but we also go over this assignment together in our final class meeting.
3. **Sleep Data Scoring -** Students will apply and demonstrate their knowledge of AASM scoring rules and ability to distinguish among sleep stages and recognize micro-arousals in EEG patterns, different types of scorable respiratory events, Limb movements and PLMs. This task will be assessed individually using online scoring comparison exams from the American Academy of Sleep Medicine as part of the PSG 131 Sleep Scoring course in the fall semester. It is referred to as inter-scorer reliability testing. The exams are purchased and utilized as a means of assessing how students are progressing. This online service sends individualized reports of student success/failure. Students will be compared with the AASM gold-standard scorer and must be within 85% accuracy to receive a passing score. It is worth mentioning that the AASM Gold-standard scorer has been criticized for his inaccuracy, but the AASM has hired four new gold-standard scorers to serve as scoring experts for comparison. It is also worth noting that these exams vary greatly in difficulty, depending on the patient data that is selected for each month. This service is provided online and the AASM forwards the report to VSCC. This service also complements the online format of the program well. Data scoring is also a task that graduates will be required to do as an ongoing quality assurance measure on the job. All scorers are required to pass an inter-scorer reliability exam periodically in order to continue scoring actual patient data.

   Although this goal has been consistently met, it is a job function that will continue to be monitored as an ongoing assessment. These results are extraordinarily good considering that the national average on the exam among professionals is only an 82.6% agreement on sleep stages alone. That is below the 85% cut-off score needed to pass the test. This means that our students must score better than an average experienced professional to pass the exam. I could not find a value for what percentage score above 85% nationally for comparison. However, a research analysis conducted by Rosenberg and Van Hout, S. (2013) on scoring consistency presented in the Journal of Clinical Sleep Medicine suggested a lower threshold. The conclusion was stated as follows:

"Returning to the issue that prompted the development of the AASM ISR program, center managers may wish to set a relatively low cut point (such as 75%) that reflects basic stage scoring competence. A consistent score above this threshold could be used as a criterion for entrance to the pool of scorers. It is not recommended that a high threshold, such as 95%, be used as a criterion for promotion or bonuses. Such a criterion would be in excess of the agreement between expert scorers and is likely to occur only as a result of chance." (para. 24)

Although outcomes #2 and #3 above have been assessed with very good results prior to this action plan, they will continue to be monitored because of the fluid nature of the registry exam and to ensure that we are staying up to date on the AASM guidelines and parameters. Outcome #1 above will utilize a new rubric that does not yet have a baseline set. We will monitor it over the next few years to establish trends for improvement.

**Student Success Planning for Future Cohorts**

1. The Registry exam has added more content on home sleep apnea testing, and that is a task that has become a larger part of the daily testing routine in a sleep clinic. Therefore, we want to add more hands-on practice of that task. To do this, we are purchasing some HSAT testing devices so that students can practice the actual procedure either on themselves or their lab partners. If the supply chain issues allow, this will be added to the curriculum of the summer 120C course for the 2022 cohort. This is recommended by our accrediting body and our program advisory board. It is our hope that more hands-on practice will not only make graduates more proficient at these procedures after starting their new careers, but it could also help to improve their registry exam scores. Existing funds accumulated from the collection of the health sciences fee will be utilized to purchase the necessary equipment.
2. Work with IERPA to create a dashboard that can more easily review program statistics by cohort rather than designated degree, or major. The way the data tables are set up in the Fact Book, it is currently impossible to pull a report summary regarding our entire cohort at the same time. Therefore,

it is especially difficult to convey information to others who might view this data. It gives the impression that our cohort sizes are smaller than they really are.

**Reference**

Rosenberg, R. S., & Van Hout, S. (2013). The American Academy of Sleep Medicine inter-scorer reliability program: sleep stage scoring. *Journal of clinical sleep medicine: JCSM: official publication of the American Academy of Sleep Medicine*, *9*(1), 81–87. https://doi.org/10.5664/jcsm.2350

**Evidence:**

- Email from CoAPSG

### DEAN(S) APPROVAL AND FEEDBACK.

The Dean(s) will provide feedback and approval on the completed review and action plan.

### AVPAA & VPAA Review and Feedback

The Vice President of Academic Affairs and Assistant Vice President of Academic Affairs will provide feedback and approval on the completed review and action plan. VPAA and AVPAA should consider institutional planning and budgetary details when providing their review.

### PRESENTATION FEEDBACK

The APR will be presented to a group of peers. Feedback retrieved from the presentation will be summarized and included.

## Action Plan Status Reports

### YEAR 2 ACTION PLAN REVIEW

The reviewer should review progress on action plan over the past year. Discussion should include challenges and successes. If any changes are planned due to the results, those should be discussed and an updated action plan should be included in the narrative or attached as evidence.

**Narrative:**

| YEAR 2 | | | | | |
|---|---|---|---|---|---|
| | Initiative | Achieved Milestones | Challenges | Changes if applicable | Progress Toward Performance Indicator(s) |
| 1 | Increase student performance on SLR #1 | Increasing the number of examples | The 2022 cohort was a strong group | The goal was achieved among those | The 2022 cohort class average was |

| | | | | | |
|---|---|---|---|---|---|
| | (Demonstrate critical thinking skills in the analysis of recorded sleep data) by spending more time on data interpretation during the summer clinical course. Goal was increased to 80% or more of the class earning a minimum of 14/20 on the rubric which computes to a percentage score of 70% or better. PLO 1 | appeared to have limited effect between the 2022 and 2023 cohorts. However, the goal was well achieved. | with 100% participation and a high scoring average. However, the 2023 cohort included a few students who did not complete the assignment. The ones who completed it did well, however. The main challenge in achieving 100% participation is that students are doing 3rd shift clinical rotations during this time and many are exhausted and have gotten into a routine by the time this assignment is released. This assignment is part of an online course. | completing the assignment. Still, 100% participation is desired. Students were prompted and reminded several times during the semester regarding this assignment. Making the assignment should factor more into the students' overall grade may be worth considering. However, no changes to the timeline are needed at this time. | 91.6% with a SD of 10.11, and the 2023 cohort average was 88.5% with a SD of 10.01. All scores for both cohorts (2022 & 2023) were at or above the set threshold level of three on the rubric. It is worth noting that the 2022 cohort was an exceptionally strong group academically. |
| 2 | Increase Student familiarity with polysomnographic instruments and ancillary equipment. Students need to understand how each piece of equipment works as well as why and when to use it. Assessed during PSG 120C. PLO 2 | A list of knowledge concepts related to boost the electrical concepts and ancillary equipment and a new rubric to measure understanding of students in the PSG 120C course more on a more quantifiable scale were developed. | Challenges in completing this goal are largely based on the fact that there is only one instructor, and the plan was to quiz students orally. This increases the necessary | No changes in teaching methodology needed at this time. However, we need to transition to using the online rubric tool in VIA instead of recording these grades | The 2022 and 2023 cohorts did well on this activity. All students scored at an adequate level on the rubric. |

| | | | | | |
|---|---|---|---|---|---|
| | | Familiarity with Polysomnographic instruments and ancillary equipment. | testing time greatly. Individual attention in the lab to correct or retrain a student is also time-consuming. | on hard copies. | |
| 3 | Students will demonstrate their knowledge of the American Academy of Sleep Medicine (AASM) scoring rules and ability to distinguish among sleep stages and recognize micro-arousals in EEG patterns, different types of scorable respiratory events, Limb movements, and PLMs. PSG 131 PLO 2 | The outcome goal of this assessment has continued to rise due to our students' success. Students have achieved each new milestone. Much effort has been directed at increasing student participation as well as in boosting their performance on the samples. | One area that can be challenging is getting students to begin participation early enough that they can gain the benefit of seeing multiple patients' data. Some seem apprehensive for fear of not doing well. They are also very busy during this semester. | No changes are planned for the upcoming year. However, we plan to continue applying pressure on students to participate early and consistently throughout the second half of the semester. | The students in the 2022 and 2023 cohorts have done very well on this assessment. For 2022, all but two students scored high enough to pass this exam at the professional level. One missed by two points, and the other just failed to participate. For the 2023 cohort, 13 out of 16 participated with 11 of those scoring above the professional threshold. This meets expectations, but 100% participation is desired. |
| 4 | Successful completion of the program will help prepare the graduate to take the Board of Registered | More focus on AASM guidelines and parameters seems to have boosted graduate performance on the exam. Also, | Primary challenges are attracting high caliber students to the program and | RPSGT exam content must be continually monitored and course content | The 2022 cohort currently has a passage rate of 100% from those attempting the |

| | | | | | |
|---|---|---|---|---|---|
| | Polysomnographic Technologists Registry Exam. PLO 4 | library of prep resources has been purchased. | developing the work ethic in students that is required to succeed on the exam. It is also a challenge to keep up with the exam content because it changes every four years. Lastly, there were several very strong students in the 2022 cohort that never attempted the exam. Some chose to either continue their educations or pursue another career. | adjusted in coordination. | exam. The results for the 2023 cohort are not complete at the time of this assessment. However, 4 out of 5, or 80% of those attempting have passed so far. The one who did not pass is expected to retake it shortly. These results are within our expectations and well above the national average. |

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## YEAR 3 ACTION PLAN REVIEW

The reviewer should review progress on action plan over the past year. Discussion should include challenges and successes. The reviewer should discuss the actions future and continuous in the program.

**Narrative:**

| YEAR 3 Progress | | | | | |
|---|---|---|---|---|---|
| | Initiative | Achieved Milestones | Challenges | Performance Indicator Result | Initiative Achieved? Yes/No |
| 1 | Increase student performance on SLR #1 (Demonstrate critical thinking | Increasing the number of examples appeared to have limited effect between the | The 2022 cohort was a strong group with 100% participation | Only one student out of the 3-year assessment period scored | Yes |

| | | | | | |
|---|---|---|---|---|---|
| | skills in the analysis of recorded sleep data) by spending more time on data interpretation during the summer clinical course. Goal was increased to 80% or more of the class earning a minimum of 14/20 on the rubric which computes to a percentage score of 70% or better. | 2022 and 2023 cohorts. However, the goal was well achieved for all three cohorts. | and a high scoring average. However, the 2023 cohort included a few students who did not complete the assignment. The ones who completed it did well, however. The main challenge in achieving 100% participation is that students are doing 3rd shift clinical rotations during this time and many are exhausted and have gotten into a routine by the time this assignment is released. This assignment is part of an online course. | lower than the goal. | |
| 2 | Increase Student familiarity with polysomnographic instruments and ancillary equipment. Students need to understand how each piece of equipment works as well as why and when to use it. Assessed during PSG 120C. PLO 2 | A list of knowledge concepts related to boost the electrical concepts and ancillary equipment and a new rubric to measure understanding of students in the PSG 120C course more on a more quantifiable scale were developed. Familiarity with Polysomnographic instruments and ancillary equipment. | Challenges in completing this goal are largely based on the fact that there is only one instructor, and the plan was to quiz students orally. This increases the necessary testing time greatly. Individual attention in the lab to correct or retrain a student is also | This initiative has worked well. Students have demonstrated more retention of important concepts. However, due to time constraints as well as the value of group preparation, this will become a group grade for future cohorts. | Yes |

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|---|---|---|---|---|---|
| | | | time-consuming. | | |
| 3 | Students will demonstrate their knowledge of the American Academy of Sleep Medicine (AASM) scoring rules and ability to distinguish among sleep stages and recognize micro-arousals in EEG patterns, different types of scorable respiratory events, Limb movements, and PLMs. PSG 131 | The outcome goal of this assessment has continued to rise due to our students' success. Students have achieved each new milestone. Much effort has been directed at increasing student participation as well as in boosting their performance on the samples. | One area that can be challenging is getting students to begin participation early enough that they can gain the benefit of seeing multiple patients' data. Some seem apprehensive for fear of not doing well. They are also very busy during this semester. | All three year cohorts exceeded the threshold set by the AASM for professional RPSGT's to be able to maintain scoring privileges on the job. 89% of the three combined cohorts passed the ISR on the professional threshold level. The final year, only one student did not participate. | Yes |
| 4 | Successful completion of the program will help prepare the graduate to take the Board of Registered Polysomnographic Technologists Registry Exam. | More focus on AASM guidelines and parameters seems to have boosted graduate performance on the exam. Also, library of prep resources has been purchased. | Primary challenges are attracting high caliber students to the program and developing the work ethic in students that is required to succeed on the exam. It is also a challenge to keep up with the exam content because it changes every four years. Lastly, there were several very strong students in the 2022 cohort that never attempted the | So far, all three cohorts have a passage rate of well over 80%, which is much higher than the accreditation threshold set by the CoAPSG of 65%. In addition, the cohort size has increased for 2025 which should also increase the competitive level for entry into the program. | Yes |

| | | | | | |
|---|---|---|---|---|---|
| | | | exam. Some chose to either continue their educations or pursue another career. | | |

| YEAR 3 Initiative Analysis | | | | |
|---|---|---|---|---|
| | Initiative | How has this initiative supported student learning or student success? | Will there be any further action taken on this initiative? If yes, explain. | How will this initiative inform program decisions in the future? |
| 1 | Increase student performance on SLR #1 (Demonstrate critical thinking skills in the analysis of recorded sleep data) by spending more time on data interpretation during the summer clinical course. Goal was increased to 80% or more of the class earning a minimum of 14/20 on the rubric which computes to a percentage score of 70% or better. | It has pushed students to look at patient data more critically, consider how the data was collected, and recognize when it may be flawed to gain a more accurate picture of what can be drawn from it. We have a classwide discussion of the data afterwards. I think it has been one of the most valuable assignments asked of our students. | This has worked very well. However, the sleep data used for this activity may need to be changed periodically to discourage cheating. More examples could be used as well. | We will continue to encourage students to think critically of the data they are analyzing. It appears that the more patient data students are exposed to, the better they become at interpreting it. |
| 2 | Increase Student familiarity with polysomnographic instruments and ancillary equipment. Students need to understand how each piece of equipment works as well as why and when to use it. Assessed during PSG 120C. | It has forced students to retain more of how and why lab equipment is used. This will make them better technologists by improving their troubleshooting skills and help them to explain procedures to their patients. | It is anticipated that this will be converted to a group activity. Each lab group will be tested as a group instead of individually. We have seen the benefit for groups as a whole who study together. It is expected that the group pressure will encourage students to be better prepared | Knowing the equipment well is a large part of their job, so we will continue to test students skill level and retention, making adjustments as necessary to achieve a high level of cohort. We will closely monitor the success of the group testing format as well. |

| | | | | |
|---|---|---|---|---|
| | | | so not to negatively affect the grades of their peers. The program continues to grow in size and attendance. The group testing will also help make the process more efficient. | |
| 3 | Students will demonstrate their knowledge of the American Academy of Sleep Medicine (AASM) scoring rules and ability to distinguish among sleep stages and recognize micro-arousals in EEG patterns, different types of scorable respiratory events, Limb movements, and PLMs. PSG 131 | Participating in ISR testing, students are prepared and ready to perform this duty on the job after graduation. It also provides students with proof of their scoring ability that they can provide to potential employers. In addition, the results of this assignment provided valuable feedback to our instructors and to each individual student. | This is a graded activity in which we will continue to participate and a service we will continue to provide for our students. | This initiative provides ongoing feedback to our instructors on how to adjust our scoring training in the PSG 131 course. |
| 4 | Successful completion of the program will help prepare the graduate to take the Board of Registered Polysomnographic Technologists Registry Exam. | Our student's ultimate goals are to become registered polysomnographic technologists (RPSGTs). Therefore, the pressure to maintain a high passage encourages instructors to constantly evaluate the content of our program and find innovative ways to convey the information needed bolster students' chances of passing this exam. The exam feedback is informing of which areas need more attention. | We will continue to monitor our students success and scourer the feedback data for ways to improve this ultimate outcome for each cohort. | Much has been done over the past three years to bolster students' chances of passing the RPSGT exam. A library of resources and a complete study-prep website have been developed by our program Director. This information serves as a constant source of program feedback. The program cohort size has increased for 2025, which also means that more quality |

| | | | | |
|---|---|---|---|---|
| | | | | students are choosing this profession. However, the RPSGT exam content is updated every four years requiring an ongoing monitoring effort. |

# Academic Program Review Cohort B- Cycle 1

Dental Assisting CERT

## Reflection and Mission

### PREVIOUS ACADEMIC PROGRAM REVIEW

Reflect on the past APR (if any) and respond to any recommendations that were made at that time. The reviewer should list recommendations, actions taken, and discuss implications or challenges resulting from those actions.

**Narrative:**

This is the first year for the Academic Program Review. Therefor, no previous data is available for comparison.

### Program Mission Statement

What changes have the program made to the mission statement over the course of this cycle? Why were these changes made? Are any revisions planned?

**Narrative:**

The mission of the Dental Assisting Program at Volunteer State Community College is to produce entry-level graduates that are competent assistants, registry eligible and demonstrate professional concern for their patient. The mission reflects the industry need for qualified dental assistants in the communities in the service area of the Collège.

**Evidence:**

- Dental Assisting Program Informaiton Packet with Mission Statement 2021

### Alignment to Institution Mission

How does the mission of the program align with the mission of the institution?

**Narrative:**

Volunteer State Community College is a public, comprehensive community college offering quality, innovative educational programs, support, and services. Vol State is committed to building partnerships, strengthening internal and external community engagement, and promoting diversity, cultural awareness, and economic development to prepare students for successful careers, university transfer, and meaningful civic participation in a global society.

The Dental Assisting Program aligns with the Institution's mission statement by preparing student for successful careers in field of dental assisting.

**Evidence:**

- DAST Program Mission Alignment with VSCC Mission

## Curriculum and Student Learning Outcomes Assessment

### Curriculum Map

Review the Student Learning Outcomes and course(s) where each one is addressed. For each outcome, provide the assessment method and target level of achievement.

**Evidence:**

- Dental Assissting Technical Certificate Curriculum Mapping_
- Fall 2021 DAST1510 001 Activity Assessments Aggregated Result 11-15-21 color aggregate
- Fall 2021 DAST1530 451 Activity Assessments Aggregated Result 11-30-2021 132105

## EVIDENCE OF STUDENT LEARNING

Provide evidence of student learning and the program's effectiveness in reaching both program outcomes and course learning outcomes.

Please provide the following:
1. Summary of most recent assessment results.
2. Analysis of trends
3. Ideas for how those results could be used to improve student learning.

**Narrative:**

The program utilizes the Employer Satisfaction Surveys to assess the effectiveness of Program Goals.
1. Basic skills to be a competent dental assistant
2. Ability to recognize terminology unique to dental assisting
3. Ability to illustrate comprehension of professional ethical standards for dental assisting]
4. Ability to apply and demonstrate critical thinking problem solving and decision making skills within the parameters of dental assisting.

In addition, assessing student learning assessment was conducted in two courses DAST 151o Dental Radiology and DAST 1530 General Chairside I. Each of the courses utilized the learning outcomes for hands-on exams that also incorporated Program Goals. (see evidence)

Program Response:
- **Program Outcome 1.** Students will demonstrate the basic skills to be a competent dental assistant. This assessed by using the employer satisfaction survey. It accurately demonstrates skills learned and applied while in the program and utilized in employment. The target on the Employer Satisfaction Survey is a 3.8 out of 5-point liker score. The Program meets this target of 3.8 as shown below. See DA Table A, B and C. in evidence for the survey results.

| Assessment Tools | 2018 | 2019 | 2020 | 2021 |
|---|---|---|---|---|
| Employer Survey Results | | | | |
| #3 Competency in the skills required in position | 4.11 | 4.33 | 4.00 | Data not available |

- **Program Outcome 2**. Students will recognize terminology unique to dental assisting. The target level of performance was a 3.9 out of a 5.0 Likert scale on the Employer Satisfaction Survey. See DA Table A, B, and C in the evidence for the survey results. the . The program also used the Supervised Clinical Progress Report to evaluate dental terminology. The target level of performance for the Supervised Clinical Progress is 85% of students would score competent (a 2) or higher. The program meets both of the targets. See DA Supervised Clinical Progress Report in the evidence.

| Assessment tools | 2018 | 2019 | 2020 | 2021 |
|---|---|---|---|---|
| Employer Survey Results | | | | |

| | | | | |
|---|---|---|---|---|
| # 12 Ability to understand technical information | 3.89 | 4.0 | 4.0 | No data at this time |
| #13 Ability to use technical information | 3.89 | 4.33 | 4.0 | No data at this time |
| Supervised Clinical Progress Report | 100% scored 2 (competent) | 100% scored a 2 (competent) | 100% scored a 2 (competent) | 100% scored a 2 (competent) |

- **Program Outcome 3.** Students will illustrate comprehension of professional ethical standards for dental assisting. The target level of performance is scoring a 3.5 on the Employer Survey Satisfaction survey and on the SCPR a score of 80% or higher would score a 3 were met. The program meets this target as shown in the table below. See DA Table A, B, and C and DA Supervised Clinical Progress in the evidence.

| Assessment tools | 2018 | 2019 | 2020 | 2021 |
|---|---|---|---|---|
| Employer Survey Results | | | | |
| # 6 Appropriate work ethics | 4.56 | 4.67 | 5.0 | No data |
| Supervised Clinical Progress Report | 90% scored 3 | 89% scored 3 | 92% scored 3 | 100% scored a 3 (Commendable) |

- **Program outcome 4.** Students will apply and demonstrate critical thinking, problem solving and decision-making skills within the parameters of dental assisting. The target level of performance is scoring a 4.3 or higher on the Employer Satisfaction Survey and a score of competent a 2 or higher a 3 which is commendable on the SCPR. The program did not meet the target score of 4.3 on the Employer Survey averages. See DA Table A, B, and C and DA Supervised Clinical Progress in the evidence.

| • Assessment tools | 2018 | 2019 | 2020 | 2021 |
|---|---|---|---|---|
| Employer Survey Results | | | | |
| # 10 Problem solving skills | 4.33 | 4.33 | 4.0 | No data |
| Supervised Clinical Progress Report | 91% scored 3 | 90% scored 3 | 89% scored 3 | 91% scored a 3 |

**Analysis of Trends**
The VSCC Dental Assisting program continues to meet the benchmarks set by the program outcome goals as demonstrated by the Employer Satisfaction Surveys. The student learning outcomes for 2020 varied slightly and 2020 academic year was challenging due to the pandemic and rapid changes in how the program was offered. The faculty continued to seek opportunities to engage the students.

**Student Learning Improvements**
To address the lowered scores in problem solving skills, the program has implemented an evaluation assessment tool that includes problem solving in DAST 1510 Dental Radiology. See evidence.

**Evidence:**

- DA Program Outcome 1. Table A, B, and C
- DA Supervised Clinical Progress
- DAST 1510 Dental Radiology Assessment

# Qualitative and Quantitative Program Information

## STUDENT, GRADUATE, AND ALUMNI SATISFACTION

Use student and graduate surveys, course evaluations, and any other appropriate evidence needed to explore the qualitative satisfaction results from students and graduates to provide information to the items below.

Discuss student, graduate, alumni perceptions of the program, faculty, career projection, and any other program recommendations given "during this survey".

Summarize any noted themes, trends, areas for improvement, identified strengths, and implications for the program for both students and alumni.

**Narrative:**

The Graduate and Alumni Surveys for the dental assisting program were rated a 4.7 out of a 5.0 for the 2020-2021 academic year. The Program faculty overcame and successfully delivered the curriculum during the pandemic so that students received as much hands- on experiences are were allowed.

**Positive comments:**
1. I am so happy to be in that program.
2. I believe the primary goal of the program is to give students a basic knowledge of the dental assisting career. I felt prepared as much as I could be.
3. My instructors were always one step ahead of the game when it came to things we needed to know for our clinical rotations.

**Comments for program improvement:**
1. More chairside assisting
2. I wish they (the instructors) could have prepared us on resume, cover letter and interview questions.

The faculty have reviewed recommendations made by graduates in the area of job seeking skills. Resume building, cover letters and interview skills were added to the curriculum in the Fundamentals of Dental Assisting Course in order to thoroughly cover the important job seeking needs.

**Evidence:**

- Fundamentals of Dental Assisting syllabus
- Section 4- Student, Graduate and Alumni Satisfaction

## RECRUITMENT, ENROLLMENT, RETENTION, AND GRADUATION: FACT BOOK DATA

Attach screenshots from the Fact Book of the areas below.

1. Headcounts
2. FTE Comparison
3. Student Primary Location
4. Retention Statistics
5. Graduation Statistics
6. Demographics

Using the Fact Book data listed below, analyze the follow:

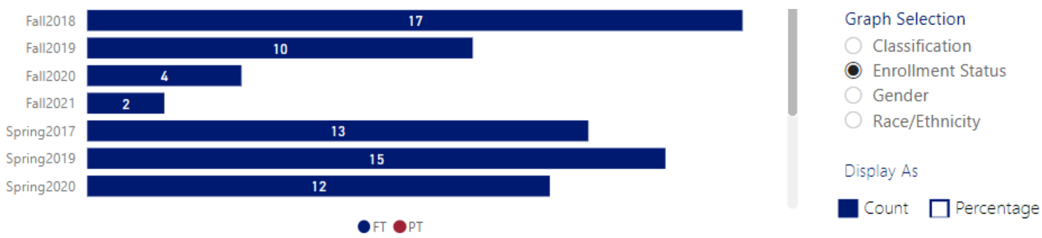
Recruitment
Enrollment
Retention

Graduation

The reviewer could discuss marketing and recruitment efforts in the community and local high schools; analyze enrollment, retention, and graduation trends; and discuss potential reasons, strengths, plans for improvement, and any other implications deemed appropriate.

**Narrative:**

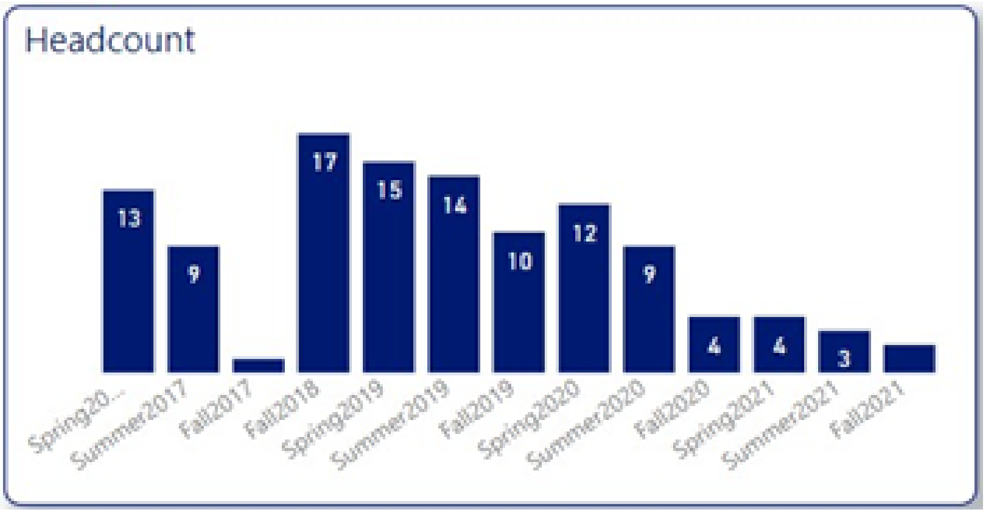
- Recruitment- Direct recruitment comes from the college. In addition, the program participates in hosting service area high school students as well as the Discovery Days launched during 2021 fall semester. Discovery Days was created by the Health Science faculty to reach out to current VSCC students and introduce them to health science careers that are not as widely publicized or known. Each program allots for time to share program information and guided tours of their faculties and to allow students to ask questions about the industry of interest.
- Enrollment- The program declined by 50%. Discrepancies could be attributed to the Program name change mandated by TBR to reflect the common curricular requirements by all Commission on Dental Accreditation (CODA) approved Programs as well as the pandemic of 2020. The former program certificate name/degree was AAS in Dental Assistant. Now the degree name is Dental Assisting. See charts below regarding the decline in numbers. There are still two separate degree names, Dental Assistant and Dental Assisting. Dental Assistant is specific to the certificate, while Dental Assisting is specific to the AAS degree. If a student chooses the degree, the number of students for the certificate drops. The faculty also felt that the pandemic may have been an additional factor in the lower enrollment numbers for the program. Although the college and the surrounding service areas have returned to a semi-normal lifestyle, students were reluctant to pursue a career in the dental field because of the close proximity of a dental assistant to the patient as well as other dental team members and the fear of the spread of the virus.
-
-




a career in dental asssiting beacuse of the nature of the job skills.

- Retention- 88.4 from 2016- 2020 excluding fall 2017 and spring 2018 -no data was available.
- Graduation- 100% graduates for 2018 and 2021 graduates; 93% for 2019 graduates and 87% for 2020 graduates. A mean average of 95% graduation rate for 2018-2021.

Although a decline in student enrollment was significant, the program faculty believe the lowered number of students enrolled in dental assisting was attributed to the pandemic. From the data provided by IEARPA, the Program maintains an 88.4 percent retainment of those students that choose dental assisting as a profession. In addition, the program has a mean average of 95% for students who stay in the program and graduate. The demographics for the program are mainly located in the VSCC service areas. Average age of students is about 22 and mostly those who identify as female are enrolled in the program.




| State/County | Fall2017 | Fall2018 | Fall2019 | Fall2020 | Fall2021 |
|---|---|---|---|---|---|
| Sumner County | 1 | 9 | 5 | 1 | 2 |
| Macon County | | 2 | 1 | 1 | |
| Robertson County | | 1 | 2 | 1 | |
| Davidson County | | 2 | 1 | | |
| Wilson County | | 1 | 1 | 1 | |
| Bedford County | | 1 | | | |
| Cheatham County | | 1 | | | |

## Fall to Spring Persistence

92.9%
Fall 2016 to Spring 2017

0.0%
Fall 2017 to Spring 2018

82.4%
Fall 2018 to Spring 2019

90.0%
Fall 2019 to Spring 2020

100.0%
Fall 2020 to Spring 2021

| Concentrations | 16-17 | 17-18 | 18-19 | 19-20 | 20-21 |
|---|---|---|---|---|---|
| Art (Studio) | 7 | 13 | 14 | 13 | 15 |
| Biology | 3 | 4 | 3 | 2 | 7 |
| Business Administration | 38 | 33 | 32 | 23 | 33 |
| Chemistry | | 4 | | 2 | 1 |
| Civil Engineering | 2 | 1 | 2 | 3 | 1 |
| Computer Info Sys Tech | 2 | | | | |
| Computer Science | | 1 | 5 | 3 | 4 |
| Criminal Justice | 37 | 34 | 35 | 41 | 36 |
| Criminal Justice (TSU) | 3 | 1 | | | |
| Criminal Justice Law Enfrcemnt | | 1 | | | |
| Cyber Defense | 6 | 11 | 18 | 21 | 35 |
| Dental Assistant | 14 | 10 | 6 | 5 | 2 |

**Evidence:**

- Section 5 Recruitment, Enrollment Retention and Graduation

## Supporting Program Information

### FACULTY USAGE AND DEVELOPMENT (OPTIONAL)

Review faculty teaching loads, adjunct to full time ratios, faculty development opportunities, faculty end of course evaluations, faculty surveys, and any other evidence deemed appropriate. The reviewer could discuss the impact of teaching, the ratio of adjunct to full time staffing, end of course evaluations relative to faculty, and any noted themes identified from the review.

### CLASS OFFERINGS (OPTIONAL)

Address scheduling effectiveness and space utilization. The reviewer could review faculty to student ratios at each location and online and discuss the impact relative to faculty and student success.

### CO-CURRICULAR ACTIVITIES (OPTIONAL)

Discuss co-curricular activities available for students in the program. These might include clubs, internships or other Work Based Learning opportunities, service learning, guest speakers, or any other activities designed to enhance student experience in the program.

**Narrative:**

Clinical experiences play a significant role in student learning. The students participate in clinical rotations in working dental offices during the fall, spring, and summer. The clinical rotations are valuable in that the foundations being taught in the classroom and laboratories are developing and increasing student competencies. Each rotation and semester, the expectations for students to participate in dental procedures are increased. Feedback from the clinical team is helpful in identifying the students' areas of weakness as well as comments about their strengths . The assessment tool used in assessing student learning for clinical experiences is via the Supervised Clinical Progress Report.

**Evidence:**

- 2018-2021 Supervised Clinical Progress Report

### INDUSTRY TRENDS AND INNOVATIONS (OPTIONAL)

Examine the program demand within the labor market including local, regional, and/or national. The reviewer could examine information from industry advisory boards and information gathered from creditable industry sources as well as from external accrediting bodies. The reviewer could also discuss any existing or emerging trends within the field as they relate to your program and employment of your student's post-graduation as well as the outlook for the field in the next 5 years. This summary could also identify best practices, trends, and innovation in the field and discuss implications for the program.

**Narrative:**

When the common core curriculum was mandated for all accredited dental assisting programs in Tennessee, the program faculty saw an opportunity to enhance the summer curriculum by adding lectures in coronal polishing, dental sealants and monitoring nitrous oxide in DAST 1650, offered in the summer semester. The addition of the coursework was a result of requests from the dental community and the TBR common curriculum mandate. In addition to following curriculum changes, in order for the Program to offer the certifications, approval by the Tennessee Board of Dentistry was sought and approved. Also because the Tennessee Board of Dentistry statues state that the certification course must be taught by a dentist in good standing with the TBD, the faculty reached out to affiliate's seeking instructors for the courses. The faculty have been able to retain two local dentists to teach the courses and the students are not only adding certification courses to their licensure, but they are also receiving life experiences from seasoned dentists.

**Evidence:**

- DAST 1650 Special topics 2021

### EMPLOYMENT AND TRANSFER FORECASTS (OPTIONAL)

Discuss where students go after graduation and what is being done to enhance the likelihood of success or assist students in reaching those goals. The reviewer could discuss employment outlook, obtainment, curriculum alignment for transfers, and the employment demands of the industries where students will work.

## Recommendations Action Plan, Review, and Feedback

### SUMMARY AND ANALYSIS

The reviewer should discuss what went right, what went wrong, and the implications of those results.

**Narrative:**

**The Program is doing well in:**

- Developing a foundational knowledge base in which a graduate is able to build on once in the dental assisting field. Based on Employer feedback and ratings, graduates have successfully met the goals implemented for basic skills in dental assisting.
- The students are understanding the importance of ethical decision making as it pertains to the field of dentistry as noted by the employer surveys scores. The ability to make decision based

upon the students' personal ethics that mirrors the ethical standards held by the profession of dentistry is expected and met.
- The additional certification courses were a positive improvement for students in their ability to obtain employment faster and with better retention in staying in the position. It increased the graduate's confidence in obtaining employment as well as confidence in their level of foundational skills learned.

**Areas the Program will improve:**

- The faculty will add components in dental software (Eaglesoft) use in Clinical Chairside I and II as well as Dental Office Management. Remarks from alumni and graduates were that there should have been more opportunities to use the software to build confidence and comfortability in navigating the icons and areas in the software.
- Also, the faculty will spend more time on employment seeking skills such as resume building, developing a cover letter and soft interviewing skills as noted in a comment results for 2019 graduates.
- The faculty will work with IERPA in collecting data for student resource surveys and program personnel resource surveys.
- The faculty will look for opportunities to increase program enrollment by attending high schools and interacting with guidance counselors to make the program more visible. The faculty will also recruit on campus by setting up information booths during the spring in order to showcase the program. The dental assisting advisory board members gave feedback on recruitment by recommending by creating a video showcasing the activities of dental assisting and to invite admissions recruiters to sit down with program faculty for ways to recruit students for the program.

**Evidence:**

- Technology and Software use graphs

## TWO YEAR RECOMMENDATION ACTION PLAN

Discuss the next steps for the program by providing a program action plan for the next two academic years. Recommendations must include at least two actions focused on student learning and one action focused on student success. The program action plan must include a timeline for each recommendation.

Feedback and approval will be provided. Refer to the APR timeline for approval deadlines.

**Narrative:**

The two-year plan of action will include:

- **Student Learning Recommendation: Increase outcome achievement for Program Outcome #1** Students will demonstrate the basic skills to be a competent dental assistant.
    - The faculty will look at evaluate the clinical assessment documents from Spring 2022.
    - **2022-2023 Actions: Review and take action accordingly, Clinical Chairside I (fall) skills assessment document**
    - **2023-2024 Actions:** Review and take action accordingly, **Clinical Chairside I (fall) skills assessment document**

- **Student Success Recommendation: Improve Student Feedback**

    - Working closely with IERPA to create and distribute a Student Resources Survey and a Program Personnel Resource Survey. Both surveys will provide the program with data to make sure that students are utilizing and receiving resources necessary for their successes as well as revealing areas lacking in each
    - **2022-2023 Actions: Create and distribute survey (collect data to evaluate and assess**
    - **2023-2024 Actions: Evaluate survey results, examine effectiveness**

- **Student Success Recommendation: Improve recruiting efforts**
    - **Attend high school visits to speak with health science students, participate in the Health Science Discovery days, Participate in the Health Sciences high school visits to VSCC Campus, Develop a Program specific video to showcase the program and the facilities**
    - **2022-2023 Actions: High School visits, participate in the Health Sciences Discovery days in the fall and spring**
    - **2023-2024 Actions: High School visits, participate in the Health Sciences Discovery days in the fall and spring**

- **Student Learning Recommendation: Increase outcome achievement for Program Outcome #4 Students will apply and demonstrate critical thinking, problem solving and decision-making skills within the parameters of dental assisting**
    - **To address the lowered scores in problem solving skills, the program has implemented an evaluation assessment tool that includes problem solving in DAST 1510 Dental Radiology. See evidence:**
    - **2022-2023 Actions: Implement new assessment,**
    - **2023-2024 Actions: evaluate effectiveness and implement new assessment if necessary for lowered scores**

- **Student Learning Recommendation: Increase outcome achievement for Program Outcome #2 Students will recognize terminology unique to dental assisting.**
    - **2022-2023 Actions: The faculty will add components in dental software (Eaglesoft) use in Clinical Chairside I and II as well as Dental Office Management. Remarks from alumni and graduates were that there should have been more opportunities to use the software to build confidence and comfortability in navigating the icons and areas in the software.**
    - **2023-2024 Actions:**

**Evidence:**

- DAST 1510 DENTAL RADIOLOGY Assessment 2021
- DAST 1530 GOALS AND ASSESSMENT 2021

## DEAN(S) APPROVAL AND FEEDBACK.

The Dean(s) will provide feedback and approval on the completed review and action plan.

**Narrative:**

The program faculty have done an effective job in producing graduates who are meeting the needs of business and industry. The addition of certification courses increased their marketability in a new profession. Introducing dental software into the program will be of great benefit to all stakeholders by increasing familiarization with technology and building confidence in maintaining adequate medical records. The College will collaborate with the program director to increase visibility with the overall goal of increasing enrollment and completion.

### AVPAA & VPAA Review and Feedback

The Vice President of Academic Affairs and Assistant Vice President of Academic Affairs will provide feedback and approval on the completed review and action plan. VPAA and AVPAA should consider institutional planning and budgetary details when providing their review.

**Narrative:**

The Office of Institutional Effectiveness and Assessment has reviewed this report and will begin supporting the report champion in preparing for the presentation of this review to the Institutional Effectiveness Committee.

### PRESENTATION FEEDBACK

The APR will be presented to a group of peers. Feedback retrieved from the presentation will be summarized and included.

## Action Plan Status Reports

### YEAR 2 ACTION PLAN REVIEW

The reviewer should review progress on action plan over the past year. Discussion should include challenges and successes. If any changes are planned due to the results, those should be discussed and an updated action plan should be included in the narrative or attached as evidence.

**Narrative:**

| YEAR 2 | | | | | |
|---|---|---|---|---|---|
| | Initiative | Achieved Milestones | Challenges | Changes if applicable | Progress Toward Performance Indicator(s) |
| 1 | Increase outcome achievement for Program Outcome #1 Students will demonstrate the basic skills to be a competent dental assistant. | PLO # 1- 80% of the students will receive a score a 2.5 or higher on the overall competency in Clinical Chairside I competency evaluation. On SLO DAST 1530 there were 50 skills evaluated. 80% of the students scored a 2.5 or higher. | No challenges in the evaluation of students' basic skills were noted. | No changes needed | On PLO # 1, 80% of the students received a 2.5 or higher on the evaluations in DAST 1530. |

| | | | | | |
|---|---|---|---|---|---|
| 2 | Improve Student Feedback | The faculty failed to work on an assessment tool in order to develop a Student Resources Survey and a Program Personnel Resources Survey. | The faculty failed to develop the survey necessary to retrieve data for the action plan. | The faculty will work with IERPA to develop a new resource survey or use the current survey with minor changes to satisfy the needs for the Program. | No data available |
| 3 | Improve recruiting efforts. | Hendersonville High School, Sumner County Middle School and Career Exploration Fair 2023 and 2024 | No challenges in recruitment | No changes | Although no students were gleaned from the high school visits who chose the dental assisting program, the faculty will continue to reach out to the high schools and will participate in the on-campus activities for recruitment. |
| 4 | Increase outcome achievement for Program Outcome #4 Students will apply and demonstrate critical thinking, problem solving and decision-making skills within the parameters of dental assisting. | PLO # 4- 80% of students will score a 2.5 or higher on problem solving activity located in DAST 1510 Dental Radiology evaluation tool # 48.<br>Outcome- a 2.18 was the overall score, which is less than the set baseline. | Developing a more comprehensive evaluation tool to assess the students' problem-solving skills in the evaluation of quality dental x-rays. | The faculty will develop an additional and separate evaluation tool for assessing diagnostically acceptable x-rays. | The score of a 2.5 or higher was not achieved. Therefore, the faculty will work on a separate assessment tool that will demonstrate the students' skill sin critical thinking and problem solving. In addition, the faculty will incorporate Clinical Practicum I and II Supervised Clinical Progress Report for the spring and summer Which addressed the evaluation of the students' ability to problem solve. |
| 5 | Increase outcome achievement for Program Outcome #2 Students will recognize terminology unique to dental assisting. | PLO # 2- 86% of the students received a 75% of higher on the EagleSoft assignment final in Dental Office Management Course.<br>92% of the students received a 75% or higher on the terminology final in Clinical Chairside I<br>83% of the students received a 75% or higher in the | The faculty will be involved in the distribution of the Employer Surveys in order to receive adequate data to analyze for student success | Contacting Employers for survey completion<br>Add the Clinical Practicum I and I to the evaluation tools in order to adequately measure the students' understanding and knowledge of dental terminology. | The faculty will continue to embed the terminology final sin both Clinical Chairside I and II as well as incorporating Eaglesoft exercise in Clinical Chairside I and Dental Office Management.<br>The Program will add the Clinical Practicum Supervised Clinical Progress Report for spring and summer.<br>Part I of the said reports # 9- |

| | | | | | |
|---|---|---|---|---|---|
| | | terminology final in Clinic Chairside II. No data from the 2023 Employer Survey results have been retrieved in regard to understanding terminology. | | | Demonstrates Knowledge of dental terminology, 80% of the students will score a 2 (competent) or higher. |

**Evidence:**

- VSCC_HealthSciences_DiscoveryDays SP22

**YEAR 3 ACTION PLAN REVIEW**

The reviewer should review progress on action plan over the past year. Discussion should include challenges and successes. The reviewer should discuss the actions future and continuous in the program.

**Narrative:**

## YEAR 3 Progress

| Initiative | Achieved Milestones | Challenges | Performance Indicator Result | Initiative Achieved? |
|---|---|---|---|---|
| PLO #1 - Basic Clinical Skills | Continued use of DAST 1530 Clinical Chairside I competency evaluation (50 skills). | None | 80% scored 2.5 or higher; consistent with Year 2. | YES |
| Improve Student Feedback | Reviewed survey tools with IERPA. | Survey implementation in progress. | No data available. | NO |
| Improve Recruiting | Continued high school outreach and career fairs. | None | No immediate enrollment impact. | YES |
| PLO #4 - Critical Thinking | Added Clinical Practicum I & II progress reports. | New tool still under development. | Below 2.5 benchmark; steady vs Year 2. | NO |
| PLO #2 - Terminology | Continued terminology assessments and Eaglesoft activities. | Low employer survey response. | 80-90% met competency benchmarks. | YES |

## YEAR 3 Initiative Analysis

| Initiative | Support of Student Learning | Further Action | Future Program Decisions |
|---|---|---|---|
| PLO #1 - Clinical Skills | Reinforced consistent measurement of core competencies. | No | Maintain current assessment approach. |
| Student Feedback | Identified need for improved data collection. | Yes | Guide future resource planning. |
| Recruiting | Increased program visibility. | Yes | Refine outreach strategy. |
| PLO #4 - Critical Thinking | Expanded assessment perspective. | Yes | Improve assessment tools. |
| PLO #2 - Terminology | Confirmed strong terminology mastery. | Yes | Seek employer validation. |

**Summary**

During Year 3, the Dental Assisting Program continued its initiatives with student performance remaining steady and consistent with Year 2 results. PLO #1 met the established benchmark, with 80% of students achieving a score of 2.5 or higher on the Clinical Chairside I competency evaluation, indicating continued effectiveness of current instructional practices.

Efforts to improve student feedback and recruitment remained in progress. Faculty reviewed survey tools with IERPA but did not collect new data during Year 3. Recruitment activities continued through high school outreach and career events to support long-term enrollment goals.

For PLO #4, critical thinking and problem-solving results remained below the benchmark, similar to Year 2. In response, faculty expanded assessment methods by incorporating Clinical Practicum I and II Supervised Clinical Progress Reports while continuing development of a more targeted evaluation tool. PLO #2 terminology outcomes remained strong, with approximately 80-90% of students meeting competency benchmarks across courses. Overall, Year 3 reflects sustained performance, ongoing assessment refinement, and continued focus on improvement.

# General Education Review Cohort B- Cycle 1

Humanities and Fine Arts GE

## Reflection and Mission

### Previous General Education Review

Reflect on the past GER (if any) and responds to any recommendations that were made at that time. The reviewer should list recommendations, actions taken, and discuss implications or challenges resulting from those actions.

**Narrative:**

This is our first round completing the general education review.

### Program Mission Statement

Have any changes been made to the general education mission statement over the course of this cycle? Why were these changes made? Are any revisions planned?

**Narrative:**

The goal of the Humanities and/or Fine Arts requirement is to enhance the understanding of students who, as citizens and educated members of their communities, need to know and appreciate their own human cultural heritage and its development in a historical and global context. Also, through study of Humanities and/or Fine Arts, students will develop an understanding, which they otherwise would not have, of the present as informed by the past.

### Alignment to Institution Mission

How does the general education mission align with the mission of the institution?

**Narrative:**

Our program mission is consistent with our institution's mission; we prepare our students for successful careers in a variety of courses, including English Composition and Communications. We foster meaningful civic participation and promote cultural awareness and diversity in all of our course offerings.

## Student Learning Outcomes Assessment and Curriculum

### Curriculum Map

Curriculum mapping ensures the alignment of TBR GenEd outcomes and student learning outcomes; which affords an opportunity to ensure course coverage and sequencing of outcomes in courses.

The reviewer should review the current mapping to ensure there is pathways to meeting the TBR GenEd outcomes for all general education students.

**Narrative:**

The current TBR General Education outcomes are not consistently written in a way to lend themselves to being mapped to unique course-level outcomes. Rather, the TBR General Education Outcomes are broad and often map to multiple course learning outcomes, especially given how the course learning outcomes are currently written. Assessments are mapped to specific TBR General Education Outcomes, but they may also be connected to multiple course learning outcomes, depending on how the course learning outcomes are structured.
Literature and Film: a review of course learning outcomes and mapping will take place over the next year, so these mappings may be adjusted during this cycle.
Philosophy:
Art:
Music: Faculty who teach MUS 1030 will meet during the spring of 2022 to review course learning outcomes and ensure course mapping aligns with TBR GenEd outcomes. If necessary, we will modify course mapping for future semesters.
Theater:

**Evidence:**

- GenEd Hummanities Course Outcomes

## Evidence of Student Learning

Evidence of Student Learning

Provide evidence of student learning, and the program's effectiveness in reaching both TBR GenEd outcomes and student learning outcomes.

Please provide the following:
1. Summary of assessment most recent results.
2. Analysis of trends.
3. Ideas of how those results could be used to improve student learning.

**Narrative:**

For all courses in humanities & fine arts: This is the first cycle with the new general education review. We do not have enough consistent data to make analyses on specific outcomes as they related to specific years, but we can draw general conclusions based on limited data.

English:
(Film | 2860): We discovered an oversight; we have not assessed our film course in recent years. This will change beginning in spring 2022.
(literature course offerings): We only have one year of data -- 2018-2019. The next two years were disrupted due to COVID and changes in our general education review process. As for the year we do have, we were able to assess 22 sections of our literature offerings. We examined three areas of student learning using an excel sheet: analysis of theme, assessment of relationships and culture, and comparative analysis. All literature courses require one essay; this was the assignment used for all faculty to assess all three areas on the rubric. Students had the highest numbers (3.32/5) on analyzing relationships and culture. Analysis of theme was close, with a score of 3.29/5. Students had the most trouble with comparative analysis, which had a mean score of 3.32. Anecdotally, our faculty agree that students tend to struggle with comparative analysis. Our off-campus sites (dual enrollment) had higher scores in all categories. The higher score in analysis of culture and relationships makes sense, as students tend to do engage with theme for every assignment in literature. Many have already analyzed theme in previous English courses as well. It is not surprising that our off-campus sites had higher scores; dual enrollment students tend to be better prepared than our traditional students and outperform our main campus students with grades as well. During the assessment process, we discovered we did not assess ENGL 2045 (introduction to literature). This is unfortunate, because 2045 is our most commonly offered course. We will plan to assess it starting in spring 2022.

Philosophy:

Art:
Comparing our scores from the writing rubrics to those of Music and Theater, it seems we may be grading a bit easier than we should, particularly in the writing. We will discuss this as a department and decide the best way to proceed, perhaps by including more short writing assignments and/or emphasizing writing skills more.
Looking at our pre- and post-quiz scores, the increases between the two fluctuate a bit from semester to semester, but are consistently fairly strong with an average increase of nearly 15% (ranging from a low of 8.44% to a high of 19.3%).

Music:
Our pre- and post-test scores frequently show improvement for most students. However, not all students take the pre- and/or post-tests, and since we are unable to treat these quizzes as graded items, students will perhaps need more incentive to complete them (extra-credit, allowing one late assignment, etc).
The concert reports consistently demonstrate what students have learned in class and how well they incorporate that knowledge into narrative form. There are two concert report per semester, approximately two pages each. More emphasis can be placed on rigorous grading to help students improve. Overall, students have received 3.97 on the grading rubric, with "Writing Skills" receiving the lowest score of 3.83 and 1/3 of the students receiving "Adequate".

Theater:
Overall, Theatre final course grades are significantly correlated with Post-test scores for students completing the assessment. However, overall, correlation strength is low to moderate, suggesting that Post-test score alone is not a strong predictor of student course outcome. The data tells us that theatre students understand the assignment but need help completing tasks on time. There is a need to improve how we explain how humanistic and artistic expression throughout the ages expresses the culture and values of its time and place. Likewise, the need to improve how theatre is has influenced our society.

**Evidence:**

- ART 1819 Rubric report tables
- ART Pre-Post VPA 2020
- ART VPA Rubric Writing Assessment Results Spring 2018
- Literature tables 18-19

# Faculty and Courses

## Faculty Usage and Development (optional)

May review faculty teaching loads, adjunct to full time ratios, faculty development opportunities, faculty end of course evaluations, faculty surveys, and/or any other evidence deemed appropriate. The reviewer could discuss the impact of teaching, the ratio of adjunct to full time staffing, end of course evaluations relative to faculty, and any noted themes identified from the review.

**Narrative:**

English (literature):
Based on the most recent data for literature courses, it is clear that our faculty assign very different numbers when assessing students; some faculty have scored categories with a mean as high as 4.8, while others have numbers as low as 2.5. While assessing is very different from grading, the numbers suggest faculty teaching literature may benefit from some grade norming sessions. Our department has

done this in the past three years for our composition courses, but it has been a while since we have had sessions geared toward literature.

Philosophy:
Music:
Art:
Theater:

### Class Offerings (optional)

May address scheduling effectiveness and space utilization. The reviewer could review faculty to student ratios at each location and online and/or discuss the impact relative to faculty and student's success.

**Narrative:**

English (Film): For the immediate future, ENGL 2860 will be assessed using the _____ for spring 2022. Faculty teaching 2860 are currently revising the model course. It will be piloted in Fall 2022, and faculty teaching the course will include an assessment of the course outcomes as indicated in the attached mapping.
English (Literature): It appears ENGL 2045 has not been assessed in recent years, which is a problem since it is one of our most popular literature courses. It is also evident we have not assessed 2310, though only 3-4 sections are usually offered each year. We plan to continue to assess 2045 sections in addition to 2055, 2110, 2120, and 2320 using our current course mapping. Our department plans to re-examine course outcomes beginning in Fall 2022, so we may re-map or wait for the new TBR outcomes.

Philosophy:
Music:
Art:
Theater:

## Recommendations and Next Actions

### Analysis and Summary

Analyze and summarize the proceeding sections. The reviewer should discuss what went right, what went wrong, and the implications of those results.

**Narrative:**

English (literature and film)
This review highlighted the need to assess our film course and to ensure all literature courses (including ENGL 2045) are assessed. For ENGL 2860, we will assess the genre analysis assignment. We will continue to assess each literature course's respective essay using the same rubric categories (analysis of theme, assessment of relationships and culture, and comparative analysis) for the time being. We will discuss ways to incorporate more comparative analysis earlier in the semester with lower-stakes assignments (such as discussion board posts) so students have more practice before writing their major essay; hopefully, this will improve assessment results. The English department has an academic audit starting in fall 2022, so we may find we need to revise the assignment we use for the general education assessment.

Philosophy:

Art:

Our writing rubrics include scores generally higher than those using the same rubric in Music and Theater courses, so we will discuss to determine if we think this is an indication that our students are performing better on the paper in question or rather we are grading too easy.
Pre- and post-quiz scores are generally good and will continue to be monitored as a means of identifying any problems that develop.

Music:
MUS 1030 instructors need to improve student participation in pre- and post-tests, particularly the latter. It may be advisable to tie these in to student participation or other graded items. The post-tests, when submitted, show student improvement.
We require two 500-word Concert Reports, and these remain excellent assessment instruments in showing students' writing skills and application of material learned in class. However, even though we have a set rubric, faculty have often vastly different methods to grading these reports, so we can better align our grading approach. Our students tend to have adequate writing skills at best.

Theater:
Students need a better understanding of how theatre classes can assist them with other courses.
Students need to discover how theatre can positively affect life outside of the classroom.

## Next Actions

The reviewer should discuss the next steps for the general education for the following two academic years. A plan of action and timeline should be included in the narrative or document library.

Feedback and approval will be provided. Refer to the GER timeline for approval deadlines.

**Narrative:**

English (literature and film):
Film: faculty teaching 2860 are currently revising the model course. Starting in spring 2022, we will assess the genre analysis assignment.
Literature: It appears ENGL 2045 has not been assessed in recent years, which is a problem since it is one of our most popular literature courses. It is also evident we have not assessed 2310, though only 3-4 sections are usually offered each year. We plan to continue to assess 2045 sections in addition to 2055, 2110, 2120, and 2320 using our current course mapping.
Our department plans to re-examine course outcomes beginning in Fall 2022, so we may re-map or wait for the new TBR outcomes. Our 2022-2023 academic audit may give us more direction.

Philosophy:

Art:
The Art Department will meet in early spring 2022 to discuss the way we are using the writing rubrics and in particular whether we should be expecting more of our students in their writing. We are beginning the process of pushing students more toward the Learning Commons for help with their writing, which will potentially counter the likely outcome of this meeting of increasing our expectations in this area. This should all be in place before anyone is grading these assignments in spring of 2022.

Music:
Music faculty will meet in spring 2022 to discuss a more consistent approach to grading the Concert Reports, and also to determine better methods to improve student participation in pre- and post-tests. The Art Department's suggestion of encouraging more students to use the Learning Commons for tutoring in writing is an approach that we will also use, since most of our students would benefit from more instruction in writing.

Theater:

- To work closely with criminal justice faculty providing them with student actors to role-play situations that law enforcement face daily. One of the benefits of having students role-play is that it will aid in training students who want to become police officers.
- Show how theater relates to other disciplines.
- To have two to three one-on-one meetings with students to discuss their Theatre Project
- Students will do a group PowerPoint presentation on theatre and social media.

**Evidence:**

- ART1030_THEA1030 TBR General Education Assessment Rubric Rubric Detail 12-17-2021 085734
- Art History Fall 2020 Rubric Detail 12-17-2021 084001
- ENGL2045 SLO Assessment Rubric Rubric Detail 12-17-2021 085741
- ENGL2055 SLO Assessment Rubric Rubric Detail 12-17-2021 085748
- ENGL2110 SLO Assessment Rubric Rubric Detail 12-17-2021 085811
- ENGL2120 SLO Assessment Rubric Rubric Detail 12-17-2021 085817
- ENGL2310 SLO Assessment Rubric Rubric Detail 12-17-2021 085825
- ENGL2320 SLO Assessment Rubric Rubric Detail 12-17-2021 085835
- PHIL 1030 Gen Ed Rubric Rubric Detail 12-17-2021 085847
- PHIL 1040 Gen Ed Rubric Rubric Detail 12-17-2021 085901
- PHIL 2430 Gen Ed Rubric Rubric Detail 12-17-2021 085907

## Feedback and Review

### Feedback (General Education Committee)

Presentation by faculty to the General Education Committee of the Program Review. The General Education Committee will provide feedback on the completed review and action plan. The feedback will be included and summarized.

### Dean(s) Approval and Feedback

The Dean(s) of the general education discipline(s) will provide feedback and approval on the completed review and action plan.

### VPAA and AVPAA

The VPAA & AVPAA will provide feedback on the completed review and action plan.

## Action Plan Reports

### Year 2 Action Plan Review

The reviewer should review progress on action plan over the past year. Discussion should include challenges and successes. If any changes are planned due to the results, those should be discussed and an updated action plan should be included in the narrative or document library.

**Narrative:**

ENGLISH:

**Film (ENGL 2860)**
Between spring 2022 and spring 2023, we began assessing the Genre Analysis assignment in ENGL 2860. This marked an important step in aligning our teaching and assessment practices with course outcomes. The initial assessment helped faculty pinpoint strengths in student engagement with genre conventions and revealed areas where students needed more support in critical analysis and writing. Building on this work, a newly revised OER-based model course was launched in spring 2023. The updated course features refreshed materials, streamlined assignment scaffolding, and more intentional connections between outcomes and major assessments. Faculty who piloted the new course have provided valuable feedback, and we plan to continue refining the materials based on those insights.
**Literature offerings**

We made steady progress in addressing gaps in literature course assessment. ENGL 2045, one of our most popular offerings, had not been assessed in recent years, so we prioritized it for inclusion in our assessment cycle. We also began assessment efforts in ENGL 2310, which, while offered less frequently, remains a key part of our literature offerings. Alongside these efforts, we continued assessing ENGL 2055, 2110, 2120, and 2320, maintaining consistency with our current course mapping.

At the same time, faculty initiated broader conversations about our course outcomes. These discussions have focused on whether our current outcomes still serve the needs of students and whether updates might be needed in light of forthcoming TBR general education revisions. By the end of spring 2023, the department was weighing whether to re-map outcomes now or wait for guidance from the new statewide framework.

## Year 3 Action Plan Review

The reviewer should review progress on action plan over the past year. Discussion should include challenges and successes. The reviewer should discuss the actions future and continuous in the program.

**Narrative:**

ENGLISH:

**Film (ENGL 2860)**
During the 2023–2024 academic year, we continued to build on our assessment work in ENGL 2860, our film course. The recently launched OER-based model course has made a positive impact by reducing costs and improving access to course materials. Faculty have observed that this change has helped students engage more deeply with the content, and early feedback has been encouraging. That said, the pass rate for film (71.3%) remains lower than in our literature courses. We believe this may be partly due to the higher number of students in learning support who enroll in ENGL 2860. As the English department revises ENGL 0810 (Learning Support Writing), we expect that changes to that curriculum may support stronger outcomes in film in future semesters.

**Literature Offerings**
In our literature courses, we've maintained a consistent approach to assessment while also working to update course outcomes across the board. ENGL 2045 continues to be our most popular literature course, with a strong pass rate of 86.6% and a low withdrawal rate of 1.8%. We also continued assessment efforts in ENGL 2310, which saw an 82.6% pass rate this spring. Both courses have been part of a broader push to refresh outcomes across our literature offerings, and those revised outcomes will officially go into effect in fall 2025.

Meanwhile, our other literature courses showed solid performance:

- ENGL 2055 had a 72% pass rate
- ENGL 2110 reached 76.8%
- ENGL 2120 stood out with a 93% pass rate
- ENGL 2320 had an 87.8% pass rate

Each of these courses continues to be mapped and assessed according to departmental plans. The literature faculty also spent time this year discussing whether our current outcomes still align with the skills and knowledge we want students to take away from these courses. These conversations have helped us prepare for the upcoming TBR general education revisions, ensuring we're ready to adapt when the new framework is finalized.

# General Education Review Cohort B- Cycle 1

Natural Science GE

## Reflection and Mission

### Previous General Education Review

```
Reflect on the past GER (if any) and responds to any recommendations that were made at that time. The reviewer should list recommendations, actions taken, and discuss implications or challenges resulting from those actions.
```

No prior GERs have been conducted. Science faculty have developed, administered, and evaluated outcomes assessments over the past several years. This review evaluates data from academic years 2018/2019, 2019/2020, and 2020/2021. We will define our departmental mission, examine the current IE assessment data, discuss the impacts of the Covid-19 pandemic, explore multiple faculty- and course-related topics, and present a plan to improve student learning and success.

Evidence:

- Academic Audit Science 2007
- Academic Audit Science 2013
- Academic Audit Science 2018

### Program Mission Statement

```
Have any changes been made to the general education mission statement over the course of this cycle? Why were these changes made? Are any revisions planned?
```

The mission of Natural Sciences General Education is to provide innovative courses and encourage active learning, communication, and decision-making skills by providing immersive and hands-on educational opportunities to help students meet degree and certificate requirements, prepare for university transfer, and meet their personal goals.

The science department created a new mission statement during Fall 2021 to align with the college's updated format. We collected faculty comments via email and Jamboard, a collaborative, online whiteboard where faculty can post comments and share ideas. Science faculty then met to review the comments and identify themes to create the new statement.

**Previous Mission Statement**
The department provides coursework and supporting services and facilities which allow students to meet the degree and certificate requirements in programs of the college or to build a foundation of skills and knowledge for pursuit of advanced degrees, a lifetime of learning, and enhanced career performance.

Evidence:

- Natural Sciences Mission Jamboard

### Alignment to Institution Mission

```
How does the general education mission align with the mission of the institution?
```

These educational opportunities contribute to the college mission by fostering lifelong learning and meaningful participation in a global society.

## Student Learning Outcomes Assessment and Curriculum

### Curriculum Map

```
Curriculum  mapping ensures the alignment of TBR GenEd outcomes and student learning outcomes; which affords an opportunity to ensure course coverage and sequencing of outcomes in courses.

The reviewer should review the current mapping to ensure there is pathways to meeting the TBR GenEd outcomes for all general education students.
```

The Tennessee Board of Regents (TBR) states that the goals of the Natural Science curriculum are to "guide students toward becoming scientifically literate. This scientific understanding gained in these courses enhances students' ability to define and solve problems, reason with an open mind, thinking critically and creatively, suspend judgment, and make decisions that may have local or global significance." (TBR Proposal for the Establishment of a Lower Division General Education Core, November 2002).

## TBR General Education Outcomes

TBR has identified five Natural Science General Education Outcomes.
Students will demonstrate the ability to:

- **Outcome 1:** Conduct an experiment, collect and analyze data, and interpret results in a laboratory setting.
- **Outcome 2:** Analyze, evaluate and test a scientific hypothesis
- **Outcome 3:** Use basic scientific language and processes and be able to distinguish between scientific and non-scientific explanations.
- **Outcome 4:** Identify unifying principles and repeatable patterns in nature, the values of natural diversity, and apply them to problems or issues of a scientific nature.
- **Outcome 5:** Analyze and discuss the impact of scientific discovery on human thought and behavior.

## Vol State Science Department Outcomes

Our faculty created the following four outcomes by modifying the TBR General Education Outcomes to better align with the needs of the department.

Students will demonstrate the ability to:

- **Outcome 1:** Apply established scientific methods to the interpretation of experimental results. (TBR 1 and 2)
- **Outcome 2:** Use scientific principles to solve problems of a scientific nature. (TBR 1 and 2)
- **Outcome 3:** Demonstrate knowledge of scientific principles, processes and terminology. (TBR 3 and 4)
- **Outcome 4:** Discuss the impact of scientific discovery on human thought and behavior. (TBR 5)

## Course-level Learning Outcomes

Each general education natural science course has developed a set of course learning outcomes that can be linked to the above departmental outcomes. The attached curriculum map links course-level learning outcomes to our four departmental Learning Outcomes, which are then linked to the five TBR Learning Outcomes.

## Course Learning Objectives

Several courses have also developed a more detailed list of course topics, called Learning Objectives (LOs). The list of LOs for BIOL 2010 are attached as an example.

## TBR General Education Program Revisioning

TBR is conducting a comprehensive review of the General Education Core. The mission is to create a core "framework that best meets the needs of all student populations, disciplines, and pathways including a clear structure of general education global learning outcomes" (TBR General Education Core, 2021). TBR has asked schools to provide input on the highest level, global outcomes called Very Important Learning Outcomes (VILOs).

The figure below outlines the Gen Ed Revisioning process, with the goal to launch the new Learning Outcomes in 2025.

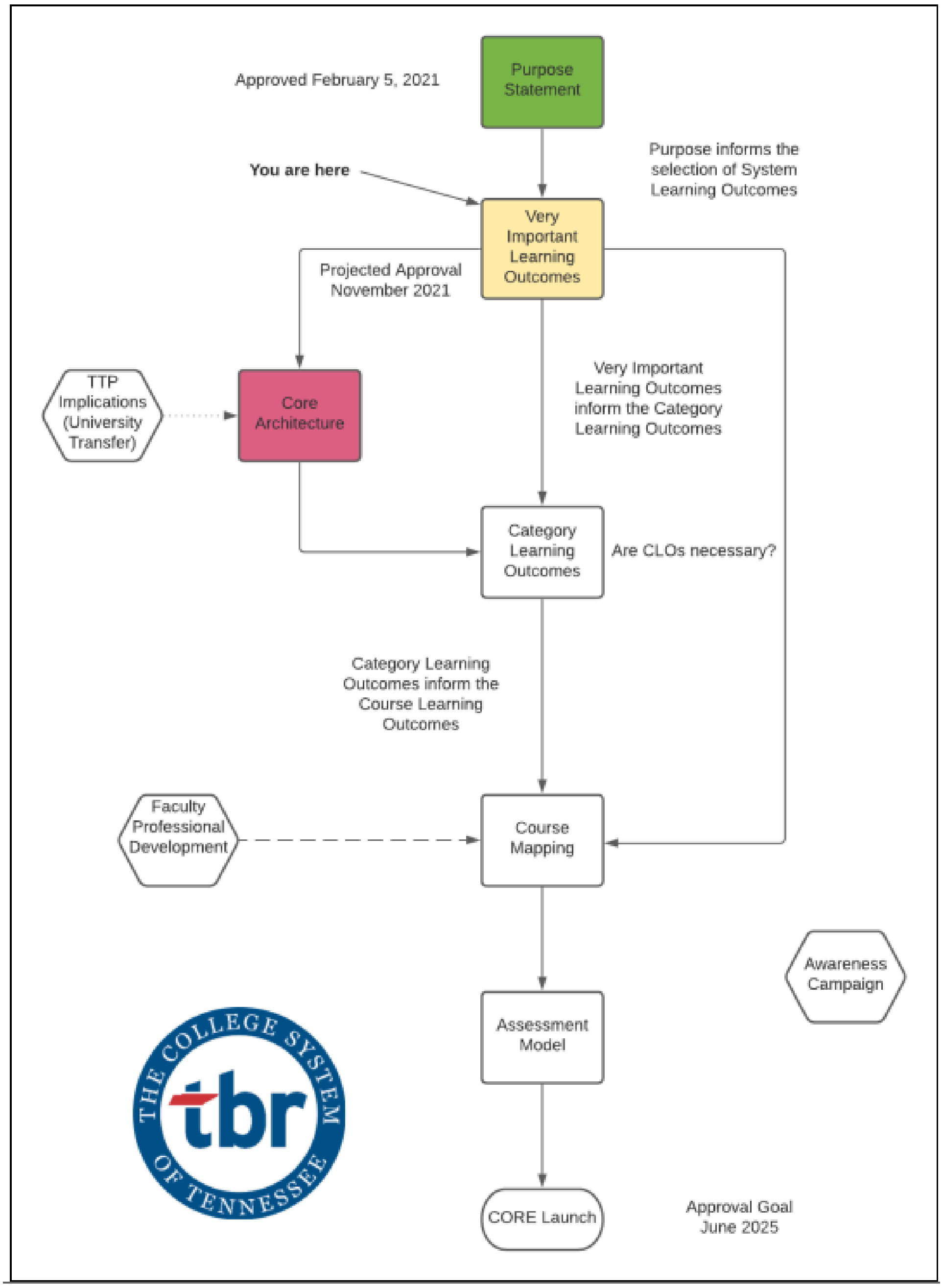



Figure 1. TBR General Education Core Revisioning Process overview.

Evidence:

- BIOL 2010 Learning Objectives
- Gen Ed Natural Science Course Outcome
- TBR General Education Outcomes

## Evidence of Student Learning

```
Evidence of Student Learning

Provide evidence of student learning, and the program's effectiveness in reaching both TBR GenEd outcomes and student learning outcomes.

Please provide the following:
1.      Summary of assessment most recent results.
2.      Analysis of trends.
3.      Ideas of how those results could be used to improve student learning.
```

General Education science faculty created outcome assessment tools for their courses and have been using them for several years. Course coordinators collect and analyze the data and post the results in a common folder on the T drive. In the past, each course was not required to assess every outcome. Additionally, the years under review were impacted by Covid-19, so the data in this review cycle is incomplete. We have identified this as an area for improvement and dedicated our first Student Learning Goal to collecting assessment data for all four learning outcomes in each section of every general education science course.

## Covid-19

The last two years under review were affected by the Covid-19 pandemic. The college announced that all classes were moving online during spring break in 2020, and faculty were given one additional week to prepare their classes before they resumed in an online or virtual format. Science faculty placed emphasis on maintaining our high quality of classes and academic rigor. We wanted to ensure that students would receive the specific skill sets they would learn through traditional on-ground teaching methods.

Laboratories were a challenge for science faculty to move online, especially in classes that teach hands-on skills and STEM classes. Instructors employed a variety of methods, including lab kits, online simulations, and videos demonstrating the lab activities. Some utilized "kitchen chemistry," where students conduct experiments at home using common ingredients. For example, DNA extraction from fruit using dish soap and salt in a biology class. Chemistry faculty recorded videos performing laboratory experiments. Students then used the data from the video and pictures for calculations as if they had performed the experiment. These videos were highlighted on the local news to demonstrate creative methods our faculty utilized to adapt their courses during the pandemic. ("Chemistry classes continue virtually." WSMV News. Sept 2020.) Our BIOL 1050 and 1060 did virtual field trips and invited guests to speak over Zoom. This includes doing Citizen Science Projects virtually and remotely in place of in person.

Faculty collaborated with many other departments, including Distributed Education, Media Services, and Information Technology to prepare their classes for online and virtual delivery. Distributed Education created an Online/Virtual faculty mentor position and a "faculty fellows" position to support faculty during this time.

Testing was a concern for many science faculty. Honorlock and the testing center were available for proctoring, but the capacity of the testing center is limited, and many faculty were skeptical about online proctoring services. Most faculty pivoted to giving exams on eLearn. The loss of the use of Honorlock during Spring 2022 is a significant concern for many faculty.

Access to the internet was a challenge for many students and faculty, especially at our satellite locations where internet service is not available in some areas. Information Technology provided faculty and students with laptops, Wi-Fi hot spots, video cameras, and other items needed to participate in their online courses. Faculty could also request instructional tools such as document cameras and smartpens enabled for touchscreen 2-in-1 laptops.

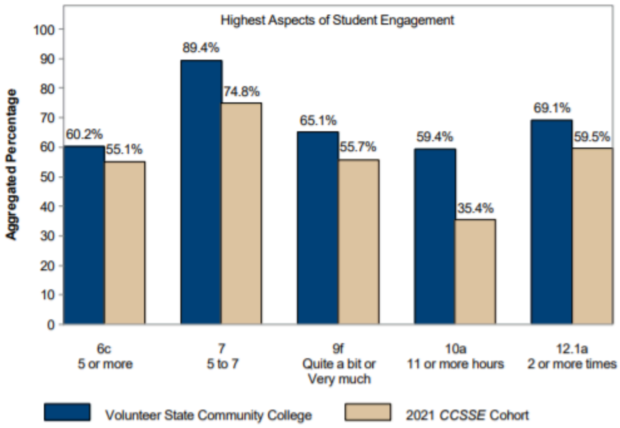
## Continuing effects of Covid-19

In a survey of more than 120,000 students across 273 colleges, greater than one-third of students described their financial situation as "worse than it was before the COVID-19 pandemic." More students are struggling to pay for college, with Asian, Hispanic/Latino, American Indian/Alaskan Native, and Black/African American individuals most affected. (CCCSE. (2021). The Continued Impact of COVID-19 on Community College Students.)

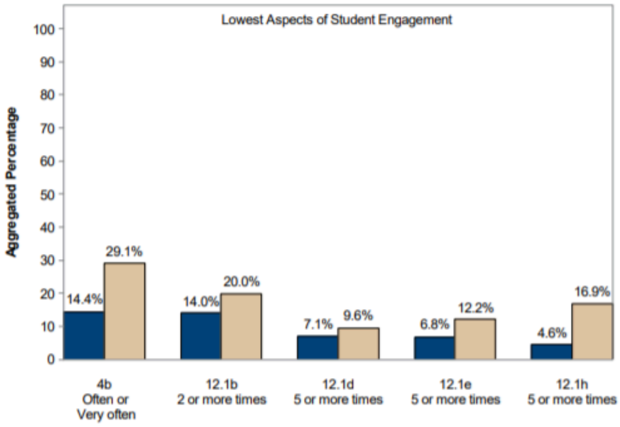
The financial stress of the pandemic may cause additional challenges for our students, with 29% of Vol State students reporting a lack of finances as "very likely" to cause them to withdraw. Approximately 36% reported working more than 30 hours per week. (CCSSE. (2021). Survey of Student Engagement, Volunteer State Community College).

Safety continues to be a significant concern for faculty and students. 59% percent of students reported trying to avoid situations on campus in which they are unable to stay 6 feet away from another person, with nontraditional-age students and women expressing the highest levels of concern. (CCCSE. (2021). The Continued Impact of COVID-19 on Community College Students.) Some faculty have utilized cohorts to reduce the number of students in the room at any given time, and adjustments were made to the schedule to maximize the time between classes for rooms to be cleaned. TBR recently changed the mask policy, and we will no longer be allowed to require students on campus to wear masks.

College enrollment has fallen since the pandemic, and we are making additional efforts to increase student success and retention. Student engagement is a factor that can increase success and retention. The graphs in Figure 2 represent Vol State's highest and lowest aspects of student engagement relative to a CCSSE Cohort. The lowest aspects could be areas for improvement. Class presentations could be a good idea for some courses looking for capstone projects for their learning outcome assessments.




| Item | Benchmark |
|---|---|
| 6c. Number of written papers or reports of any length | Academic Challenge |
| 7. Extent: examinations challenged you to do your best work | Academic Challenge |
| 9f. Providing the financial support you need to afford your education | Support for Learners |
| 10a. Preparing for class (studying, reading, writing, rehearsing, doing homework, or other activities related to your program) | Student Effort |
| 12.1a. Frequency: Academic advising/planning | Support for Learners |




| Item | Benchmark |
|---|---|
| 4b. Made a class presentation | Active and Collaborative Learning |
| 12.1b. Frequency: Career counseling | Support for Learners |
| 12.1d. Frequency: Peer or other tutoring | Student Effort |
| 12.1e. Frequency: Skill labs (writing, math, etc.) | Student Effort |
| 12.1h. Frequency: Computer lab | Student Effort |

Figure 2. Vol State's highest and lowest aspects of student engagement relative to a CCSSE Cohort. (CCSSE. (2021). Volunteer State Community College CCSSE Executive Summary of Results.)

Evidence:

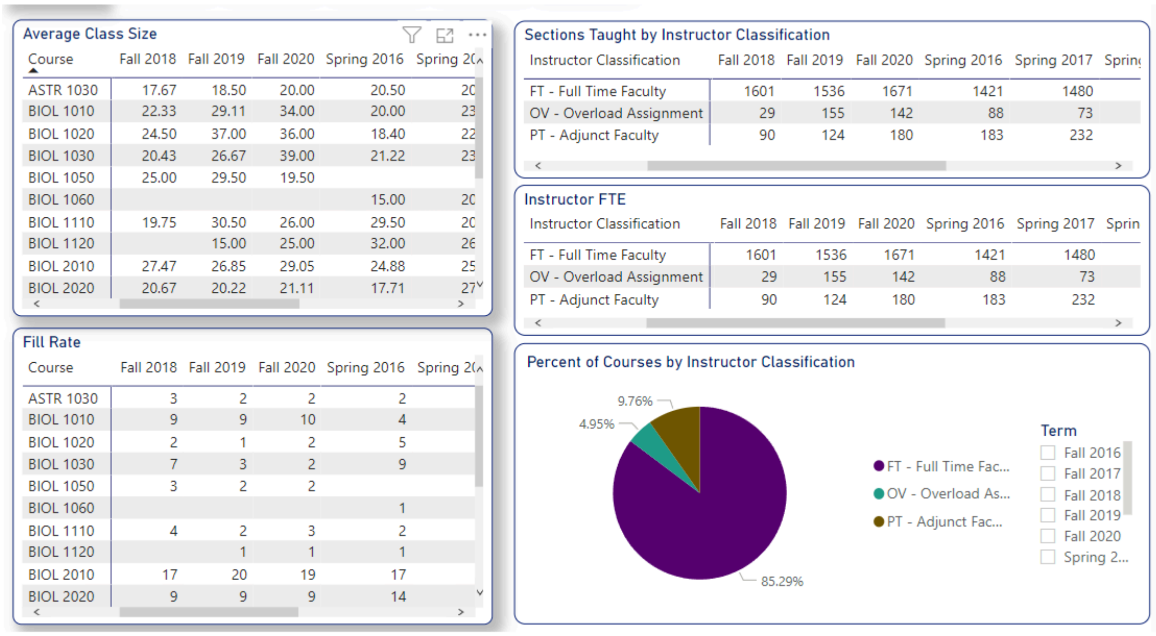
- CCSSE 2021 Executive Summary
- CCSSE Student Engagement 2021 Vol State
- Implementation Plan 2013 vs now

# Faculty and Courses

## Faculty Usage and Development (optional)

```
May review faculty teaching loads, adjunct to full time ratios, faculty development opportunities, faculty end of
course evaluations, faculty surveys, and/or any other evidence deemed appropriate. The reviewer could discuss the
impact of teaching, the ratio of adjunct to full time staffing, end of course evaluations relative to faculty, and
any noted themes identified from the review.
```

We have 25 full-time faculty and 10 adjunct instructors in the Science Department. We have several new additions to our faculty, with 16% our full-time instructors and 70% of our adjunct instructors hired during the review period. Full-time faculty members have an average of 15 TLE and 127 students. Adjunct instructors have an average of 7 TLE and 63 students. Figures 3 and 4 include the average class size, sections taught by instructor classification, and percent courses by instructor classification for General Education science courses.



Average Class Size

| Course | Fall 2018 | Fall 2019 | Fall 2020 | Spring 2016 | Spring 20 |
|---|---|---|---|---|---|
| ASTR 1030 | 17.67 | 18.50 | 20.00 | 20.50 | 20 |
| BIOL 1010 | 22.33 | 29.11 | 34.00 | 20.00 | 23 |
| BIOL 1020 | 24.50 | 37.00 | 36.00 | 18.40 | 22 |
| BIOL 1030 | 20.43 | 26.67 | 39.00 | 21.22 | 23 |
| BIOL 1050 | 25.00 | 29.50 | 19.50 | | |
| BIOL 1060 | | | | 15.00 | 20 |
| BIOL 1110 | 19.75 | 30.50 | 26.00 | 29.50 | 20 |
| BIOL 1120 | | 15.00 | 25.00 | 32.00 | 26 |
| BIOL 2010 | 27.47 | 26.85 | 29.05 | 24.88 | 25 |
| BIOL 2020 | 20.67 | 20.22 | 21.11 | 17.71 | 27 |

Fill Rate

| Course | Fall 2018 | Fall 2019 | Fall 2020 | Spring 2016 | Spring 20 |
|---|---|---|---|---|---|
| ASTR 1030 | 3 | 2 | 2 | 2 | |
| BIOL 1010 | 9 | 9 | 10 | 4 | |
| BIOL 1020 | 2 | 1 | 2 | 5 | |
| BIOL 1030 | 7 | 3 | 2 | 9 | |
| BIOL 1050 | 3 | 2 | 2 | | |
| BIOL 1060 | | | | 1 | |
| BIOL 1110 | 4 | 2 | 3 | 2 | |
| BIOL 1120 | | 1 | 1 | 1 | |
| BIOL 2010 | 17 | 20 | 19 | 17 | |
| BIOL 2020 | 9 | 9 | 9 | 14 | |

Sections Taught by Instructor Classification

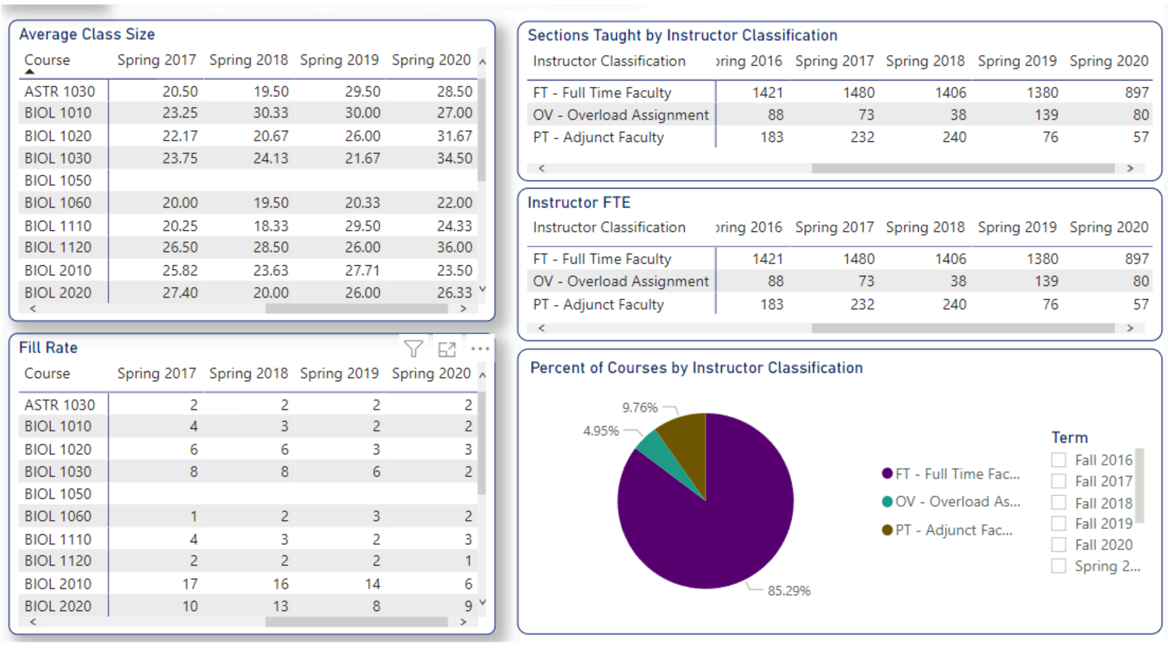
| Instructor Classification | Fall 2018 | Fall 2019 | Fall 2020 | Spring 2016 | Spring 2017 | Sprin |
|---|---|---|---|---|---|---|
| FT - Full Time Faculty | 1601 | 1536 | 1671 | 1421 | 1480 | |
| OV - Overload Assignment | 29 | 155 | 142 | 88 | 73 | |
| PT - Adjunct Faculty | 90 | 124 | 180 | 183 | 232 | |

Instructor FTE

| Instructor Classification | Fall 2018 | Fall 2019 | Fall 2020 | Spring 2016 | Spring 2017 | Sprin |
|---|---|---|---|---|---|---|
| FT - Full Time Faculty | 1601 | 1536 | 1671 | 1421 | 1480 | |
| OV - Overload Assignment | 29 | 155 | 142 | 88 | 73 | |
| PT - Adjunct Faculty | 90 | 124 | 180 | 183 | 232 | |



Figure 3. General education science course and faculty loads. All locations, fall semesters displayed.



Average Class Size

| Course | Spring 2017 | Spring 2018 | Spring 2019 | Spring 2020 |
|---|---|---|---|---|
| ASTR 1030 | 20.50 | 19.50 | 29.50 | 28.50 |
| BIOL 1010 | 23.25 | 30.33 | 30.00 | 27.00 |
| BIOL 1020 | 22.17 | 20.67 | 26.00 | 31.67 |
| BIOL 1030 | 23.75 | 24.13 | 21.67 | 34.50 |
| BIOL 1050 | | | | |
| BIOL 1060 | 20.00 | 19.50 | 20.33 | 22.00 |
| BIOL 1110 | 20.25 | 18.33 | 29.50 | 24.33 |
| BIOL 1120 | 26.50 | 28.50 | 26.00 | 36.00 |
| BIOL 2010 | 25.82 | 23.63 | 27.71 | 23.50 |
| BIOL 2020 | 27.40 | 20.00 | 26.00 | 26.33 |

Fill Rate

| Course | Spring 2017 | Spring 2018 | Spring 2019 | Spring 2020 |
|---|---|---|---|---|
| ASTR 1030 | 2 | 2 | 2 | 2 |
| BIOL 1010 | 4 | 3 | 2 | 2 |
| BIOL 1020 | 6 | 6 | 3 | 3 |
| BIOL 1030 | 8 | 8 | 6 | 2 |
| BIOL 1050 | | | | |
| BIOL 1060 | 1 | 2 | 3 | 2 |
| BIOL 1110 | 4 | 3 | 2 | 3 |
| BIOL 1120 | 2 | 2 | 2 | 1 |
| BIOL 2010 | 17 | 16 | 14 | 6 |
| BIOL 2020 | 10 | 13 | 8 | 9 |

Sections Taught by Instructor Classification

| Instructor Classification | ring 2016 | Spring 2017 | Spring 2018 | Spring 2019 | Spring 2020 |
|---|---|---|---|---|---|
| FT - Full Time Faculty | 1421 | 1480 | 1406 | 1380 | 897 |
| OV - Overload Assignment | 88 | 73 | 38 | 139 | 80 |
| PT - Adjunct Faculty | 183 | 232 | 240 | 76 | 57 |

Instructor FTE

| Instructor Classification | ring 2016 | Spring 2017 | Spring 2018 | Spring 2019 | Spring 2020 |
|---|---|---|---|---|---|
| FT - Full Time Faculty | 1421 | 1480 | 1406 | 1380 | 897 |
| OV - Overload Assignment | 88 | 73 | 38 | 139 | 80 |
| PT - Adjunct Faculty | 183 | 232 | 240 | 76 | 57 |



Figure 4. General education science course and faculty loads. All locations, spring semesters displayed.

Full-time instructors teach approximately 85% of general education science courses, compared to approximately 59% for the entire college, as indicated in figures 5 and 6. Adjunct instructors give our department flexibility to respond to changing enrollment demands. They bring an interesting perspective to our students, and the instructors grow both professionally and personally. We have developed great relationships with many dedicated adjunct instructors over the years who are a great asset to our students and our department.

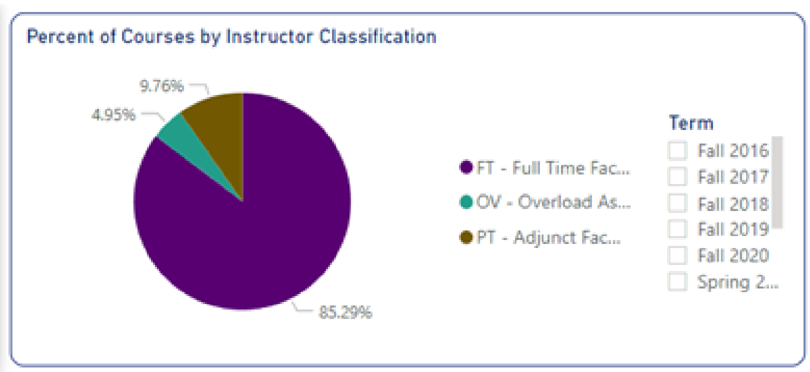



Figure 5. Distribution of classes by instructor classification. Science General Education.

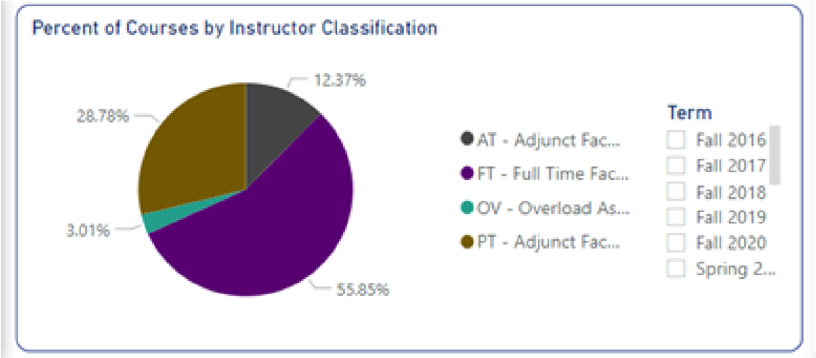


Figure 6. Distribution of classes by instructor classification. Entire college.

BIOL 1010 utilized the greatest number of adjunct instructors, with four online sections, and one conventional section at each of our Upper Cumberland locations, Cookeville and Livingston. Biology 2010 utilized two new adjunct instructors this semester to teach online, virtual, and conventional course formats. Figure 7 shows the number of adjunct instructors utilized in general education science courses during Fall 2021.

| Course | Adjunct Instructors |
|---|---|
| ASTR 1030 | 1 |
| BIOL 1010 | 3 |
| BIOL 2010 | 2 |
| CHEM 1030 | 1 |
| GEOL 1030 | 1 |

Figure 7. General education science courses utilizing adjunct instructors. Fall 2021.

## Course coordinators

All general education science courses have a faculty member assigned to perform and/or delegate tasks required for the maintenance of that course (Course coordinators job description, 2017). Course coordinators responsibilities include the following:

- Prepare and distribute an ADA-compliant syllabus with updated college statements and policies.
- Create a laboratory schedule and coordinate with the lab technician regarding the preparation and ordering of materials.
- Maintain eLearn model shells and course materials including, but not limited to, power points, laboratory guides, and comprehensive exams.
- Manage Institutional Effectiveness plans, assessments, and compilation of results from all faculty.
- Communicate with textbook publishers and the bookstore regarding course materials, textbook edition updates, and IncludEd, when applicable.
- In courses that utilize online materials such as Mastering, course coordinators are responsible for working with the publishers to ensure student and faculty access.
- Course coordinators may also be asked to create course guides, handbooks, or other materials.

Course coordinators are the primary support for new and adjunct instructors, and they were especially helpful during the pandemic, assisting faculty with the online transition by developing new course materials, eLearn shells, laboratories, and other resources.

The course coordinator position does not offer any release time or stipend. The department could consider providing benefits to course coordinators of highly demanding or high enrollment courses, such as A&P, or those who coordinate multiple courses. A reduced teaching load could provide the coordinators with the time needed to meet the various demands of the position. Additional job duties and goals of the BIOL 2010 and 2020 course coordinators are discussed below to demonstrate the need for release time:

During Fall 2021, Human Anatomy & Physiology had two course coordinators to manage more than 1500 students in 27 lecture and 40 laboratory sections. They work with 12 different instructors (10 full-time, 2 adjunct) and maintain eLearn models shells for online, hybrid, and traditional lecture and laboratory courses. During the pandemic, one of our A&P course coordinators requested CARES funding for two different online laboratory simulation programs. They built assignments using these programs and shared their resources with faculty. They also coordinated with the publishers to create faculty accounts and trained interested faculty on how to use these programs.

The A&P course coordinators are currently managing the creation of updated assessment tools to match the needs of our GER. They are also involved with the Gateway program and have identified student preparation as an indicator of student success in BIOL 2010. They are leading conversations regarding the addition of prerequisites, advisor training, and the addition of language in DegreeWorks to help students and advisors across campus determine if students are prepared for BIOL 2010.

Human Anatomy & Physiology needs a curriculum review. The laboratories should be updated with new activities and technologies. The faculty would like to communicate with our stakeholders to determine how well-prepared our students are for health science and nursing programs and identify areas for improvement to help guide our curriculum updates.

During the Fall 2021 semester, our two Human Anatomy & Physiology coordinators had 16.5 and 19.5 TLE, with one being asked to take an overload due to the need for A&P coverage. In the first year under review, Human A&P had only one course coordinator for both BIOL 2010 and 2020. A reduced teaching load could provide the course leaders with the time needed to accomplish our goals.

## Professional Development:

There are various professional development opportunities offered through the Teaching and Learning Center (along with co-sponsored programming with the Office of Diversity and Inclusion). In addition to traditional professional development opportunities offered by the TLC, science faculty are also encouraged to attend various conferences at the state and nationwide level, in an effort to broaden their expertise in their respective fields and their teaching. Faculty also have the opportunity to participate in learning communities to discuss various best practices regarding pedagogy in culturally responsive teaching (which is a part of our Gateway Course Academy Initiative through TBR). Along with all of that, we have several science faculty (5 to be exact – one being the Coordinator of the TLC) that serve on the TLC Advisory Board to develop and implement the various initiatives mentioned earlier (and some that will be listed later).

In regard to traditional professional development sessions, the sessions listed below are just a few of the opportunities our science faculty have participated in over the past 4 semesters. In the Fall of 2020, we had over 20 sessions attended by Math and Science Division faculty, and the same number was attended in the Spring of 2021. On average we have 3-5 science faculty in attendance at those sessions throughout the academic year. We also have some science faculty that present various professional development sessions through the TLC, which are also listed below.

In an effort to help faculty pivot to virtual and online teaching, the TLC hosted one professional development session almost every day in the Summer of 2020 (called the VSCC's 2020 Summer Series), ranging over a variety of topics – 15 science faculty participated. Along with that, we had 10 science faculty participate in VSCC's 2021 Summer Institute, which was a three-day intensive program covering topics on culturally responsive teaching, technology-enhanced learning, and "getting back to the basics".

**Selection of PD Sessions Attended by Science Faculty**

- Teaching Virtually and Online
- Safe Zone Training (all levels)
- Trauma-Informed Instruction
- Interactivity Tools and Project-Based Learning
- Student Success for Non-Traditional Students
- EdTech Workshops on Nearpod, Padlet, and Kahoot
- Learning Support Math Advising Session
- Faculty Advisors and Appreciative Advising
- Building Strong Brains
- Culturally Responsive Teaching in the Online Classroom
- Deconstructing Implicit Bias
- Identifying Microaggressions
- Successful Students Programming – Interdependence; Personal Responsibility; Connection and Belonging
- (and a ton more!)

**Selection of PD Sessions Presented by Science Faculty**

- Erin Bloom presented several sessions at the 2020 VSCC Summer Series
- Chrysa Malosh has presented various trainings on Teaching Virtually and Online since Spring 2020
- Shane Kelly offered a session on Calibrated Peer Review in Spring 2021
- Billy Dye presented a seminar on COVID 19 Vaccines in Spring 2021
- Erin Bloom presented several sessions and Girija Shinde presented one session at VSCC 2021 Virtual Summer Institute
- Erin Bloom has co-presented the Safe Zone: Basic training Spring 2021 and Fall 2021, and is slated to co-present again in Spring 2022
- Shane Talbott offered a session on Faculty Advisors and Appreciative Advising in Fall 2021
- Erin Bloom is slated to offer a session on Successful Students: Life-long Learning; an EdTech Workshop on Padlet; in Spring 2022
- Shane Talbott is also slated to offer a session on Student Services and Academic Affairs in Spring 2022

**Conferences attended**

- We All Rise in Fall 2020 – Erin Bloom and Maryam Farsian

- TBR Summer Institute in Summer 2021 – Erin Bloom (did present as well)
- IUPUI Assessment Institute in Fall 2020 and 2021 – Erin Bloom
- Transforming STEM Higher Education in Fall 2021 – Erin Bloom

Our goals are to increase participation in TLC-sponsored professional development throughout the year and to increase the number of science faculty that are hosting TLC-sponsored professional development sessions on our campus. Also increasing participation in VSCC's Summer Institute and Learning Communities. As the TLC has moved more/most PD virtually, we have been able to increase faculty participation from our other campuses. We have gotten great feedback and the TLC has really enjoyed being able to see faculty at these locations having the chance to participate. While we have seen a LARGE increase in participation across the college and within our Division and Department, we want to continue to gather momentum and provide support to our faculty, which includes getting them in the door at some of these programs.

As faculty continue to attend state and national conferences, plans are being made to host "lunch and learn" sessions to hear more about what our other science faculty have learned and what can they use at VSCC. The Coordinator of the TLC, the Science Department Chair, and the Assistant Chair are working on a program to coach science faculty to reflect on how culturally responsive their courses are and provide support to help faculty develop and eventually implement changes they would like to make.

Also in Spring 2022, our Gateway Course (Chem 1030 and Biol 2010) Coordinators will begin piloting small changes in those courses in the effort to close equity gaps, as well as continue to analyze data for future course changes. This is being done with the support of the TLC and the Gateway Course Academy Task Force.

## Learning Commons

The science department began offering free tutoring through the Learning Commons in Fall 2019. Prior to Covid, all tutoring was done on campus, leaving students at the satellite locations with limited opportunities for tutoring. On the Gallatin campus, Human Anatomy & Physiology held "open lab" study sessions for students. Some attempts were made with chemistry, but there was very little student participation.

In response to the pandemic, we followed the Math department's model of a virtual learning commons to increase the number of tutoring sessions we could offer and the number of students we could reach. Figure 8 indicates the number of students who visited the learning commons for each course. Future goals include marketing improvements to reach more students and to appear more welcoming for all students who need help.

| Course | Fall 2020 | Spring 2021 |
|---|---|---|
| BIOL 1010 | 4 | 0 |
| BIOL 1020 | 1 | 0 |
| BIOL 1120 | 8 | 0 |
| BIOL 2010 | 52 | 65 |
| BIOL 2020 | 2 | 50 |
| CHEM 1030 | 23 | 26 |
| CHEM 1110 | 30 | 13 |
| CHEM 1120 | 0 | 22 |
| GEOL 1030 | 4 | 1 |
| PHYS 1030 | 10 | 6 |
| **Total** | **134** | **183** |

Figure 8. Learning Commons student use data, Fall 2020 and Spring 2021.

Evidence:

- Science Dept Course Coordinator Job Description

## Class Offerings (optional)

```
May address scheduling effectiveness and space utilization. The reviewer could review faculty to student ratios at
each location and online and/or discuss the impact relative to faculty and student's success.
```

General education science classes currently have approximately 3500 students enrolled in 148 lecture and laboratory sections during Fall 2021. Biology is the highest-enrollment subject, with more than 4x the number of students than chemistry, the next highest subject. Human Anatomy & Physiology and Introduction to Biology are the highest enrollment courses.

| Subject | Lecture Students | Lecture Sections |
|---|---|---|
| Astronomy | 26 | 1 |

| | | |
|---|---|---|
| Biology | 1273 | 49 |
| Chemistry | 285 | 14 |
| Geology | 87 | 4 |
| Integrated Science | 15 | 1 |
| Physics | 50 | 4 |
| Physical Science | 27 | 1 |

Figure 9. The number of students and sections per subject. Lecture courses only, general education, Fall 2021.

The average class size for most general education science course sections is between 20-30 students per class. The college average is 21 students per class. Class sizes vary according to the subject and teaching facility, but the science department consistently offers the largest class sizes at the college. The faculty handbook states that web-based (online) courses should have a generic maximum of 25 students, but online science courses have a maximum capacity of 40. (Vol State Faculty Handbook, 2021). The college could consider smaller class sizes to align with the other departments and so faculty can provide science students with more individualized support.

### Online and Virtual education

The number and variety of non-traditional course offerings increased during the pandemic, with demands for online, virtual, and hybrid courses remaining high. Figure 10 shows the number of sections offered by format in Fall 2018 and 2021.

| Semester | Conventional | Virtual | Online |
|---|---|---|---|
| Fall 2021 | 29 | 26 | 22 |
| Fall 2018 | 81 | 0 | 0 |

Figure 10. The number of lecture sections offered with conventional, virtual, online, and hybrid formats. General education science.

Starting in Spring 2022, all online classes must go through review by Distributed Education. The process is described on our Distributed Education New Course Development page. Several of our general education classes have already gone through review. One of our online course developers had the following feedback regarding Distributed Education and the use of open educational resources (OER):

"The development of the open educational resource based BIOL 1050 Master Course was productive. The Distributed Education Instructional Designer did an outstanding job working with both course developers via Zoom meetings over the summer 2021 semester. The Instructional Designer was able to pull the strengths from all involved to help create a course that offers the richness of current issues in the field of environmental science and maintain the academic rigor requirements using open educational resources. The Instructional Designer was especially helpful with various technical issues such as building HTML files from OER material, imbedding videos, and providing congruency among the lessons."

### Scheduling

Scheduling was especially challenging during the pandemic. We monitored the schedule closely and made changes as needed to respond to enrollment demands, administrative requests, and faculty needs. This meant that schedules were changed frequently, and classes were sometimes changed after students had already registered. Faculty and students were flexible and accommodating as we attempted to balance the schedule, but schedule changes will increase confusion and should be reserved for rare and special circumstances.

Suggestions to improve scheduling include more involvement of course coordinators and an improved STEM schedule that allows students to take each course recommended by their program of study within each semester. New AdAstra software was implemented in Fall 2021 to project and track enrollment across the college. This may eliminate the need for a STEM schedule, and it should help us meet student needs regarding the schedule.

### Satellite campus locations

We currently offer science classes at three satellite locations – Cookeville (CHEC), Livingston (LIV), and Springfield (SPR). We have full-time faculty who teach permanently at these locations and those who split their time between locations. Livingston and CHEC regularly utilize adjunct instructors, but the Springfield campus rarely does, as indicated in the figures below.

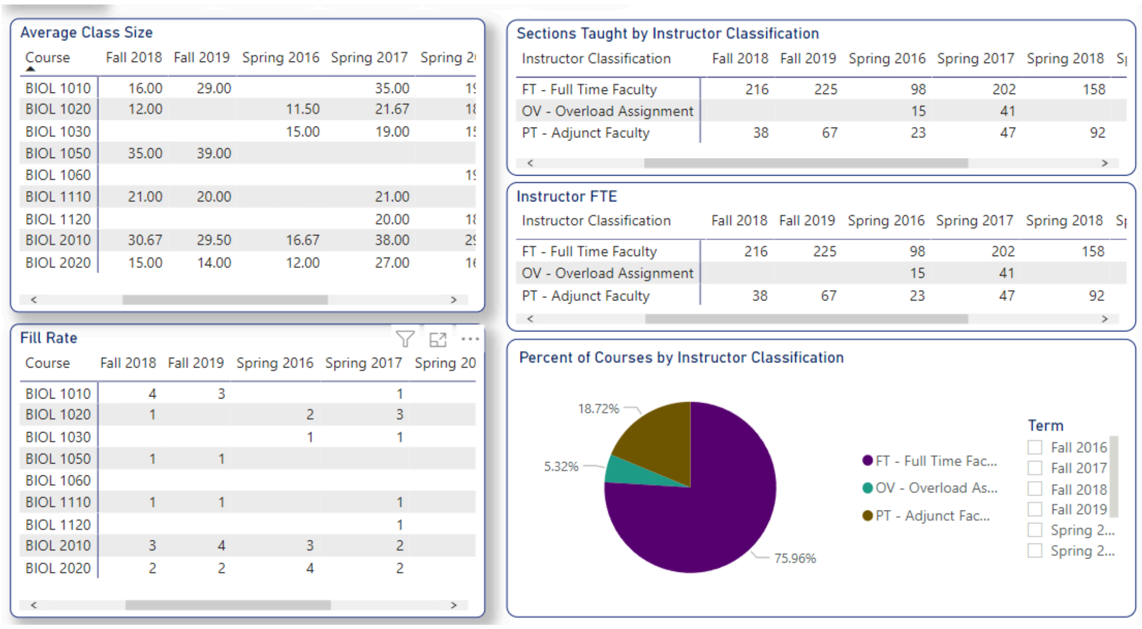



Figure 11: Course and faculty loads, CHEC and Livingston locations. Fall displayed.

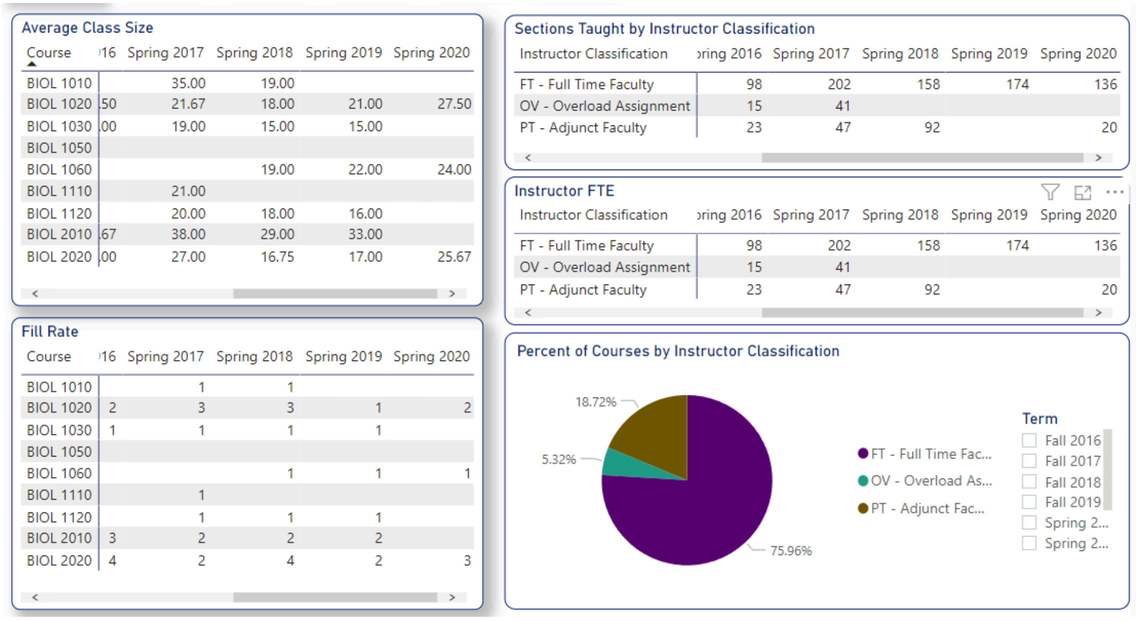



Figure 12: Course and faculty loads, CHEC and Livingston locations. Spring displayed.

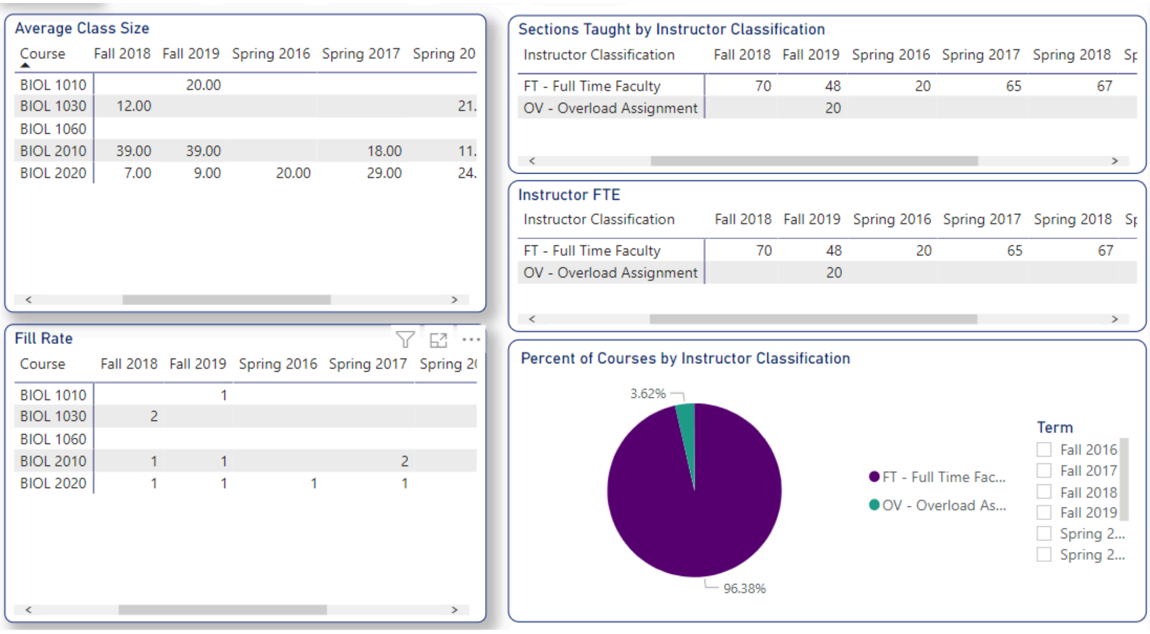



Figure 13: Course and faculty loads, Springfield location.

**Facilities**

The Warf building on the Gallatin campus was recently renovated. The renovations improve laboratory spaces, as well as provide more space for instruction, research, and storage. The renovations improve some but not all scheduling challenges, as we still have a limited number of rooms, and we must share some of our lecture space with the Mathematics Department. The renovations also provide a maximum class size of 35 in the Warf building, but science lecture courses usually have a maximum of 40, further limiting our use of these rooms.

Faculty report that the satellite location facilities meet the basic needs of students and faculty, but each location has areas for improvement. The diversity of course offerings is limited because each location is not equipped to offer all science courses. For example, the Livingston lab is not properly equipped for chemistry classes, so they can only be offered from CHEC. The CHEC lab is deficient in many areas, not only for chemistry but also biology courses. For example, there is no electricity at the lab tables, therefore microscopes can only be used on the counterspace around the perimeter of the room, requiring students to face the wall while instructors talk to their back. Microbiology cannot be offered at CHEC, but it is a regular offering at LIV. We are offering our first microbiology course at Springfield during Spring 2022.

Office space is limited at the satellite campuses. Many LIV and CHEC faculty teach at both locations but only have office space at one location. Most Springfield faculty share a common office space, so they may not always have access to a quiet workspace and a computer. Prior to the pandemic, faculty at our satellite locations were able to access their offices after hours, but now they lose access whenever the site is closed. This includes weekends and any weekday after 4:30 pm if there are no on-campus classes scheduled that night. Faculty cannot access their offices after 4:30 pm Friday until Monday morning. This creates a significant challenge for faculty at these locations, especially during the pandemic when classes are taught primarily online, and some faculty live in areas with limited access to the internet.

LIV and CHEC faculty do not have reliable copier/printer access. These devices malfunction frequently, and there are no backup options (other copiers that can be used). If a copier has an issue at Livingston, for example, then faculty may not be able to print or make copies for that day, or in some cases, a full work week. There is only one fully functional copier on the Livingston campus. This is the same for CHEC. If each campus had at least two fully functional copiers/printers, this would help resolve the issue.

As we return to on-campus instruction, not only are printers and copiers utilized more often, but we also rely on a scantron machine to grade some objective assessments. However, there is only a scantron machine at the Livingston campus and it is not fully functional. It will grade 50 questions at a time. If more than 50 questions are to be graded, any questions over 50 will be graded alone and not incorporated in the first 50. This results in two scores for the test instead of one. Instructors must implement manual grading to justify this issue. If both CHEC and Livingston could be furnished with new scantron machines, this would be an area of improvement. Currently, if our chemistry professor who is stationed at CHEC uses scantron for testing, they must drive to Livingston to grade the tests.

Additionally, the front door at the Livingston campus is labeled as accessible, yet it does not consistently open when the automatic button is pushed and must be opened manually. This is the only entrance for student use at the campus. If someone using a wheelchair needs to enter through that door, it may or may not work.

**Scheduling and room availability at our satellite locations**

The addition of the CHEC facility to the Upper Cumberland created a need for the assistant chair position to coordinate schedules between the two Upper Cumberland campuses. This has resulted in an increased ability to offer more science courses and optimal schedules for both faculty and students.

Classroom availability does not currently seem to be an issue at the Livingston and Springfield locations, but their ability to grow will be limited by the constraint of having only one laboratory room at each site. For example, one lab room is used for BIOL 1010, 1020, 1030, 2010, 2020, and 2230 at both sites. Adding an additional lab space would allow for increased science offerings at each location.

Room availability is an issue at CHEC, as TCAT and TTU continue to progressively acquire rooms at the facility each semester, reducing Vol State's options for room use.

**Adjunct instructors at the satellite locations**

It can be difficult to acquire and retain adjunct instructors at our satellite locations. The Upper Cumberland lost all except one adjunct during the onset of the pandemic. This one adjunct is versatile and willing to teach day or night. As some low enrollment courses are attempted to be placed back on campus, this adjunct has been assigned to them. However, we are concerned if these courses are canceled, we may lose our last, experienced adjunct. One way to prevent this may be to consider assigning this adjunct to a virtual synchronous course if possible. This could ensure that the adjunct instructor will continue to work through next semester and hopefully more often, as enrollment increases in the following semesters. The Upper Cumberland may also need a chemistry adjunct instructor in the future.

**Accomplishments and improvements at the satellite locations**

There are several areas where improvements have been made in the Upper Cumberland in the past years. During the last additions to the Livingston campus, each full-time faculty member received their own office. Previously, science faculty utilized the laboratory prep room as an office and shared this space. As well, science faculty at CHEC now each have their own office space. This is an improvement from the first few years CHEC was open, where all science faculty shared one office.

Another major improvement to the Upper Cumberland locations was the hiring of our laboratory technician, Patrick Kent. He manages all lab supplies and preps most labs. Prior to the addition of his position, science faculty had to prepare each lab, which was very difficult for faculty teaching multiple different courses and/or at multiple locations. For example, a faculty member may teach BIOL 1030, 2010, and 2230 in the same semester and at multiple campus locations within the same day. The addition of a lab tech to serve the LIV and CHEC locations has been a great benefit to the faculty and students. Springfield currently does not have a laboratory technician, and faculty must prepare all their labs.

Upper Cumberland Science faculty say they feel more included in discussions that affect the department within the past few years. They have made a "wish list" to summarize their needs regarding facilities and course offerings:

LIV/CHEC faculty wish list:

1. Two fully functional copiers on each campus
2. Fully functional scantron machines on each campus
3. Afterhours access to offices on each campus
4. The front door at Livingston is not accessible for someone in a wheelchair, most of the time. The door doesn't work consistently.
5. Chemistry lab at Livingston campus
6. One more Biology lab at Livingston campus
7. Fully functional chemistry and Biology labs at Cookeville campus

## International Education

We have offered a variety of opportunities for international education with general education courses. We have a BIOL 1060 course planned for Japan during summer 2022, and we have a section of BIOL 1030 at the Livingston location that will be tied to a special topics course where the students will travel to Ecuador and the Galapagos Islands during spring break.

Travel was not possible during the pandemic, and it remains limited, but our faculty were still able to provide students with international connections through virtual exchange programs. Students from Vol State worked with students from Wuxi Health School and Hangzhou Normal University in China to solve problems for their classes as described below:

Santino Ladogana's CHEM 1030, 1110, and ISCI 1030 students participated in a Green Chemistry Project in which they wrote a 3-4 page paper and gave a poster presentation on a selected topic. The students from Vol State experienced global exposure and an international perspective on the future of sustainable living. They acquired skills on how to communicate with different cultures. For example, they had to adjust to the time difference and to speak clearly so a non-native speaker can understand them. The new environment was daunting at first, but most came out of the exchange program with a new feeling of confidence in interacting with people from different cultures.

Girija Shinde's BIOL 2020 students worked with 70 health-science students from the Chinese schools. Dr. Shinde provided the following comments about the Chinese exchange program and her teaching abroad experiences.

"The purpose of this exchange was to interact and learn several aspects (culture, health care, diseases, education, nursing profession, Covid 19 response, holistic medicine) with Chinese students, to explore the health issues in China, and to collaborate and develop friendships with the Chinese students. The objectives of this exchange were to identify the major health issues in the USA and compare the same with China, recognize the similarities and differences of the same health issue in different countries, learn the methods of treatment in both countries, explore any holistic methods of cure /prevention of the diseases, and expose students to solving problems on health issues at the international level. This exchange program gave a broader perspective about the same health issue in a different country. They learned how the same health issue is treated differently in another culture. In addition, some of my rural students also had email conversations about the lifestyle and challenges with resources in rural America. The students also shared each other's hobbies and schooling experiences. Some students also exchanged recipes like they requested kimchi recipes from their Chinese counterparts! Overall, it was an amazing experience for my students, but there were some challenges like time differences, technology issues, and language barriers, but the students gained innovative ways to communicate."

"Though I have offered the same course in each of my study abroad trips, I have always tailored it to match the environmental issues in that country. All the assignments, field trips, field reports were different for each place. So, it was exciting to be innovative and resourceful with all the available opportunities in each country. It has enriched my teaching and life experience which I share with all students at home."

"I am directing the study abroad trip to The Galapagos Islands during the spring break of 2022. I will be teaching a one-credit Special Topics in Biology course which deals with topics carefully selected to meet the needs of this specific group. The program site which is The Galapagos Islands will be a perfect place as a living laboratory for the students of Biology to study selected topics that are relevant to this specific program site. The students will study topics that will include the study of native and invasive flora and fauna, survival of the native species, factors responsible for thriving of invasive species, and the study of endangered and endemic species. This course will enable the students to gain firsthand knowledge about all the selected topics at one location. This would be a perfect location for the discussion of less genetically diverse species, the effect of the environment on the survival of the flora and fauna of the Islands. Finally, the students will also be able to study Darwin's theory of evolution firsthand. The Galapagos Islands is one of the dream destinations of biologists to study the invasive and indigenous flora and fauna in real-time. That is why it is called the "Mecca" of Biologists."

## Recommendations and Next Actions

### Analysis and Summary

`Analyze and summarize the proceeding sections. The reviewer should discuss what went right, what went wrong, and the implications of those results.`

The science department has a robust plan for assessing our general education learning outcomes. We utilize established assessment tools and have been collecting data for many years. The college has a new plan for learning outcome assessments, with an increased focus on individual student scores and reporting through the Watermark/Via system. All general education courses must also assess each student learning outcome, which was not a requirement in the past. These changes in reporting and the GER self-study have provided us with a valuable opportunity to re-evaluate our strategies and identify areas for improvement. Several courses chose to make significant changes to their assessment tools, and we look forward to seeing the results of their efforts and how individualized student reporting can be used to improve success and retention.

**Science GER Committee members:**
Maryam Farsian, Department Chair, Associate Professor of Biology
Erin Bloom, Associate Professor of Biology
Robert Carter, Professor of Biology
Clark Cropper, Professor of Geology
Santino Ladogana, Assistant Professor of Chemistry
Chrysa Malosh, Instructor of Chemistry

The GER committee consisted of faculty volunteers who met weekly from October 6 to December 7. We worked with course coordinators in October and November to introduce them to the new reporting system and learning outcome assessment requirements. The Science Department met to review the results of our self-study and discuss our GER plans on December 3rd (Meeting recording). Faculty submitted assessment rubrics to the IE Officer during the last weeks of Fall 2021 and the first weeks of Spring 2022. Our first assessments will be reported in Spring 2022.

**Thank you to the following faculty contributors to this self-study:**

- Erin Bloom - mission statement, professional development, and editing
- Linda Cathey - satellite campus locations (CHEC, LIV)
- Rufus Darden - Assistant Chair of the Science Department, satellite campus locations (CHEC/LIV)
- Mark Green - online course development
- Santino Laodgana - virtual travel abroad
- Chrysa Malosh - learning commons and editing
- Jonathan Martin - satellite campus locations (SPR)
- Girija Shinde - virtual travel abroad, international education, and satellite campus locations (CHEC/LIV)
- Brenda Wolff - course coordinators and online course development

Thank you to our Institutional Effectiveness and Assessments Officer, Jonna Alexander. She guided us through the GER process, attending meetings with our department, the GER committee, course coordinators, and individual faculty. We appreciate all her efforts to help the science department conduct our self-analysis and develop a sustainable plan.

### Next Actions

`The reviewer should discuss the next steps for the general education for the following two academic years. A plan of action and timeline should be included in the narrative or document library.`

`Feedback and approval will be provided. Refer to the GER timeline for approval deadlines.`

### Two Year Recommendation Action Plan

**Student Learning Goal 1:** Every general education science course will assess all four TBR learning outcomes. Currently, 76% of our general education courses are assessing some of the outcomes. 38% are assessing all four outcomes. We are setting a goal of 100% compliance in science courses, and 80% compliance in course sections.
Strategy 1: Course coordinators will develop assessment activities.
Strategy 2: Faculty will receive training pertinent to assessment tools.
Strategy 3: Course coordinators will work with course faculty to evaluate the data and decide if changes to the course should be implemented.

**Student Learning Goal 2:** Science department will implement a sustainable process to utilize assessment data. Departmental faculty will meet to discuss assessment results and determine what steps should be taken to improve student learning and to close the assessment loop. The strategic plan should include efforts to improve communication and collaboration between faculty members, which will improve the alignment between courses, subjects, and within the entire department.
Strategy 1: A survey will be administered to faculty to assess their feelings regarding collaboration and identify areas for improvement. We will compare the results to the past faculty surveys in 2007 and 2012 (results attached).
Strategy 2: Based upon our survey results, we may initiate efforts to improve collaboration among faculty to make the department and division work more cohesively.
Strategy 3: Department/Division leadership will work with faculty and course coordinators to ensure the assessments are meeting our needs and to guide discussions regarding improving student learning.

**Student Success Goal 1:** Analyze factors that bear upon student success and identify potential curriculum changes.
Strategy 1: Evaluate data from IERPA related to student success factors and consider if changes should be made to prerequisites, catalog descriptions, advising, etc. to ensure students are fully prepared to succeed in our class at the time of registration. This effort will be tied to our Gateway Course Initiative.

**Student Success Goal 2:** Science department will ensure campus facilities meet the curriculum requirements and student needs at all campus locations. Special attention will go to the satellite locations that have the greatest need.

**Additional efforts to improve the Science Department and our Gen Ed courses:**

- Deployment of AdAstra scheduling software, improvement of our STEM schedule, and efforts to maximize enrollment and space utilization while remaining sensitive to student and faculty comfort levels, as well as compliant with any mask or social distancing guidelines.
- Offer a variety of course formats and times to meet student needs.
- Improve support and consider possible release time for course coordinators of highly demanding courses or those who coordinate multiple courses.
- Support online education with the Distributed Education course development process.
- Continue to offer excellent professional development opportunities. The science department will make specific efforts to increase inclusion, accessibility, and cultural sensitivity within our course materials.
- Continue to support our satellite locations and find solutions to problems including enrollment challenges, limited office space and facilities, and access to campus and instructional resources.
- Continue to support international education opportunities and find new ways to expand the worldview and cultural knowledge and skills of our students.

# Feedback and Review

## Feedback (General Education Committee)

Presentation by faculty to the General Education Committee of the Program Review. The General Education Committee will provide feedback on the completed review and action plan. The feedback will be included and summarized.

## Dean(s) Approval and Feedback

The Dean(s) of the general education discipline(s) will provide feedback and approval on the completed review and action plan.

## VPAA and AVPAA

The VPAA & AVPAA will provide feedback on the completed review and action plan.

# Action Plan Reports

## Year 2 Action Plan Review

The reviewer should review progress on action plan over the past year. Discussion should include challenges and successes. If any changes are planned due to the results, those should be discussed and an updated action plan should be included in the narrative or document library.

### Year 3 Action Plan Review

The reviewer should review progress on action plan over the past year. Discussion should include challenges and successes. The reviewer should discuss the actions future and continuous in the program.

## Narrative:

### Two Year Recommendation Action Plan

**Student Learning Goal 1:** Every general education science course will assess all four TBR learning outcomes. Currently, 76% of our general education courses are assessing some of the outcomes. 38% are assessing all four outcomes. We are setting a goal of 100% compliance in science courses, and 80% compliance in course sections.
Strategy 1: Course coordinators will develop assessment activities.
Strategy 2: Faculty will receive training pertinent to assessment tools.
Strategy 3: Course coordinators will work with course faculty to evaluate the data and decide if changes to the course should be implemented.

**Student Learning Goal 2:** Science department will implement a sustainable process to utilize assessment data. Departmental faculty will meet to discuss assessment results and determine what steps should be taken to improve student learning and to close the assessment loop. The strategic plan should include efforts to improve communication and collaboration between faculty members, which will improve the alignment between courses, subjects, and within the entire department.
Strategy 1: A survey will be administered to faculty to assess their feelings regarding collaboration and identify areas for improvement. We will compare the results to the past faculty surveys in 2007 and 2012 (results attached).
Strategy 2: Based upon our survey results, we may initiate efforts to improve collaboration among faculty to make the department and division work more cohesively.
Strategy 3: Department/Division leadership will work with faculty and course coordinators to ensure the assessments are meeting our needs and to guide discussions regarding improving student learning.

**Student Success Goal 1:** Analyze factors that bear upon student success and identify potential curriculum changes.
Strategy 1: Evaluate data from IERPA related to student success factors and consider if changes should be made to prerequisites, catalog descriptions, advising, etc. to ensure students are fully prepared to succeed in our class at the time of registration. This effort will be tied to our Gateway Course Initiative.

**Student Success Goal 2:** Science department will ensure campus facilities meet the curriculum requirements and student needs at all campus locations. Special attention will go to the satellite locations that have the greatest need.

**Additional efforts to improve the Science Department and our Gen Ed courses:**

- Deployment of AdAstra scheduling software, improvement of our STEM schedule, and efforts to maximize enrollment and space utilization while remaining sensitive to student and faculty comfort levels, as well as compliant with any mask or social distancing guidelines.
- Offer a variety of course formats and times to meet student needs.
- Improve support and consider possible release time for course coordinators of highly demanding courses or those who coordinate multiple courses.
- Support online education with the Distributed Education course development process.
- Continue to offer excellent professional development opportunities. The science department will make specific efforts to increase inclusion, accessibility, and cultural sensitivity within our course materials.
- Continue to support our satellite locations and find solutions to problems including enrollment challenges, limited office space and facilities, and access to campus and instructional resources.
- Continue to support international education opportunities and find new ways to expand the worldview and cultural knowledge and skills of our students.

**REVIEW OF ACTION PLAN:**

The overall strategies of the science department were to reach a place where not only assessment was being done, but that the assessment was also meaningful and impactful for the faculty to improve the learning outcomes of their courses. The science department made great strides but will use the findings of this action plan to move forward into our next General Education Review (GER). The course coordinators worked with their faculty to develop assessment tools. The faculty identified test questions and course activities that measured the four leaning outcomes to "close the loop" on the data collected. What the faculty found was that while the assessments were good that the assessments may need to be further revisited.

One course sequence to highlight during this GER was the Anatomy and Physiology courses. Over the years our student success data had shown that students were not prepared for the course. As we reviewed the data, we began to see a pattern where students with an ACT composite of 19 or higher improved success as well as completing CHEM 1030 or a biology course improved success. Using this data, the science department changed the pre-requisite for the course to include either an ACT of 19 or higher or completion of CHEM 1030 or BIOL 1030 with a C or higher. The goal is to review this course over time to determine if the new pre-requisites do improve the success rates for students. This process really connects back to Student Learning Goals 1 and 2 and Student Success Goal 2. The work here connects to Student Success Goal 1 because anatomy tied to the previous Gateway Course Initiative as well as CHEM 1030 tied to that initiative.

While we were not able to implement Strategies 1 and 2 under Student Learning Goal 2, we have worked together as a science department and as a division to determine how we can connect student learning goals across the departments in our division. While no survey was conducted, informally improved connections happened organically as assessment data began to be utilized more intentionally. For instance, the Anatomy and Physiology faculty consulted with the chemistry and other biology faculty to arrive at the courses added to the Anatomy and Physiology I pre-requisite. While the science department sees the need to further improve collaboration, efforts have been made to improve collaboration. Furthermore, the science department has collaborated with the math department to improve course offerings related to the STEM schedule to diminish the number of scheduling conflicts to improve degree completion for students.

Moreover, the area where the science department felt they closed the looped was under the "**Additional efforts to improve the Science Department and our Gen Ed courses".** The science department chair, the dean, and the assistant dean began utilizing AdAstra to impact some improvements in scheduling. We also have made great strides in not only returning to more on-ground from the pandemic, but we have worked with Star Boe and her team to improve or develop model courses to teach asynchronous options and hybrid options of our courses. While we have seen success in improving student needs related to scheduling, we hope to monitor success in both the improved asynchronous and hybrid courses to improve further to positively impact student success. These novel course offering strategies have allowed us to offer courses for our regional campus needs as those enrollments grow after the pandemic.

We have begun a remodeling effort to improve the chemistry lab at the Cookeville campus beginning our plans to improve facilities at the regional campuses. Moreover, the Maddox Building remodel on the Gallatin campus will create a multi-use laboratory for mostly geology use but could be useful for general education biology courses.

While we were not able to add information in our Year 2 review due to some changes in personnel, changes in IERPA, and movement of academic assessment to academic affairs which impacted some of our work on the project. However, the science department has demonstrated some successes with our action plan in several areas. Moreover, this GER cycle has allowed the department to make significant strides to improvement of student success, modalities of courses, collaboration between faculty, and changes in some course pre-requisites by following the data. The science department plans to close the loop further as we develop our next GER in our improved data culture.

# General Education Review Cohort B- Cycle 1

Social Behavioral Sciences GE

## Reflection and Mission

### Previous General Education Review

Reflect on the past GER (if any) and responds to any recommendations that were made at that time. The reviewer should list recommendations, actions taken, and discuss implications or challenges resulting from those actions.

**Narrative:**

The information for this report is based on a compilation of input from three different departments within the Social Science and Education Division; the Social Sciences Department the Behavioral Sciences Department, and the Physical and Health Education Department. The Social Sciences Department reported on Geography, Economics, and Political Science courses. The Behavioral Sciences Department reported on Psychology and Sociology courses, The Physical and Health Education Department reported on Wellness courses.

The Social and Behavioral Science courses at Vol State have not undergone a general education review. The departments underwent an academic audit in the 2015-2016 school year. Between 2017 -2019 the departments conducted institutional effectiveness reports. This is the first General Education review cycle. This resulted in reworking some course alignments and dropping some courses from the catalog. The previous self-study produced data on learning outcomes, but since so much time has passed, it is believed this data is of little use to us now. See Section 3 for a discussion of recent assessments of student learning.

### Program Mission Statement

Have any changes been made to the general education mission statement over the course of this cycle? Why were these changes made? Are any revisions planned?

**Narrative:**

Our mission is to foster the development of critical thinking in students. To encourage them to seek an understanding of the relationship between the individual and society as it affects personal behavior, social development, and the quality of life of individuals, the family, and the community; to examine the impact of behavioral and social scientific research on major contemporary issues in behavioral sciences.

**Evidence:**

- Mission Statement SSE 2024

### Alignment to Institution Mission

How does the general education mission align with the mission of the institution?

**Narrative:**

VSCC Mission:

Volunteer State Community College is a public, comprehensive community college offering associate degrees, certificates, continuing education, and service to our constituencies. The College is committed to providing quality innovative educational programs; strengthening community and workforce partnerships; promoting diversity, and cultural awareness, and economic development; inspiring lifelong learning; and preparing students for successful careers, university transfer, and meaningful civic participation in a global society.

The Social and Behavioral Science mission statement aligns with the college's mission by promoting diversity, cultural awareness, and meaningful participation in a global society. Furthermore, the department's and college's mission statements are both focused on preparing students to be critical thinkers and lifelong learners.

# Student Learning Outcomes Assessment and Curriculum

## Curriculum Map

Curriculum mapping ensures the alignment of TBR GenEd outcomes and student learning outcomes; which affords an opportunity to ensure course coverage and sequencing of outcomes in courses.

The reviewer should review the current mapping to ensure there is pathways to meeting the TBR GenEd outcomes for all general education students.

**Narrative:**

Program outcomes are derived from the Tennessee Board of Regents (TBR) list of Social and Behavioral Sciences learning outcomes. The Volunteer State's learning outcomes are mapped to align wiith the TBR outcomes. TBR provides 7 Learning Outcomes but only 4 are required per course.

**Evidence:**

- GenEd Social Behavioral Sciences Course Outcome

## Evidence of Student Learning

Evidence of Student Learning

Provide evidence of student learning, and the program's effectiveness in reaching both TBR GenEd outcomes and student learning outcomes.

Please provide the following:
1. Summary of assessment most recent results.
2. Analysis of trends.
3. Ideas of how those results could be used to improve student learning.

**Narrative:**

The assessments made between 2018-2020 demonstrated that the department was successful in meeting its goals for 70% proficiency for a majority of the outcomes. Outcome #1 was met by all assessed sections. Due to the switch to fully online classes during the COVID-19 outbreak, the proficiency average did drop across all courses, but the department was still able to meet the goal of 70% proficiency. With this in mind, and the return to more face-to-face classes, we expect the next round of assessment testing will result in a higher success rate.

After reviewing the data for this round of measurements, it has been decided that the next set of assessments will use more similar writing assignments across all courses to measure the proficiency of students.

The scores from the assessments for psychology, sociology, political science, and wellness were all at or above 70% for all outcomes.

Economics 2100, did not meet 70% proficiency for outcomes #1 & 2. The averages were 69 and 68 respectively. In an effort to improve students' proficiency scores, the instructors reviewed the material and spent additional time covering the material. Upon a follow-up assessment, the ECON students improved their scores to 76%.

Geography 1015 was the other course that failed to meet proficiency for outcomes #1 (68%), #6 (68%), & #7 (67%). As with the ECON classes, after the assessments scores were recorded, the instructor recorded a video that reviewed the information associated with the outcomes not met. An assignment given later in the term produced higher average scores (81%) using the same rubric.

Comparing the results from such different courses as geography, economics, psychology, wellness, political science, and sociology, has shown us that our next assessment tool needs to focus on the similarities among the courses.

Not every course was assessed during this period. However, the department did assess 11 of 15 courses and did assess 4 outcomes within 11 of the courses.

Upon a further review, the results indicate that the social and behavioral sciences have been effective in assuring students are meeting the course learning outcomes. The use of six different assessments tools was not the best plan. For the next assessment period, the department will use writing assignments and an improved rubric. The intention is to align our assessments to provide valid and reliable measurements.

**Evidence:**

- SS data template (3)

# Faculty and Courses

## Faculty Usage and Development (optional)

May review faculty teaching loads, adjunct to full time ratios, faculty development opportunities, faculty end of course evaluations, faculty surveys, and/or any other evidence deemed appropriate. The reviewer could discuss the impact of teaching, the ratio of adjunct to full time staffing, end of course evaluations relative to faculty, and any noted themes identified from the review.

**Narrative:**

We have 15 full-time faculty and 12 adjunct instructors in the Social and Behavioral Sciences Departments teaching the involved courses. Full time faculty carry a load of five courses and adjunct instructors are limited to four courses per term.

Approximately 55% of the involved general education Social and Behavioral Science courses were taught by full time faculty.

The use of social science adjunct instructors increased during the years under review. This was due to additional online sections during COVID. Adjunct instructors help our department in being more responsive to students' needs and to changing enrollment demands. Adjunct instructors bring a different perspective to our students. These dedicated professionals benefit from their service to the college as well as see personal growth in their fields. We continually work to strengthen the relationships with our adjunct instructors.

The Social and Behavioral Science Departments used the last two years to develop new courses and professional development opportunities for the college's faculty. Open Education Resources (OER) courses are classes designed using fully digital textbooks. These courses provide no-cost textbooks to the students. Psychology 1010, 2130, Sociology 1010 and 1040 have all adopted an OER platform.

It is worth mentioning, Dr. Krista Mazza-Carter's work with the Learning center (TLC) has impacted faculty within our department and across the college. Dr. Mazza-Carter was able to develop, and conduct trainings on topics including: *Culturally Responsive Teaching and the Brain* and *Wellness at Work.*

### Class Offerings (optional)

May address scheduling effectiveness and space utilization. The reviewer could review faculty to student ratios at each location and online and/or discuss the impact relative to faculty and student's success.

## Recommendations and Next Actions

### Analysis and Summary

Analyze and summarize the proceeding sections. The reviewer should discuss what went right, what went wrong, and the implications of those results.

**Narrative:**

The Social and Behavioral Sciences Departments completed their last academic audit in 2018. The reviewers made recommendations that align well with the GER and the two-year action plan.

1. Faculty should be more collaborative when designing assessment tools.
2. More focus should be given to connecting the results from the assessment tools to improve teaching and learning.

The reviewers approved of the assessment instruments used, but they had suggestions on how the faculty could collaborate more effectively. The reviewers suggested that the departments develop the next strategic plan to better align with the assessment tool and learning outcomes. They also want to see the documentation used to develop and design the instrument for measuring proficiency of content.

The auditors were complimentary to the faculty, but did note that some courses were more aligned with learning outcomes than others. This has become one of the priorities for the next cycle; collaborate on developing assessments so we can work together in planning which skills might focus on which courses for the term.

During this review period, we (as with the rest of the college) dealt with COVID-19 and the consequences of a lockdown, isolation, and the complete switch to online courses from March 2020, through August 2021. Faculty and students alike were pushed into learning, adapting, and becoming proficient with a 100% virtual environment.

Many instructors had experience with online education, so there were plenty of faculty to assist those moving into this platform for the first time. Faculty began using Zoom, Microsoft Teams, and created numerous pathways to success for our students.
For Fall 2020 and Spring 2021, several courses started using a synchronous approach. The choice of asynchronous or synchronous online learning was a good option and was well received by students.

A review of student grades did not reveal any significant difference in grade distribution between synchronous and asynchronous sections.

## Next Actions

The reviewer should discuss the next steps for the general education for the following two academic years. A plan of action and timeline should be included in the narrative or document library.

Feedback and approval will be provided. Refer to the GER timeline for approval deadlines.

**Narrative:**

The next steps for the Social and Behavioral Sciences departments is the Spring 2022 implementation of a newly developed set of writing assignments. These assignments have been designed to measure the mastery of both course and general education learning outcomes. Collaboration among the faculty has led to the creation of rubrics strongly linked to Learning Outcomes that align with departmental recommendations and professional organizations such as American Psychology Association (APA), American Sociological Association (AS), American Geographical Society (AGS), American Political Science Association (APSA), American Economic Association (AEA), American Alliance for Health, Physical Education, Recreation, and Dance (AAHPERD), and American Association of Colleges and Universities (AAC&U).

1. The new assessment will be assigned during the middle third of each course.
2. Overall goals for improvement include: higher proficiency averages for Geography courses.
3. Increase Geography course proficiency to 70% for all sections and all outcomes
4. Future improvements include: assessing all general education courses within the department and improving the effectiveness of assessments made. We will also review assessment data to become more aware of trends and patterns to improve curriculum.
5. Our expectations also include providing feedback on students progress to increase retention. This would be accomplished through increasing faculty usage of the ALERTS system and increasing the number of faculty that use the ALERTS system.

For academic year 2022-2023:

1. Improve Geography courses: the plan to achieve this mark is through increasing class assignments and discussions.
2. Increase the number of courses assessed to 100% of the departments. This will be accomplished by monitoring progress in VIA.
3. Increase the number of faculty using the ALERTS system. This will be accomplished by monitoring faculty use of the ALERTS system and communicating with those who do not use the system.

For academic year 2023-2024:

1. Provide opportunities for faculty to discuss assessments and assessment data.
2. Increase number of ALERTS faculty submit. This will be done by tracking the number of ALERTS each faculty member sends for each recording period.

**Evidence:**

- ECON2100 General Education SLO Rubric _VALUE_ Rubric Detail 12-17-2021 083132
- ECON2200 General Education SLO Rubric _VALUE_ Rubric Detail 12-17-2021 083139
- GEOG1012 Student Learning Outcomes Assessment Rubric Detail 12-17-2021 083152
- GEOG1015 Student Learning Outcome Assessment Rubric Detail 12-17-2021 083158
- GEOG2010 Student Learning Outcomes Assessment Rubric Detail 12-17-2021 083205
- HED120 General Education SLO Rubric _VALUE_ Rubric Detail 12-17-2021 083211
- PSYC1030 General Education SLO Rubric _VALUE_ Rubric Detail 12-17-2021 083553
- PSYC2010 General Education SLO Rubric _VALUE_ Rubric Detail 12-17-2021 083225
- PSYC2130 General Education SLO Rubric _VALUE_ Rubric Detail 12-17-2021 083233
- SOCI1010 General Education SLO Rubric _VALUE_ Rubric Detail 12-17-2021 083256
- SOCI1040 General Education SLO Rubric _VALUE_ Rubric Detail 12-17-2021 083302

# Feedback and Review

## Feedback (General Education Committee)

Presentation by faculty to the General Education Committee of the Program Review. The General Education Committee will provide feedback on the completed review and action plan. The feedback will be included and summarized.

## Dean(s) Approval and Feedback

The Dean(s) of the general education discipline(s) will provide feedback and approval on the completed review and action plan.

**Narrative:**

I have reviewed this General Education Review (GER) report for Cohort B (Geography, Economics, Political Science, Psychology, Sociology, and Wellness) and have the following comments and suggestions:

- Philip Williams has done an excellent job with a challenging task. His report was complete and thorough. It did require some minor editing, but this was because of the potential confusion related to his having to report on multiple programs; some of them not in his department.
- The Social Science and Education Division has six departments. Three of those departments have general education courses. They are the Social Sciences Department, the Behavioral Sciences Department, and the Physical and Health Education Department.
  - The Social Sciences Department includes the following general education courses:
    - History
    - Economics
    - Geography
    - Political Science
  - The Behavioral Sciences Department includes the following general education courses:
    - Psychology
    - Sociology
  - The Physical and Health Education Department includes the following general education course:
    - Wellness
- The Tennessee Board of Regents (TBR) prescribes two different sets of student learning outcomes (SLOs); one for History and one for the other general education courses mentioned above.
- Logically, it made sense to have one GER report for History and have it address the related TBR SLOs and have a second GER report for the remaining courses and have it address the related TBR SLOs.
  - It was this division of the work that caused some of the confusion in the current report.

At some point in the future, it might be advisable to consider a different approach to who prepares these reports. I would suggest the following:

- Continue to have the Social Sciences Chair complete the History GER report and address the related TBR SLOs.
- Because the Economics, Geography, and Political Science courses are in the Social Sciences Department, have the Chair complete that GER report and address the related TBR SLOs.
  - This would have the GER report being completed by the Chair who best knows the courses and the faculty who teach them.
  - Since the TBR SLOs are the same, this report should be completed collaboratively with the involved faculty of both departments and the Behavioral Sciences Chair.
- Have the Behavioral Sciences Chair complete the GER report for Psychology, Sociology, and Wellness and address the related TBR SLOs.

Overall, I was very pleased with this report. The above observations and suggestions are offered in the spirit of making the next report(s) even better.

### VPAA and AVPAA

The VPAA & AVPAA will provide feedback on the completed review and action plan.

## Action Plan Reports

### Year 2 Action Plan Review

The reviewer should review progress on action plan over the past year. Discussion should include challenges and successes. If any changes are planned due to the results, those should be discussed and an updated action plan should be included in the narrative or document library.

**Narrative:**

Year 2 (2022-2023) was not have updated since there was a transition in the research office and reminders did not go out. See Year 3 Action Plan Review for final review and results.

### Year 3 Action Plan Review

The reviewer should review progress on action plan over the past year. Discussion should include challenges and successes. The reviewer should discuss the actions future and continuous in the program.

**Narrative:**

The Social and Behavioral Sciences (SBS) departments introduced, in the Spring 2022, implementation of a newly developed set of writing assignments. These assignments had been designed to measure the mastery of both course and general education learning outcomes. Collaboration among the faculty had led to the creation of rubrics strongly linked to Learning Outcomes that aligned with departmental recommendations and professional organizations such as the American Psychology Association (APA), American Sociological Association (ASA), American Geographical Society (AGS), American Political Science Association (APSA), American Economic Association (AEA), American Alliance for Health, Physical Education, Recreation, and Dance (AAHPERD), and American Association of Colleges and Universities (AAC&U).

The new assessment was assigned during the middle third of each course. Overall goals for improvement included higher proficiency averages for Geography courses and a goal of course proficiency to 70% for all sections and all outcomes.

Future improvements included assessing all general education courses within the SBS area and enhancing the effectiveness of the assessments made. Another step was to review assessment data to become more aware of trends and patterns to improve curriculum. Lastly, a goal was to increase faculty usage of the ALERTS system. These expectations included providing feedback on students' progress to increase retention.

For academic year 2023-2024:

**1. Improve Geography courses: the plan to achieve this mark is through increasing high-impact class assignments and discussions.**
**Results:** The geography courses aimed to improve proficiency through a strategy of increasing high-impact class assignments and discussions. By implementing more interactive and engaging activities, the courses sought to enhance student understanding and retention of geographical concepts. The success of this approach was evaluated by measuring student performance and proficiency, with the goal of reaching 70% mastery rate for all sections and outcomes on the assessment. Increased assignments and discussions were intended to provide more opportunities for students to apply their knowledge, receive feedback, and engage deeply with the material, ultimately leading to improved educational outcomes. Faculty reported some improvement in classroom engagement and an overall increase in students' preparedness. However, the goal of 70% mastery was not achieved.

**2. Increase the number of gen ed SBS courses assessed to 100%. This will be accomplished by monitoring progress in VIA.**
**Results:** There were increases in total number of courses monitored, however Political Science/Government classes were not assessed in 2023-2024. The data collected has indicted the rubrics used by each course's assessment tool need to be reviewed and updated to reflect a concise and comprehensive analysis.

**3. Increase the number of faculty using the ALERTS system. This will be accomplished by monitoring faculty use of the ALERTS system and communicating with those who do not use the system.**
**Results:** Between 2020-2023 AY, the total number of Alerts sent by faculty decreased. The total number of faculty using the Alerts system also decreased. Analysis does show that some faculty did show a consistence use of the Alert system. Also noted are the lack of Alerts from specific faculty members. Follow-up to this report will involve debriefings with faculty to discuss the results and to make plans for improvement.

**Evidence:**

- GEOG 1015 SLO Report AY21-22-23
- GEOG 2010 SLO Report AY21-22-23
- alerts 22-24

# Academic Program Review Cohort B- Cycle 1

Associate of Science in Teaching- All Concentrations

## Reflection and Mission

### PREVIOUS ACADEMIC PROGRAM REVIEW

Reflect on the past APR (if any) and respond to any recommendations that were made at that time. The reviewer should list recommendations, actions taken, and discuss implications or challenges resulting from those actions.

**Narrative:**

Our Associate of Science of Teaching Institutional Effectiveness outcomes have changed, and no Academic Program Review has been conducted. Tennessee Board of Regents Associate of Science Teaching Academic audit completed in 2017. The Academic Audit fulfills reporting requirements for the Tennessee Education Commission's Quality Assurance Funding Standard 3.

### Program Mission Statement

What changes have the program made to the mission statement over the course of this cycle? Why were these changes made? Are any revisions planned?

**Narrative:**

The Associate of Science Teaching program prepares a diverse student population to transfer to a Tennessee Board of Regents four-year institution. For students pursuing the associate degree, the program is designed to prepare the students with substantial knowledge about children ages five through twelve, an overview of theoretical orientations that have shaped the education discipline, and a base knowledge of empirical teaching and learning strategies. Co-curriculum courses in the general education requirements develop the student's baseline of knowledge to prepare the student for employment in the field of teaching.
The Associate of Science Teaching program is an important part of Volunteer State Community College's mission to provide quality innovative education programs, to strengthen community and workforce partnerships, to promote diversity, and to address cultural and economic development. Also, another part of the mission is to inspire lifelong learning and to prepare students for successful careers, university transfers, and meaningful civic participation in a global society.
The Associate of Science Teaching program links to the university mission addressing community engagement, promoting diversity, economic development to prepare students for successful careers in teaching.
A change will be made after reviewing the old mission statement, and the new institutional mission statement. A revision to clearly align with the institutional mission will be made by adding cultural awareness in the Associate of Science Teaching program mission statement.

### Alignment to Institution Mission

How does the mission of the program align with the mission of the institution?

**Narrative:**

The Associate of Science Teaching program is an important part of Volunteer State Community College's mission to provide quality innovative education programs, to strengthen community and workforce partnerships, to promote diversity, and to address cultural and economic development. Also, another part of the mission is to inspire lifelong learning and to prepare students for successful careers, university transfers, and meaningful civic participation in a global society.

The Associate of Science Teaching program links to the university mission addressing community engagement, promoting diversity, economic development to prepare students for successful careers in teaching.

# Curriculum and Student Learning Outcomes Assessment

## Curriculum Map

Review the Student Learning Outcomes and course(s) where each one is addressed. For each outcome, provide the assessment method and target level of achievement.

**Narrative:**

This curriculum map was implemented in fall 2021

| **Curriculum Chart: Sample learning opportunities and rubric assessments for each outcome** | | | |
|---|---|---|---|
| **Education Outcomes** | **Learning Opportunities (activity, course number)** | **Rubric assessments of performance** | **Achievement Target** |
| The students will be able to explain historical, political, social, economic factors, diverse cultures, special education and integrating technology currently affect American education. | **EDU 101:**<br>• Collaborate through authentic problem based learning activities.<br>• Demonstrate and apply technology as an integrated tool in the teaching and learning process.<br>• Create a multicultural lesson, and align to culturally responsive teaching.<br>• Collaborate to develop a historical timeline.<br>• Discuss and debate political, social, special education and economic factors in American education.<br>• Explore the role of the K-12 teacher and decide whether or not to pursue a career in education. | **EDU 101:**<br>Portfolio Presentation- Personal Teaching Philosophy and Narrative Reflection of Observation 1 & 2.<br>Students will create an education portfolio in power point demonstrating the value the learning has provided that aligns to their teaching career.<br>Portfolio Rubric embedded in eLearn Evaluation and Oral Presentation Rubric (face-to-face modality) | 90% of the students will meet this outcome. Scoring- "Meets Expectations" on all levels of the rubric. |
| The students will be able to create his/her own philosophy of teaching including current issues, tends, legal liabilities/responsibilities and reform in public education. | **EDU 101:**<br>• Design his/her own philosophy of teaching and learning in a project- based learning group. (Examining peers philosophy and providing constructive feedback. Receive constructive feedback | **EDU 101:**<br>Philosophy Paper- Relevant Aspects, Evident of Reflection and Examples (section of rubric)<br>Students will formulate their philosophy in relation to the role of | 90% of the students will meet this outcome. Scoring: "Meets Expectations" on a three level rubric. |

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| | to make improvements).<br>• Examine and explore the role of the students, teacher, classroom and curriculum.<br>• Debate and compare legal liabilities and responsibilities in the teaching profession.<br>• Discuss current issues, trends, and reform in public education. | students, teachers, and environment, and instruction that support them. Philosophy Paper Rubric embedded in eLearn in Evaluation | |
| The students will be able to write a lesson plan including the alignment of curriculum standards, educational outcomes, and adaptations for disabilities, predominant theorists and assessment. | **EDUC 2210:**<br>• Explore Tennessee curriculum standards and the use of these in lesson planning.<br>• Collaborative objective writing utilizing Bloom's taxonomy verbs.<br>• Create a chart to Identify predominant theorist and apply their various learning theories within educational modalities.<br>• Describe alternative learning assessments to measure predetermined outcomes, explain procedures for interpretation, and evaluating various tools of assessment.<br>• Develop adaptations to differentiate learning due to the complexities of diversity. | **EDUC 2210:** Lesson Plan- Objectives, Standard, Alignment to Activity, Adaptations for Children with Special Needs and Assessment (section of rubric)<br><br>Create a lesson plan including; objectives, standards, adaptations for special needs and assessment- scores.<br><br>Lesson Plan Rubric- embedded in eLearn in Evaluation | 90% of the students will meet this outcome. Scoring: "Meets Expectations" on a three level rubric. |
| The students will be able to compare and contrast a variety of K-12 classroom settings (including special education) by observing and conferring to pursue a career in education | **EDU-264:**<br>• Compare and contrast a K-12 regular education setting to a special education setting.<br>• Identify the accommodations and modifications for all exceptionalities, including assistive technology observed in the classrooms.<br>• Examine and explore the role of the students, | **EDU 264:** Field Observation- Description of Classroom, Description of Classroom Management and Teach-Student Relationships, Description of Teacher Qualities Observed and Student Reflection (sections of rubric) | 90% of the students will meet this outcome. Scoring: "Meets Expectations" on a three level rubric. |

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| | teacher, classroom and curriculum.<br>• Describe the teaching strategies to meet the needs of all students observed in the classrooms.<br>• Apply information from observations; relate the classroom field experience to expectations and outlook for a career in education.<br>• Identify classroom management strategies and examples of teacher-student relationship techniques observed in the classrooms. | Students will observe classrooms and write a description of the environment, behavior management, teacher-student relationships, and teacher qualities. Field Observation Rubric-embedded in eLearn | |

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| The students will identify characteristics of exceptional children, multi-disciplinary teaming, special education laws and legislation, accommodations and modifications to address classroom | **EDU 264:**<br>• Define the importance of the process of working effectively with a multi-disciplinary team to coordinate appropriate related services.<br>• Analyze the laws and legislation with regards to exceptionalities, basic terms, concepts of special needs including confidentiality, appropriate assessment, documentation, service delivery, and ethical conduct.<br>• Identify the causes, categories, characteristics of exceptionalities addressed in special education teacher interview.<br>• Describe the accommodations and modifications for all exceptionalities, including assistive technology from the special education teacher interview. | **EDU 264:**<br>Special Education Teacher Interview-Focused Topic, Content (sections of rubric)<br><br>Students will interview a special education teacher and write a narrative paper.<br><br>Special Education Teacher Rubric embedded in eLearn | 90% of the students will meet this outcome. Scoring: “Meets Expectations” on a three level rubric. |
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## EVIDENCE OF STUDENT LEARNING

Provide evidence of student learning and the program’s effectiveness in reaching both program outcomes and course learning outcomes.

Please provide the following:

1. Summary of most recent assessment results.
2. Analysis of trends
3. Ideas for how those results could be used to improve student learning.

**Narrative:**

The evidence of student learning and the program's effectiveness is indicated in the continuous improvement of the student Core for Educators Math Praxis exam scores demonstrated a rise from 156. 5 to 176.3. The analysis to these trends indicate the learning opportunities are helping the students to improve their scores and be successful. The program will add strategies to improve this outcome and change learning opportunities to increase success rate. The philosophy paper outcome will be continued based on the new outcome assessment method, instructional opportunities will be changed to increase student success. The other outcomes listed in the chart will not be continued.

| **Data Chart: 2018- summer 2021** | | | |
|---|---|---|---|
| **Education Outcomes** | **Learning Opportunities** | **Achievement Target** | **Three years of trend data** |
| The students will pass the Praxis Core for Educators Exam in Reading/Writing/Math. PLO 3 | • Assign a grade for completion of the Praxis Core for Educators Math practice test 1-at the beginning of the semester.<br>• Provide the Praxis Core for Educators Reading/Writing and Math practice tests on eLearn.<br>• Students who score low, will be directed to the Learning Commons to get assistance to improve the score.<br>• Provide students handouts and general information about the Praxis Exam, share these materials with Social Science and Education completion coach and the advising office. | Improve the Math mean score to 160. | **2018-2019**: 156.5 mean on Math<br>**2019-2020:** 162.67 mean on Math<br>**2020-2021:** 176.3 mean on Math<br>Continue these learning opportunities to increase student success rate. Add strategies to improve in two year action plan. |
| The students will be able to create his/her own | • Examine and explore the role | 90% of the students will meet this outcome. | **2018-2019:** |

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| philosophy of teaching including current issues, tends, legal liabilities/responsibilities and reform in public education. PLO 2 | of the students, teacher, classroom and curriculum.<br>• Lecture and discussion related to examining and comparing philosophies of education (essentialism, perennialism, progressivism, social reconstructivism, existentialism)<br>• Instructor will demonstrate how students will incorporate the teaching philosophy in their teacher portfolio | | 89% of students met this outcome<br>**2019-2020:**<br>84% of students met this outcome<br>**2020-2021:**<br>87% of the students met this outcome<br>This outcome will be continued based on new assessment plan to increase student learning outcome achievement. |
| Provide opportunities for students with multiple partner universities. | • Faculty contact universities to meet and share course information required for AST<br>• Discuss advising options with universities to align a seamless transfer for students<br>• Share materials with students to help with transfer<br>• Contact and invite local private universities to visit education classes, communicate | Contact 4-5 universities to discuss a teacher preparation program transfer agreement so that students have multiple opportunities/choices to transfer. | **2017-2018:**<br>3 universities visited Vol State campus.<br>**2018-2019:**<br>3 universities visited Vol State campus<br>Outcome was not continued. Collaboration with the universities continue to improve transfer opportunities for students. |

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| | their Bachelors Teaching Education program, and scholarship opportunities. | | |
| Develop embedded assessment items to track education data related to curriculum effectiveness. PLO 1<br><br>Embedded question:<br><br>The students will identify major historical, political, social, economic, and psychological trends | <ul><li>Embed questions in fall and spring midterm and/or final exams</li><li>Contact Institutional Effectiveness to get assistance with tracking the data</li><li>Submit data from assessments to Institutional Effectiveness office each semester</li></ul> | 90% average or higher on the 20 embedded questions from each education course<br><br>85% of the students will meet this outcome. | **2018-2019:** There is no student data at this time. Faculty made curriculum changes required by the State of Tennessee and online courses were fully redeveloped. Given the curriculum changes and update to the online sections, the embedded assessments were not developed and added to course assessment.<br><br>**2019-2020:**<br><br>76% of students met this outcome. Online courses significantly lower. This outcome is changed based on new |

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| that currently affect the field of education. | | | assessment plan to increase student learning outcome achievement. |

## Qualitative and Quantitative Program Information

### STUDENT, GRADUATE, AND ALUMNI SATISFACTION

Use student and graduate surveys, course evaluations, and any other appropriate evidence needed to explore the qualitative satisfaction results from students and graduates to provide information to the items below.

Discuss student, graduate, alumni perceptions of the program, faculty, career projection, and any other program recommendations given "during this survey".

Summarize any noted themes, trends, areas for improvement, identified strengths, and implications for the program for both students and alumni.

**Narrative:**

Schools within our community partnership participated in our 2020 Employer Survey of Results of education graduates (managed evidence). All of the schools indicated that they would hire another education graduate from the education program. The employer survey indicates a overall 66.7% satisfaction with education graduates, with a mean of 4.33. The employers indicate that students/teachers strengths include that they work very hard to find fun activities and prepared well for students. Additional comments that would help to better prepare future graduates include; more skills and adaptation to working with children with individualized needs. This comment clarifies a need for implementing more strategies to help teachers be better equipped to work with students with disabilities. educational activities in the classroom.
According to the Teaching placement report (managed evidence), there were 31 graduates with 18 placement indicates a 58% of students completed the program and reported placement. Tracking of student placement within community partnerships is a need to improve the success of the program and was identified as a need for improvement. Community College Survey of Student Engagement (managed evidence) results share the highest and lowest aspects of student engagement. The highest benchmark for Volunteer State Community College was in Academic Challenge 89%, with a standardize score of 52.5- students reported examinations challenged them to do their best work. This report aligns well with education employers indicating that students work very hard to develop fun and educational activities with their students. This benchmark data and employer data demonstrates effective quality programming and students are successfully prepared for the workforce.
Education faculty completed a survey developed by the chair for fall 2021, 100% of faculty report effective collaboration with education faculty members, sharing of education resources and campus student services activities. 4/5 indicate that liability insurance would benefit students, and 60%

reported that background checks haven't been reported necessary by students in schools districts. The Middle Tennessee Associate of Science in Teaching taskforce (AST Taskforce) group have discussed these requirements for several years. Some Community Colleges do require and some do not, Vol State teacher education program has been partnering with local districts for many years. This has not been a problem, however, when reviewing the program it is necessary to collaborate, discuss requirements, and make changes if necessary. Education adjuncts indicated the Vol State Teaching and Learning center professional development opportunities, current teaching strategies, current trends, and virtual learning engagement tools are beneficial for training. In addition, training was reported as needed on current changes in the education courses to help improve communication with students. An eLearn education page to improve the communication of current course changes by semester and updated regularly.
Spring student course evaluations indicate education faculty scored a 4.6 out of 5.0 possible score on all courses. "The content of the course challenged me to think" (4.88) was the highest score reported by students. Moreover, "the instructor provided feedback on my progress in the course" was reportedly higher (4.78) than the overall Social Science and Education division mean. The lowest was "the instructor's use of technology helped me to understand and use the material in this course" (4.43). The comments include; " I enjoyed the videos for this course, they were very thought provoking" and "this instructor was amazing and kind- very helpful and patient throughout my injury and made this class as stress-free as possible- very quick to answer emails and phone calls as well"- each of these align with the scores reflective of the data. The online content of each course is on a current redevelopment cycle for approval every 3 years, and updated every semester. 10 hours of field observations are a requirement for licensing purposes. Continuous training improvement for education adjunct faculty to address use of eLearn materials.

**Evidence:**

- CCSSE_2021_5D858BF002_ExecSum
- Teach Grads and Progress Update
- VSCC AST Faculty Survey
- VSCC Alumni Survey 2021 (2019 Graduates)
- VSCC Continuing Student Satisfaction Survey
- VSCC Teaching A.S.T 2020 Employer Survey Results

## RECRUITMENT, ENROLLMENT, RETENTION, AND GRADUATION: FACT BOOK DATA

Attach screenshots from the Fact Book of the areas below.

1. Headcounts
2. FTE Comparison
3. Student Primary Location
4. Retention Statistics
5. Graduation Statistics
6. Demographics

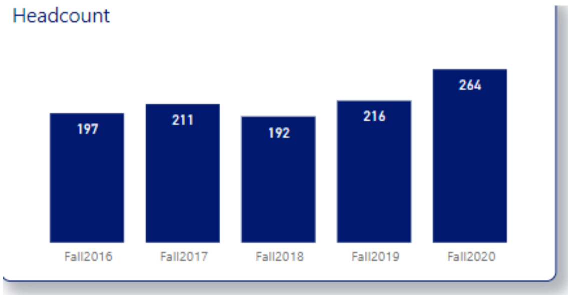
Using the Fact Book data listed below, analyze the follow:

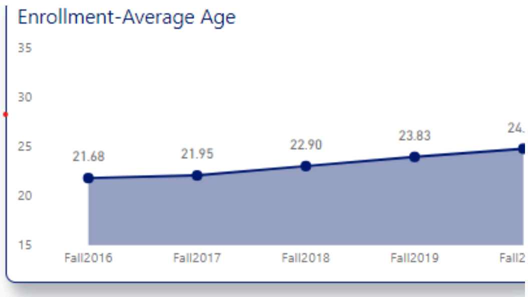
Recruitment
Enrollment
Retention
Graduation

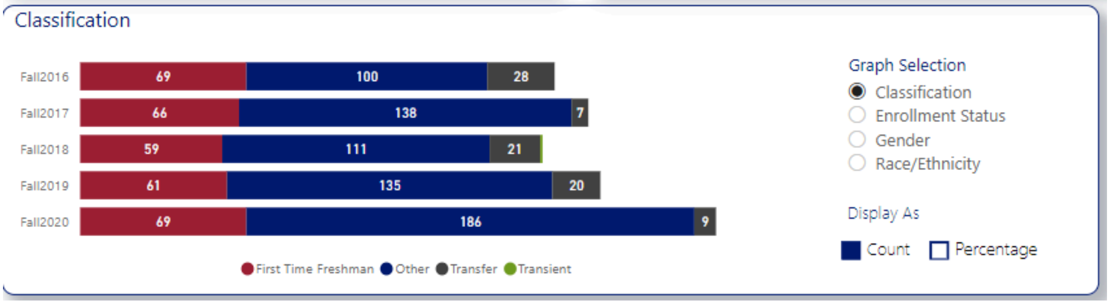
The reviewer could discuss marketing and recruitment efforts in the community and local high schools; analyze enrollment, retention, and graduation trends; and discuss potential reasons, strengths, plans for improvement, and any other implications deemed appropriate.

**Narrative:**

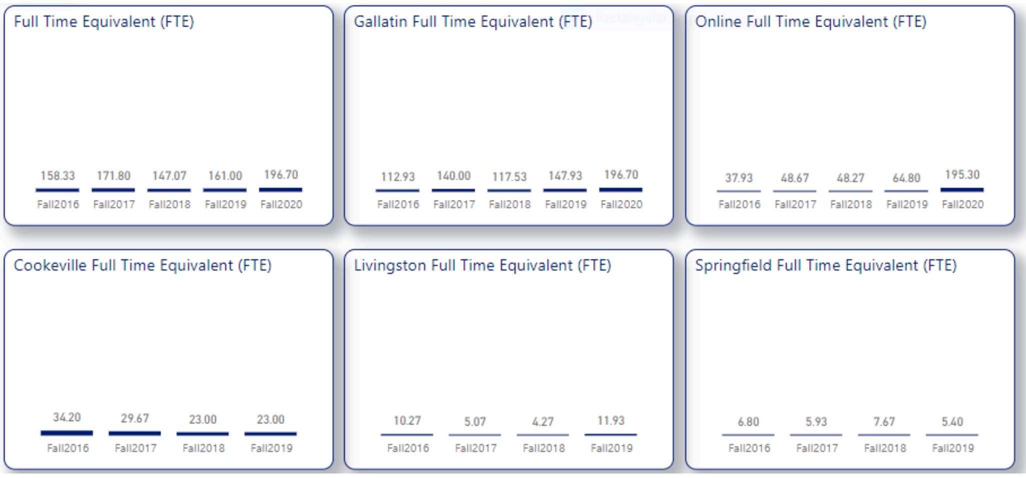
Despite the pandemic, the Vol State teacher education enrollment data indicates an increase in the last five years. However, lower enrollment trends in Community Colleges and teacher retention throughout Tennessee and across the nation is a concern. https://files.eric.ed.gov/fulltext/ED608848.pdf Recent data indicates enrollment in Tennessee Community College is down 19% overall. One of Governor Lee's Tennessee teacher education preparation initiative to address this concern is called the "Grow Your Own" partnership. This State of Tennessee grant, supports partnerships between educator preparation programs (EPP) and local Education Agencies (LEAs) to provide innovative, no-cost pathways to the teaching profession by increasing educator preparation programs enrollment and growing the supply of qualified teachers. Vol State teacher education program is currently running a cohort of students through a partnership with Austin Peay State University and Robertson County Schools. Vol State education program faculty are continuously pursuing partnerships with schools to recruit and improve enrollment. Headcount increased data demonstrates a 45 student increase from Fall 2019 to Fall 2020. The enrollment-average age has increased over the past 3 years, this could be related to Tennessee Reconnect state program, which began in the fall semester of 2018, which allows eligible adult student to earn a college degree tuition-free. The age increased from 22.90 in fall 2018 to 24.67 by fall 2020. Faculty share fliers and information at community events to recruit non-traditional students. Incoming freshman data indicates a increase over the last five years. The teacher education program faculty shares degree information with high schools to recruit more incoming freshman to declare the AST as their intended major. The program recruitment strengths include; recruitment materials that are shared periodically with area high school guidance counselors participating in high schools who hold annual job fairs. The education website is updated regularly, with degree options and programs discussed in a video- created by faculty. Over 2,500 8th grade students from Sumner County attended virtually the P16 Council Career Exploration Fair hosted by Volunteer State Community College last spring 2021. Industry representatives shared their future career opportunities through live, virtual meetings. The Career Exploration Fair promotes an avenue for eighth graders preparing to enter local high schools to explore a breadth of career areas and make an informed decision when choosing their courses. After meeting with career and education college representatives, students are better equipped to identify potential career pathways in teaching. Pre-COVID, Social Science and Education faculty set-up a table to promote career pathways in education and distribute degree information pamphlets. In addition, education faculty set-up a table to recruit new students during the 3rd Annual Hometown Hero's Festival honoring the local school teachers and staff. Education faculty presented program information to perspective high schoolers interested in teaching during Tennessee Department of Education Commissioners Teach Today tour. The education department is working closely with the Associate of Science (AST) taskforce group of Middle Tennessee Community Colleges to collect new ideas to form new opportunities to improve recruitment and enrollment.



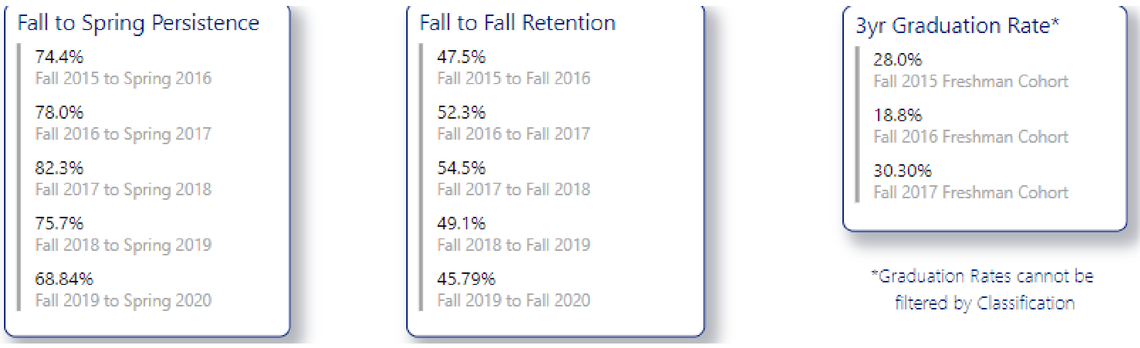




The Associate of Science in Teaching program's FTE enrollment improved significantly for fulltime students attending the Gallatin campus. There was a drastic increase on online enrollment- FTE of 147.93 in 2019 to 196.70 in 2020. This increase is pandemic related- due to most all course offering methods going online. Our Livingston satellite campus increased in FTE from 4.7 in 2018 to 11.93 2019.This enrollment increase could be related to a new education partnership program started on this campus. Our education "Grow Your Own" partnership program with Montgomery County Schools and Austin Peay State University started in Fall 2019. A cohort of  10 teacher assistants from Montgomery County Schools began in Fall 2019 and are scheduled to graduate in Spring 2022. Vol State Vice-President Dr. Brezina, Austin Peay State University, Social Science and Education Dean, and education faculty started collaboration on the new program in spring 2019. A five semester schedule was created, including method of delivery- designed for the newly cohort of assistant teachers.




The Retention rates decline to 69% from fall to spring, 46% from fall to fall, and 30% 3 year graduation rate. Student persistence/retention rates are an overall concern for the college at this time. Strengths to help AST students include; offering the Praxis Core for Educators math workshops, utilizing Vol State campus alert system, and sharing Vol State Learning Commons resources. Education faculty meet with the division completion coach and collaborate on ways to help prevent dropout and increase completion rates. Education faculty collaborate regularly with adjuncts to share advising tools and pamphlets suggesting adjuncts add the resources to eLearn announcement pages for every education course.  Education faculty offered advising zoom sessions for education students, and advising times during class. The education department program chair is working with  Vol State campus faculty to develop strategies to improve the success of students.




The Associate of Science Teaching degree graduation rate data indicates growth in graduation rates from 14 students in 2018 to 21 students in 2019 (7 more students) and 18 in 2020. University representatives visit every semester to speak to the students face-to-face courses and share transfer offerings and assistance. In addition,  the Universities offered zoom links and meeting opportunities were posted in eLearn online courses.  University representative include; Tennessee Tech, Tennessee State University, Welch College, Austin Peay University, and Middle Tennessee State University.  Education faculty have a 2+2 articulation agreement with Tennessee State University. It is offered on the Gallatin campus and students report that they love the convenience and instructors.  The partnerships and visits with the universities removes the uncertainty of graduation and transfer confusion for students. These opportunities are provided to help students graduate, complete transfer paperwork to be admitted to a teacher education program and transfer smoothly. In addition, the State of Tennessee has added a variety of licensure requirement changes and EPP program options for students. The data is a reflection of the continued trend in fallen community college enrollment and persistence/retention nationwide.

| Degree | 2016 | 2017 | 2018 | 2019 | 2020 |
|---|---|---|---|---|---|
| Associate of Teaching | 19 | 27 | 14 | 21 | 18 |
| **Total** | **19** | **27** | **14** | **21** | **18** |

| Major | 2016 | 2017 | 2018 | 2019 | 2020 |
|---|---|---|---|---|---|
| Web Designer Level I | 1 | 2 | | | |
| Web App Developer Level II | 23 | 19 | 1 | | |
| Video Production | | 1 | | | |
| Veterinary Technology | 23 | 21 | 17 | 21 | 18 |
| Veterinary Assistant | 10 | 10 | 9 | 16 | 15 |
| University Parallel | 461 | 458 | 508 | 552 | 560 |
| Tennessee Transfer Path | 191 | 232 | 282 | 268 | 310 |
| Teaching | 19 | 27 | 14 | 21 | 18 |
| Sleep Diagnostic Technology | 23 | 18 | 30 | 23 | 31 |
| Respiratory Care Technology | 14 | 30 | 17 | 14 | 11 |
| Radiologic Technology | 28 | 31 | 24 | 28 | 25 |
| Professional Studies | 1 | 2 | | | |
| Professional Music | | | | | 2 |
| Physical Therapist Assistant | 32 | 31 | 34 | 32 | 32 |
| Paramedic | | 44 | 33 | 44 | 48 |

**Core Praxis for Educators**
**Scores 2020-2021**

| | Reading | Writing | Math | Mean |
|---|---|---|---|---|
| **Mean Score** | 181 | 172 | 176 | 176.3 |
| **Passing Score** | 156 | 150 | 150 | |

Prepared by: Office of Institutional Effectiveness, Research, Planning and Assessment Oct 22, 2021

Vol State educating students scored higher on the Praxis Core Academic Skills for Educators (Core) Reading, Writing, and Math test in 2020-2021- 176.3 mean average in all three require tests. In the recent data indicates students scored; 2018-2019- 166.3 and 2019-2020- 162.67 (mean average). Math and Writing are the lowest shows 150- with a required passing score of 150. A passing score is required for graduation of the AST program and entering into a universities teaching education preparation (EPP) program or an ACT score of 21 or higher.

Plans to improve recruitment include; attending Gallatin High School's Career Technical Education instructor's education advisory meetings, a currently new advisory board created by the CTE instructor. Moreover, the high school students enrolled their "Teaching as a Profession I, II, and III program will participate in a field trip to Vol State and TSU in the spring. Vol State Student Services and Social Sciences and Education division will conduct the tour, and speak to the students about the AST program options. Dual enrollment courses will be offered at the local high schools, the EDU 101 course is scheduled to start fall 2022 at Gallatin high school. FTE data indicated a rise in the Livingston campus AST data, could be related to the pilot of our "Grow Your Own" partnership program with Montgomery County Schools and APSU. Teacher assistants/students will complete spring 2022. Vol State Springfield site director is recruiting students with Montgomery County Schools for the fall 2022. The education faculty are collaborating with the partners to create a new schedule, troubleshooting solutions for Learning Support requirements and delivery methods. Continuous improvements will be made to towards the new "cohort" selection and requirements and the program is working with another university (TTU) to begin the process to secure another partnership. Plans for improve the persistence/retention rates include collaboration with the Vol State math instructor who teaches MATH 1410 and 1420 (Math for Elementary Teachers) to improve success and plan a fall math and spring workshop to prepare for the Praxis Core Academic Skills for Educators exam. In addition, to better equip students in writing, education faculty will post https://www.tutor.com/ in Learn and suggest students visit the Vol State Learning commons online or via zoom.

A passing score is required for graduation of the AST program and entering into a universities teaching education preparation (EPP) program. In addition, the AST program department chair has a partnership with a local high school to establish an education student mentoring program paired with Teaching as a Profession III students. Students who have completed EDU 101 and are enrolled in EDUC 22210 will participate in the mentorship program to help with persistence/retainment of the high school future educators.

**Evidence:**

- APSU K-5 2+2 Final
- Copy of Praxis Scores for AST 2020-21 Graduates
- EducationPrograms-Brochure
- Grow Your Own Grants in Tennessee
- Praxis Core Academic Sk9lls for Educators (Core) Qualifying Scores
- Praxis Scores for AST 2018-19 Graduates
- Praxis Scores for AST 2019-20 Graduates
- Teacher Retention in Tennessee
- Tutoring for Praxis Core Academic Skills for Educators (Core) Tests

# Supporting Program Information

## FACULTY USAGE AND DEVELOPMENT (OPTIONAL)

Review faculty teaching loads, adjunct to full time ratios, faculty development opportunities, faculty end of course evaluations, faculty surveys, and any other evidence deemed appropriate. The reviewer could discuss the impact of teaching, the ratio of adjunct to full time staffing, end of course evaluations relative to faculty, and any noted themes identified from the review.

**Narrative:**

The education teaching faculty continually seek ways to set students up for success and impactful, high-engagement teaching faculty has been identified as one of these ways. A model course map was created to secure consistency between program outcomes, course objectives, delivery content and textbooks. Course manuals are updated adding the new course objectives determined by the AST Taskforce group. Course rubrics were redeveloped to align with new course outcomes to track key assessment data that lead to continuous improvements in the program. The course map, rubrics and data is shared regularly with education faculty.

Faculty teaching loads include; 2 fulltime education (one with doctorate degree and with a Master's degree- working on a doctorate) and 4 education (2 with a doctorate degree) adjuncts. The satellite Cookeville program, employs 2 of the adjuncts and 2 are located at the Gallatin campus. As education enrollment continues to grow, another education adjunct will be hired to assist with face-to-face and online course education courses. In the spring, a education adjunct's hiring paperwork will be processed to teach a new dual enrollment course offered at Gallatin High School. This opportunity had fostered through the community partnership developed with the Career Technical Education (CTE) teacher . Vol State education faculty has met with various local high school career coaches to discuss an education dual enrollment offering for the past, and plans to contact again for fall 2022.

Full-time faculty professional development include the bi-annually Tennessee Association of Colleges for Teacher Education conference and Higher Education Institute conference. These are each held every fall and spring semester, bring Tennessee teacher education program faculty together to update on licensure, state requirements, initiatives and trends in the field. In addition, the "High Impact Practices" conference is regularly attended by education faculty. The faculty share new information with the Vice President of Academic Affairs, Dean of Social Science and Education and education faculty. Education faculty attend professional development offered through the Teaching and Learning Center at Vol State and presented several trainings. These include; Adverse Childhood Experiences in Higher Education" 'Alternative Assessments, "Self-Motivation" and "Thinking Outside the Box". In addition, a presentation at the 2021 Virtual Tennessee Conference on Volunteerism and Service Learning on "Improving Community Literacy through Collaboration and Service'. Education faculty member partnered with Vol State distance learning to present on "Digital Student Engagement" at the Tennessee Early Childhood Training Alliance conference. Faculty department chair served on Columbia State Community College's Tennessee Board of Regents academic audit team in 2021. Being an auditor for another school's academic audit was a great way to learn more about the process and what is expected when is it Vol State's turn. This faculty service will benefit the education program because the department chair is familiar with the self-study report process and the underlying quality principles as foundations of good educational practice. The education program will be up for review in 2022 and have experience with the Quality Assurance Funding Academic Audit Rubric.

The Teaching and Learning Center is a faculty-led center, designed to support and provide professional development and build community with other teaching colleagues. Several notable speakers such as; author, Rich Milner (These Kids are Out of Control) author, Flower Darby (Small Teaching Online) and author, James Lang (Small Teaching) and book studies, "Cultural Responsive Teaching and the Brain" and "Over-Coming Bias" were attended by education faculty to improve teaching practices and self-awareness of these topics.

All full time faculty go through a peer evaluation by 2 assigned peers, complete classroom observations and required documents to division secretary. In addition, all adjuncts are evaluated by fulltime faculty, including a classroom observation and required documents signed and completed and sent to the division secretary. All evaluations are considered constructive feedback to adjust and improve teaching modalities, schedule, classroom climate and to benefit the teacher and student.

**Evidence:**

- AST Degree Curriculum Map
- EDU 101 Course Manual
- EDU 264 Course Manual
- EDUC 2210 Course Manual
- Fall_2021_AST_Faculty_Survey

## CLASS OFFERINGS (OPTIONAL)

### Address scheduling effectiveness and space utilization. The reviewer could review faculty to student ratios at each location and online and discuss the impact relative to faculty and student success.

**Narrative:**

The teacher education program offers a variety of course scheduling and modality to meet the needs of students at every campus. In the fall of 2020, online courses included synchronous offerings at an average of 21 students and asynchronous courses showing an average of 25 students. Spring 2021 data reflected an average of 20 students in all asynchronous offerings. Student course evaluation comments and education faculty input suggested only offering the asynchronous offerings for the spring. Students and faculty reflections indicated difficulty and adding more stress and pressure of COVID- thus, required to sit in a number of synchronous classes. Faculty and future teacher educators reported this format of learning was not the best use of time or align with pedagogy/best practice to meet student learning styles. Summer asynchronous course average was 17, typically, every summer 1 education course is offered. Fall 2021 data indicated an average of 15 students in face-to-face courses and 21 in asynchronous. All of the education courses are offered every semester either online or face-to-face. Cookeville satellite campus offerings include daytime and night face-to-face , and Gallatin campus offerings are offered various days, morning, and late afternoon time frames. The scheduling and modality are considered and assessed each semester based on student course evaluations and faculty collaboration. These will continue to be improved to reflect options that best meet the needs of students to ensure completion. Often, a student might have missed a course or have a transfer that needs another hour of course work. The faculty will offer an independent study to ensure the student graduates and/or offer substitutions upon credibility of transferring college and approval from the dean of Social Science and Education.

## CO-CURRICULAR ACTIVITIES (OPTIONAL)

Discuss co-curricular activities available for students in the program. These might include clubs, internships or other Work Based Learning opportunities, service learning, guest speakers, or any other activities designed to enhance student experience in the program.

**Narrative:**

Faculty work collaboratively to provide quality, expansive workforce experience and incorporating best practices in instruction. Teaching and learning opportunities are rich. Service Learning, the use of problem solving strategies as a teaching method, the use of technology, and the availability to learn through international programs are components of our program.

The program is driven by the Department of Education teacher licensing requirements, which require 30 hours of field observations (10 in each education course). Vol State has a network of community partnerships who allow students to complete these field observations. A list of schools partnering is provided to the students, as a resource if needed. Through this network of school partnerships Vol State faculty have initiated guest speakers, service learning and work based learning opportunities. Collaborating across departments and divisions to support student success has been addressed through an on-going evolving service-learning project to promote early literacy called "Read Across America" changed to " The project this year included the library, english, education, foreign language and communication faculty,  Over 500 new books were donated and delivered to 5 childcares in Sumner County. An event was hosted by the library and various social-distancing activities were arranged throughout the library, and the public was invited. In the past, Vol State students have held a used book drive on campus, and distributed the books as they visit early childhood centers and read to classes of students. It was certainly a modified version of this project, but was a great success! This project was selected to present at "2021 Tennessee Conference on Volunteerism and Service-Learning." Education faculty. Guest speakers include; Open Arms Care facility director, Sumner County Schools behavioral analyst, and Metro Schools special education teacher. Students are required to write questions for the speakers and reflect and apply the experience to their teaching career.

A "Future Educators" student group was started in 2018. A few of the projects include; collected books for the "Read Across America" project, read books to students and donated items to Vol State "The Feed". In addition, decorating of the teacher education boards in the Noble Hall Caudill building. International education students collaborated with Ireland students on several study abroad trips. A partnership with " The Presentation School" in Thurles, Ireland led to students from Ireland visiting on campus and meeting with new education students. International Education offers students a global perspective and offers an opportunity of education students to compare and contrast various education programs and schools to broaden and add to pedagogy of teaching.

In the past, education faculty have partnered with various schools to arrange opportunities for students designed to enhance student experience in classrooms. Two special education programs: Harris-Hillman (Nashville) and R. T. Fisher Alternative (Gallatin) both offer students a tour and presentation on each facility and the offering to students who have disabilities.

**Evidence:**

- Dr Seuss Flier
- Future Educators Student Group flier
- Literacy, Service-learning and HIPS...OH My! video
- Read Across America 2021
- SL Conf Pres 2021

## INDUSTRY TRENDS AND INNOVATIONS (OPTIONAL)

Examine the program demand within the labor market including local, regional, and/or national. The reviewer could examine information from industry advisory boards and information gathered from creditable industry sources as well as from external accrediting bodies. The reviewer could also discuss any existing or emerging trends within the field as they relate to your program and employment of your student's post-graduation as well as the outlook for the field in the next 5 years. This summary could also identify best practices, trends, and innovation in the field and discuss implications for the program.

**Narrative:**

The labor market for kindergarten and elementary school teachers are expected to grow 7 percent from 2020 to 2030, about as fast as the average for all occupations. About 124,300 opening for kindergarten and elementary school teachers each year, on average, over the decade.
https://www.bls.gov/ooh/education-training-and-library/kindergarten-and-elementary-school-teachers.htm (visited *October 28, 2021*)
The United States Department of Education indicates a teacher shortage in most areas of secondary education; math, science, social studies, and all areas of special education and English language learners. TSA (ed.gov)
Emerging trends include a framework of primary partnerships with high schools to help more high school graduates prepare for, attend and complete college. The goals for this joint venture are to expand college access, to boost college persistence and completion. A particular focus on closing all achievement, equity and opportunity gaps to expand student opportunities and to build our education workforce. Gallatin high school and Vol State are working together to enroll education students in dual enrollment education courses. This early college experience helps illustrate college to help increase enrollment, retention and builds post secondary success- persistence in this degree. Vol State faculty and education students complete a presentation on how to to prepare for college and share personal experiences. High school students come on campus- a field trip to Vol State in the spring. In addition, education faculty visit and share degree programs to the Teaching as a Profession (TAP) courses that are offered through the Tennessee Department of Educations Career and Technical Clusters. In hopes of motivating more young people to pursue teaching as a profession. Another tend, as mentioned before- is the "Grow Your Own" program, as a means to increase workforce diversity. The grant is granted to public school systems and colleges to encourage more aspiring educators to pursue the profession and develop a local pipeline of well-qualified teachers who are ready for the classroom. Vol State currently partners with Montgomery County Schools , through matching scholarship funds with Austin Peay State University. This has fully funded 8-9 teaching assistants to become a licensed teacher. Vol State is excited to be a part of an initiative that we believe will help remove barriers and transform the important profession of teaching.
motivate more young people to pursue teaching as a profession.
Education Commissioner Penny Schwinn's strategic plan for Tennessee educator prioritizes three key areas of providing quality academic programs, serving the whole child, and developing and supporting teachers and leaders. The outlook for the field at Vol State aligns with the this plan, in the next 5 years faculty will recruit partnerships for the "Grow Your Own" program with various prospective educators in our service area. In addition, Vol State plans to ensure our students are prepared to be successful in the classroom upon completion of the their educator preparation programs- attending every Tennessee Associate of Colleges for Teacher Education higher education conferences bi-annually and Community Colleges Program Directors (job alike) sessions. Staying current on current licensure, state policy changes, and new state education initiatives in the field and assess the implications for change and continuous improvement in the education program.

**Evidence:**

- Department of Education-Teacher Shortages
- U.S Bureau of Labor Statistics- Elementary Teaching

### EMPLOYMENT AND TRANSFER FORECASTS (OPTIONAL)

Discuss where students go after graduation and what is being done to enhance the likelihood of success or assist students in reaching those goals. The reviewer could discuss employment outlook, obtainment, curriculum alignment for transfers, and the employment demands of the industries where students will work.

**Narrative:**

The education degree is designed to create a smooth transition to all public universities offering a Baccalaureate degree in this major by guaranteeing the transferability of all courses. There are no course substitutions for the program.
The education program has a 2+2 partnership program with Tennessee State University offered on the Gallatin Vol State campus- with some online and hybrid method options for students. This degree leads to a Bachelor of Science Degree in Arts and Science; Child Development and Learning K-5 from Tennessee State University with clinical placements in Sumner County. Education faculty collaborate with local university representatives where students go after graduation to assist students in reaching their goals and enhance the likelihood of success.

## Recommendations Action Plan, Review, and Feedback

### SUMMARY AND ANALYSIS

The reviewer should discuss what went right, what went wrong, and the implications of those results.

**Narrative:**

In the last 3 years, here are some trends we found; effective faculty collaboration, positive and high scores from students on our course evaluations and improved enrollment. Education faculty at Vol State collaboration and faculty involvement in the AST Taskforce has made continuous improvement and changes to the program. Student evaluations indicated that faculty build strong student-teacher relationships, quick to communicate feedback, and course content was challenging. The feedback from students aligned with the employer survey results. The positive implications of this data indicate a program strength in faculty usage and development. Although, Community College and teacher retention through Tennessee is a concern. The enrollment has increased in the education program in the last 3 years. Recruiting efforts, The Grow Your Own Program, and partnering with local community organizations and schools efforts are some of the positive efforts for possible implications of the program increase.
Retention rates have declined to 69% in the last 3 years indicating that student success rate is a weakness of the program. Strategies to improve student persistence and retention will be addressed in the 2 year plan. Efforts include; Praxis Core for Educators workshops, collaboration with division completion coach, utilizing alert system and Learning Common resources are positive- yet, just not working. One of our student outcomes addressed the Praxis Core for Educators Test and the need for improvement on the Math scores and successful passing of this exam for completion. The scores increased from 162.67 in fall 2019-2020 to 176.3 in 2020-2021. Identifying specific program course

student success rates such as; Math 1410 and Math 1420 will be a new plan of action to help students complete the program. Implications of the teacher placement report indicate 58% of students completed and reported placement. Tracking of student placement is an area weakness in the program and identified as a need for improvement. Building strong community partners in local schools and emphasizing the benefits of reporting/completing student student placement surveys to demonstrate student success rates and placement data results.

A new assessment plan to increase student learning outcome achievement has been developed. The embedded tests placed in the education courses were removed and changed to alternative assessments in each course to improve success student and make continuous improvements in the program.

**Evidence:**

- AST Data Chart 2018-2021

## TWO YEAR RECOMMENDATION ACTION PLAN

Discuss the next steps for the program by providing a program action plan for the next two academic years. Recommendations must include at least two actions focused on student learning and one action focused on student success. The program action plan must include a timeline for each recommendation.

Feedback and approval will be provided. Refer to the APR timeline for approval deadlines.

**Narrative:**

**2 Year Recommendation Action Plan**

**Student Success Recommendation**

1. Increase Job Placement rates for education majors.

- Faculty monitor first of each month
- Communicate to future educators the purpose of completing the Graduated Satisfaction and 6 months later the Job Placement survey (contacted 3 times within 6 months)
- Share a copy of the Job Placement survey to showcase to the classroom

**Timeline: October 31, 2021 - October 31, 2022**

- Is the number of Job Placement survey completions improved implementing these action steps? If not, what steps can be determined to improve?

**Timeline: October 31,2021 - October 31, 2023**

- Steps necessary to make continuous improvement in tracking student success

**Student Success Recommendation**

2. Increase Praxis for Educators exam support for education majors

- Schedule Math Praxis Core for Educators workshops every mid-semester
- Share information with division completion coach, faculty and the advising center
- Offer ZOOM for satellite campus students

**Timeline: November 1, 2021 - July 1, 2022**

- Track attendance in the workshops and gather student feedback from education faculty. Make changes necessary for continuous improvement

**Timeline: July 1, 2022 - July 1, 2023**

- Steps necessary to make continuous improvement in student success

**Student Success Recommendation**

3. Increase success in MATH 1410 and MATH 1420 for education majors

- Obtain data each semester from IERPA office to review- to track success
- Contact math faculty teaching to brainstorm improving student success
- Increase usage of the Learning Center for these courses

**Timeline: July 1, 2021 - July 1, 2022**

- Can the student success rates improved as a result of these action step? If not, what steps can be determined to improve?

**Timeline: July 1, 2022 - July 1, 2023**

- Steps necessary for improve student success in these courses

**Student Learning Recommendation**

1. Increase Grow Your Own partnerships, and improve the Springfield satellite campus current program

- Explore partnerships with Tennessee Tech University
- Collaborate with Springfield director and faculty to make accommodations for students, alter schedule and methods of delivery

**Timelines: August 2021 - July 1, 2022**

- Meet with TTU to discuss grant partnership and plan
- Make changes necessary to recruit and improve Springfield satellite new cohort

**Timelines: July 1, 2022 - July 1, 2023**

- Develop a Grow Your Own program with TTU, Putnam County Schools and Vol State
- Continuous improvements to Grow Your Own program located on Springfield satellite campus

**Student Learning Recommendation**

2. Increase Student Learning Outcome achievement based on the new assessment plan

- Collaborate with faculty to change or alter assessments to improve student scores
- Analyze student course evaluations to make improvements to course assessments
- Develop new assessment when students master student learning outcomes

**Timelines: August 2021 - August 2022**

- Meet with education faculty bi-annually to discuss assessment results
- Analyze data to make continuous improvements

**Timelines: August 2022 - August 2023**

- Steps necessary to make continuous improvements on the Student Learning Outcomes for the education courses and/or develop new ones to monitor

**Evidence:**

- 2021 Job Placement Survey
- 2021_Graduate_Job_Placement_Survey
- APSU K-5 2+2 Final
- Praxis Math Workshop

### DEAN(S) APPROVAL AND FEEDBACK.

The Dean(s) will provide feedback and approval on the completed review and action plan.

**Narrative:**

I have reviewed the information and documentation provided for the AST APR. Not only do I approve, I commend the Education Department's faculty for providing such a thorough, accurate, and comprehensive analysis of their AST programs. This no doubt took a significant commitment of time and energy. It is an example of a job done well..

### AVPAA & VPAA Review and Feedback

The Vice President of Academic Affairs and Assistant Vice President of Academic Affairs will provide feedback and approval on the completed review and action plan. VPAA and AVPAA should consider institutional planning and budgetary details when providing their review.

**Narrative:**

The Office of Institutional Effectiveness and Assessment has reviewed this report and will begin supporting the report champion in preparing for the presentation of this review to the Institutional Effectiveness Committee.

### PRESENTATION FEEDBACK

The APR will be presented to a group of peers. Feedback retrieved from the presentation will be summarized and included.

## Action Plan Status Reports

### YEAR 2 ACTION PLAN REVIEW

The reviewer should review progress on action plan over the past year. Discussion should include challenges and successes. If any changes are planned due to the results, those should be discussed and an updated action plan should be included in the narrative or attached as evidence.

**Narrative:**

**YEAR 2 ACTION PLAN REVIEW**

The reviewer should review progress on action plan over the past year. Discussion should include challenges and successes. If any changes are planned due to the results, those should be discussed, and an updated action plan should be included in the narrative or attached as evidence.

| YEAR 2 | | | | | |
|---|---|---|---|---|---|
| | Initiative | Achieved Milestones | Challenges | Changes if applicable | Progress Toward Performance Indicator(s) |
| 1 | Increase the job placement rates for education majors | Communicated with current students the teacher shortage and job placement is in high demands | The fact book has been in the process of being reconstructed and improved- data was not provided. | Faculty results are monitored ad shared with current students in March. | Due to a national teacher shortage, the field of teaching is in high demand. The PL1 has been met. The job placement rates were 100%. |
| 2 | Increase Praxis Core for Educators Math score. | A Math Praxis Core workshop held in February by Math department (Shelly Fenton) to practice/study. | There were 2-3 students who participated and indicated that it was very helpful- scheduling is a challenge. | Due to lack of attendance would change the ground workshop to providing online resources and study guides for students. In addition, Learning Commons resources. Develop a handout with the Praxis steps to study and prepare resources. | Math scores for spring and fall 2023 indicate a mean score of 162- above the passing score requirement of 150. Writing scores for spring and fall 2023 indicate a mean score of 168.5- above the passing score requirement of 156. Reading scores for spring and fall 2023 indicate a mean score of 175- above the passing score of 162. Continue PL2 |
| 3 | Increase Math 1410 and 1420 grades for education students | Worked with Math & Science chairs and arranged the AST schedule and methods of offerings. In addition, collaborated with Learning Commons director to assign students to take the EdReady to prepare them for Math. | Students need to take face-to-face Math classes and seek the learning commons resources to improve grades. | Continue to work with the math chair and learning commons. | Math 1010: 60% pass<br>Math 1410: 72% pass<br>Continue the PL3 |
| 4 | Increase Grow Your Own partnerships, and improve the Springfield satellite campus current program | Held an interest meeting with Robertson County Schools and Putnam County Schools. Developed a MOU with both school districts. Completed an Appendix A form with Tennessee Grow Your Own center to | Advising the cohort is time consuming. I have scheduled ZOOM meetings on a regular semester schedule to schedule at one time. The admittance process was difficult- created a handout of | Schedule and meet with students in a group on ZOOM more often, and additional help advising. Collaborate with department chairs to develop the | 19 students are currently enrolled in the Robertson County AST cohort, with 5 more additions. 10 students are currently admitted into the Putnam County cohort to start in the Fall. An interest meeting will be |

| | | | | | |
|---|---|---|---|---|---|
| | | become a registered apprenticeship college. | steps, MOU and contacted the business office and bookstore. | schedule for each semester. | scheduled for Overton County in April or May. Continue to expand and develop the GYO cohort partnerships in PLO4. |

**Evidence:**

- 2021 Job Placement Survey
- 2021_Graduate_Job_Placement_Survey
- Praxis Math Workshop

## YEAR 3 ACTION PLAN REVIEW

The reviewer should review progress on action plan over the past year. Discussion should include challenges and successes. The reviewer should discuss the actions future and continuous in the program.

**Narrative:**

| YEAR 3 Progress | | | | | |
|---|---|---|---|---|---|
| | Initiative | Achieved Milestones | Challenges | Performance Indicator Result | Initiative Achieved? Yes/No |
| 1 | Increase the job placement rates for education majors | Communicated with current students the teacher shortage and job placement is in high demands | There was a RASI change the data was not provided. | Faculty results are monitored ad shared with current students in March | No There was no job placement data. A principal survey was created with RASI to be sent to school districts Continue PL! |
| 2 | Increase Praxis Core for Educators Math score. | Department Chair shared Praxis Core ZOOM links and information for workshop held at Chattanooga State by STEA to practice/study. ETS Praxis Study Materials provided to advising and faculty to share with AST students | There were a few students who participated and indicated that it was very helpful- scheduling is a challenge. Grow YOUR OWN students indicated a need to more study materials and in person tutoring. | VSCC Learning Commons offered a new resource "Peterson's Test and Career Prep Praxis" In addition, shared ZOOM and ground one-on-one tutoring schedule. Developed an additional handout and | Math scores for spring and fall 2023 indicate a mean score of 162- above the passing score requirement of 150.<br>Writing scores for spring and fall 2023 indicate a mean score of 168.5- above the passing score requirement of 168.5<br>Reading scores for spring and fall |

| | | | | | |
|---|---|---|---|---|---|
| | | | | shared with advising and education faculty. | 2023 indicate a mean score of 175- above the passing score of 162. Continue PL2 |
| 3 | Increase Math 1410 and 1420 grades/success for education students | Worked with Math & Science chairs and arranged the AST schedule and methods of offerings. In addition, collaborated with Learning Commons director to assign students to take the EdReady to prepare them for Math. | Students need to take face-to-face Math classes and seek the learning commons resources to improve grades | Continue to work with the math chair and learning commons. | Collaborated with the Math Instructor to methods and instruction to help education students be successful. Align with Tennessee Academic Learning Standards. Continue PL3 |
| 4 | Increase Grow Your Own partnerships, and improve all satellite campus current program | Held an interest meeting with Robertson County Schools and Putnam County Schools. Developed a MOU with both school districts. Completed an Appendix A form with Tennessee Grow Your Own center to become a registered apprenticeship college. | Advising the cohort is time consuming. I have scheduled ZOOM meetings on a regular semester schedule to schedule at one time. The admittance process was difficult- created a handout of steps, MOU and contacted the business office and bookstore. | Schedule and meet with students in a group on ZOOM more often, and additional help advising. Collaborate with department chairs to develop the schedule for each semester. Collaborate with Workforce Development, Marketing and Learning Commons | 19 students are graduating in the Robertson County first AST cohort, with 8 more additions. 8 students are currently admitted into the first Putnam County cohort. GYO Interest meeting was held in January 2025 to add 10 more students in cohort 2. Overton County has 5 in cohort 1. April or May. Continue to expand and develop the GYO cohort partnerships in PLO4. |
| 4 | | | | | |

# Academic Program Review Cohort B- Cycle 1

Early Childhood Education CERT

## Reflection and Mission

### PREVIOUS ACADEMIC PROGRAM REVIEW

Reflect on the past APR (if any) and respond to any recommendations that were made at that time. The reviewer should list recommendations, actions taken, and discuss implications or challenges resulting from those actions.

**Narrative:**

Our Associate of Science of Teaching Institutional Effectiveness outcomes have changed, and no Academic Program Review has been conducted. Tennessee Board of Regents Associate of Science Teaching Academic audit completed in 2017.

### Program Mission Statement

What changes have the program made to the mission statement over the course of this cycle? Why were these changes made? Are any revisions planned?

**Narrative:**

This career program is designed to prepare students for entry and advancement in the early childhood education profession. Students can work for an associate of applied science degree or a technical certificate in early childhood education. Graduates of this degree program may also be able to transfer to a four year university and complete the baccalaureate degree in early childhood education.

### Alignment to Institution Mission

How does the mission of the program align with the mission of the institution?

**Narrative:**

The Early Childhood Education technical certificate is an important part of Volunteer State Community College's mission to provide quality innovative education programs, to strengthen community and workforce partnerships, to promote diversity, and to address cultural and economic development. Also, another part of the mission is to inspire lifelong learning and to prepare students for successful careers, university transfers, and meaningful civic participation in a global society. The Associate of Science Teaching program links to the university mission addressing community engagement, promoting diversity, economic development to prepare students for successful careers in teaching.

## Curriculum and Student Learning Outcomes Assessment

### Curriculum Map

Review the Student Learning Outcomes and course(s) where each one is addressed. For each outcome, provide the assessment method and target level of achievement.

**Narrative:**

This curriculum map was implemented in Fall of 2021 Early Childhood Technical Certificate

| Education Outcomes | Learning Opportunities | Achievement Target | Three years of data trend |
|---|---|---|---|
| The students will describe the historical, philosophical, and social aspects of early childhood education. | • Students in ECED 1310 will examine various components of early childhood education (history, philosophies, etc.)<br>• Instructor will facilitate lectures and activities directly related to these topics.<br>• Students will complete discussions and journal activities.<br>• Report on websites related to multiple influences on development and learning. (NAEYC, Zero to Three).<br>• Develop a narrative of the historical, philosophical, and social aspects of the early childhood site observ | 85% of students meet expectations | |
| The students will define the basic components of setting up an early childhood education environment. | • Students in ECED 1310 will complete an early childhood learning environment.<br>• Instructor will lecture and provide activities that support an understanding of setting up an early childhood environment.<br>• Lecture and modeling of creating an early childhood activity plan. | 90% of students meet expectations | |

| | | | |
|---|---|---|---|
| | <ul><li>Define basic components of the site set-up, play activities, schedule, and early childhood stages.</li></ul> | | |
| The students will examine and compare different types of early childhood programs and align with various theoretical curriculum models. | <ul><li>Students in ECED 2315 will complete an observation and write a narrative of an early childhood program of their choice.</li><li>Instructor will lecture and provide learning activities on the various early childhood curriculums.</li><li>Students will complete discussions and journal activities.</li><li>Examine and compare the different types of early childhood programs (Head Start, Montessori, Public, Private).</li></ul> | 95% of students meet expectations | |
| | | | |

## EVIDENCE OF STUDENT LEARNING

Provide evidence of student learning and the program's effectiveness in reaching both program outcomes and course learning outcomes.

Please provide the following:
1. Summary of most recent assessment results.
2. Analysis of trends
3. Ideas for how those results could be used to improve student learning.

**Narrative:**

Due to this being the first Academic Program Review plan of study a summary of most recent assessment results is not available.
At this time students enrolled in ECED 1310 will complete an assessment (Site Observation Assignment) that is linked to the outcomes for the course. Students will be introduced to the outcomes in this course. The knowledge learned in all other courses will build upon this information.

Once the results of the assessment are made available, the information can be used to determine if the information should be shared in various ways. All data will have input to how student learning can be improved.

| 2019-2020 IE Plan Results and Future Planning | | | | | | |
|---|---|---|---|---|---|---|
| Outcome # | Outcome Statement | Outcome Final Status | List of Evidence (Provide in Email Attachments) | Results of the Institutional Effectiveness Effort What does the evidence show regarding the Outcome? To what extent was the Outcome met? As a result of the success of the Outcome, what Continuous Improvement need will have been met? Provide all supporting evidence. | Future Planning How will this outcome inform future planning? Will the results of this Outcome lead to completing the same or a similar Strategic Plan item at a higher level? | Funds Budgeted to Support This Outcome Document/attach any evidence showing if additional funds are needed for this initiative and/or if funding from existing budgets is to be dedicated to this Outcome. |
| 1 | Phase out the Early Childhood Education AAS degree program. | | 1. Letter of Teach Out Plan approved by college. 2. Letter of | 1. Received approval from college curriculum committee. 2. Received approval from IERPA to implement Teach Out Plan. 3. Received approval from TBR to phase out program. | This is the last year this outcome will be used. | No funding needed. |
| 2 | Increase the number of Early Childhood Education Technical Certificate graduates. | Complete - Met | Evidence from IERPA | The evidence indicates that there was an enrollment increase in the ECED Technical Certificate program. The enrollment goal was 15 students, and there were 17 total students for fall 2019 and spring 2020. Contiuous improvement will be made to visit | This is the last year this outcome will be used. | No funding needed. |

| Data Chart: 2018- summer 2021 | | | |
|---|---|---|---|
| **Education Outcomes** | **Learning Opportunities** | **Achievement Target** | **Three years of trend data** |
| Increase graduation of students with Technical Certificate. | N/A | Increase program by 15 students | **2018-2019**<br>3 graduates<br>**2019-2020**<br>Graduate information not available in Factbook Dashboard. |
| Knowing about and upholding ethical standards and other early childhood professional guidelines. | The rubric score results will indicate instructional changes such as:<br>Instruction provided (lectures, videos) on the NAEYC Code of Ethics. A copy given to the students.<br>Instruction provided (textbook readings, lectures, panel discussions, video demonstrations) on how to apply these ethical standards and other early childhood professional guidelines to young children and families. | Students will earn a score 90% on the science study and the ethics case study assignments | **2017-2018**<br>62.7% of students scored 90%<br>**2018-2019**<br>44% of students earned a 90%<br>**2019-2020**<br>This outcome was not continued. |
| Students will engage in continuous collaborative learning to inform practice; using technology effectively with young children. | The rubric score results will indicate instructional changes such as:<br>Lesson plan writing and ways to create a science study.<br>Examples provided to incorporate technology in the classroom. | Students will score a 90% or above on the: Science Study rubric and the Ethics Case Study rubric. | **2017-2018**<br>62.7% of students scored 90%<br>**2018-2019**<br>44% of students earned a 90%<br>**2019-2020**<br>This outcome was not continued. |

# Qualitative and Quantitative Program Information

## STUDENT, GRADUATE, AND ALUMNI SATISFACTION

Use student and graduate surveys, course evaluations, and any other appropriate evidence needed to explore the qualitative satisfaction results from students and graduates to provide information to the items below.

Discuss student, graduate, alumni perceptions of the program, faculty, career projection, and any other program recommendations given "during this survey".

Summarize any noted themes, trends, areas for improvement, identified strengths, and implications for the program for both students and alumni.

**Narrative:**

Schools within our community partnership participated in our 2020 Employer Survey of education graduates (managed evidence). All of the schools indicated that they would hire another education graduate from the education program. The employer survey indicates a overall 66.7% satisfaction with education graduates, with a mean of 4.33. The employers indicate that students/teachers strengths include that they  work very hard to find fun and educational activities in the classroom. Additional comments that would help to better prepare future graduates include; more skills and adaptation to working with children with individualized needs. This comment clarifies a need for implementing more strategies to help teachers be better equipped to work with students with disabilities.
Spring student course evaluations indicate early childhood faculty scored a 4.5 out of 5.0 possible score on all courses. Students comments include; "engagement throughout the course, and "let us know- if we had questions- she was available". Each of these align with the positive reflective data scores.

The academic year 2019-2020 graduate satisfaction survey results indicated that an 86% job placement rate with seven of the eight graduates being placed in a position in early childhood education. This is a 61% increase from the previous year. However, the employer satisfaction rate is at 50%.

The increase in job placement is an interesting trend that needs to be watched. If the job placement increases over the next year for students earning a technical certificate it should be examined.

## RECRUITMENT, ENROLLMENT, RETENTION, AND GRADUATION: FACT BOOK DATA

Attach screenshots from the Fact Book of the areas below.

1. Headcounts
2. FTE Comparison
3. Student Primary Location
4. Retention Statistics
5. Graduation Statistics
6. Demographics

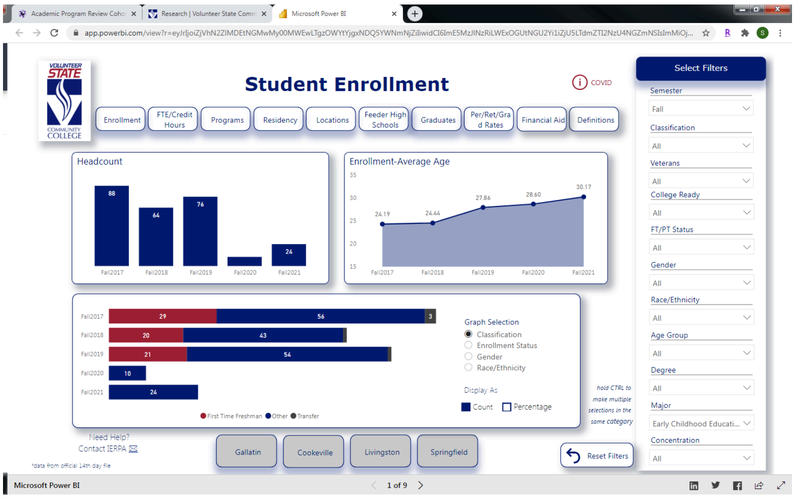
Using the Fact Book data listed below, analyze the follow:

Recruitment
Enrollment
Retention
Graduation

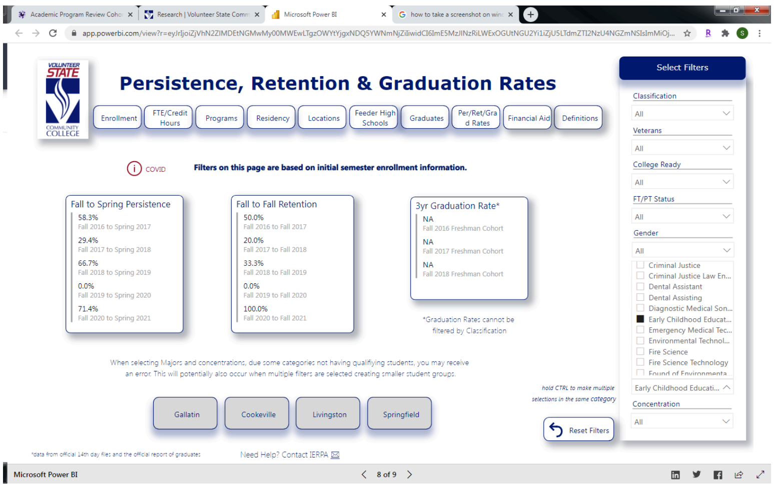
The reviewer could discuss marketing and recruitment efforts in the community and local high schools; analyze enrollment, retention, and graduation trends; and discuss potential reasons, strengths, plans for improvement, and any other implications deemed appropriate.

**Narrative:**

The Early Childhood Technical Certificate (ECTC) has seen a slight increase in enrollment of students on the Gallatin campus despite the pandemic changing the modality of course offerings. The enrollment in the ECTC has increased from 7 students in fall of 2019 to 16 students in fall of 2021. This increase is related to the states new early childhood program licensure requirements for early childhood educators.




The retention rates for the ECTC is listed as 0%for both fall 2019 to spring 2020 and fall 2019 to spring 2020. This is due to a lack of reported data for this year. However, the fall 2019 to spring 2020 persistence rate for the technical certificate is reported at 71.4% and a 100% retention rate for fall to fall. While persistence and retention rates are a concern for the college the rates reported are not a concern for the technical certificate and show a steady increase over the past five years. The Tennessee Early Childhood Training Alliance (TECTA) funds ECED courses for eligible ECED students. Vol State faculty partners with this agency to help students be successful and complete degrees. This funding opportunity removes a financial burden and faculty assist with advising, retention and completion.




Graduation rates for the Early Childhood Technical Certificate illustrate a slight increase of 56% from the academic year 2019-2020 with 25 graduates to 39 graduates for the 2020-2021 academic year. This increase in certificate recipients suggests more early childhood programs are aligning to the state program licensing rules and having staff pursue academic coursework.
Recruitment includes

The early childhood program shares degree information with high schools to recruit more incoming freshman to declare the ECED technical certificate as their intended major. The program recruitment strengths include: recruitment materials shared periodically with high school guidance counselors, presenting to high school classrooms, and participating in high school job fairs. In addition, attending the local Sumner County directors meeting, sending a bi-annual Vol State newsletter to local childcare agencies and directors, and holding annual Vol State ECED advisory meetings with community partners.

**Evidence:**

- Advisory Board Members
- ECED Advisory Board Agenda 2020

# Supporting Program Information

## FACULTY USAGE AND DEVELOPMENT (OPTIONAL)

Review faculty teaching loads, adjunct to full time ratios, faculty development opportunities, faculty end of course evaluations, faculty surveys, and any other evidence deemed appropriate. The reviewer could discuss the impact of teaching, the ratio of adjunct to full time staffing, end of course evaluations relative to faculty, and any noted themes identified from the review.

**Narrative:**

The early childhood education teaching faculty (1 full time and 2 adjunct instructors at the Gallatin location and 2 adjuncts at the Cookeville satellite location) continually seek ways to set students up for success and high-engagement teaching faculty has been identified as one. A model course map was created to secure consistency between program outcomes, course objectives, delivery content and textbooks.

All full-time faculty participate in a peer evaluation completed by two assigned peers, complete classroom observations and required documents to division secretary. In addition, all adjuncts are evaluated by full-time faculty, including a classroom observation and required documents signed and complete and sent to the division secretary. All evaluations are considered constructive feedback to adjust and improve teaching modalities, schedule, classroom climate and to benefit the teacher and student.

Course rubrics were redeveloped to align with new course outcomes to track key assessment data that lead to continuous improvements in the program. The course map, rubrics and data is shared regularly between early childhood education faculty and adjuncts.

Faculty teaching loads include; one full-time early childhood education faculty ( working on a doctoral degree) and two early childhood education (one with a doctorate degree) adjuncts. The two adjuncts are located at the Gallatin campus.

**Evidence:**

- ECED 1310 Course Manual
- ECED 2310 Course Manual
- ECED 2315 Course Manual
- ECED 2320 Course Manual
- ECED 2335 Course Manual
- ECED 2340 Course Manual
- ECED 2365 Course Manual

- ECED 2370 Course Manual (002)
- ECED 2380 Course Manual
- ECED 2385 Course Manual
- Technical Certificate Data Chart 2018-2021[87]

## CLASS OFFERINGS (OPTIONAL)

Address scheduling effectiveness and space utilization. The reviewer could review faculty to student ratios at each location and online and discuss the impact relative to faculty and student success.

**Narrative:**

The early childhood education technical certificate offers a variety of course scheduling and modality to meet the needs of students at every campus. In the fall of 2019, it was determined that to best meet student needs the program will convert to a fully online program. This allows students to take the 24 credit credential in an online modality. The course conversion was completed in the summer of 2020.

In the past, the courses were offered in fast track hybrid format. The courses would meet once a week for seven weeks. The students were able to take four courses in the fall and four courses in the spring to complete the technical certificate. However, with the pandemic causing courses to move to an online format. This allowed for the early childhood technical certificate to progress to the online modality. In the fall of 2019 there were 13 students taking courses fully online. That dropped in the fall of 2021 by 2 students. The online content of each course is on a current redevelopment cycle for approval every 3 years, and updated every semester. Each course has a required minimum of 2 or more clinical field hours, students completed assigned projects. The ECED faculty have childcare community partners who allow students to complete this requirement. In addition, the Initial Practicum ECED 2335 has 45 clock hours of required clinical field work. Students may choose to complete half of these hours in the childcare they are employed with.

In Cookeville, the course offerings are determined by enrollment and student needs in this area.  One of the locations of the TECTA office is at the Cookeville satellite campus. This agency is convenient for students to complete the eligibility requirements and seek advising from Vol State faculty.

While discussing the spring 2022 course offerings, it was determined to try a different hybrid format for the early childhood education courses. The courses are required for the technical certificate and for the early childhood AST (TTP). In order to provide face to face courses a new hybrid format is to be offered. The two courses will meet once a week for the full semester. One course being held on Monday and the other on Wednesday in the same time slot. This will allow for students to have a full semester rather than the compresses version of the past "fast track" modality. Students will be surveyed to determine if this is a popular offering.

## CO-CURRICULAR ACTIVITIES (OPTIONAL)

Discuss co-curricular activities available for students in the program. These might include clubs, internships or other Work Based Learning opportunities, service learning, guest speakers, or any other activities designed to enhance student experience in the program.

**Narrative:**

Faculty work collaboratively to provide quality, expansive workforce experience and incorporating best practices in instruction. Teaching and learning opportunities are rich. Service Learning, the use of problem solving strategies as a teaching method, the use of technology, and the availability to learn through international programs are components of our program.

The program coursework is driven by a workforce development mentality. Students have an average of eight hours of hands-on, fieldwork assignments for all courses in the technical certificate. The students also have two practicum courses with total of 135 hours of fieldwork.

Collaborating across departments and divisions to support student success has been addressed through an on-going evolving service-learning project to promote early literacy called "Read Across America" changed to " The project this year included the library, english, education, foreign language and communication faculty,  Over 500 new books were donated and delivered to 5 childcares in Sumner County. An event was hosted by the library and various social-distancing activities were arranged throughout the library, and the public was invited. In the past, Vol State students have held a used book drive on campus, and distributed the books as they visit early childhood centers and read to classes of students. It was certainly a modified version of this project, but was a great success! This project was selected to present at "2021 Tennessee Conference on Volunteerism and Service-Learning." Education faculty. Guest speakers include; Open Arms Care facility director, Sumner County Schools behavioral analyst, and Metro Schools special education teacher. Students are required to write questions for the speakers and reflect and apply the experience to their teaching career.

A "Future Educators" student group was started in 2018. A few of the projects include; collected books for the "Read Across America" project, read books to students and donated items to Vol State "The Feed". In addition, decorating of the teacher education boards in the Noble Hall Caudill building.

## INDUSTRY TRENDS AND INNOVATIONS (OPTIONAL)

Examine the program demand within the labor market including local, regional, and/or national. The reviewer could examine information from industry advisory boards and information gathered from creditable industry sources as well as from external accrediting bodies. The reviewer could also discuss any existing or emerging trends within the field as they relate to your program and employment of your student's post-graduation as well as the outlook for the field in the next 5 years. This summary could also identify best practices, trends, and innovation in the field and discuss implications for the program.

**Narrative:**

The field of early childhood is a dynamic field with trends and innovations evolving with the times.

In the state of Tennessee, early childhood educators are a part of the ever changing the landscape of child care. One program that stands out as an innovative program is the Child Care Wage$ Tennessee. This program rewards Early Childhood Educators with financial incentives based on their education and continuity of employment. By increasing teacher retention, this program provides the children of Tennessee with stable relationships with caregivers. Also, research indicates the teachers with more academic training and course work provide a higher level of care.

The National Association for the Education of Young Children (NAEYC) is at the forefront of early childhood trends and innovation. This is evident in their initiative "Power to the Profession". This national collaboration defines the early childhood education profession. It has established a Unifying Framework of recommendations of educator roles and responsibilities, aligned preparation and pathways, profession compensation, and a supportive infrastructure with shared accountability. The goal of the initiative is to create equitable learning opportunities for young children.

Another trend in Early Childhood Education is a focus on expanding the Voluntary Pre-K program in the state. Voluntary Pre-K promotes a high-quality learning environment for children. The program fosters a the joy of learning and promotes school readiness through developmentally appropriate practices.

### EMPLOYMENT AND TRANSFER FORECASTS (OPTIONAL)

Discuss where students go after graduation and what is being done to enhance the likelihood of success or assist students in reaching those goals. The reviewer could discuss employment outlook, obtainment, curriculum alignment for transfers, and the employment demands of the industries where students will work.

## Recommendations Action Plan, Review, and Feedback

### SUMMARY AND ANALYSIS

The reviewer should discuss what went right, what went wrong, and the implications of those results.

**Narrative:**

In the last 3 years, here are some trends we found; effective faculty collaboration, positive and high scores from students on our course evaluations and improved enrollment. Student evaluations indicated that faculty build strong student-teacher relationships, quick to communicate feedback, and course content was challenging.

The positive implications of this data indicate a program strength in faculty usage and development. The enrollment has increased in the early childhood technical certificate has had a steady increase over the last three years. The largest increase was 6 students (fall 2020 enrollment was 10 students and fall of 2021 enrollment was 16 students). The TN Wages program is having a positive impact on enrollment and retention.

The academic year 2019-2020 graduate satisfaction survey results indicated that an 86% job placement rate with seven of the eight graduates being placed in a position in early childhood education. This is a 61% increase from the previous year. However, the employer satisfaction rate is at 50%. The high placement rate along with the low employer satisfaction rate are trends to watch.

The increase in job placement is an interesting trend that needs to be watched. If the job placement increases over the next year for students earning a technical certificate it should be examined.

Retention rates for the early childhood technical certificate have increased from 20% the fall of 2018 to 100% in the fall of 2021. The data illustrates an 80% increase. Strategies to improve student persistence and retention will be addressed in the 2 year plan. Persistence rates (fall to spring) have shown an increase of 41% over the last three years.

The program advisory committee meetings will continue going into the next program year. This allows for participates to learn more about the technical certificate program at Volunteer State Community College. It also allows for the department chair and program faculty to hear about the needs of early childhood education learning programs.

A new assessment plan to increase student learning outcome achievement has been developed. The embedded tests placed in the early childhood education courses were removed and changed to alternative assessments in each course to improve success student and make continuous improvements in the program.

### TWO YEAR RECOMMENDATION ACTION PLAN

Discuss the next steps for the program by providing a program action plan for the next two academic years. Recommendations must include at least two actions focused on student learning and one action focused on student success. The program action plan must include a timeline for each recommendation.

Feedback and approval will be provided. Refer to the APR timeline for approval deadlines.

**Narrative:**

**Narrative:**
The two-year plan of action will include:

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### DEAN(S) APPROVAL AND FEEDBACK.

The Dean(s) will provide feedback and approval on the completed review and action plan.

**Narrative:**

I have reviewed the information provided for the Early Childhood Education CERT APR and approve it. Again, I found this to be a thorough report about an important program. Kudos to the Education faculty for doing such fine work.

### AVPAA & VPAA Review and Feedback

The Vice President of Academic Affairs and Assistant Vice President of Academic Affairs will provide feedback and approval on the completed review and action plan. VPAA and AVPAA should consider institutional planning and budgetary details when providing their review.

**Narrative:**

The Office of Institutional Effectiveness and Assessment has reviewed this report and will begin supporting the report champion in preparing for the presentation of this review to the Institutional Effectiveness Committee.

### PRESENTATION FEEDBACK

The APR will be presented to a group of peers. Feedback retrieved from the presentation will be summarized and included.

## Action Plan Status Reports

### YEAR 2 ACTION PLAN REVIEW

The reviewer should review progress on action plan over the past year. Discussion should include challenges and successes. If any changes are planned due to the results, those should be discussed and an updated action plan should be included in the narrative or attached as evidence.

**Narrative:**

| YEAR 2 | | | | | |
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| | Initiative | Achieved Milestones | Challenges | Changes if applicable | Progress Toward Performance Indicator(s) |
| EX. | Increase the achievement of PLO #1 by developing and implementing an online tutorial resource to complement teaching the XYZ assignment in VSCC1234 | The tutorial video created. Tutorial video presented at the November department meeting. Tutorial video piloted in 10 sections. | Delays in development occurred and additional support from campus media services improved the video quality. | No changes to the timeline. | The assignment success rate for pilot sections was 78% PLO #1 achievement of pilot sections was 83% |
| | Increase the number of students enrolled in the technical certificate. | Increase the program by 15 students. | Pandemic reduced the number of students enrolling in the program. | | Fall 2020- 10 students<br>Fall 2021- 10 students<br>Fall 2022- 12 students<br>*Some students enroll as "Adult Special" but are actively pursuing the Technical Certificate. The number of students enrolled is higher. |
| 2 | Strengthen student understanding and application of ethical standards and professional guidelines in early childhood education. PLO 2 | Students will earn a score of 90% on the science study and ethics case study. | | The ethics case study was in ECED 2365 which is no longer required. | 90% of students met expectation or exceeded expectation on the science study 2021-2022, 2022-2023. |
| 3 | Apply technology-based tools to inform instructional practice with young children.PLO 2 | Students will score a 90% or above on the: Science study and ethics case study | | The ethics case study was in ECED 2365 which is no longer required. | 90% of students earned met or exceeded expectations. Students who earned below did not complete the assignment. |

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## YEAR 3 ACTION PLAN REVIEW

The reviewer should review progress on action plan over the past year. Discussion should include challenges and successes. The reviewer should discuss the actions future and continuous in the program.

**Narrative:**

| YEAR 3 Progress | | | | | |
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| | Initiative | Achieved Milestones | Challenges | Performance Indicator Result | Initiative Achieved? Yes/No |
| EX. | Increase the achievement of PLO #1 by developing and implementing an online tutorial resource to complement teaching the XYZ assignment in VSCC1234 | The tutorial video was launched in the model shell. Full-time faculty were also encouraged to use the new tool The tutorial video was launched in a total of 24 sections | None | The assignment success rate for pilot sections was 80% PLO #1 achievement of pilot sections was 86% | YES |
| 1 | Increase the number of students enrolled in the technical certificate. | Increase the number by 15 students. | The TECTA students from TTU are not enrolling as technical certificate students at the start of the program. These students enroll as adult special lowering the number of students enrolled. | | Yes. The 2023-24 academic year there were 55 students enrolled. The fall of 2024 there were 28 enrolled and another 34 enrolled as "adult special". |
| 2 | Strengthen student | Students will earn a score of | Not all students completed the | In Spring 2024, 9 of the 15 students | Yes, but not all students |

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| | understanding and application of ethical standards and professional guidelines in early childhood education. PLO 2 | 90% on the science study. | assignment. | completed the assignment with all of them earning a 90% or more. In Fall of 2024 the program added zoom seminars to help students with the transition to asynchronous online courses. Spring of 2025 the assignment is due on May 2, 2025. | completed the assignment. |
| 3 | Apply technology-based tools to inform instructional practice with young children. PLO 2 | Students will earn a 90% or above on the science study. | Not all students submitted the assignment. | In the spring of 2024, 9 of the 15 students completed the assignment with a 90% or above. In Fall of 2024 the program added zoom seminars to help students with the transition to asynchronous online courses. science study is due on May 2, 2025. | Yes, but not all students completed the assignment. |
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| YEAR 3 Initiative Analysis | | | | |
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| | Initiative | How has this initiative supported student learning or student success? | Will there be any further action taken on this initiative? If yes, explain. | How will this initiative inform program decisions in the future? |
| EX. | Increase the achievement of PLO #1 by developing and | | None | The assignment success rate for pilot sections was 80% PLO #1 achievement of pilot sections |

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| | implementing an online tutorial resource to complement teaching the XYZ assignment in VSCC1234 | | | was 86% |
| 1 | Increase the number of students enrolled in the technical certificate. | An increase in students this year will increase the number of CRT2 graduates. | Yes, we want to continue to increase the number of students enrolled. | The increased number of students enrolled could allow for the ECE AAS to be reinstated. |
| 2 | Knowing about and upholding ethical standards and other early childhood professional guidelines. | This initiative will increase the employment satisfaction survey results. Students have more insight on how to handle ethical dilemmas in the classroom or program. | None. | • We added zoom seminars to all online ECED courses. This has helped students connect with the instructor and peers. The instructors use this meeting to guide students through project instructions. It also provided students a place to ask questions about projects.<br>• Going forward, the program is adding key assessments to all courses in the technical certificate. This will allow for more detailed assessment of PLOs and CLOs for the next few program years. |
| 3 | Learning to inform practice; using technology effectively with young children. | Informed practices and using technology will prepare students for success in their classrooms. | None. | • We added zoom seminars to all online ECED courses. This has helped students connect with the instructor and peers. The instructors use this meeting to guide students through project instructions. It also provided students a place to ask questions about projects. |

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| | | | | • Going forward, the program is adding key assessments to all courses in the technical certificate. This will allow for more detailed assessment of PLOs and CLOs for the next few program years. |
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